

SMITHIAN FOLD THEORY V3

The Quantum Fold Machine

An Exact, Parameter-Free and Machine-Closed Derivation of Reversible and Quantum
Computation from Smithian Fold Theory

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An Exact, Parameter-Free and Machine-Closed Derivation of Reversible and Quantum Computation from Smithian Fold Theory

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in the same concept DOI lineage. The correction changes no scientific claim,

evidence classification, mathematical value, source identity or machine record.

***Lean verification integration.** The current SFT ordered census passed the*

independent Lean 4 verification layer on 2 August 2026: 2,777 accepted claims,

898,902 candidate records and decisions, 11,108 controls, seventeen branches

and no reported issue. Lean natively proves the two-class operational root and

its unique survivor and constructs proof-bearing acceptance-gate certificates.

The remaining claim content is checked as the complete registered artifact

graph; that distinction is retained throughout this successor.

Abstract

This successor presents the complete current `quantum_computation` paper surface: 290 live model-admitted claims, 74,240 generated candidates, 290 unique survivors and 1,160 passed controls. It integrates 2 claim section(s) not present in the preceding abstract's dated surface and remains complete only to this versioned registered census, with lawful extension open. The independent Lean 4 layer returned PASS for this surface as part of all 2,777 current claims.

The reconstruction derives reversible computation; quantum information units; state preparation and composition; complete-word superposition-equivalent support; finite held phase and exact interference; nonfactorable entangling composition; measurement with retained records; transformations, gates, circuits, compilation and universality; thirty quantum-algorithm laws; twenty-six resource and complexity laws; twenty-four channel and communication laws; thirty-two coding, error-correction and fault-tolerance laws; twenty-four simulation and verification laws; twenty-two learning laws; and twenty-two classical-quantum correspondence and limit laws. The quantum and classical modes are two exact regimes of one Fold machine rather than unrelated formalisms.

No conventional quantum postulate, Hilbert-space axiom, complex amplitude, irrational normalization, imaginary proof value, stochastic-collapse premise, ontic randomness, fitted parameter, physical benchmark, pretrained model or authority label selects a law. Displayed 0 denotes structural absence rather than a numerical proof magnitude. Error correction forces width $2t+1$ for every supplied positive finite fault order t ; quantum speedup, device thresholds, timing, energy and natural measurement distributions remain explicitly assigned to their empirical physical or engineering owners.

Version 1.4 preserves the full version 1.3 foundation, public scientific mission and 22 extensive foundational derivations, then documents all 266 executed full-field extensions claim by claim. No protected engine or verification-authority source was modified. A preserved first-attempt halt and separately versioned retry demonstrate that the engine remains allowed to reject the work.

Keywords: Smithian Fold Theory; complete-field quantum computation; reversible computation; quantum information; superposition; interference; entanglement; measurement; quantum circuits; quantum algorithms; quantum complexity; quantum communication; quantum coding; fault tolerance; quantum simulation; quantum learning; open science.

Formal admission, implementation validation, observation, measurement, empirical support, confirmation, adverse evidence, unresolved evidence and publication status remain separate classifications. Lean does not reclassify an empirical result, repair an adverse observation or convert a later comparison into an earlier prediction. Lawful criticism, falsification and versioned extension remain open.

Successor headline findings

1. **Independent whole-model result.** Lean 4 returned PASS for 2,777/2,777 current claims, 898,902 candidates and decisions, 11,108 controls and all 17 branches, with no issue.
2. **Operational root.** The exact registered two-class grammar is formalised natively, and presentedOccurrence is proved to be its unique survivor without an imported or user-declared axiom in the exported theorem's axiom audit.
3. **Typed validation.** The root and generic acceptance implications are native Lean theorems; the remaining scientific claims are checked as complete registered artifacts. Empirical truth is not inferred from proof-assistant acceptance alone.
4. **Fail-closed custody.** A six-file byte-identity mismatch halted the first run. Exact registered bytes were restored, all 385 source bindings were re-audited, and the unchanged verifier then passed.
5. **Current paper surface.** This successor accounts for all 290 current claim(s) in its declared branch ownership boundary; 2 later claim section(s) are integrated in the successor supplement.
6. **Newly integrated results.** SFT-QUANTUM-OAI26-PARALLEL-REPETITION-001 - SFT reconstruction of distribution-uniform quantum parallel repetition;
SFT-QUANTUM-OAI26-PARALLEL-REPETITION-VALIDITY-001 - SFT disproof of source-artifact validity:
QuantumParallelRepetition.distributionUniformExponential.

These findings supplement rather than erase the branch-specific headline results retained below. Any adverse, corrected, unresolved, unavailable or chronology-bound result in the preceding scientific record remains governed by its current claim package.

1. Central scientific claim and exact boundary

Within the frozen SFT V3 Reversible and Quantum Computation census dated 29 July 2026, all 288 obligations in all 13 registered families are individually admitted through the untouched engine, independently reconstructed, adversely controlled, observed under their declared custody boundary and exactly reconciled. The current closed count is 288, the current open count is structural absence, and completion remains open to lawful versioned extension.

The frozen census identity is

sha256:a7f855b7e9fc6ffe4ae9485faf26f1be44bd672b843ecbfbad42b0f523726e3c. The final reconciliation identity is

sha256:ac4acee07a49cbde19eee7baacd7be7d8c87c93e4d75a47467160fb31499e650. The branch contains 73,728 completely decided candidate structures, 288 unique survivors and 1,152 passing controls. Every full-field registration has an empty axiom list and an empty free-parameter list. Every dependency chain reaches SFT-ROOT-THERE-IS-NO-NOTHING through actual registered receipts rather than narrative assertion.

Closure means that every exact kernel named by this dated census has passed its declared grammar and evidence conditions. It does not convert a formal quantum computation law into an unperformed physical measurement. The theorem $2t+1$ is unbounded over supplied positive finite fault orders; it is not an imported stochastic hardware threshold. Physical amplitudes, device errors, timing, energy, experimental speedup and natural outcome distributions remain measurement handoffs. The branch is complete to known registered scope, not permanently closed to lawful discoveries, counterexamples or stronger versioned extensions.

2. Headline findings and results at a glance

Result	Exact executed result	Scientific meaning and boundary
Complete-field quantum computation	288/288 obligations , 13/13 families, 73,728 candidates, 288 unique survivors, 1,152 passed controls and 288 independent reconstructions.	The published roadmap is now an executed corpus, complete to its dated census and open to lawful extension.
One exact classical-quantum machine	Classical reversible execution, complete quantum support, phase-sensitive merging, joint composition and observation share one generated Fold process.	Quantum computation is derived as an exact operational extension, not installed as a second axiomatic universe.
Quantum state and measurement	One distinction supplies two held labels; b^k words supply complete support; phase is period- b held action; entanglement is nonfactorable joint support; measurement retains its closed distinctions and reconstruction record.	Superposition, interference, entanglement and measurement require no imported complex proof scalar or stochastic-collapse postulate.
Gates, circuits and universality	Reversible branch bijections, phase actions, controlled composition, causal circuits, inverse traces, compilation and a universal interpreter are separately forced and admitted.	Circuit semantics joins the already exact resource ledger while preserving branchwise provenance.
Algorithms and complexity	30 algorithm laws and 26 complexity laws cover query, phase, transform, search, counting, simulation, verification, upper/lower witnesses and correspondence limits.	Every claimed resource is tied to generated descriptions and exact traces; physical speedup remains an empirical handoff.
Communication and security	24 communication laws derive exact channels, memory, transfer, entanglement assistance, teleportation/dense-coding correspondences, no-cloning, disturbance, key and adversary boundaries.	Operational and information custody is explicit; no practical security claim outruns its registered adversary or physical owner.
Coding and fault tolerance	32 coding laws derive logical/physical carriers, syndrome distinction, recovery, degeneracy correspondence, fault propagation and the unique first correcting repetition width $2t+1$.	Widths 3, 5 and 7 exhaust all registered masks; the positive-finite successor law excludes an arbitrary maximum without claiming a measured hardware threshold.
Simulation, verification and learning	24 simulation/verification laws and 22 learning laws preserve exact target encoding, update, witness, observation, hypothesis, training and evaluation custody.	No black-box score, pretrained oracle or hidden benchmark can replace the premise-to-result trace.
Correspondence and limits	22 laws derive classical embedding, probabilistic-support correspondence, quantum decoding, bidirectional simulation, measurement disturbance, no-cloning, halting, undecidability, incompleteness and no-hypercomputation limits.	Apparent randomness is deterministic complete support plus observation-relative uncertainty; no uncaused transition is introduced.
Grand Lock and adverse evidence	Twelve validation laws replay the whole branch. One first-attempt <code>VALID-001</code> label-binding submission halted and remains preserved; the corrected separately versioned protocol binding later admitted without changing the law, target, engine, verifier or condition.	The validation Grand Lock is a dated reconciliation layer, not immunity from criticism and not permission to reward a halt.

Historical quantum names enter only after the Fold carriers, alternatives and operations have been forced. They are correspondence labels and tests; they do not select a survivor.

Current status, evidence language and reader map

Status field	Current position
Branch and completion boundary	Reversible and Quantum Computation: 288/288 frozen obligations; 288 live claims. This is dated, versioned completion and remains open to lawful extension.
Formal status	Every live claim represented here has a current model-admitted receipt. Formal admission does not by itself imply empirical confirmation.
Empirical status	Claim-specific. Blind, non-blind, development-observed, holdout, adverse, unresolved, unavailable and formal-only distinctions remain exactly as recorded in the claim ledger and evidence packages.
Chronology	Claim-specific registration, seal, custody and observation order governs. Later evidence does not retroactively become an earlier prediction.
Publication status	Version 1.5.1 is published open access at 10.5281/zenodo.21761654 in the existing Zenodo concept DOI lineage.
Existing record lineage	Current record DOI 10.5281/zenodo.21627748 ; concept DOI 10.5281/zenodo.21518312 . Publication is permitted only through the existing record's new-version route.
What is not claimed	Permanent closure of science, universal empirical confirmation, transfer of another branch's ownership or permission to replace an adverse or unresolved record.

Corpus-wide terminology and evidence key

Term	Reserved meaning in this paper
Theorem	A formally closed proposition at its declared grammar and dependency boundary.
Law	An admitted branch relation with its declared carrier, boundary, dependencies and extension rule.
Claim	The registered unit judged by the engine and bound to one immutable receipt.
Constitution	The rules governing admissible objects, derivations, evidence and publication; not an empirical result.
Derivation	Generation and elimination of the declared candidate space to its surviving structure.
Prediction	A consequence sealed before the matching target is released under the registered custody protocol.
Observation	A source-bound external record; it does not enter candidate generation.
Measurement	An instrument-, method-, condition- and uncertainty-bound observed value.
Reconstruction	A separately implemented regeneration or an explicitly identified inference of a retained state or history.
Exact numerical correspondence	Equality or registered interval relation between exact numerical objects at the declared boundary.
Structural correspondence	A post-derivation relation between forms without a claim of exact numerical prediction.
Boundary correspondence	Agreement restricted to the named interface, limit or ownership boundary.
Compatibility	Non-adverse but non-discriminating evidence.
Support	Relevant evidence that is not unique confirmation.
Confirmation	Used only when the current registered evidence protocol warrants that classification.
Validation	Successful execution of the specified formal, computational or empirical test; its kind must be named.
Adverse result	A registered test result that conflicts with the tested claim at its declared boundary.
Unresolved result	Required evidence remains unavailable, incomplete, disputed or insufficiently classified.
Implementation identity	The hash-bound identity of executable material; implementation success is not empirical confirmation.
Formal, empirical and publication status	Three independent classifications. None silently substitutes for another.
Foundational closure	Completion of the registered foundation only. It does not imply field-wide closure.
Field-wide closure	Completion of the frozen or registered dated field census named in this paper.
Current-evidence closure	Closure only to the evidence surface presently registered and preserved.
Extension openness	New questions may be added by a later version without rewriting existing receipts or history.

Proof language is reserved for formal closure; derivation for generated structure; implementation for executable demonstration; prediction for a correctly sealed prospective consequence; observation and measurement for source-bound external records; reconstruction for separately regenerated or inferred states; correspondence for post-derivation relationships; compatibility and support for non-unique evidence; confirmation only where the registered protocol authorises it; adverse for conflict; and unresolved where required evidence remains incomplete.

Three reading levels

1. **Conceptual paper.** The abstract, headline findings, branch narrative, major derivations, evidence synthesis, limitations and conclusion provide the readable scientific argument.
2. **Scientific audit layer.** Family and claim sections preserve exact status, dependencies, candidate and survivor counts, controls, chronology, sources, corrections, adverse evidence and reconciliation.
3. **Machine archive.** The cited repository packages preserve complete candidates, decisions, hashes, receipts, executable traces, source snapshots and certificates. Those files remain authoritative where prose abbreviates their display.

Editorial change control

This version may improve expression, order, typography and navigation. It may not change scientific meaning, scope, chronology, evidence class, ownership, claim status or machine identity. Any apparent conflict between authoritative records must be traced and either resolved by explicit supersession or recorded for Maria Smith; prose alone cannot manufacture agreement.

Public scientific mission and admission boundary

Ernos Labs is an open-source science movement, verification platform and public tree of knowledge founded by Maria Smith. Its purpose is not to replace one authority with another. It is to make scientific authority narrow, inspectable and revocable: a claim is admitted only through its complete derivation chain, generated alternatives, eliminations, unique survivor, adverse controls, measurement custody where applicable and unchanged-engine receipt. Open criticism is unrestricted and necessary; scientific admission is the separate machine-checked act of satisfying that public standard.

Maria Smith developed Smithian Fold Theory outside formal academic education, institutional research employment and conventional grant funding. That fact is not offered as evidence for a theorem; the derivations and observations carry the entire scientific burden. It is evidence about access. A credential-first system loses more than individual opportunity: it loses unknown questions, methods and discoveries from minds that capital and status never authorise. This work therefore treats Maria Smith's authorship neither as exceptionalism nor as a reason for dismissal, but as an indictment of every scientific contribution lost when financial gatekeeping is presented as rigour.

The institutional argument is empirical, not a claim that every institution or funded researcher acts in bad faith. Published studies document sponsor-linked differences in outcomes and conclusions, commercial influence over research agendas, limits and sensitivities in grant review, underpublication of null results, and inequalities created by paywalls and article-processing charges. Those findings establish that funding, prestige, publication and consensus are selection systems with incentives and failure modes. They cannot substitute for a public proof and evidence chain. Expertise, measurement and adversarial review remain indispensable; institutional permission does not select a fundamental law.

For Quantum Computation, inaccessible hardware summaries, proprietary simulation, fitted amplitude tables and authority labels cannot admit a quantum law. Every support branch, phase action, merge, joint cell, observation record, correction mask and resource must remain inspectable. Hardware measurements may test a separately sealed physical claim, but cannot flow backward to select the formal quantum-computation structure.

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Independent replications, lawful extensions, corrections and attempted invalidations are invited. Credentials cannot rescue a failed gate and lack of credentials cannot prevent a reproducible result from being evaluated. Contact Maria.Smith.Sftoe@gmail.com, submit through <https://discord.gg/ucwGryVxGr>, or inspect the public project at <https://github.com/MettaMazza>.

Evidence-boundary inventory identity:

sha256:d421a26e77e356ebb227bbcab026dfe6aaedcd3019dc55d7e70adf2503cfd542.

3. Fold-native quantum constitution

One Fold distinction supplies the native information unit. Complete held-label words across generated positions supply branch support. A branch carries an exact held phase label from a generated finite cycle. Joint states are complete pair-cell words retaining component identity. A transformation is a total branch-word bijection plus a period-held phase action. None of these definitions requires a real or complex amplitude, irrational normalization, imaginary proof value or ungenerated continuum.

4. Superposition, interference and entanglement

Superposition-equivalent means complete simultaneous support of every generated branch before an observation class is retained; it does not install an ontic stochastic mixture. Phase is a reversible action on held cycle labels. Interference is a complete branch-to-image relation whose predecessor fibres preserve every merging branch and its phase. Joint support is factorable exactly when it equals the Cartesian product of its marginals; a strict correlated subset is nonfactorable and supplies the entangling class.

5. Measurement and reversibility

Measurement is a total observation relation over complete branch support. The selected observation class becomes the retained state, while the measurement record keeps every pre-observation branch, phase and image label. Support reduction is therefore exact and reconstructible at the record boundary. A gate is reversible only when its branch map is bijective and its phase action has an exact inverse; any predecessor merge requires a retained predecessor label.

6. Coding, multi-error correction and unbounded finite fault order

A generated positive fault-order trace of length t forces repetition width $2t+1$: no more than t labels change and at least $t+1$ retain the source, forcing the strict majority. Every width through $2t$ has a constructive counterexample formed by changing $\text{ceil}(w/2)$ positions. The executable census exhausts every mask through one error at width 3, two errors at width 5 and three errors at width 7. It executes every predecessor-width and successor certificate through $t=14$ and round-trips all 128 depth-seven source words at $t=14$ before exact circuit evaluation. The successor t to $t+1$ adds exactly two labels and constructively defeats every fixed positive finite ceiling. No completed infinite-width object, measured error rate or physical hardware threshold is asserted.

7. Classical-quantum correspondence and limits

Classical computation embeds as the single-held-branch, phase-insensitive reversible submodel. Quantum execution extends that carrier with complete branch support, phase-sensitive predecessor merging and nonfactorable joint support. Observation decodes exact classical records. Because every quantum circuit remains a generated finite description executed by an admitted universal process, self-reference and halting boundaries transfer: quantum computation does not acquire an undeclared oracle or hypercomputational escape.

8. Complete execution census

Order	Claim	Candidates	Closure
1	SFT-QUANTUM-REVERSIBLE-MODEL-001	256	depth-independent
2	SFT-QUANTUM-INFORMATION-UNIT-001	256	depth-independent
3	SFT-QUANTUM-STATE-COMPOSITION-001	256	depth-independent
4	SFT-QUANTUM-SUPERPOSITION-001	256	depth-independent
5	SFT-QUANTUM-PHASE-INTERFERENCE-001	256	depth-independent
6	SFT-QUANTUM-ENTANGLEMENT-001	256	depth-independent
7	SFT-QUANTUM-MEASUREMENT-001	256	depth-independent
8	SFT-QUANTUM-GATE-001	256	depth-independent
9	SFT-QUANTUM-CIRCUIT-001	256	depth-independent
10	SFT-QUANTUM-UNIVERSALITY-001	256	depth-independent
11	SFT-QUANTUM-ALGORITHMS-001	256	depth-independent
12	SFT-QUANTUM-COMPLEXITY-001	256	depth-independent
13	SFT-QUANTUM-COMMUNICATION-001	256	depth-independent
14	SFT-QUANTUM-CODING-001	256	depth-independent
15	SFT-QUANTUM-ERROR-CORRECTION-001	256	depth-independent
16	SFT-QUANTUM-FAULT-TOLERANCE-001	256	depth-independent
17	SFT-QUANTUM-SIMULATION-001	256	depth-independent
18	SFT-QUANTUM-VERIFICATION-001	256	depth-independent
19	SFT-QUANTUM-LEARNING-001	256	depth-independent
20	SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001	256	depth-independent
21	SFT-QUANTUM-LIMITS-001	256	depth-independent
22	SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002	256	depth-independent

The branch total is **5,632 candidates, 22 survivors, 88 adverse controls** and **22 independent validations**.

Family results, complete claim inventory and claim-level audit records

The dependency-ordered ledger below is the complete human-readable Reversible and Quantum Computation claim inventory for this version. Each claim section exposes its claim identity, family, formal and empirical status, exact statement, dependency route, carrier and boundary, candidate grammar and count, unique survivor, elimination logic, falsification condition, controls, source and custody records, chronology, scientific meaning and receipt identities. Where a generic clause is referenced, its full unchanged wording is retained in the shared claim-record appendix. The machine packages remain authoritative for complete candidates, decisions and executable traces.

9. Derivation 1: Complete Fold reversible-computation model

Claim: SFT-QUANTUM-REVERSIBLE-MODEL-001

9.1 Scientific necessity and exact theorem

Complete Fold reversible-computation model must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced reversible computation kernel retains canonical configurations; bijective transition rows; retained held labels; exact inverse traces; universal reversible interpretation; complete resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

9.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: canonical configurations; bijective transition rows; retained held labels; exact inverse traces; universal reversible interpretation; complete resource ledger.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for reversible computation.

All finite reversible computation structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. This theorem is formal and does not assign thermodynamic energy without a later physical measurement law.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

9.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced reversible computation kernel retains canonical configurations; bijective transition rows; retained held labels; exact inverse traces; universal reversible interpretation; complete resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

9.4 Operational laws and consequences

- Support law: canonical configurations; bijective transition rows; retained held labels; exact inverse traces; universal reversible interpretation; complete resource ledger.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

9.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact reversible computation record.

Shared claim-record clause QUANT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

9.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-REVERSIBLE-MODEL-001; result: PASS.

- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-REVERSIBLE-MODEL-001;
result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-REVERSIBLE-MODEL-001;
result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-REVERSIBLE-MODEL-001;
result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; This theorem is formal and does not assign thermodynamic energy without a later physical measurement law.	PASS

9.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: reversible computation, Bennett machine.

This theorem is formal and does not assign thermodynamic energy without a later physical measurement law.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- This theorem is formal and does not assign thermodynamic energy without a later physical measurement law.

9.8 Exact evidence identities

- Source manifest: sha256:c5af094fe52290461356bf39e9d76f13ba2b6202e285508bc50ef96cd4b8e86f
- Complete-census certificate:
sha256:dd9994b76e30d066ca995523aa7528f47f985b73200c8e677750fa7f652c5033
- Derivation seal: sha256:37e9625bca1b99f14969057b963782c6b1cb62efb98ec317178cb32901e2f48c
- Independent implementation:
sha256:088141be2c397e4c3f7e640dbc6ba3850bc936eb3433537b058e95a25e98e2df
- Independent certificate:
sha256:33e527c37a50844de7b1b6228f92b22f4eb6a5279fe576a97b952c28885c5ed1
- External validation:
sha256:ded9f371a7b58164c55628cc9684d6944f8083be860caea6cb43f4f0a629bb60
- Model-admitted engine receipt:
sha256:068b389a38d47c06118600324bc263c951e6e19c6f591f257c601f6125525298

- Receipt path: receipts/engine/model_admitted/SFT-QUANTUM-REVERSIBLE-MODEL-001-068b389a38d47c06.json

10. Derivation 2: Fold quantum information-unit law

Claim: SFT-QUANTUM-INFORMATION-UNIT-001

10.1 Scientific necessity and exact theorem

Fold quantum information-unit law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum information units kernel retains one Fold distinction; two held fibre labels; complete word support; exact observation classes; retained phase-label carrier; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-REVERSIBLE-MODEL-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

10.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: one Fold distinction; two held fibre labels; complete word support; exact observation classes; retained phase-label carrier.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum information units.

All finite quantum information units structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. The unit is structural support, not an imported complex vector or conventional qubit.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

10.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced quantum information units kernel retains one Fold distinction; two held fibre labels; complete word support; exact observation classes; retained phase-label carrier; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

10.4 Operational laws and consequences

- Support law: one Fold distinction; two held fibre labels; complete word support; exact observation classes; retained phase-label carrier.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

10.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum information units record.

Shared claim-record clause QUANT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

10.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-INFORMATION-UNIT-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-INFORMATION-UNIT-001; result: PASS.

- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-INFORMATION-UNIT-001;
result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-INFORMATION-UNIT-001;
result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-INFORMATION-UNIT-001;
result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-INFORMATION-UNIT-001;
result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-INFORMATION-UNIT-001;
result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-INFORMATION-UNIT-001;
result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; The unit is structural support, not an imported complex vector or conventional qubit.	PASS

10.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: qubit, quantum information unit.

The unit is structural support, not an imported complex vector or conventional qubit.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- The unit is structural support, not an imported complex vector or conventional qubit.

10.8 Exact evidence identities

- Source manifest: sha256:a27e2cd1d555c89028a74a2426b74986adf84a83e8fac584985522f90e8b0c83
- Complete-census certificate:
sha256:76b580cc45da482f4fdbd7ad583257f4d3d7719b8062af69b5ac96098cc2f01c
- Derivation seal: sha256:8dd1b7f8a5b144885d8926f1d724e9f1ecc767f71ce3110a1884d03297673cdb
- Independent implementation:
sha256:3a82e4d5098bfd4d453af17516b932f696f816a15c6bc718253472c762d0ee10
- Independent certificate:
sha256:c7ca84cd3a8c47cd8a5e345697f2648d3861372f3c6a72c96a1288ce660f8ef5

- External validation:
sha256:7e138a1d7e2f5ce7784a9eab262e857118f8efce2a96a7ec20b6674541cc60a5
- Model-admitted engine receipt:
sha256:3cae970ebb8336d464bcbcb64e17d0f50049da6f018b95f98b53b04a46873bf72
- Receipt path: receipts/engine/model_admitted/SFT-QUANTUM-INFORMATION-UNIT-001-3cae970ebb8336d4.json

11. Derivation 3: Fold quantum state-composition law

Claim: SFT-QUANTUM-STATE-COMPOSITION-001

11.1 Scientific necessity and exact theorem

Fold quantum state-composition law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum state composition kernel retains complete branch support; exact pair-cell joint words; component identities; phase-label composition; marginal reconstruction; product and correlated classes; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-INFORMATION-UNIT-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

11.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: complete branch support; exact pair-cell joint words; component identities; phase-label composition; marginal reconstruction; product and correlated classes.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum state composition.

All finite quantum state composition structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Only finite generated joint supports are admitted.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

11.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced quantum state composition kernel retains complete branch support; exact pair-cell joint words; component identities; phase-label composition; marginal reconstruction; product and correlated classes; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

11.4 Operational laws and consequences

- Support law: complete branch support; exact pair-cell joint words; component identities; phase-label composition; marginal reconstruction; product and correlated classes.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

11.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum state composition record.

Shared claim-record clause QUANT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

11.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-STATE-COMPOSITION-001; result: PASS.

Control	Expected	Observed	Result
false_promise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Only finite generated joint supports are admitted.	PASS

11.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: tensor product, composite state.

Only finite generated joint supports are admitted.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Only finite generated joint supports are admitted.

11.8 Exact evidence identities

- Source manifest: sha256:e19a28acfd158a2adc8faa62d1c641983e613fd02abd4f5703fd4df71b5a2e32
- Complete-census certificate:
sha256:d10f6ac72430d0adbf42711c2e21033c6ef6a3fd14d43e9b6aed60ef5336f9d4

- Derivation seal: sha256:07accac25aaa131af96d31bdce98536c3ff39c0c0504a69b146f4a02240441c5
- Independent implementation:
sha256:45b6bcd57a43ea420e30db2d983e862b2206a61b2483270be469b035b5263ee0
- Independent certificate:
sha256:47fa45ac826615121aeec5d2ea41548aebd01dbac38c96e8e2ee8b359cb507e4
- External validation:
sha256:7ac24331062df574d62dbb00a5237cdf8bcd928212a5d0baf95a6cef91b867c1
- Model-admitted engine receipt:
sha256:a358610587ee2c7c6dfcba8a782e46f89a1668e2c86739eac0e8c88d6af719c2
- Receipt path: receipts/engine/model_admitted/SFT-QUANTUM-STATE-COMPOSITION-001-a358610587ee2c7c.json

12. Derivation 4: Superposition-equivalent Fold support law

Claim: SFT-QUANTUM-SUPERPOSITION-001

12.1 Scientific necessity and exact theorem

Superposition-equivalent Fold support law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced superposition-equivalent structure kernel retains complete b^k Fold-word support; one record per branch; held phase label per branch; no selected actual branch before observation; exact support provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-STATE-COMPOSITION-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

12.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: complete b^k Fold-word support; one record per branch; held phase label per branch; no selected actual branch before observation; exact support provenance.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for superposition-equivalent structure.

All finite superposition-equivalent structure structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. No amplitude magnitude, normalization constant or stochastic ontology is imported.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

12.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced superposition-equivalent structure kernel retains complete b^k Fold-word support; one record per branch; held phase label per branch; no selected actual branch before observation; exact support provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

12.4 Operational laws and consequences

- Support law: complete b^k Fold-word support; one record per branch; held phase label per branch; no selected actual branch before observation; exact support provenance.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

12.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact superposition-equivalent structure record.

Shared claim-record clause QUANT-S006 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

12.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-SUPERPOSITION-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-SUPERPOSITION-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-SUPERPOSITION-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-SUPERPOSITION-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-SUPERPOSITION-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-SUPERPOSITION-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-SUPERPOSITION-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-SUPERPOSITION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No amplitude magnitude, normalization constant or stochastic ontology is imported.	PASS

12.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: superposition, basis support.

No amplitude magnitude, normalization constant or stochastic ontology is imported.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law

- No amplitude magnitude, normalization constant or stochastic ontology is imported.

12.8 Exact evidence identities

- Source manifest: sha256:0d1eb45fe6d4e484dfcaa6ef334a3ab671ddd0da9f6f153b901cc8351f7603e3
- Complete-census certificate:
sha256:95ed3e715d21daabab6325f0d1dbeffdd25e7fdaf0c8c481c2bee0c61bd2e0a7
- Derivation seal: sha256:3d8ee27d90c7ddfea4b88ece679df71f39ba1602da046edef08f8e3ed4a8bcdd
- Independent implementation:
sha256:945ffdb45c2324c21427547d49b7e14fbe31aeaca26deee602a921d72088a911
- Independent certificate:
sha256:0de2881ba9754d71a5ad9d18c7283eafe2a63e31b131dc545c7234b8b668bc63
- External validation:
sha256:5fd70c68356f3013863b86feacd8f2a221a4002161cc5be3074f29e0c6b719e1
- Model-admitted engine receipt:
sha256:54c09712a4a53c16054fc2c8de1e247007b718e1f6f0cd9cd38a14b1a5cbe7d1
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-SUPERPOSITION-001-54c09712a4a53c16.json

13. Derivation 5: Fold phase and interference operation law

Claim: SFT-QUANTUM-PHASE-INTERFERENCE-001

13.1 Scientific necessity and exact theorem

Fold phase and interference operation law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced phase and interference kernel retains period-b held phase cycle; reversible phase action; total branch-to-image relation; complete predecessor fibres; same-image merge ledger; retained phase provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-SUPERPOSITION-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

13.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: period-b held phase cycle; reversible phase action; total branch-to-image relation; complete predecessor fibres; same-image merge ledger; retained phase provenance.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for phase and interference.

All finite phase and interference structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Interference is exact predecessor merging with phase labels; no complex exponential is assumed.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

13.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced phase and interference kernel retains period-b held phase cycle; reversible phase action; total branch-to-image relation; complete predecessor fibres; same-image merge ledger; retained phase provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

13.4 Operational laws and consequences

- Support law: period-b held phase cycle; reversible phase action; total branch-to-image relation; complete predecessor fibres; same-image merge ledger; retained phase provenance.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.

- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

13.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact phase and interference record.

Shared claim-record clause QUANT-S006 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

13.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-PHASE-INTERFERENCE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Interference is exact predecessor merging with phase labels; no complex exponential is assumed.	PASS

13.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: phase, interference.

Interference is exact predecessor merging with phase labels; no complex exponential is assumed.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate

- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Interference is exact predecessor merging with phase labels; no complex exponential is assumed.

13.8 Exact evidence identities

- Source manifest: sha256:e7c78294725fc5b8365f70ee9be63ba92165aa712a7a686ecd999e74a8db7393
- Complete-census certificate:
sha256:3125c54031dc65d7e5e4963aa4b031acf83c57e009e731ee188d7806d07b7684
- Derivation seal: sha256:da60087f0965652227de2f3c71b7584d22e2f16a2b794c4a3fa4152bc27d186a
- Independent implementation:
sha256:5b0728c303020c2658658987b7794ae08beb9b00f6e2f8c6b48c1cc5251db79b
- Independent certificate:
sha256:09d8b5b79e90ad34e20b35f399aafd7bc0c8948366962366431e8411290575a0
- External validation:
sha256:3d7e92436970f0002b6da43d04afbba1374c525fb6b98c43a99f1ab40c60d44f
- Model-admitted engine receipt:
sha256:c1c7e3f2c71ea7de6d801c397cff01ae8a005651deddccc7acd83bddc857f12
- Receipt path: receipts/engine/model_admitted/SFT-QUANTUM-PHASE-INTERFERENCE-001-c1c7e3f2c71ea7de.json

14. Derivation 6: Fold entangling-composition law

Claim: SFT-QUANTUM-ENTANGLEMENT-001

14.1 Scientific necessity and exact theorem

Fold entangling-composition law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced entanglement kernel retains joint word support; exact marginals; product-support test; nonfactorable pair-cell classes; retained component observations; compositional trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-PHASE-INTERFERENCE-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

14.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: joint word support; exact marginals; product-support test; nonfactorable pair-cell classes; retained component observations; compositional trace.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for entanglement.

All finite entanglement structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Entanglement is structural nonfactorability of registered joint support, not a metaphysical postulate.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampld-joint-states	sampld-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

14.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampld-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced entanglement kernel retains joint word support; exact marginals; product-support test; nonfactorable pair-cell classes; retained component observations; compositional trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

14.4 Operational laws and consequences

- Support law: joint word support; exact marginals; product-support test; nonfactorable pair-cell classes; retained component observations; compositional trace.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

14.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact entanglement record.

Shared claim-record clause QUANT-S006 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

14.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-ENTANGLEMENT-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-ENTANGLEMENT-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-ENTANGLEMENT-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-ENTANGLEMENT-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-ENTANGLEMENT-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-ENTANGLEMENT-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-ENTANGLEMENT-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-ENTANGLEMENT-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Entanglement is structural nonfactorability of registered joint support, not a metaphysical postulate.	PASS

14.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: entanglement, nonseparable state.

Entanglement is structural nonfactorability of registered joint support, not a metaphysical postulate.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Entanglement is structural nonfactorability of registered joint support, not a metaphysical postulate.

14.8 Exact evidence identities

- Source manifest: sha256:afb46ed17a896275cc6922c9f10115e35b92520aa8f0e38eb4bf67a7399cd1c8
- Complete-census certificate:
sha256:3dccccf9a2fd85f710d29322e1e2200163c35004bfa3d73c701a6520b7404c6d
- Derivation seal: sha256:d7fe096c012e1467e6d8253f0d04d02de4bc66f76b479cca2190386dc4298cec
- Independent implementation:
sha256:ebd4b53c6a0c5b77e9d0f9fc9f7f031c0d9f1f9a77ba9fd886f9153df0e39966
- Independent certificate:
sha256:cfeb4e7a31d71a6197bc757243682f2fa66e4ce558a1284d2d35c0fd76a7dc43
- External validation:
sha256:57031064581dc6356636b1f6f65f980795d963109bd56945287e91e13dfb9657
- Model-admitted engine receipt:
sha256:6bfa64c6f896c5f7eaa3a55872f31cd487923853353a3069ec7a839afa1d3d0d
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-ENTANGLEMENT-001-6bfa64c6f896c5f7.json

15. Derivation 7: Fold quantum measurement-semantics law

Claim: SFT-QUANTUM-MEASUREMENT-001

15.1 Scientific necessity and exact theorem

Fold quantum measurement-semantics law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced measurement kernel retains total observation relation; retained observation class; closed branch distinctions; complete pre-observation record; selected-class state; exact reconstruction boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-ENTANGLEMENT-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001

- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

15.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: total observation relation; retained observation class; closed branch distinctions; complete pre-observation record; selected-class state; exact reconstruction boundary.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for measurement.

All finite measurement structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. No stochastic collapse postulate is imported; outcome frequencies require external empirical evidence.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

15.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced measurement kernel retains total observation relation; retained observation class; closed branch distinctions; complete pre-observation record; selected-class state; exact reconstruction boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

15.4 Operational laws and consequences

- Support law: total observation relation; retained observation class; closed branch distinctions; complete pre-observation record; selected-class state; exact reconstruction boundary.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

15.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact measurement record.

Shared claim-record clause QUANT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

15.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-MEASUREMENT-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-MEASUREMENT-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-MEASUREMENT-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-MEASUREMENT-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-MEASUREMENT-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-MEASUREMENT-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-MEASUREMENT-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-MEASUREMENT-001; result: PASS.

Control	Expected	Observed	Result
false_promise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS

Control	Expected	Observed	Result
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No stochastic collapse postulate is imported; outcome frequencies require external empirical evidence.	PASS

15.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum measurement, collapse.

No stochastic collapse postulate is imported; outcome frequencies require external empirical evidence.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- No stochastic collapse postulate is imported; outcome frequencies require external empirical evidence.

15.8 Exact evidence identities

- Source manifest: sha256:13904de9dde6457d62a60404ddea441608fa18f0c92f149958bb94b8333e32cf
- Complete-census certificate:
sha256:2f837ffe8611aa00de3dde2d72e76ac5b16ec836ee79b879b5d3379af0046e71
- Derivation seal: sha256:7d99cd7ce254a64f69ce210b7b3883177227978c792faeeb846094c4856dc9fa
- Independent implementation:
sha256:09615ec981ffda13c0fd1ea5163e580708612ce2202d5b23d98d1c3b806d66c7
- Independent certificate:
sha256:59226eca7492c0aebb48b4ab677e9b24e1bcfc38d8438d37385b00ed1ab13d1f
- External validation:
sha256:0b403b2a032ee01b1b8bbc6ae03c9a26b51f564dd38d11f1bc0d775672d89026
- Model-admitted engine receipt:
sha256:e29a71f7dff97835e28850ee8d352ac80aa9ceabb4399e07105d892175776a06
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-MEASUREMENT-001-e29a71f7dff97835.json

16. Derivation 8: Fold reversible transformation and gate law

Claim: SFT-QUANTUM-GATE-001

16.1 Scientific necessity and exact theorem

Fold reversible transformation and gate law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum transformations and gates kernel retains bijective branch-word map; period-held phase action; complete domain and image support; exact inverse; source-bound gate identity; composition interface; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-MEASUREMENT-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

16.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: bijective branch-word map; period-held phase action; complete domain and image support; exact inverse; source-bound gate identity; composition interface.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum transformations and gates.

All finite quantum transformations and gates structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Physical gate fidelity remains an empirical engineering question.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

16.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.

Eliminated candidates	Exact first-failure reason
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced quantum transformations and gates kernel retains bijective branch-word map; period-held phase action; complete domain and image support; exact inverse; source-bound gate identity; composition interface; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

16.4 Operational laws and consequences

- Support law: bijective branch-word map; period-held phase action; complete domain and image support; exact inverse; source-bound gate identity; composition interface.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

16.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum transformations and gates record.

Shared claim-record clause QUANT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

16.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-GATE-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-GATE-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-GATE-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-GATE-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-GATE-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-GATE-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-GATE-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-GATE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS

Control	Expected	Observed	Result
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Physical gate fidelity remains an empirical engineering question.	PASS

16.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum gate, unitary transformation.

Physical gate fidelity remains an empirical engineering question.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Physical gate fidelity remains an empirical engineering question.

16.8 Exact evidence identities

- Source manifest: sha256:42bb13c0646eea4cd1dbb2822a4cc213980cbb6bf622d108985528f6b540117f
- Complete-census certificate:
sha256:9c9d3340df70fc9aa032104c6f83309f96ba6ce28721c8c3187e1dd29749e38e
- Derivation seal: sha256:d93e94a7c4c89f79904d0068dc37fb48db7040bb45533a02cb96f8c45cb617c1
- Independent implementation:
sha256:629fa6869c3efd87e733736338d25b83adf646fe360ce5210876449f86aa18d5
- Independent certificate:
sha256:8c737f9cb9e579c043660fde438e8405e4f6288e9d65f74e470be69d59549c89
- External validation:
sha256:9c8ee6c1d374120eab1601d55d5d859b2391a968447e023a1f818301f7f114b7
- Model-admitted engine receipt:
sha256:495704206fd6c701e1baf8d1ec0b4bcfd430a002aea0bb76453a091e01ccfdaf
- Receipt path: receipts/engine/model_admitted/SFT-QUANTUM-GATE-001-495704206fd6c701.json

17. Derivation 9: Fold quantum circuit syntax and semantics law

Claim: SFT-QUANTUM-CIRCUIT-001

17.1 Scientific necessity and exact theorem

Fold quantum circuit syntax and semantics law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum circuits kernel retains generated wires; information units; reversible gates; causal circuit order; complete branchwise evaluation; inverse circuit; observation boundary; resource ledger; branch support, phase,

transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-GATE-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

17.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: generated wires; information units; reversible gates; causal circuit order; complete branchwise evaluation; inverse circuit; observation boundary; resource ledger.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum circuits.

All finite quantum circuits structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. This derives circuit semantics; hardware timing and noise are external.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

17.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.

Eliminated candidates	Exact first-failure reason
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced quantum circuits kernel retains generated wires; information units; reversible gates; causal circuit order; complete branchwise evaluation; inverse circuit; observation boundary; resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

17.4 Operational laws and consequences

- Support law: generated wires; information units; reversible gates; causal circuit order; complete branchwise evaluation; inverse circuit; observation boundary; resource ledger.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

17.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum circuits record.

Shared claim-record clause QUANT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

17.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-CIRCUIT-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-CIRCUIT-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS

Control	Expected	Observed	Result
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; This derives circuit semantics; hardware timing and noise are external.	PASS

17.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum circuit, circuit semantics.

This derives circuit semantics; hardware timing and noise are external.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- This derives circuit semantics; hardware timing and noise are external.

17.8 Exact evidence identities

- Source manifest: sha256:d4825331a10928d077303fda7e766c916140bf5346464aedd390634a321b7586
- Complete-census certificate:
sha256:23cba4f14d69066c4576001a3d04ea7f054118d48815312b06b97cb9d14db74c
- Derivation seal: sha256:8fb9825eeec2daaff4511f6aaf81f866fe575a23d070bbfbbef9d4d5d59837990
- Independent implementation:
sha256:1d370c1f6d10c8b50b5f4ce8980a2557dead4382a9a3565990df366ab519fdd6
- Independent certificate:
sha256:613d70fcb5954bd320bae00fe784bb03726d6ab1b3cc0c287a6df616d4746618
- External validation:
sha256:0a859bd98ada50cc93ff7f774518d1fa57ec325f78cda38db2219c74eaa48ceb
- Model-admitted engine receipt:
sha256:63e751cf9ab382cdc5c4b5dfd0c6c2f01ad90659bf2c9189357f141475d9e8aa
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-CIRCUIT-001-63e751cf9ab382cd.json

18. Derivation 10: Fold quantum universality law

Claim: SFT-QUANTUM-UNIVERSALITY-001

18.1 Scientific necessity and exact theorem

Fold quantum universality law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum universal computation kernel retains frozen gate-description grammar; exact encoded circuits; one interpreter; branchwise trace simulation; phase and observation preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-CIRCUIT-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

18.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: frozen gate-description grammar; exact encoded circuits; one interpreter; branchwise trace simulation; phase and observation preservation; overhead ledger.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum universal computation.

All finite quantum universal computation structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Universality is exact for every generated description in the frozen grammar.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

18.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.

Eliminated candidates	Exact first-failure reason
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced quantum universal computation kernel retains frozen gate-description grammar; exact encoded circuits; one interpreter; branchwise trace simulation; phase and observation preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

18.4 Operational laws and consequences

- Support law: frozen gate-description grammar; exact encoded circuits; one interpreter; branchwise trace simulation; phase and observation preservation; overhead ledger.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

18.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum universal computation record.

Shared claim-record clause QUANT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

18.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.

- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-UNIVERSALITY-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Universality is exact for every generated description in the frozen grammar.	PASS

18.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: universal quantum computer, universal gate set.

Universality is exact for every generated description in the frozen grammar.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Universality is exact for every generated description in the frozen grammar.

18.8 Exact evidence identities

- Source manifest: sha256:59bfa7dc2618cb23f1897460501e00e4fdd1f0a0b1fab31fc0ffc9e06de8c549
- Complete-census certificate:
sha256:4af4128a42fc2e6efbb404ece4464adde28e77475392686199b67aa7a00117be
- Derivation seal: sha256:7f480fe1655f32f7ff3f640e9e0f508c501d21b0b44430f24ff41fcc84d35c0a
- Independent implementation:
sha256:00dc7bb66a94f5a7f0ced9c5b08dea0b489c312288e3cc0bf250f86bc8a7d89f
- Independent certificate:
sha256:0d501674df0431ac2f24be59fafd160e2c225807ab1b000013632f09ddcd1214
- External validation:
sha256:43e4644556a5eb410b174b4cdc7d6993d965e7f1b5636773dd41e3c67ba148d1
- Model-admitted engine receipt:
sha256:28b6a73e02450c70bf3ee49f3fcdcf2e28b30c463e05218b99cdc8cda723e86a
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-UNIVERSALITY-001-28b6a73e02450c70.json

19. Derivation 11: Fold quantum algorithm-family law

Claim: SFT-QUANTUM-ALGORITHMS-001

19.1 Scientific necessity and exact theorem

Fold quantum algorithm-family law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum algorithms kernel retains complete input support; reversible branch generation; phase action; interference merge; observation decoder; correctness trace per branch; classical comparison boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-UNIVERSALITY-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

19.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: complete input support; reversible branch generation; phase action; interference merge; observation decoder; correctness trace per branch; classical comparison boundary.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum algorithms.

All finite quantum algorithms structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Named speedups require their own exact problem family and resource proof.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

19.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced quantum algorithms kernel retains complete input support; reversible branch generation; phase action; interference merge; observation decoder; correctness trace per branch; classical comparison boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

19.4 Operational laws and consequences

- Support law: complete input support; reversible branch generation; phase action; interference merge; observation decoder; correctness trace per branch; classical comparison boundary.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

19.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum algorithms record.

Shared claim-record clause QUANT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

19.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.

- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-ALGORITHMS-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Named speedups require their own exact problem family and resource proof.	PASS

19.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum algorithm, quantum speedup.

Named speedups require their own exact problem family and resource proof.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Named speedups require their own exact problem family and resource proof.

19.8 Exact evidence identities

- Source manifest: sha256:779315c063473730553e1b94b47294d97acd1023e130e685d2aee25fd1c2ef67
- Complete-census certificate:
sha256:63c84d1546a1f83c51991b85670ab300de70ad6532104733532a81dc371c7282
- Derivation seal: sha256:cb94fae7c67d50383d283b028c580568c6d7e1fafb4a919217df581529703a87
- Independent implementation:
sha256:89bb200d9b5c23e863521d13d47d326f8df6f7e4ace5605cd4e62929121a8065
- Independent certificate:
sha256:8eac37074eb7bcf0579cb97aedc91fa70d056d0828515070ae6df1f157b8d22c
- External validation:
sha256:fe1b4e628eb13b88f9e401532d3f00a562cb8264986a662d0716f9bf766d6409
- Model-admitted engine receipt:
sha256:a33b13375d76ebf039fcc5711c0810ab99ac0b9456295bc7881d76c94558a86b
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-ALGORITHMS-001-a33b13375d76ebf0.json

20. Derivation 12: Fold quantum complexity law

Claim: SFT-QUANTUM-COMPLEXITY-001

20.1 Scientific necessity and exact theorem

Fold quantum complexity law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum computational resources kernel retains canonical input size; gate count; causal depth; live branch support; query and communication rows; measurement records; lower and upper witness boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-ALGORITHMS-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

20.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: canonical input size; gate count; causal depth; live branch support; query and communication rows; measurement records; lower and upper witness boundary.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum computational resources.

All finite quantum computational resources structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. No arbitrary separation between conventional complexity classes is claimed without a separate certificate.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampld-joint-states	sampld-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

20.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced quantum computational resources kernel retains canonical input size; gate count; causal depth; live branch support; query and communication rows; measurement records; lower and upper witness boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

20.4 Operational laws and consequences

- Support law: canonical input size; gate count; causal depth; live branch support; query and communication rows; measurement records; lower and upper witness boundary.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

20.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum computational resources record.

Shared claim-record clause QUANT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

20.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.

- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-COMPLEXITY-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No arbitrary separation between conventional complexity classes is claimed without a separate certificate.	PASS

20.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum complexity, BQP.

No arbitrary separation between conventional complexity classes is claimed without a separate certificate.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- No arbitrary separation between conventional complexity classes is claimed without a separate certificate.

20.8 Exact evidence identities

- Source manifest: sha256:8eac4910a233e30d537d3cc6dd92cfa0af9524abccff1bb6dae3c57759bf0f57f
- Complete-census certificate:
sha256:43fdd77acc2f1d28c1d909f806c2bf5d1bfd03d84effc0c51387915d78f83c24
- Derivation seal: sha256:2dabda188246d3553502bbfb24cebd18a4513e169b7291aa5ecbf9ca5fd66fe9
- Independent implementation:
sha256:81b3d49b18170b958f74bfe4f4a5cbd47bb4b0712c750027a7554d3576523419
- Independent certificate:
sha256:94b58ef5a03dc1e832cdf2ee0a5158ca6d8d9626c6d514f939aee795a7d0eee6

- External validation:
sha256:15ad7c676afff75fe9d9d991b0843f388b9061ed86b67929008d295eef2a2cd1
- Model-admitted engine receipt:
sha256:d6307d961273c2a383fec3b919c56319b910bccb58dbbcf5402a2a76a9ca8b6
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-COMPLEXITY-001-d6307d961273c2a3.json

21. Derivation 13: Fold quantum communication law

Claim: SFT-QUANTUM-COMMUNICATION-001

21.1 Scientific necessity and exact theorem

Fold quantum communication law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum communication kernel retains sender and receiver supports; joint state composition; local transformations; channel relation; retained correlations; observation records; resource and loss ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-COMPLEXITY-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

21.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: sender and receiver supports; joint state composition; local transformations; channel relation; retained correlations; observation records; resource and loss ledger.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum communication.

All finite quantum communication structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Physical transmission capacity and security require registered media experiments.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

21.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced quantum communication kernel retains sender and receiver supports; joint state composition; local transformations; channel relation; retained correlations; observation records; resource and loss ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

21.4 Operational laws and consequences

- Support law: sender and receiver supports; joint state composition; local transformations; channel relation; retained correlations; observation records; resource and loss ledger.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

21.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum communication record.

Shared claim-record clause QUANT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

21.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-COMMUNICATION-001; result: PASS.

Control	Expected	Observed	Result
false_promise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Physical transmission capacity and security require registered media experiments.	PASS

21.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum communication, teleportation.

Physical transmission capacity and security require registered media experiments.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Physical transmission capacity and security require registered media experiments.

21.8 Exact evidence identities

- Source manifest: sha256:8cbc3dcc0ee105c719a4c9019ba6f993d6bd51d32427286b18f5317b9a8ab96c
- Complete-census certificate:
sha256:288320813c5011753ac302d9c4d463a58e58623ae71d344625eea30eafcafe70

- Derivation seal: sha256:2b321b0b8c520885563b2c770bed8e0ec83e5aa0720a05a2f679125c5d1f63cc
- Independent implementation:
sha256:c98029cec76386f0bea9768e3536a822803d2c7fb1eb3e16616272c1e0e97bec
- Independent certificate:
sha256:b8460f43cb576317854272866636f450a0a837fd34929ae4b947d681dd95eb91
- External validation:
sha256:a91fd78dd1739e147a926a006477eb6fa4bef506490fedcedb86827c0b56fd91
- Model-admitted engine receipt:
sha256:01059b1cbe86b894ae6137de5625e1dfcd1cd640ac5634c0ca2260977993b97a
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-COMMUNICATION-001-01059b1cbe86b894.json

22. Derivation 14: Fold quantum coding law

Claim: SFT-QUANTUM-CODING-001

22.1 Scientific necessity and exact theorem

Fold quantum coding law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum coding kernel retains logical branch support; redundant joint words; reversible encoder; declared error actions; disjoint syndrome supports; decoder and reconstruction trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-COMMUNICATION-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

22.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: logical branch support; redundant joint words; reversible encoder; declared error actions; disjoint syndrome supports; decoder and reconstruction trace.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum coding.

All finite quantum coding structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. The code theorem applies only to the registered generated error family.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
composition	sampld-joint-s tates	sampld-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitu de-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irre versible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-b ound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-withou t-record	collapse-without-record removes the required exact observation structure.	retained-class-and-comple te-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-e xamples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successo r-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantu m-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

22.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampld-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced quantum coding kernel retains logical branch support; redundant joint words; reversible encoder; declared error actions; disjoint syndrome supports; decoder and reconstruction trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

22.4 Operational laws and consequences

- Support law: logical branch support; redundant joint words; reversible encoder; declared error actions; disjoint syndrome supports; decoder and reconstruction trace.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

22.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum coding record.

Shared claim-record clause QUANT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause `QUANT-S007` **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

22.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-CODING-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-CODING-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-CODING-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-CODING-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-CODING-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-CODING-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-CODING-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-CODING-001; result: PASS.

Control	Expected	Observed	Result
false_promise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; The code theorem applies only to the registered generated error family.	PASS

22.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum code, encoding.

The code theorem applies only to the registered generated error family.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- The code theorem applies only to the registered generated error family.

22.8 Exact evidence identities

- Source manifest: sha256:85e7aabcdad3f77f050dd76d25a562d3156936ff78a453bbf1f62cfd528e8a8
- Complete-census certificate:
sha256:93f1b6cf418d33752789ee30da4e3691204d71062d6e49138d4a6f9a292996aa
- Derivation seal: sha256:ea64d9cb56689b01f32e45805b41e053214e1e665f68f83c9fac52f894205efb
- Independent implementation:
sha256:3e3936d7ac16b4f76240d81294e3dab60dc4110611c44429cea46df1444590f2
- Independent certificate:
sha256:736fdb0c5a526c7b542e4eaf8fdf98681aae185d448844b242b1f371688b7746

- External validation:
sha256:e7785a525808797f14dba72e1e20acbd6dbb176c7799c1e71afca1bca34b271f
- Model-admitted engine receipt:
sha256:b79afc51951dfbebb30cd70fba329965b700fa14a16fb45b5754568ec7407a3e
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-CODING-001-b79afc51951dfbeb.json

23. Derivation 15: Fold quantum error-correction law

Claim: SFT-QUANTUM-ERROR-CORRECTION-001

23.1 Scientific necessity and exact theorem

Fold quantum error-correction law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum error correction kernel retains forced repetition width $2t+1$; exact fault-depth trace; all error masks through t ; majority held-label decoder; syndrome ledger; recovery certificate; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-CODING-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

23.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: forced repetition width $2t+1$; exact fault-depth trace; all error masks through t ; majority held-label decoder; syndrome ledger; recovery certificate.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum error correction.

All finite quantum error correction structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. The completed census covers generated fault depths one, two and three and the successor law for every positive finite t .

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

23.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced quantum error correction kernel retains forced repetition width $2t+1$; exact fault-depth trace; all error masks through t ; majority held-label decoder; syndrome ledger; recovery certificate; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

23.4 Operational laws and consequences

- Support law: forced repetition width $2t+1$; exact fault-depth trace; all error masks through t ; majority held-label decoder; syndrome ledger; recovery certificate.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

23.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum error correction record.

Shared claim-record clause QUANT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

23.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-ERROR-CORRECTION-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-ERROR-CORRECTION-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-ERROR-CORRECTION-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-ERROR-CORRECTION-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-ERROR-CORRECTION-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-ERROR-CORRECTION-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-ERROR-CORRECTION-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-ERROR-CORRECTION-001; result: PASS.

Control	Expected	Observed	Result
false_promise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; The completed census covers generated fault depths one, two and three and the successor law for every positive finite t.	PASS

23.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum error correction, repetition code.

The completed census covers generated fault depths one, two and three and the successor law for every positive finite t.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- The completed census covers generated fault depths one, two and three and the successor law for every positive finite t.

23.8 Exact evidence identities

- Source manifest: sha256:5ac864ccd27449cb7c1a756b5ed53c4094709160f00cf0e8f19e58f01a20a2d9

- Complete-census certificate:
sha256:9712241740820a883217d33d982426a39c21903950f917eafa9fef38d970389
- Derivation seal: sha256:61cc113c2d425a9f9c9d2e13bb1a983bd94e88220a6d91a4f1815d41b5059ee6
- Independent implementation:
sha256:f3e41dd58f96c808ce4cadab817f8a4ee86d6369ef8e6e43ddfaacc792071af9
- Independent certificate:
sha256:721e7c933eb6f760790d3ee15cd54171a46f8ffa24c67f5b59e62872b12fb16b
- External validation:
sha256:c5784b25becf3d48609c4a3447a3cff73cd2822418dd3800282d1948dbe6d3ef
- Model-admitted engine receipt:
sha256:a4791dcbf8753c32e8304c5d5e80f9bf7f21a7356b6adb9422ea60d5eaf59656
- Receipt path: receipts/engine/model_admitted/SFT-QUANTUM-ERROR-CORRECTION-001-a4791dcbf8753c32.json

24. Derivation 16: Fold quantum fault-tolerance law

Claim: SFT-QUANTUM-FAULT-TOLERANCE-001

24.1 Scientific necessity and exact theorem

Fold quantum fault-tolerance law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced fault-tolerant quantum computation kernel retains encoded logical support; declared component faults; transversal or contained transformations; recovery after each location; malignant-fault boundary; induction over circuit locations; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-ERROR-CORRECTION-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

24.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: encoded logical support; declared component faults; transversal or contained transformations; recovery after each location; malignant-fault boundary; induction over circuit locations.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for fault-tolerant quantum computation.

All finite fault-tolerant quantum computation structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. No hardware threshold constant is claimed; unlimited thresholds remain dependent on a separately derived physical fault model.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

24.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced fault-tolerant quantum computation kernel retains encoded logical support; declared component faults; transversal or contained transformations; recovery after each location; malignant-fault boundary; induction over circuit locations; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

24.4 Operational laws and consequences

- Support law: encoded logical support; declared component faults; transversal or contained transformations; recovery after each location; malignant-fault boundary; induction over circuit locations.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

24.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact fault-tolerant quantum computation record.

Shared claim-record clause QUANT-S006 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

24.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-FAULT-TOLERANCE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No hardware threshold constant is claimed; unlimited thresholds remain dependent on a separately derived physical fault model.	PASS

24.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: fault tolerance, threshold theorem.

No hardware threshold constant is claimed; unlimited thresholds remain dependent on a separately derived physical fault model.

Version 1.1 reconciliation: the sealed 001 wording uses 'unlimited thresholds' for physical hardware thresholds. It must not be read as leaving the exact code-order theorem open. The separately generated and admitted

SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 closes every supplied positive finite fault order by the forced width $2t+1$, while stochastic device thresholds remain empirical.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity

- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- No hardware threshold constant is claimed; unlimited thresholds remain dependent on a separately derived physical fault model.

24.8 Exact evidence identities

- Source manifest: sha256:3260e928de52083476ff03669bb6321999c2e7042d32a55027e98040c2d16c4e
- Complete-census certificate:
sha256:1e9b2a7fc565525f5353d4590607e2f6d1450e18f5c3db873f9f8b61c229960f
- Derivation seal: sha256:1816e6b491362e7cc5b5d306e833227bdf765559d55693d0b18553e2911f1a57
- Independent implementation:
sha256:ef9893933f17387358f0d7512e05451d206c5ce7a11fe348bbc1052bd42ee0dd
- Independent certificate:
sha256:0a28a0751ef1e16a701eb0fdf84d8d4a2583c0e4314f9cd1d4badfd2e5a10c7a
- External validation:
sha256:367a4648636683bd04852d04c27eb6360aede4d53948a7002d5d0feea0a37ea2
- Model-admitted engine receipt:
sha256:9f2526c5bd9f1625a3cee65b0cba4e9a600ecdc9a8031ac51367e35c37308b16
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-FAULT-TOLERANCE-001-9f2526c5bd9f1625.json

25. Derivation 17: Fold quantum simulation law

Claim: SFT-QUANTUM-SIMULATION-001

25.1 Scientific necessity and exact theorem

Fold quantum simulation law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum simulation kernel retains encoded target state support; reversible update law; phase and interference traces; observation map; exact simulator correspondence; resource ledger; falsification boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-FAULT-TOLERANCE-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

25.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: encoded target state support; reversible update law; phase and interference traces; observation map; exact simulator correspondence; resource ledger; falsification boundary.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum simulation.

All finite quantum simulation structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. A simulator derives consequences of its registered model and cannot establish a natural quantum law by itself.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

25.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced quantum simulation kernel retains encoded target state support; reversible update law; phase and interference traces; observation map; exact simulator correspondence; resource ledger; falsification boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

25.4 Operational laws and consequences

- Support law: encoded target state support; reversible update law; phase and interference traces; observation map; exact simulator correspondence; resource ledger; falsification boundary.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

25.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum simulation record.

Shared claim-record clause QUANT-S006 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

25.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-SIMULATION-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-SIMULATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; A simulator derives consequences of its registered model and cannot establish a natural quantum law by itself.	PASS

25.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum simulation, Hamiltonian simulation.

A simulator derives consequences of its registered model and cannot establish a natural quantum law by itself.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- A simulator derives consequences of its registered model and cannot establish a natural quantum law by itself.

25.8 Exact evidence identities

- Source manifest: sha256:a376c3321387216b559a5d2543ece347699b17421f174831037f93ee4012ba23
- Complete-census certificate:
sha256:0036a9974eaf22533d40dbeffc708a9b259140447ebb99dfd5406040e63c341b
- Derivation seal: sha256:bb1d2c15251c769fbcf1692edc85f6253cb6f103dbffe04680cedb4d492cbff8
- Independent implementation:
sha256:72f00440c8b0cfeaa26940b8aef2728b3e74896b435207b9b8f9426348064df9
- Independent certificate:
sha256:fcdb1c83f2a12bee60d57319cf49492523c03723aee2e0aafa22e3a88abff900
- External validation:
sha256:2e19399541f4035708283c1c043842f1c5b8ffef1330ec004109a14ff3fdcbd0
- Model-admitted engine receipt:
sha256:daf61c89738fb9f6370bc244c3a7538a3c4037ab408dfd5d15e3301c57e817a6
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-SIMULATION-001-daf61c89738fb9f6.json

26. Derivation 18: Fold quantum verification law

Claim: SFT-QUANTUM-VERIFICATION-001

26.1 Scientific necessity and exact theorem

Fold quantum verification law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum verification kernel retains claim and witness supports; verifier circuit; challenge schedule; branchwise acceptance; soundness and completeness ledgers; measurement records; tamper controls; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-SIMULATION-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001

- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

26.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: claim and witness supports; verifier circuit; challenge schedule; branchwise acceptance; soundness and completeness ledgers; measurement records; tamper controls.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum verification.

All finite quantum verification structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Security and correctness are exact only for the registered prover and resource grammar.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

26.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced quantum verification kernel retains claim and witness supports; verifier circuit; challenge schedule; branchwise acceptance; soundness and completeness ledgers; measurement records; tamper controls; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

26.4 Operational laws and consequences

- Support law: claim and witness supports; verifier circuit; challenge schedule; branchwise acceptance; soundness and completeness ledgers; measurement records; tamper controls.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

26.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum verification record.

Shared claim-record clause QUANT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

26.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-VERIFICATION-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-VERIFICATION-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS

Control	Expected	Observed	Result
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Security and correctness are exact only for the registered prover and resource grammar.	PASS

26.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum verification, QMA.

Security and correctness are exact only for the registered prover and resource grammar.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Security and correctness are exact only for the registered prover and resource grammar.

26.8 Exact evidence identities

- Source manifest: sha256:d17c41d57b8e1a861e03983b2a5a7808ec8756e28c7361cb95bb1f5a159635ee
- Complete-census certificate:
sha256:c54587141b8c1a489ee2b3d83d53548c435f0746abda0ab11358bbacblf5a159635ee
- Derivation seal: sha256:62b1704737e141ef52eab6b9687ec06ab6de072c1ba2dd8cc678880fe50e3ce0
- Independent implementation:
sha256:00bb831c0de419ea512cd89dd478e15c3804c348f3b094de0d7321006f9f408f
- Independent certificate:
sha256:fa884fd0ca86e8ae875b76054696f32cd5ef731fbbd1986b9a02b84140ece03a
- External validation:
sha256:753bc09d36d4c45faa75aa9f02e0780bc6c49a45126630495deb18d72c74cd77
- Model-admitted engine receipt:
sha256:6ca70e1a9487b428bc6e04e5e3b1a11907b79d1ed92f7d03a013a49380ecce54
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-VERIFICATION-001-6ca70e1a9487b428.json

27. Derivation 19: Fold quantum learning law

Claim: SFT-QUANTUM-LEARNING-001

27.1 Scientific necessity and exact theorem

Fold quantum learning law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced quantum learning kernel retains classical or joint example support; reversible hypothesis process; phase/interference search; observation decoder; retained alternatives; sample and gate resources; sealed evaluation; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-VERIFICATION-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

27.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: classical or joint example support; reversible hypothesis process; phase/interference search; observation decoder; retained alternatives; sample and gate resources; sealed evaluation.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for quantum learning.

All finite quantum learning structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. No pretrained quantum model, fitted amplitude or unexplained speedup is admitted.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampld-joint-states	sampld-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

27.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.

Eliminated candidates	Exact first-failure reason
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced quantum learning kernel retains classical or joint example support; reversible hypothesis process; phase/interference search; observation decoder; retained alternatives; sample and gate resources; sealed evaluation; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

27.4 Operational laws and consequences

- Support law: classical or joint example support; reversible hypothesis process; phase/interference search; observation decoder; retained alternatives; sample and gate resources; sealed evaluation.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

27.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact quantum learning record.

Shared claim-record clause QUANT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

27.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-LEARNING-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-LEARNING-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-LEARNING-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-LEARNING-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-LEARNING-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-LEARNING-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-LEARNING-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-LEARNING-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS

Control	Expected	Observed	Result
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No pretrained quantum model, fitted amplitude or unexplained speedup is admitted.	PASS

27.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum learning, quantum machine learning.

No pretrained quantum model, fitted amplitude or unexplained speedup is admitted.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- No pretrained quantum model, fitted amplitude or unexplained speedup is admitted.

27.8 Exact evidence identities

- Source manifest: sha256:720947592f997269012cdac1186615d307e7b3a553f0ff0d1e42eac0daedd840
- Complete-census certificate:
sha256:4f049e2fe5f2c3499b0b6012f93625016627146fccbfc47dc3bc5dfcca22831c
- Derivation seal: sha256:1d75a66403b8bd208fad0ca8cdd0de260f54865c8eb4e1b1984a25888845b435
- Independent implementation:
sha256:de8da3d2eb9a1c0c7cf06c6f851c311161c5e967d182c53c1d2d4f61d54045e3
- Independent certificate:
sha256:301937de6bfe10c6fc95c62a92efa37c27d5866795d316932652ba6dd953bd5d
- External validation:
sha256:2b0d54edaf033b569004c5d6e52026adaf3d9d5e8ad1129ff7ec246db2cebb42
- Model-admitted engine receipt:
sha256:c4388e69ff7412fe09e53b7e4b2a4eda12168ff6665ca351758c185ac7e018b7
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-LEARNING-001-c4388e69ff7412fe.json

28. Derivation 20: Full operational classical-quantum correspondence law

Claim: SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001

28.1 Scientific necessity and exact theorem

Full operational classical-quantum correspondence law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced classical and quantum operational correspondence kernel retains common generated descriptions; embedding of classical states; reversible classical submodel; branchwise quantum execution; measurement decoder; bidirectional result preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-LEARNING-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

28.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: common generated descriptions; embedding of classical states; reversible classical submodel; branchwise quantum execution; measurement decoder; bidirectional result preservation; overhead ledger.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for classical and quantum operational correspondence.

All finite classical and quantum operational correspondence structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Correspondence does not identify the models where phase-sensitive joint support changes the operational trace.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampled-joint-states	sampled-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

28.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

The forced classical and quantum operational correspondence kernel retains common generated descriptions; embedding of classical states; reversible classical submodel; branchwise quantum execution; measurement decoder; bidirectional result preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

28.4 Operational laws and consequences

- Support law: common generated descriptions; embedding of classical states; reversible classical submodel; branchwise quantum execution; measurement decoder; bidirectional result preservation; overhead ledger.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

28.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact classical and quantum operational correspondence record.

Shared claim-record clause QUANT-S006 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

28.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.

- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; result: PASS.

Control	Expected	Observed	Result
false_premise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Correspondence does not identify the models where phase-sensitive joint support changes the operational trace.	PASS

28.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: classical-quantum correspondence, deferred measurement.

Correspondence does not identify the models where phase-sensitive joint support changes the operational trace.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Correspondence does not identify the models where phase-sensitive joint support changes the operational trace.

28.8 Exact evidence identities

- Source manifest: sha256:a784ae4b69af81c489a660162333bca6e55fb56d401856af1d8e14516e863595
- Complete-census certificate:
sha256:922645a8e17e57036ddadbb6d3a35bf1e0a4a691864cb54ddfac3fb3f0978b59
- Derivation seal: sha256:d10d9d064e2a63254c2cae2d5d7ea26173427927650eb375fbf698f69cf61f86
- Independent implementation:
sha256:1cd0dd23ea4b5b984d9a1ebe768005d0c09258970fd241aa5a02215cfdae7d2b
- Independent certificate:
sha256:74d501babc031290bfc0b5a041bed885d7ccb65d700b20676a8260f75fd91519
- External validation:
sha256:4c2be0efb0edeb4392351b788cd58b9331c850577e49ae1b0699ad2ad420445f
- Model-admitted engine receipt:
sha256:e1712589e1aba3ce6536a3ba33d599e06bdf793a335079d1a775975bc234b614
- Receipt path: receipts/engine/model_admitted/SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001-e1712589e1aba3ce.json

29. Derivation 21: Limits of Fold quantum computation law

Claim: SFT-QUANTUM-LIMITS-001

29.1 Scientific necessity and exact theorem

Limits of Fold quantum computation law must be forced from admitted Fold and computation laws before conventional quantum models may be compared.

The engine-admitted theorem is:

The forced limits of quantum computation kernel retains classical computability dependency; finite description grammar; exact quantum execution; self-description support; halting and undecidability transfer; resource and observation boundaries; no hypercomputational oracle; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
- SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
- SFT-COMP-FORM-CIRCUIT-001
- SFT-COMP-FORM-UNIVERSALITY-001
- SFT-COMP-CPLX-REVERSIBILITY-COST-001
- SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
- SFT-COMP-LEARN-CLASSICAL-LEARNING-001
- SFT-INFO-QUANTUM-CORRESPONDENCE-001

29.2 Generated grammar and exact boundary

Complete Fold-word support, exact information ledgers, reversible classical processes and pair-cell composition force the registered kernel: classical computability dependency; finite description grammar; exact quantum execution; self-description support; halting and undecidability transfer; resource and observation boundaries; no hypercomputational oracle.

Generation rule: Generate the literal product of eight registered Fold-quantum structural axes for limits of quantum computation.

All finite limits of quantum computation structures generated from admitted Fold words, distinction ledgers, classical computation and exact held phase labels. Quantum computation does not escape the admitted computability and self-reference boundaries; physical claims remain empirical.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
support	partial-or-selected-branch-support	partial-or-selected-branch-support removes the required exact support structure.	complete-generated-branch-support	complete-generated-branch-support retains the required exact support structure.
unit	imported-bit-or-qubit	imported-bit-or-qubit removes the required exact unit structure.	one-Fold-distinction-unit	one-Fold-distinction-unit retains the required exact unit structure.
composition	sampld-joint-states	sampld-joint-states removes the required exact composition structure.	complete-pair-cell-joint-support	complete-pair-cell-joint-support retains the required exact composition structure.
phase	complex-amplitude-postulate	complex-amplitude-postulate removes the required exact phase structure.	period-held-phase-action	period-held-phase-action retains the required exact phase structure.
transformation	partial-or-irreversible-map	partial-or-irreversible-map removes the required exact transformation structure.	total-reversible-source-bound-map	total-reversible-source-bound-map retains the required exact transformation structure.
observation	collapse-without-record	collapse-without-record removes the required exact observation structure.	retained-class-and-complete-record	retained-class-and-complete-record retains the required exact observation structure.
generality	fixed-circuit-examples	fixed-circuit-examples removes the required exact generality structure.	branch-and-depth-successor-certificate	branch-and-depth-successor-certificate retains the required exact generality structure.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
addition	imported-quantum-model	imported-quantum-model removes the required exact addition structure.	no-extra-quantum-premise	no-extra-quantum-premise retains the required exact addition structure.

Shared claim-record clause QUANT-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

29.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-selected-branch-support removes the required exact support structure.
64	imported-bit-or-qubit removes the required exact unit structure.
32	sampled-joint-states removes the required exact composition structure.
16	complex-amplitude-postulate removes the required exact phase structure.
8	partial-or-irreversible-map removes the required exact transformation structure.
4	collapse-without-record removes the required exact observation structure.
2	fixed-circuit-examples removes the required exact generality structure.
1	imported-quantum-model removes the required exact addition structure.

Shared claim-record clause QUANT-S003 applies. The full, unchanged clause is preserved in the shared-clause appendix.

The forced limits of quantum computation kernel retains classical computability dependency; finite description grammar; exact quantum execution; self-description support; halting and undecidability transfer; resource and observation boundaries; no hypercomputational oracle; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

29.4 Operational laws and consequences

- Support law: classical computability dependency; finite description grammar; exact quantum execution; self-description support; halting and undecidability transfer; resource and observation boundaries; no hypercomputational oracle.
- Phase law: period-held label actions transform branch provenance without imaginary or irrational proof values.
- Reversibility law: transformations are total bijections or retain the exact predecessor label needed for inversion.
- Observation law: selected support and every closed distinction remain reconstructible from the measurement record.
- Composition law: joint pair-cell words preserve component identity and explicitly classify factorable and nonfactorable support.

Shared claim-record clause QUANT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

29.5 Depth-independent closure

Base: One Fold distinction supplies the minimal two-label support and one exact limits of quantum computation record.

Shared claim-record clause QUANT-S006 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Shared claim-record clause QUANT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

29.6 Executed witnesses and adverse controls

- quantum-witness-1 - exact Fold quantum operational witness 1 for SFT-QUANTUM-LIMITS-001; result: PASS.
- quantum-witness-2 - exact Fold quantum operational witness 2 for SFT-QUANTUM-LIMITS-001; result: PASS.
- quantum-witness-3 - exact Fold quantum operational witness 3 for SFT-QUANTUM-LIMITS-001; result: PASS.
- quantum-witness-4 - exact Fold quantum operational witness 4 for SFT-QUANTUM-LIMITS-001; result: PASS.
- quantum-witness-5 - exact Fold quantum operational witness 5 for SFT-QUANTUM-LIMITS-001; result: PASS.

- quantum-witness-6 - exact Fold quantum operational witness 6 for SFT-QUANTUM-LIMITS-001; result: PASS.
- quantum-witness-7 - exact Fold quantum operational witness 7 for SFT-QUANTUM-LIMITS-001; result: PASS.
- quantum-witness-8 - exact Fold quantum operational witness 8 for SFT-QUANTUM-LIMITS-001; result: PASS.

Control	Expected	Observed	Result
false_promise	reject incomplete state support	partial-or-selected-branch-support removes the required exact support structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Quantum computation does not escape the admitted computability and self-reference boundaries; physical claims remain empirical.	PASS

29.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: limits of quantum computing, quantum computability.

Quantum computation does not escape the admitted computability and self-reference boundaries; physical claims remain empirical.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinity or ungenerated continuum
- no free, fitted, learned or hardware parameter
- no application output or physical measurement may select the law
- Quantum computation does not escape the admitted computability and self-reference boundaries; physical claims remain empirical.

29.8 Exact evidence identities

- Source manifest: sha256:f111f69d04fe049625cc50c9378bfca7d08fd3c2e6cc82cfbc7151cf183223db
- Complete-census certificate:
sha256:d9059ae5cf371742769379c802668b1bc83493f42cf636184e5357c3e3279f95
- Derivation seal: sha256:4abd56778c0176f6426706806e75fdf866dedbbe2e78345585c0f9d0c6f360f
- Independent implementation:
sha256:f1ed114f18db0541db98fc2966784fabca2116c53921fb473b68c2e866a204c5
- Independent certificate:
sha256:e018eb19348a191249d0fa680329d0cf71d1c2e1d62b0734732426f74afb2cf5
- External validation:
sha256:2eaff27d824980b4d9b2b6cfade5b6a24002915fd4faa1ba47c521331707c176
- Model-admitted engine receipt:
sha256:332e3b1f6df5725ae3e5cf97d9080daa26fd501e7138f1b9d1c7b4837469c545
- Receipt path:
receipts/engine/model_admitted/SFT-QUANTUM-LIMITS-001-332e3b1f6df5725a.json

30. Derivation 22: Unbounded finite Fold quantum fault-tolerance law

Claim: SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002

30.1 Scientific necessity and exact theorem

The later V2 result removes the fixed-order frontier and requires a constructive depth-independent theorem for every supplied positive finite fault allowance.

The engine-admitted theorem is:

For every supplied positive finite fault-order trace t , the unique first repetition width correcting every mask through t is $2t+1$: at least $t+1$ source labels remain against at most t changes, every width through $2t$ has a constructive tie or wrong-majority counterexample, redundancy is exactly $2t$, and the successor fault order adds exactly two labels. The corrected word composes with exact Fold circuit semantics. This is an unbounded finite fault-order theorem, not a stochastic hardware-rate or physical threshold constant.

The derivation cites only these earlier model-admitted receipts:

- SFT-QUANTUM-ERROR-CORRECTION-001
- SFT-QUANTUM-FAULT-TOLERANCE-001
- SFT-QUANTUM-CIRCUIT-001
- SFT-COMP-CPLX-ARBITRARY-CIRCUIT-LOWER-BOUND-002

30.2 Generated grammar and exact boundary

Strict-majority preservation forces the width: t changed positions require strictly more than t retained source positions, so the least total is $t+(t+1)=2t+1$. The generated counterexample at every predecessor width proves minimality, and the fault-order successor proves unbounded positive-finite closure.

Generation rule: Generate the literal product of fault order, width, mask coverage, minimality, recovery, circuit composition, successor and no-extra-parameter coordinates.

Every supplied positive finite Fold fault-order trace, its generated repetition widths in increasing order, every fault mask through that order by exact subset class, and composition with admitted finite Fold circuit semantics.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
fault_order	fixed-one-error-model	fixed-one-error-model removes the required exact fault_order structure.	supplied-positive-finite-fault-trace	supplied-positive-finite-fault-trace retains the required exact fault_order structure.
width	selected-or-even-width	selected-or-even-width removes the required exact width structure.	first-strict-majority-width-2t-plus-1	first-strict-majority-width-2t-plus-1 retains the required exact width structure.
coverage	sampled-fault-patterns	sampled-fault-patterns removes the required exact coverage structure.	all-masks-through-t-by-subset-count	all-masks-through-t-by-subset-count retains the required exact coverage structure.
minimality	shorter-widths-unchecked	shorter-widths-unchecked removes the required exact minimality structure.	counterexample-for-every-width-through-2t	counterexample-for-every-width-through-2t retains the required exact minimality structure.
recovery	decoded-label-without-record	decoded-label-without-record removes the required exact recovery structure.	exact-syndrome-and-recovery-trace	exact-syndrome-and-recovery-trace retains the required exact recovery structure.
circuit	isolated-code-example	isolated-code-example removes the required exact circuit structure.	corrected-word-feeds-exact-circuit-semantics	corrected-word-feeds-exact-circuit-semantics retains the required exact circuit structure.
generality	fixed-t-table	fixed-t-table removes the required exact generality structure.	fault-order-successor-and-ceiling-defeat	fault-order-successor-and-ceiling-defeat retains the required exact generality structure.
addition	imported-hardware-threshold	imported-hardware-threshold removes the required exact addition structure.	no-rate-device-or-noise-parameter	no-rate-device-or-noise-parameter retains the required exact addition structure.

Shared claim-record clause QUANT-S001 **applies**. The full, unchanged clause is preserved in the shared-clause appendix.

30.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause QUANT-S002 applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	fixed-one-error-model removes the required exact fault_order structure.
64	selected-or-even-width removes the required exact width structure.
32	sampled-fault-patterns removes the required exact coverage structure.
16	shorter-widths-unchecked removes the required exact minimality structure.
8	decoded-label-without-record removes the required exact recovery structure.
4	isolated-code-example removes the required exact circuit structure.
2	fixed-t-table removes the required exact generality structure.
1	imported-hardware-threshold removes the required exact addition structure.

The unique all-preserving member is supplied-positive-finite-fault-trace__first-strict-majority-width-2t-plus-1__all-masks-through-t-by-subset-count__counterexample-for-every-width-through-2t__exact-syndrome-and-recovery-trace__corrected-word-feeds-exact-circuit-semantics fault-order-successor-and-ceiling-defeat no-rate-device-or-noise-parameter.

For every supplied positive finite fault-order trace t , the unique first repetition width correcting every mask through t is $2t+1$: at least $t+1$ source labels remain against at most t changes, every width through $2t$ has a constructive tie or wrong-majority counterexample, redundancy is exactly $2t$, and the successor fault order adds exactly two labels. The corrected word composes with exact Fold circuit semantics. This is an unbounded finite fault-order theorem, not a stochastic hardware-rate or physical threshold constant.

Shared claim-record clause QUANT-S004 applies. The full, unchanged clause is preserved in the shared-clause appendix.

30.4 Operational laws and consequences

- Width law: the first strict-majority carrier at fault order t contains exactly $2t+1$ held positions.
- Correction law: at most t changes leave at least $t+1$ original labels, so the unique strict majority is retained.
- Minimality law: every width w through $2t$ is defeated by changing $\text{ceil}(w/2)$ positions, producing a tie or wrong majority.
- Successor law: replacing t by its next generated fault order adds exactly two carrier positions and defeats every fixed positive finite ceiling.
- Composition law: recovery supplies the exact source word to the already-admitted reversible circuit semantics.

Shared claim-record clause QUANT-S005 applies. The full, unchanged clause is preserved in the shared-clause appendix.

30.5 Depth-independent closure

Base: At the first positive fault order, widths one and two have explicit failure rows and width three retains two source labels against one change.

Successor: If width $2t+1$ is the first survivor at fault order t , the next order requires one further possible change and one further retained source majority; exactly two added positions produce $2(t+1)+1$, while every shorter successor width has the same constructive counterexample.

Shared claim-record clause QUANT-S007 applies. The full, unchanged clause is preserved in the shared-clause appendix.

30.6 Executed witnesses and adverse controls

- unbounded-fault-witness-1 - one-error order forces width three for SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002; result: PASS.
- unbounded-fault-witness-2 - separate two- and three-error orders force widths five and seven for SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002; result: PASS.
- unbounded-fault-witness-3 - fault order fourteen forces width twenty-nine for SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002; result: PASS.

- unbounded-fault-witness-4 - every mask through orders one, two and three decodes exactly for SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002; result: PASS.
- unbounded-fault-witness-5 - every predecessor width through fault order fourteen has a constructive failure for SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002; result: PASS.
- unbounded-fault-witness-6 - the fault-order successor adds exactly two labels for SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002; result: PASS.
- unbounded-fault-witness-7 - every supplied finite ceiling has a generated successor fault order for SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002; result: PASS.
- unbounded-fault-witness-8 - all depth-seven words round-trip at fault order fourteen before circuit evaluation for SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002; result: PASS.

Control	Expected	Observed	Result
false_promise	reject incomplete state support	fixed-one-error-model removes the required exact fault_order structure.	PASS
tampered_source	reject changed source identity	changed source differs	PASS
tampered_artifact	reject altered survivor support	one literal survivor	PASS
boundary	halt complex amplitude, stochastic collapse or imported quantum postulate	no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinite-width code or completed infinity; no measured error rate, hardware threshold, fitted noise model or device parameter; no application output or physical benchmark may select the code law	PASS

30.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: quantum error correction, fault tolerance, threshold theorem boundary.

The theorem is unbounded over supplied positive finite code fault orders. It does not estimate a stochastic physical device threshold, correlated hardware noise rate or thermodynamic implementation cost.

The following imports remain prohibited at this boundary:

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no complex amplitude, Hilbert-space axiom or stochastic collapse postulate
- no completed infinite-width code or completed infinity
- no measured error rate, hardware threshold, fitted noise model or device parameter
- no application output or physical benchmark may select the code law

30.8 Exact evidence identities

- Source manifest: sha256:46d41cc55c7f4d0fba44e1a189bebcab0183453b75f53b24e8fd9a8959e6996e
- Complete-census certificate:
sha256:64561c32af9d020a6344cf3c2b92a2a332ee102671e713ead0acfe8e10bdc58b
- Derivation seal: sha256:3f4c7dd00b70b80685d2182f1c66afb4d2ed6ff33d7fb2e1e0c574afeaffd2e3
- Independent implementation:
sha256:08f284371749dfe9c00b2f28d468d9006d7cebfccd62ec6d6fa8a54b4d1b45d2
- Independent certificate:
sha256:8260f560843df345b17ec1cc8d7ff891a3486f59dfcdf9d28fa9a1313e9fed93
- External validation:
sha256:64d49f73d9b50bee7174034718386bbe89ed8d292ee33fe37580a61bdf55acf3

- Model-admitted engine receipt:
sha256:97e061e60929c595adbc3706113e920f80a7b9ea224b3606893edd4c244d445b
- Receipt path: receipts/engine/model_admitted/SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002-97e061e60929c595.json

31. Branch synthesis

The 22 laws establish one operational machine family in which classical and quantum modes share descriptions, traces, resources and verification. Quantum distinctions arise from complete branch support, held phase action, predecessor merging and nonfactorable composition; they are not imported as a separate mathematical universe.

32. Complete V1/V2 reconstruction audit

The categorical-owner audit reads all 763 V1/V2 census entries. It identifies 9 formal Quantum Computation source entries, decomposes them into 29 obligations, closes 29 at same strength and leaves 0 open. V1's quantum-named natural phenomena are not omitted: they are explicitly assigned to Physics because physical quantum effects and measured constants are not formal computation laws.

Source	Entry	Atomic obligation	Disposition	V3 receipt
V2	351	Same-strength reconstruction of Fold quantum complexity law: The forced quantum computational resources kernel retains canonical input size; gate count; causal depth; live branch support; query and communication rows; measurement records; lower and upper witness boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-COMPLEXITY-001
V2	372	Same-strength reconstruction of Fold quantum algorithm-family law: The forced quantum algorithms kernel retains complete input support; reversible branch generation; phase action; interference merge; observation decoder; correctness trace per branch; classical comparison boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-ALGORITHMS-001
V2	397	Same-strength reconstruction of Complete Fold reversible-computation model: The forced reversible computation kernel retains canonical configurations; bijective transition rows; retained held labels; exact inverse traces; universal reversible interpretation; complete resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-REVERSIBLE-MODEL-001

Source	Entry	Atomic obligation	Disposition	V3 receipt
V2	397	Same-strength reconstruction of Fold quantum information-unit law: The forced quantum information units kernel retains one Fold distinction; two held fibre labels; complete word support; exact observation classes; retained phase-label carrier; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-INFORMATION-UNIT-001
V2	397	Same-strength reconstruction of Fold quantum state-composition law: The forced quantum state composition kernel retains complete branch support; exact pair-cell joint words; component identities; phase-label composition; marginal reconstruction; product and correlated classes; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-STATE-COMPOSITION-001
V2	397	Same-strength reconstruction of Superposition-equivalent Fold support law: The forced superposition-equivalent structure kernel retains complete b^k Fold-word support; one record per branch; held phase label per branch; no selected actual branch before observation; exact support provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-SUPERPOSITION-001
V2	397	Same-strength reconstruction of Fold phase and interference operation law: The forced phase and interference kernel retains period-b held phase cycle; reversible phase action; total branch-to-image relation; complete predecessor fibres; same-image merge ledger; retained phase provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-PHASE-INTERFERENCE-001
V2	397	Same-strength reconstruction of Fold entangling-composition law: The forced entanglement kernel retains joint word support; exact marginals; product-support test; nonfactorable pair-cell classes; retained component observations; compositional trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-ENTANGLEMENT-001

Source	Entry	Atomic obligation	Disposition	V3 receipt
V2	397	Same-strength reconstruction of Fold quantum measurement-semantics law: The forced measurement kernel retains total observation relation; retained observation class; closed branch distinctions; complete pre-observation record; selected-class state; exact reconstruction boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-MEASUREMENT-001
V2	398	Same-strength reconstruction of Fold reversible transformation and gate law: The forced quantum transformations and gates kernel retains bijective branch-word map; period-held phase action; complete domain and image support; exact inverse; source-bound gate identity; composition interface; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-GATE-001
V2	398	Same-strength reconstruction of Fold quantum circuit syntax and semantics law: The forced quantum circuits kernel retains generated wires; information units; reversible gates; causal circuit order; complete branchwise evaluation; inverse circuit; observation boundary; resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-CIRCUIT-001
V2	398	Same-strength reconstruction of Fold quantum universality law: The forced quantum universal computation kernel retains frozen gate-description grammar; exact encoded circuits; one interpreter; branchwise trace simulation; phase and observation preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-UNIVERSALITY-001

Source	Entry	Atomic obligation	Disposition	V3 receipt
V2	398	Same-strength reconstruction of Fold quantum algorithm-family law: The forced quantum algorithms kernel retains complete input support; reversible branch generation; phase action; interference merge; observation decoder; correctness trace per branch; classical comparison boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-ALGORITHMS-001
V2	398	Same-strength reconstruction of Fold quantum complexity law: The forced quantum computational resources kernel retains canonical input size; gate count; causal depth; live branch support; query and communication rows; measurement records; lower and upper witness boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-COMPLEXITY-001
V2	399	Same-strength reconstruction of Fold quantum communication law: The forced quantum communication kernel retains sender and receiver supports; joint state composition; local transformations; channel relation; retained correlations; observation records; resource and loss ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-COMMUNICATION-001
V2	399	Same-strength reconstruction of Fold quantum coding law: The forced quantum coding kernel retains logical branch support; redundant joint words; reversible encoder; declared error actions; disjoint syndrome supports; decoder and reconstruction trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-CODING-001
V2	399	Same-strength reconstruction of Fold quantum error-correction law: The forced quantum error correction kernel retains forced repetition width $2t+1$; exact fault-depth trace; all error masks through t ; majority held-label decoder; syndrome ledger; recovery certificate; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-ERROR-CORRECTION-001

Source	Entry	Atomic obligation	Disposition	V3 receipt
V2	399	Same-strength reconstruction of Fold quantum fault-tolerance law: The forced fault-tolerant quantum computation kernel retains encoded logical support; declared component faults; transversal or contained transformations; recovery after each location; malignant-fault boundary; induction over circuit locations; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-FAULT-TOLERANCE-001
V2	400	Same-strength reconstruction of Fold quantum simulation law: The forced quantum simulation kernel retains encoded target state support; reversible update law; phase and interference traces; observation map; exact simulator correspondence; resource ledger; falsification boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-SIMULATION-001
V2	400	Same-strength reconstruction of Fold quantum verification law: The forced quantum verification kernel retains claim and witness supports; verifier circuit; challenge schedule; branchwise acceptance; soundness and completeness ledgers; measurement records; tamper controls; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-VERIFICATION-001
V2	400	Same-strength reconstruction of Fold quantum learning law: The forced quantum learning kernel retains classical or joint example support; reversible hypothesis process; phase/interference search; observation decoder; retained alternatives; sample and gate resources; sealed evaluation; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-LEARNING-001

Source	Entry	Atomic obligation	Disposition	V3 receipt
V2	400	Same-strength reconstruction of Full operational classical-quantum correspondence law: The forced classical and quantum operational correspondence kernel retains common generated descriptions; embedding of classical states; reversible classical submodel; branchwise quantum execution; measurement decoder; bidirectional result preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001
V2	400	Same-strength reconstruction of Limits of Fold quantum computation law: The forced limits of quantum computation kernel retains classical computability dependency; finite description grammar; exact quantum execution; self-description support; halting and undecidability transfer; resource and observation boundaries; no hypercomputational oracle; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-LIMITS-001
V2	402	The complete b-way quantum support successor is reconstructed at every supplied finite depth.	closed	SFT-QUANTUM-SUPERPOSITION-001
V2	402	Quantum circuit support, size, width and depth retain exact base/successor resource recurrences.	closed	SFT-QUANTUM-CIRCUIT-001, SFT-QUANTUM-COMPLEXITY-001
V2	402	Each fresh quantum observation or reversible Fold layer adds its exact retained inverse record.	closed	SFT-QUANTUM-REVERSIBLE-MODEL-001, SFT-QUANTUM-MEASUREMENT-001
V2	403	Same-strength reconstruction of Fold quantum error-correction law: The forced quantum error correction kernel retains forced repetition width $2t+1$; exact fault-depth trace; all error masks through t ; majority held-label decoder; syndrome ledger; recovery certificate; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-ERROR-CORRECTION-001

Source	Entry	Atomic obligation	Disposition	V3 receipt
V2	403	Same-strength reconstruction of Fold quantum fault-tolerance law: The forced fault-tolerant quantum computation kernel retains encoded logical support; declared component faults; transversal or contained transformations; recovery after each location; malignant-fault boundary; induction over circuit locations; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.	closed	SFT-QUANTUM-FAULT-TOLERANCE-001
V2	407	Same-strength reconstruction of Unbounded finite Fold quantum fault-tolerance law: For every supplied positive finite fault-order trace t , the unique first repetition width correcting every mask through t is $2t+1$: at least $t+1$ source labels remain against at most t changes, every width through $2t$ has a constructive tie or wrong-majority counterexample, redundancy is exactly $2t$, and the successor fault order adds exactly two labels. The corrected word composes with exact Fold circuit semantics. This is an unbounded finite fault-order theorem, not a stochastic hardware-rate or physical threshold constant.	closed	SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002

The audit also corrects the former wording around 'unlimited thresholds'. The current theorem is unbounded over supplied positive finite code fault orders. A measured stochastic hardware threshold is a different empirical object; it is neither denied nor claimed here and must be derived and tested in its correct Physics or engineering branch.

33. Quantum knowledge, open science and institutional accountability

Quantum science is especially vulnerable to authority by opacity: proprietary devices, inaccessible calibration data, result-only cloud interfaces and mathematical notation presented without the full route from premise to observation. Such systems may be useful instruments, but a reported score or vendor label cannot close a law when reviewers cannot reconstruct the support, transformation, measurement record, alternatives and failure boundary. This release exposes each of those objects.

The institutional critique is evidential, not an allegation that every funded scientist or institution acts in bad faith. Reviews report associations between commercial sponsorship and favourable outcomes, influence on research agendas, limits in grant-review reliability and fairness, and underpublication of null results. Capital, prestige, paywalls and consensus can therefore shape what becomes visible; none substitutes for an inspectable branch trace and independently reproducible receipt.

Maria Smith developed this programme without conventional academic credentials, appointment or a funding gate. That biography proves no quantum theorem; the public derivations must bear the whole scientific burden. It demonstrates why credentials cannot be permission to create knowledge and why every excluded mind is a scientific loss. Expert criticism is invited on the same evidential terms as outsider criticism.

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34. Reproduction and invalidation

Run the repository-wide one-command verifier. A reviewer can invalidate a result by producing an omitted same-boundary coordinate, a second survivor, a nonminimal form, an induction counterexample, a failed inverse, an incomplete observation record, an uncorrected registered error mask or a mismatched receipt.

35. Scope and next branch

The next branch is Physics. It must test whether independently derived Fold relations correspond to mechanics, fields, spacetime, thermodynamics, quantum phenomena, gravitation, matter, waves, fluids, plasmas and condensed structures. Formal quantum computation cannot select the natural laws; sealed observation must do that work.

36. Complete-field execution - version 1.4

Version 1.3 froze the complete-field roadmap. Version 1.4 executes it. The 22 extensive foundational derivations above remain part of this paper; this section exposes every current full-field obligation, law, unique survivor, observation, boundary and receipt. Hashes identify evidence but never substitute for the scientific result.

36.1 Complete family census

Order	Family	Scientific scope	Claims	Candidates	Controls	Status
1	BASE	reversible and quantum computational foundation	22	5,632	88	complete, receipt-backed, extension-open
2	REXX	complete reversible computation	18	4,608	72	complete, receipt-backed, extension-open
3	QSTATEX	quantum information units, states, support and composition	28	7,168	112	complete, receipt-backed, extension-open
4	GATEX	transformations, gates, circuits, compilation and universality	22	5,632	88	complete, receipt-backed, extension-open
5	QALGX	quantum algorithms and correctness	30	7,680	120	complete, receipt-backed, extension-open
6	QCPLXX	quantum resources, complexity and bounds	26	6,656	104	complete, receipt-backed, extension-open
7	QCOMM	quantum channels, communication and cryptography	24	6,144	96	complete, receipt-backed, extension-open
8	QCODEX	quantum coding, error correction and fault tolerance	32	8,192	128	complete, receipt-backed, extension-open
9	QSIMX	quantum simulation and verification	24	6,144	96	complete, receipt-backed, extension-open
10	QLEARNX	quantum learning and intelligence	22	5,632	88	complete, receipt-backed, extension-open
11	QLIMITX	classical-quantum correspondence and computational limits	22	5,632	88	complete, receipt-backed, extension-open
12	VALID	dated full-field validation and Grand Lock	12	3,072	48	complete, receipt-backed, extension-open
13	HAND	single-owner downstream handoffs and open extension	6	1,536	24	complete, receipt-backed, extension-open
Total	13 families	complete frozen field census	288	73,728	1,152	288/288

36.2 Admission constitution applied to every obligation

Each entry below reports the registered scientific question, forced law, unique survivor, exhaustive enumeration, adverse controls, observation, source custody, prohibited imports, dependency route and immutable receipt. The machine-readable packages are controlling and independently replayable.

36.3 Family BASE - 22/22 complete

Scope. Reversible and quantum computational foundation. This family decides 5,632 candidates, retains 22 unique survivors and passes 88 adverse controls.

SFT-QUANTUM-OBL-BASE-001 - Complete Fold reversible-computation model

- **Claim and ownership:** SFT-QUANTUM-REVERSIBLE-MODEL-001; Quantum Computation family BASE; Complete Fold reversible-computation model.

- **Forced law:** The forced reversible computation kernel retains canonical configurations; bijective transition rows; retained held labels; exact inverse traces; universal reversible interpretation; complete resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise. Exact result: The forced reversible computation kernel retains canonical configurations; bijective transition rows; retained held labels; exact inverse traces; universal reversible interpretation; complete resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; This theorem is formal and does not assign thermodynamic energy without a later physical measurement law..
- **Evidence identities:** derivation
sha256:37e9625bca1b99f14969057b963782c6b1cb62efb98ec317178cb32901e2f48c; independent implementation sha256:088141be2c397e4c3f7e640dbc6ba3850bc936eb3433537b058e95a25e98e2df; independent certificate
sha256:33e527c37a50844de7b1b6228f92b22f4eb6a5279fe576a97b952c28885c5ed1; measurement preserved in package; engine receipt
sha256:068b389a38d47c06118600324bc263c951e6e19c6f591f257c601f6125525298 at receipts/engine/model_admitted/SFT-QUANTUM-REVERSIBLE-MODEL-001-068b389a38d47c06.json.

SFT-QUANTUM-OBL-BASE-002 - Fold quantum information-unit law

- **Claim and ownership:** SFT-QUANTUM-INFORMATION-UNIT-001; Quantum Computation family BASE; Fold quantum information-unit law.
- **Forced law:** The forced quantum information units kernel retains one Fold distinction; two held fibre labels; complete word support; exact observation classes; retained phase-label carrier; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise. Exact result: The forced quantum information units kernel retains one Fold distinction; two held fibre labels; complete word support; exact observation classes; retained phase-label carrier; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach

- **Dependency and root trace:** SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; The unit is structural support, not an imported complex vector or conventional qubit..
- **Evidence identities:** derivation
`sha256:8dd1b7f8a5b144885d8926f1d724e9f1ecc767f71ce3110a1884d03297673cdb`; independent implementation
`sha256:3a82e4d5098bfd4d453af17516b932f696f816a15c6bc718253472c762d0ee10`; independent certificate
`sha256:c7ca84cd3a8c47cd8a5e345697f2648d3861372f3c6a72c96a1288ce660f8ef5`; measurement preserved in package; engine receipt
`sha256:3cae970ebb8336d464bcbcb64e17d0f50049da6f018b95f98b53b04a46873bf72` at `receipts/engine/model_admitted/SFT-QUANTUM-INFORMATION-UNIT-001-3cae970ebb8336d4.json`.

SFT-QUANTUM-OBL-BASE-003 - Fold quantum state-composition law

- **Claim and ownership:** SFT-QUANTUM-STATE-COMPOSITION-001; Quantum Computation family BASE; Fold quantum state-composition law.
- **Forced law:** The forced quantum state composition kernel retains complete branch support; exact pair-cell joint words; component identities; phase-label composition; marginal reconstruction; product and correlated classes; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`. Exact result: The forced quantum state composition kernel retains complete branch support; exact pair-cell joint words; component identities; phase-label composition; marginal reconstruction; product and correlated classes; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum;

no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Only finite generated joint supports are admitted..

- **Evidence identities:** derivation

sha256:07accac25aaa131af96d31bdce98536c3ff39c0c0504a69b146f4a02240441c5; independent implementation sha256:45b6bcd57a43ea420e30db2d983e862b2206a61b2483270be469b035b5263ee0; independent certificate sha256:47fa45ac826615121aee5d2ea41548aebd01dbac38c96e8e2ee8b359cb507e4; measurement preserved in package; engine receipt sha256:a358610587ee2c7c6dfcba8a782e46f89a1668e2c86739eac0e8c88d6af719c2 at receipts/engine/model_admitted/SFT-QUANTUM-STATE-COMPOSITION-001-a358610587ee2c7c.json.

SFT-QUANTUM-OBL-BASE-004 - Superposition-equivalent Fold support law

- **Claim and ownership:** SFT-QUANTUM-SUPERPOSITION-001; Quantum Computation family BASE; Superposition-equivalent Fold support law.
- **Forced law:** The forced superposition-equivalent structure kernel retains complete b^k Fold-word support; one record per branch; held phase label per branch; no selected actual branch before observation; exact support provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise. Exact result: The forced superposition-equivalent structure kernel retains complete b^k Fold-word support; one record per branch; held phase label per branch; no selected actual branch before observation; exact support provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No amplitude magnitude, normalization constant or stochastic ontology is imported..
- **Evidence identities:** derivation sha256:3d8ee27d90c7ddfea4b88ece679df71f39ba1602da046edef08f8e3ed4a8bcdd; independent implementation sha256:945ffdb45c2324c21427547d49b7e14f3e31aeaca26deee602a921d72088a911; independent certificate sha256:0de2881ba9754d71a5ad9d18c7283eafe2a63e31b131dc545c7234b8b668bc63; measurement preserved in package; engine receipt sha256:54c09712a4a53c16054fc2c8de1e247007b718e1f6f0cd9cd38a14b1a5cbe7d1 at receipts/engine/model_admitted/SFT-QUANTUM-SUPERPOSITION-001-54c09712a4a53c16.json.

SFT-QUANTUM-OBL-BASE-005 - Fold phase and interference operation law

- **Claim and ownership:** SFT-QUANTUM-PHASE-INTERFERENCE-001; Quantum Computation family BASE; Fold phase and interference operation law.
- **Forced law:** The forced phase and interference kernel retains period-b held phase cycle; reversible phase action; total branch-to-image relation; complete predecessor fibres; same-image merge ledger; retained phase provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise. Exact result: The forced phase and interference kernel retains period-b held phase cycle; reversible phase action; total branch-to-image relation; complete predecessor fibres; same-image merge ledger; retained phase provenance; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-SUPERPOSITION-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Interference is exact predecessor merging with phase labels; no complex exponential is assumed..
- **Evidence identities:** derivation
 sha256:da60087f0965652227de2f3c71b7584d22e2f16a2b794c4a3fa4152bc27d186a; independent implementation sha256:5b0728c303020c2658658987b7794ae08beb9b00f6e2f8c6b48c1cc5251db79b; independent certificate
 sha256:09d8b5b79e90ad34e20b35f399aafd7bc0c8948366962366431e8411290575a0; measurement preserved in package; engine receipt
 sha256:c1c7e3f2c71ea7de6d801c397cff01ae8a005651deddccb7acd83bddc857f12 at receipts/engine/model_admitted/SFT-QUANTUM-PHASE-INTERFERENCE-001-c1c7e3f2c71ea7de.json.

SFT-QUANTUM-OBL-BASE-006 - Fold entangling-composition law

- **Claim and ownership:** SFT-QUANTUM-ENTANGLEMENT-001; Quantum Computation family BASE; Fold entangling-composition law.
- **Forced law:** The forced entanglement kernel retains joint word support; exact marginals; product-support test; nonfactorable pair-cell classes; retained component observations; compositional trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise. Exact result: The forced entanglement kernel retains joint word support;

exact marginals; product-support test; nonfactorable pair-cell classes; retained component observations; compositional trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Entanglement is structural nonfactorability of registered joint support, not a metaphysical postulate..
- **Evidence identities:** derivation
`sha256:d7fe096c012e1467e6d8253f0d04d02de4bc66f76b479cca2190386dc4298cec`; independent implementation `sha256:ebd4b53c6a0c5b77e9d0f9fc9f7f031c0d9f1f9a77ba9fd886f9153df0e39966`; independent certificate
`sha256:cfeb4e7a31d71a6197bc757243682f2fa66e4ce558a1284d2d35c0fd76a7dc43`; measurement preserved in package; engine receipt
`sha256:6bfa64c6f896c5f7eaa3a55872f31cd487923853353a3069ec7a839afa1d3d0d` at `receipts/engine/model_admitted/SFT-QUANTUM-ENTANGLEMENT-001-6bfa64c6f896c5f7.json`.

SFT-QUANTUM-OBL-BASE-007 - Fold quantum measurement-semantics law

- **Claim and ownership:** SFT-QUANTUM-MEASUREMENT-001; Quantum Computation family BASE; Fold quantum measurement-semantics law.
- **Forced law:** The forced measurement kernel retains total observation relation; retained observation class; closed branch distinctions; complete pre-observation record; selected-class state; exact reconstruction boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`. Exact result: The forced measurement kernel retains total observation relation; retained observation class; closed branch distinctions; complete pre-observation record; selected-class state; exact reconstruction boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.

- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No stochastic collapse postulate is imported; outcome frequencies require external empirical evidence..
- **Evidence identities:** derivation
`sha256:7d99cd7ce254a64f69ce210b7b3883177227978c792faeeb846094c4856dc9fa`; independent implementation `sha256:09615ec981ffda13c0fd1ea5163e580708612ce2202d5b23d98d1c3b806d66c7`; independent certificate
`sha256:59226eca7492c0aebb48b4ab677e9b24e1bcfc38d8438d37385b00ed1ab13d1f`; measurement preserved in package; engine receipt
`sha256:e29a71f7dff97835e28850ee8d352ac80aa9ceabb4399e07105d892175776a06` at `receipts/engine/model_admitted/SFT-QUANTUM-MEASUREMENT-001-e29a71f7dff97835.json`.

SFT-QUANTUM-OBL-BASE-008 - Fold reversible transformation and gate law

- **Claim and ownership:** SFT-QUANTUM-GATE-001; Quantum Computation family BASE; Fold reversible transformation and gate law.
- **Forced law:** The forced quantum transformations and gates kernel retains bijective branch-word map; period-held phase action; complete domain and image support; exact inverse; source-bound gate identity; composition interface; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`. Exact result: The forced quantum transformations and gates kernel retains bijective branch-word map; period-held phase action; complete domain and image support; exact inverse; source-bound gate identity; composition interface; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-MEASUREMENT-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Physical gate fidelity remains an empirical engineering question..
- **Evidence identities:** derivation
`sha256:d93e94a7c4c89f79904d0068dc37fb48db7040bb45533a02cb96f8c45cb617c1`; independent implementation `sha256:629fa6869c3efd87e733736338d25b83adf646fe360ce5210876449f86aa18d5`; independent certificate

```
sha256:8c737f9cb9e579c043660fde438e8405e4f6288e9d65f74e470be69d59549c89; measurement
preserved in package; engine receipt
sha256:495704206fd6c701e1baf8d1ec0b4bcfd430a002aea0bb76453a091e01ccfdaf at
receipts/engine/model_admitted/SFT-QUANTUM-GATE-001-495704206fd6c701.json.
```

SFT-QUANTUM-OBL-BASE-009 - Fold quantum circuit syntax and semantics law

- **Claim and ownership:** SFT-QUANTUM-CIRCUIT-001; Quantum Computation family BASE; Fold quantum circuit syntax and semantics law.
- **Forced law:** The forced quantum circuits kernel retains generated wires; information units; reversible gates; causal circuit order; complete branchwise evaluation; inverse circuit; observation boundary; resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise. Exact result: The forced quantum circuits kernel retains generated wires; information units; reversible gates; causal circuit order; complete branchwise evaluation; inverse circuit; observation boundary; resource ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-GATE-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; This derives circuit semantics; hardware timing and noise are external..
- **Evidence identities:** derivation

```
sha256:8fb9825eeec2daff4511f6aaf81f866fe575a23d070bbfbbe9d4d5d59837990; independent
implementation sha256:1d370c1f6d10c8b50b5f4ce8980a2557dead4382a9a3565990df366ab519fdd6;
independent certificate
sha256:613d70fcb5954bd320bae00fe784bb03726d6ab1b3cc0c287a6df616d4746618; measurement
preserved in package; engine receipt
sha256:63e751cf9ab382cdc5c4b5dfd0c6c2f01ad90659bf2c9189357f141475d9e8aa at
receipts/engine/model_admitted/SFT-QUANTUM-CIRCUIT-001-63e751cf9ab382cd.json.
```

SFT-QUANTUM-OBL-BASE-010 - Fold quantum universality law

- **Claim and ownership:** SFT-QUANTUM-UNIVERSALITY-001; Quantum Computation family BASE; Fold quantum universality law.
- **Forced law:** The forced quantum universal computation kernel retains frozen gate-description grammar; exact encoded circuits; one interpreter; branchwise trace simulation; phase and observation preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

- **Unique surviving structure:** complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise. Exact result: The forced quantum universal computation kernel retains frozen gate-description grammar; exact encoded circuits; one interpreter; branchwise trace simulation; phase and observation preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-CIRCUIT-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Universality is exact for every generated description in the frozen grammar..
- **Evidence identities:** derivation
sha256:7f480fe1655f32f7ff3f640e9e0f508c501d21b0b44430f24ff41fcc84d35c0a; independent implementation sha256:00dc7bb66a94f5a7f0ced9c5b08dea0b489c312288e3cc0bf250f86bc8a7d89f; independent certificate
sha256:0d501674df0431ac2f24be59fafd160e2c225807ab1b000013632f09ddcd1214; measurement preserved in package; engine receipt
sha256:28b6a73e02450c70bf3ee49f3fcdcf2e28b30c463e05218b99cdc8cda723e86a at receipts/engine/model_admitted/SFT-QUANTUM-UNIVERSALITY-001-28b6a73e02450c70.json.

SFT-QUANTUM-OBL-BASE-011 - Fold quantum algorithm-family law

- **Claim and ownership:** SFT-QUANTUM-ALGORITHMS-001; Quantum Computation family BASE; Fold quantum algorithm-family law.
- **Forced law:** The forced quantum algorithms kernel retains complete input support; reversible branch generation; phase action; interference merge; observation decoder; correctness trace per branch; classical comparison boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise. Exact result: The forced quantum algorithms kernel retains complete input support; reversible branch generation; phase action; interference merge; observation decoder; correctness trace per branch; classical comparison boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-UNIVERSALITY-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 ->

SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 ->
 SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 ->
 SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream
 receipts. Imported axioms: 0; free parameters: 0.

- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status:
 independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no
 complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum;
 no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Named
 speedups require their own exact problem family and resource proof..
- **Evidence identities:** derivation
 sha256:cb94fae7c67d50383d283b028c580568c6d7e1fafb4a919217df581529703a87; independent
 implementation sha256:89bb200d9b5c23e863521d13d47d326f8df6f7e4ace5605cd4e62929121a8065;
 independent certificate
 sha256:8eac37074eb7bcf0579cb97aedc91fa70d056d0828515070ae6df1f157b8d22c; measurement
 preserved in package; engine receipt
 sha256:a33b13375d76ebf039fcc5711c0810ab99ac0b9456295bc7881d76c94558a86b at
 receipts/engine/model_admitted/SFT-QUANTUM-ALGORITHMS-001-a33b13375d76ebf0.json.

SFT-QUANTUM-OBL-BASE-012 - Fold quantum complexity law

- **Claim and ownership:** SFT-QUANTUM-COMPLEXITY-001; Quantum Computation family BASE; Fold quantum
 complexity law.
- **Forced law:** The forced quantum computational resources kernel retains canonical input size; gate count; causal depth; live
 branch support; query and communication rows; measurement records; lower and upper witness boundary; branch support,
 phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported
 amplitudes or postulates.
- **Unique surviving structure:** complete-generated-branch-support__one-Fold-distinction-unit__
 complete-pair-cell-joint-support__period-held-phase-action__total-reversible-sourc
 e-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certif
 icate__no-extra-quantum-premise. Exact result: The forced quantum computational resources kernel retains
 canonical input size; gate count; causal depth; live branch support; query and communication rows; measurement records;
 lower and upper witness boundary; branch support, phase, transformation, observation, composition and resources remain
 exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls
 passed; closure is depth_independent. Falsification condition: a second survivor, omitted coordinate, failed control,
 trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-ALGORITHMS-001 ->
 SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 ->
 SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 ->
 SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 ->
 SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream
 receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status:
 independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no
 complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum;
 no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No
 arbitrary separation between conventional complexity classes is claimed without a separate certificate..

- **Evidence identities:** derivation

sha256:2dabda188246d3553502bbfb24cebd18a4513e169b7291aa5ecbf9ca5fd66fe9; independent implementation sha256:81b3d49b18170b958f74bfe4f4a5cbd47bb4b0712c750027a7554d3576523419; independent certificate sha256:94b58ef5a03dc1e832cdf2ee0a5158ca6d8d9626c6d514f939aee795a7d0eee6; measurement preserved in package; engine receipt sha256:d6307d961273c2a383fecdb919c56319b910bccb58dbbcf5402a2a76a9ca8b6 at receipts/engine/model_admitted/SFT-QUANTUM-COMPLEXITY-001-d6307d961273c2a3.json.

SFT-QUANTUM-OBL-BASE-013 - Fold quantum communication law

- **Claim and ownership:** SFT-QUANTUM-COMMUNICATION-001; Quantum Computation family BASE; Fold quantum communication law.
- **Forced law:** The forced quantum communication kernel retains sender and receiver supports; joint state composition; local transformations; channel relation; retained correlations; observation records; resource and loss ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise. Exact result: The forced quantum communication kernel retains sender and receiver supports; joint state composition; local transformations; channel relation; retained correlations; observation records; resource and loss ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Physical transmission capacity and security require registered media experiments..
- **Evidence identities:** derivation sha256:2b321b0b8c520885563b2c770bed8e0ec83e5aa0720a05a2f679125c5d1f63cc; independent implementation sha256:c98029cec76386f0bea9768e3536a822803d2c7fb1eb3e16616272c1e0e97bec; independent certificate sha256:b8460f43cb576317854272866636f450a0a837fd34929ae4b947d681dd95eb91; measurement preserved in package; engine receipt sha256:01059b1cbe86b894ae6137de5625e1dfcd1cd640ac5634c0ca2260977993b97a at receipts/engine/model_admitted/SFT-QUANTUM-COMMUNICATION-001-01059b1cbe86b894.json.

SFT-QUANTUM-OBL-BASE-014 - Fold quantum coding law

- **Claim and ownership:** SFT-QUANTUM-CODING-001; Quantum Computation family BASE; Fold quantum coding law.

- **Forced law:** The forced quantum coding kernel retains logical branch support; redundant joint words; reversible encoder; declared error actions; disjoint syndrome supports; decoder and reconstruction trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise. Exact result: The forced quantum coding kernel retains logical branch support; redundant joint words; reversible encoder; declared error actions; disjoint syndrome supports; decoder and reconstruction trace; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; The code theorem applies only to the registered generated error family..
- **Evidence identities:** derivation
 sha256:ea64d9cb56689b01f32e45805b41e053214e1e665f68f83c9fac52f894205efb; independent implementation sha256:3e3936d7ac16b4f76240d81294e3dab60dc4110611c44429cea46df1444590f2; independent certificate
 sha256:736fdb0c5a526c7b542e4eaf8fdf98681aae185d448844b242b1f371688b7746; measurement preserved in package; engine receipt
 sha256:b79afc51951dfbebb30cd70fba329965b700fa14a16fb45b5754568ec7407a3e at receipts/engine/model_admitted/SFT-QUANTUM-CODING-001-b79afc51951dfbeb.json.

SFT-QUANTUM-OBL-BASE-015 - Fold quantum error-correction law

- **Claim and ownership:** SFT-QUANTUM-ERROR-CORRECTION-001; Quantum Computation family BASE; Fold quantum error-correction law.
- **Forced law:** The forced quantum error correction kernel retains forced repetition width $2t+1$; exact fault-depth trace; all error masks through t ; majority held-label decoder; syndrome ledger; recovery certificate; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise. Exact result: The forced quantum error correction kernel retains forced repetition width $2t+1$; exact fault-depth trace; all error masks through t ; majority held-label decoder; syndrome ledger; recovery certificate; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: a second survivor, omitted coordinate, failed control,

trace mismatch or boundary breach

- **Dependency and root trace:** SFT-QUANTUM-CODING-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; The completed census covers generated fault depths one, two and three and the successor law for every positive finite t..
- **Evidence identities:** derivation
`sha256:61cc113c2d425a9f9c9d2e13bb1a983bd94e88220a6d91a4f1815d41b5059ee6`; independent implementation `sha256:f3e41dd58f96c808ce4cadab817f8a4ee86d6369ef8e6e43ddfaacc792071af9`; independent certificate
`sha256:721e7c933eb6f760790d3ee15cd54171a46f8ffa24c67f5b59e62872b12fb16b`; measurement preserved in package; engine receipt
`sha256:a4791dcbf8753c32e8304c5d5e80f9bf7f21a7356b6adb9422ea60d5eaf59656` at `receipts/engine/model_admitted/SFT-QUANTUM-ERROR-CORRECTION-001-a4791dcbf8753c32.json`.

SFT-QUANTUM-OBL-BASE-016 - Fold quantum fault-tolerance law

- **Claim and ownership:** SFT-QUANTUM-FAULT-TOLERANCE-001; Quantum Computation family BASE; Fold quantum fault-tolerance law.
- **Forced law:** The forced fault-tolerant quantum computation kernel retains encoded logical support; declared component faults; transversal or contained transformations; recovery after each location; malignant-fault boundary; induction over circuit locations; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`. Exact result: The forced fault-tolerant quantum computation kernel retains encoded logical support; declared component faults; transversal or contained transformations; recovery after each location; malignant-fault boundary; induction over circuit locations; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`; all rows preserved: `True`.

- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No hardware threshold constant is claimed; unlimited thresholds remain dependent on a separately derived physical fault model..
- **Evidence identities:** derivation
`sha256:1816e6b491362e7cc5b5d306e833227bdf765559d55693d0b18553e2911f1a57; independent implementation sha256:ef9893933f17387358f0d7512e05451d206c5ce7a11fe348bbc1052bd42ee0dd; independent certificate sha256:0a28a0751ef1e16a701eb0fdf84d8d4a2583c0e4314f9cd1d4badfd2e5a10c7a; measurement preserved in package; engine receipt sha256:9f2526c5bd9f1625a3cee65b0cba4e9a600ecdc9a8031ac51367e35c37308b16 at receipts/engine/model_admitted/SFT-QUANTUM-FAULT-TOLERANCE-001-9f2526c5bd9f1625.json.`

SFT-QUANTUM-OBL-BASE-017 - Fold quantum simulation law

- **Claim and ownership:** SFT-QUANTUM-SIMULATION-001; Quantum Computation family BASE; Fold quantum simulation law.
- **Forced law:** The forced quantum simulation kernel retains encoded target state support; reversible update law; phase and interference traces; observation map; exact simulator correspondence; resource ledger; falsification boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise.` Exact result: The forced quantum simulation kernel retains encoded target state support; reversible update law; phase and interference traces; observation map; exact simulator correspondence; resource ledger; falsification boundary; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; A simulator derives consequences of its registered model and cannot establish a natural quantum law by itself..
- **Evidence identities:** derivation
`sha256:bb1d2c15251c769fbcf1692edc85f6253cb6f103dbffe04680cedb4d492cbfff8; independent implementation sha256:72f00440c8b0cfeaa26940b8aef2728b3e74896b435207b9b8f9426348064df9; independent certificate sha256:fcdb1c83f2a12bee60d57319cf49492523c03723aee2e0aafa22e3a88abfff900; measurement preserved in package; engine receipt sha256:daf61c89738fb9f6370bc244c3a7538a3c4037ab408dfd5d15e3301c57e817a6 at`

receipts/engine/model_admitted/SFT-QUANTUM-SIMULATION-001-daf61c89738fb9f6.json.

SFT-QUANTUM-OBL-BASE-018 - Fold quantum verification law

- **Claim and ownership:** SFT-QUANTUM-VERIFICATION-001; Quantum Computation family BASE; Fold quantum verification law.
- **Forced law:** The forced quantum verification kernel retains claim and witness supports; verifier circuit; challenge schedule; branchwise acceptance; soundness and completeness ledgers; measurement records; tamper controls; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise. Exact result: The forced quantum verification kernel retains claim and witness supports; verifier circuit; challenge schedule; branchwise acceptance; soundness and completeness ledgers; measurement records; tamper controls; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-SIMULATION-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Security and correctness are exact only for the registered prover and resource grammar..
- **Evidence identities:** derivation
sha256:62b1704737e141ef52eab6b9687ec06ab6de072c1ba2dd8cc678880fe50e3ce0; independent implementation sha256:00bb831c0de419ea512cd89dd478e15c3804c348f3b094de0d7321006f9f408f; independent certificate
sha256:fa884fd0ca86e8ae875b76054696f32cd5ef731fbbd1986b9a02b84140ece03a; measurement preserved in package; engine receipt
sha256:6ca70e1a9487b428bc6e04e5e3b1a11907b79d1ed92f7d03a013a49380ecce54 at receipts/engine/model_admitted/SFT-QUANTUM-VERIFICATION-001-6ca70e1a9487b428.json.

SFT-QUANTUM-OBL-BASE-019 - Fold quantum learning law

- **Claim and ownership:** SFT-QUANTUM-LEARNING-001; Quantum Computation family BASE; Fold quantum learning law.
- **Forced law:** The forced quantum learning kernel retains classical or joint example support; reversible hypothesis process; phase/interference search; observation decoder; retained alternatives; sample and gate resources; sealed evaluation; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certif

icate__no-extra-quantum-premise. Exact result: The forced quantum learning kernel retains classical or joint example support; reversible hypothesis process; phase/interference search; observation decoder; retained alternatives; sample and gate resources; sealed evaluation; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; No pretrained quantum model, fitted amplitude or unexplained speedup is admitted..
- **Evidence identities:** derivation
sha256:1d75a66403b8bd208fad0ca8cdd0de260f54865c8eb4e1b1984a25888845b435; independent implementation sha256:de8da3d2eb9a1c0c7cf06c6f851c311161c5e967d182c53c1d2d4f61d54045e3; independent certificate
sha256:301937de6bfe10c6fc95c62a92efa37c27d5866795d316932652ba6dd953bd5d; measurement preserved in package; engine receipt
sha256:c4388e69ff7412fe09e53b7e4b2a4eda12168ff6665ca351758c185ac7e018b7 at receipts/engine/model_admitted/SFT-QUANTUM-LEARNING-001-c4388e69ff7412fe.json.

SFT-QUANTUM-OBL-BASE-020 - Full operational classical-quantum correspondence law

- **Claim and ownership:** SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001; Quantum Computation family BASE; Full operational classical-quantum correspondence law.
- **Forced law:** The forced classical and quantum operational correspondence kernel retains common generated descriptions; embedding of classical states; reversible classical submodel; branchwise quantum execution; measurement decoder; bidirectional result preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise. Exact result: The forced classical and quantum operational correspondence kernel retains common generated descriptions; embedding of classical states; reversible classical submodel; branchwise quantum execution; measurement decoder; bidirectional result preservation; overhead ledger; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-LEARNING-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 ->

SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.

- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Correspondence does not identify the models where phase-sensitive joint support changes the operational trace..
- **Evidence identities:** derivation
`sha256:d10d9d064e2a63254c2cae2d5d7ea26173427927650eb375fbf698f69cf61f86`; independent implementation `sha256:1cd0dd23ea4b5b984d9a1ebe768005d0c09258970fd241aa5a02215cfdae7d2b`; independent certificate
`sha256:74d501babc031290bfc0b5a041bed885d7ccb65d700b20676a8260f75fd91519`; measurement preserved in package; engine receipt
`sha256:e1712589e1aba3ce6536a3ba33d599e06bdf793a335079d1a775975bc234b614` at `receipts/engine/model_admitted/SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001-e1712589e1aba3ce.json`.

SFT-QUANTUM-OBL-BASE-021 - Limits of Fold quantum computation law

- **Claim and ownership:** SFT-QUANTUM-LIMITS-001; Quantum Computation family BASE; Limits of Fold quantum computation law.
- **Forced law:** The forced limits of quantum computation kernel retains classical computability dependency; finite description grammar; exact quantum execution; self-description support; halting and undecidability transfer; resource and observation boundaries; no hypercomputational oracle; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Unique surviving structure:** `complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificate__no-extra-quantum-premise`. Exact result: The forced limits of quantum computation kernel retains classical computability dependency; finite description grammar; exact quantum execution; self-description support; halting and undecidability transfer; resource and observation boundaries; no hypercomputational oracle; branch support, phase, transformation, observation, composition and resources remain exact, finite, source-bound and free of imported amplitudes or postulates.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-INFO-QUANTUM-CORRESPONDENCE-001. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinity or ungenerated continuum; no free, fitted, learned or hardware parameter; no application output or physical measurement may select the law; Quantum computation does not escape the admitted computability and self-reference boundaries; physical claims remain empirical..

- **Evidence identities:** derivation

sha256:4abd56778c0176f6426706806e75fdff866dedbbe2e78345585c0f9d0c6f360f; independent implementation sha256:f1ed114f18db0541db98fc2966784fabca2116c53921fb473b68c2e866a204c5; independent certificate
sha256:e018eb19348a191249d0fa680329d0cf71d1c2e1d62b0734732426f74afb2cf5; measurement preserved in package; engine receipt
sha256:332e3b1f6df5725ae3e5cf97d9080daa26fd501e7138f1b9d1c7b4837469c545 at receipts/engine/model_admitted/SFT-QUANTUM-LIMITS-001-332e3b1f6df5725a.json.

SFT-QUANTUM-OBL-BASE-022 - Unbounded finite Fold quantum fault-tolerance law

- **Claim and ownership:** SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002; Quantum Computation family BASE; Unbounded finite Fold quantum fault-tolerance law.
- **Forced law:** For every supplied positive finite fault-order trace t , the unique first repetition width correcting every mask through t is $2t+1$: at least $t+1$ source labels remain against at most t changes, every width through $2t$ has a constructive tie or wrong-majority counterexample, redundancy is exactly $2t$, and the successor fault order adds exactly two labels. The corrected word composes with exact Fold circuit semantics. This is an unbounded finite fault-order theorem, not a stochastic hardware-rate or physical threshold constant.
- **Unique surviving structure:** supplied-positive-finite-fault-trace__first-strict-majority-width-2t-plus-1__all-masks-through-t-by-subset-count__counterexample-for-every-width-through-2t__exact-syndrome-and-recovery-trace__corrected-word-feeds-exact-circuit-semantics__fault-order-successor-and-ceiling-defeat__no-rate-device-or-noise-parameter. Exact result: For every supplied positive finite fault-order trace t , the unique first repetition width correcting every mask through t is $2t+1$: at least $t+1$ source labels remain against at most t changes, every width through $2t$ has a constructive tie or wrong-majority counterexample, redundancy is exactly $2t$, and the successor fault order adds exactly two labels. The corrected word composes with exact Fold circuit semantics. This is an unbounded finite fault-order theorem, not a stochastic hardware-rate or physical threshold constant.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: a second survivor, omitted coordinate, failed control, trace mismatch or boundary breach
- **Dependency and root trace:** SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-COMP-CPLX-ARBITRARY-CIRCUIT-LOWER-BOUND-002. Registered root: reached through admitted upstream receipts. Imported axioms: 0; free parameters: 0.
- **Observation:** the foundational package preserves its admitted observation and implementation-distinct reconstruction
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no numerical zero, negative, irrational, imaginary or floating proof quantity; no complex amplitude, Hilbert-space axiom or stochastic collapse postulate; no completed infinite-width code or completed infinity; no measured error rate, hardware threshold, fitted noise model or device parameter; no application output or physical benchmark may select the code law.
- **Evidence identities:** derivation
sha256:3f4c7dd00b70b80685d2182f1c66afb4d2ed6ff33d7fb2e1e0c574afeaffd2e3; independent implementation sha256:08f284371749dfe9c00b2f28d468d9006d7cebfcccd62ec6d6fa8a54b4d1b45d2; independent certificate
sha256:8260f560843df345b17ec1cc8d7ff891a3486f59dfcdf9d28fa9a1313e9fed93; measurement preserved in package; engine receipt
sha256:97e061e60929c595adbc3706113e920f80a7b9ea224b3606893edd4c244d445b at receipts/engine/model_admitted/SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002-97e061e60929c595.json.

36.4 Family REVX - 18/18 complete

Scope. Complete reversible computation. This family decides 4,608 candidates, retains 18 unique survivors and passes 72 adverse controls.

SFT-QUANTUM-OBL-REVS-001 - Reversible configuration and transition identity

- **Claim and ownership:** SFT-QUANTUM-REVS-CONFIGURATION-TRANSITION-001; Quantum Computation family REVX; Reversible configuration and transition identity.
- **Forced law:** A reversible transition is a source-complete one-to-one relation over canonical configurations with an exact inverse row and retained forward and inverse traces.
- **Unique surviving structure:** complete-canonical-configuration__source-complete-one-to-one-transition__complete-bijective-configuration-transition__complete-forward-and-inverse-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-logical-physical-handoff. Exact result: REVX-001 uniquely retains complete-bijective-configuration-transition, complete inverse custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete reversible grammar, exact forward/inverse execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-COMP-HAND-CLASSICAL-QUANTUM-004 -> SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** REVX-001: configuration_transition_round_trip; exact reversible execution: {"image":"returned","inverse_image":"held","one_predecessor":true,"source":"held"}; complete reversible-computation custody: 18 records; all rows preserved; complete finite forward and inverse traces supply the observation; physical energy, heat, timing, noise and hardware performance remain downstream measurements and cannot select the formal law
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-REVERSIBLE-COMPUTATION-OBSERVER, SFT-V1-V2-QUANTUM-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported reversible or quantum machine selects the law; host absence is not a semantic numerical-zero object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, selected branch, stochastic cause or unregistered oracle; no physical energy or hardware value is inferred without its owning measurement branch; no failed route retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:29993022cf6633ec563bdac8587b51bda1900b2c65b2f82e05f5ee643234f121; independent implementation sha256:31f72fafeca75c4ddd14a1c36917d892f89814a1a0ac6b285131d2a4445d12d3; independent certificate
sha256:acbaef32ec774809c3a955f1488b8ad3ee8d22c16618b06bb146c53ec03757d6; measurement sha256:ae089cd4ecab0f3f46fc0e9a1133ab0f6eaac8b49da549e9cb1609c9648d4bfe; engine receipt sha256:c1a6af8987b1c3ec5ebf02d54926aadb94976b59321379bedea165ff11e677d3 at receipts/engine/model_admitted/SFT-QUANTUM-REVS-CONFIGURATION-TRANSITION-001-c1a6af8987b1c3ec.json.

SFT-QUANTUM-OBL-REVS-002 - Injective, surjective and bijective process distinction

- **Claim and ownership:** SFT-QUANTUM-REVS-MAP-CLASSIFICATION-002; Quantum Computation family REVX; Injective, surjective and bijective process distinction.
- **Forced law:** Injectivity, surjectivity and bijectivity are decided from the complete declared source, image and predecessor-fibre ledgers; only a bijection supplies an inverse without an added predecessor record.
- **Unique surviving structure:** complete-canonical-configuration__source-complete-one-to-one-transition__exact-map-fibre-and-image-classification__complete-forward-and-inverse-

trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-logical-physical-handoff. Exact result: REVX-002 uniquely retains exact-map-fibre-and-image-classification, complete inverse custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete reversible grammar, exact forward/inverse execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-COMP-HAND-CLASSICAL-QUANTUM-004 -> SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-REXX-CONFIGURATION-TRANSITION-001. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** REVX-002: map_classification; exact reversible execution: {"bijection":true,"image_count":2,"predecessor_fibre_widths":[1,1],"source_count":2}; complete reversible-computation custody: 18 records; all rows preserved; complete finite forward and inverse traces supply the observation; physical energy, heat, timing, noise and hardware performance remain downstream measurements and cannot select the formal law
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-REVERSIBLE-COMPUTATION-OBSERVER, SFT-V1-V2-QUANTUM-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported reversible or quantum machine selects the law; host absence is not a semantic numerical-zero object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, selected branch, stochastic cause or unregistered oracle; no physical energy or hardware value is inferred without its owning measurement branch; no failed route retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:010c86914f700e3ff0233f024fd021c11e84e89b0590340e90e05033cd853c9a; independent implementation sha256:31f72fafeca75c4ddd14a1c36917d892f89814a1a0ac6b285131d2a4445d12d3; independent certificate
sha256:acbaef32ec774809c3a955f1488b8ad3ee8d22c16618b06bb146c53ec03757d6; measurement sha256:744b48311c341894c6dfa6fe6149a24e741751861966629a90658340624aa6d1; engine receipt sha256:d57a9265e397e3ce1308c7873bec25654a19faf28d8ff9c0c0609824f0a3d58e at receipts/engine/model_admitted/SFT-QUANTUM-REXX-MAP-CLASSIFICATION-002-d57a9265e397e3ce.json.

SFT-QUANTUM-OBL-REXX-003 - Reversible language and grammar relation

- **Claim and ownership:** SFT-QUANTUM-REXX-LANGUAGE-GRAMMAR-003; Quantum Computation family REXX; Reversible language and grammar relation.
- **Forced law:** A reversible language process retains a complete forward derivation and a unique inverse derivation for every generated word in its declared grammar.
- **Unique surviving structure:** complete-canonical-configuration__source-complete-one-to-one-transition__bidirectional-language-derivation__complete-forward-and-inverse-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-logical-physical-handoff. Exact result: REXX-003 uniquely retains bidirectional-language-derivation, complete inverse custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete reversible grammar, exact forward/inverse execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-COMP-HAND-CLASSICAL-QUANTUM-004 -> SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-REXX-MAP-CLASSIFICATION-002. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** REVX-003: reversible_language; exact reversible execution: {"forward_words":["returned","held"],"generated_words":["held","returned"],"round_trip_complete":true}; complete reversible-computation custody: 18 records; all rows preserved; complete finite forward and inverse traces supply the observation; physical energy, heat, timing, noise and hardware performance remain downstream measurements and cannot select the formal law
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-REVERSIBLE-COMPUTATION-OBSERVER, SFT-V1-V2-QUANTUM-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported reversible or quantum machine selects the law; host absence is not a semantic numerical-zero object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, selected branch, stochastic cause or unregistered oracle; no physical energy or hardware value is inferred without its owning measurement branch; no failed route retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:c2dd331327c56f87d387f6c75fe10f30547080eb5875f53e3de0ab0c8e0e2f6e; independent implementation sha256:31f72fafeca75c4ddd14a1c36917d892f89814a1a0ac6b285131d2a4445d12d3; independent certificate
sha256:acbaef32ec774809c3a955f1488b8ad3ee8d22c16618b06bb146c53ec03757d6; measurement sha256:62b821e3801f8628ac81957fb07e3b93961b6c6fd71807524bb145425e1b62aa; engine receipt sha256:54259a3fe2986ed640b13e38d61df6b5ba1c284f65af2c9445b0a423cb7782c8 at receipts/engine/model_admitted/SFT-QUANTUM-REXX-LANGUAGE-GRAMMAR-003-54259a3fe2986ed6.json.

SFT-QUANTUM-OBL-REXX-004 - Reversible automaton and transducer construction

- **Claim and ownership:** SFT-QUANTUM-REXX-AUTOMATON-TRANSDUCER-004; Quantum Computation family REXX; Reversible automaton and transducer construction.
- **Forced law:** A reversible automaton or transducer has one predecessor and one successor for every declared state-symbol row while retaining every emitted label needed for inverse execution.
- **Unique surviving structure:** complete-canonical-configuration__source-complete-one-to-one-transition__reversible-state-output-transduction__complete-forward-and-inverse-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-logical-physical-handoff. Exact result: REVX-004 uniquely retains reversible-state-output-transduction, complete inverse custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete reversible grammar, exact forward/inverse execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-COMP-HAND-CLASSICAL-QUANTUM-004 -> SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-REXX-LANGUAGE-GRAMMAR-003. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** REVX-004: reversible_automaton; exact reversible execution: {"forward_steps":2,"inverse_trace_retained":true,"state_rows":2,"terminal_state":"held"}; complete reversible-computation custody: 18 records; all rows preserved; complete finite forward and inverse traces supply the observation; physical energy, heat, timing, noise and hardware performance remain downstream measurements and cannot select the formal law
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-REVERSIBLE-COMPUTATION-OBSERVER, SFT-V1-V2-QUANTUM-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported reversible or quantum machine selects the law; host absence is not a semantic numerical-zero object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, selected branch, stochastic cause or unregistered oracle; no physical energy or hardware value is inferred without its owning measurement branch; no failed route retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:2868580ed065dd51628f57d4edbf9a0a73a5f5179e990d556deeb143c51affa3; independent implementation sha256:31f72fafeca75c4ddd14a1c36917d892f89814a1a0ac6b285131d2a4445d12d3; independent certificate
sha256:acbaef32ec774809c3a955f1488b8ad3ee8d22c16618b06bb146c53ec03757d6; measurement sha256:d67c0b6ae843f43f88181d1a05c2406dd4c31e81b3f7f766d8169f0ad44927fe; engine receipt sha256:25bf5a2ab341a8b3399ea18a694936140752d2ce9f0ee174d8003d34d734fac2 at receipts/engine/model_admitted/SFT-QUANTUM-REXX-AUTOMATON-TRANSDUCER-004-25bf5a2ab341a8b3.json.

SFT-QUANTUM-OBL-REXX-005 - Reversible rewriting and retained predecessor custody

- **Claim and ownership:** SFT-QUANTUM-REXX-REWRITING-PREDECESSOR-005; Quantum Computation family REXX; Reversible rewriting and retained predecessor custody.
- **Forced law:** A rewrite is reversible exactly when its rule, held position and predecessor word are uniquely recoverable from the successor or from one explicit retained predecessor record.
- **Unique surviving structure:** complete-canonical-configuration__source-complete-one-to-one-transition__inverse-rewrite-and-predecessor-record__complete-forward-and-inverse-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-logical-physical-handoff. Exact result: REXX-005 uniquely retains inverse-rewrite-and-predecessor-record, complete inverse custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete reversible grammar, exact forward/inverse execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-COMP-HAND-CLASSICAL-QUANTUM-004 -> SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-REXX-AUTOMATON-TRANSDUCER-004. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** REXX-005: reversible_rewrite; exact reversible execution: {"restored_word":"ab","rule_identity_retained":true,"source_word":"ab","successor_word":"ba"}; complete reversible-computation custody: 18 records; all rows preserved; complete finite forward and inverse traces supply the observation; physical energy, heat, timing, noise and hardware performance remain downstream measurements and cannot select the formal law
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-REVERSIBLE-COMPUTATION-OBSERVER, SFT-V1-V2-QUANTUM-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported reversible or quantum machine selects the law; host absence is not a semantic numerical-zero object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, selected branch, stochastic cause or unregistered oracle; no physical energy or hardware value is inferred without its owning measurement branch; no failed route retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:ab7b05a446171c8cf7ac7171953696006ec7009eb1ba40de8f51252707717ecb; independent implementation sha256:31f72fafeca75c4ddd14a1c36917d892f89814a1a0ac6b285131d2a4445d12d3; independent certificate
sha256:acbaef32ec774809c3a955f1488b8ad3ee8d22c16618b06bb146c53ec03757d6; measurement sha256:9a1970ef13ca3b060b656bc67baeabcb34f679a9ac6fcd1b27bce866981fbf96; engine receipt sha256:881d51963db1786c680ef872808abf0be3aa51c5253225602e48de746087b40e at receipts/engine/model_admitted/SFT-QUANTUM-REXX-REWRITING-PREDECESSOR-005-881d51963db1786c.json.

SFT-QUANTUM-OBL-REVS-006 - Reversible tape-machine configuration law

- **Claim and ownership:** SFT-QUANTUM-REVS-TAPE-MACHINE-006; Quantum Computation family REVS; Reversible tape-machine configuration law.
- **Forced law:** A reversible tape step retains the exact prior process state, held tape label and head position so writing and movement reconstruct the unique predecessor configuration.
- **Unique surviving structure:** complete-canonical-configuration__source-complete-one-to-one-transition__reversible-tape-head-write-move__complete-forward-and-inverse-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-logical-physical-handoff. Exact result: REVS-006 uniquely retains reversible-tape-head-write-move, complete inverse custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete reversible grammar, exact forward/inverse execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-COMP-HAND-CLASSICAL-QUANTUM-004 -> SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-REVS-REWRITING-PREDECESSOR-005. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** REVS-006: reversible_tape; exact reversible execution: {"round_trip_complete":true,"source_head":0,"source_state":"held","source_word":["held","returned"],"successor_head":1}; complete reversible-computation custody: 18 records; all rows preserved; complete finite forward and inverse traces supply the observation; physical energy, heat, timing, noise and hardware performance remain downstream measurements and cannot select the formal law
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-REVERSIBLE-COMPUTATION-OBSERVER, SFT-V1-V2-QUANTUM-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported reversible or quantum machine selects the law; host absence is not a semantic numerical-zero object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, selected branch, stochastic cause or unregistered oracle; no physical energy or hardware value is inferred without its owning measurement branch; no failed route retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:c88e5f1653085a689e5becff0029af53fe7c17afb002a88e0bb4e9ae75519c4; independent implementation sha256:31f72fafeca75c4ddd14a1c36917d892f89814a1a0ac6b285131d2a4445d12d3; independent certificate
sha256:acbaef32ec774809c3a955f1488b8ad3ee8d22c16618b06bb146c53ec03757d6; measurement sha256:096378c70ca7b04ea82e3fc565c72753a3e27353964ee2c1b72cc1d615b88531; engine receipt sha256:e8e58c21f295a561303c01aef080746002f60b6e468924ed0866027ff08852eb at receipts/engine/model_admitted/SFT-QUANTUM-REVS-TAPE-MACHINE-006-e8e58c21f295a561.json.

SFT-QUANTUM-OBL-REVS-007 - Reversible universal interpreter

- **Claim and ownership:** SFT-QUANTUM-REVS-UNIVERSAL-INTERPRETER-007; Quantum Computation family REVS; Reversible universal interpreter.
- **Forced law:** A universal reversible interpreter executes every description in the frozen reversible grammar and returns the source from the terminal state plus its retained instruction history.
- **Unique surviving structure:** complete-canonical-configuration__source-complete-one-to-one-transition__description-driven-reversible-interpretation__complete-forward-and-inverse-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-logical-physical-handoff. Exact result: REVS-007 uniquely retains description-driven-reversible-interpretation, complete inverse custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete reversible grammar, exact forward/inverse execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-COMP-HAND-CLASSICAL-QUANTUM-004 -> SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-REXX-TAPE-MACHINE-006. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** REXX-007: `reversible_universal_interpreter`; exact reversible execution: `{"history_rows":2,"instruction_count":2,"source_restored":true,"terminal_word":["returned"]}`; complete reversible-computation custody: 18 records; all rows preserved; complete finite forward and inverse traces supply the observation; physical energy, heat, timing, noise and hardware performance remain downstream measurements and cannot select the formal law
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-REVERSIBLE-COMPUTATION-OBSERVER, SFT-V1-V2-QUANTUM-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no imported reversible or quantum machine selects the law; host absence is not a semantic numerical-zero object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, selected branch, stochastic cause or unregistered oracle; no physical energy or hardware value is inferred without its owning measurement branch; no failed route retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:cda6cda6d4044b8c0bf05ad372e221d052eb331c2b77b4c6d7454b15592efe21`; independent implementation `sha256:31f72fafeca75c4ddd14a1c36917d892f89814a1a0ac6b285131d2a4445d12d3`; independent certificate
`sha256:acbaef32ec774809c3a955f1488b8ad3ee8d22c16618b06bb146c53ec03757d6`; measurement `sha256:b7d13919a30c20406f0457cec573fbc23e9a48279772d382dbfaf609340dd72e`; engine receipt `sha256:601a25567e990e88812397ae239c63cdf96c06a45dddb080de44f8bc5c6f9c` at `receipts/engine/model_admitted/SFT-QUANTUM-REXX-UNIVERSAL-INTERPRETER-007-601a25567e990e88.json`.

SFT-QUANTUM-OBL-REXX-008 - History tape and uncomputation

- **Claim and ownership:** SFT-QUANTUM-REXX-HISTORY-UNCOMPUTATION-008; Quantum Computation family REXX; History tape and uncomputation.
- **Forced law:** A retained history contains one source-bound row per forward step; inverse traversal in reverse causal order restores the initial configuration exactly.
- **Unique surviving structure:** `complete-canonical-configuration__source-complete-one-to-one-transition__complete-history-and-inverse-uncompute__complete-forward-and-inverse-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-logical-physical-handoff`. Exact result: REXX-008 uniquely retains `complete-history-and-inverse-uncompute`, complete inverse custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete reversible grammar, exact forward/inverse execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-COMP-HAND-CLASSICAL-QUANTUM-004 -> SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-REXX-UNIVERSAL-INTERPRETER-007. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** REXX-008: `history_uncomputation`; exact reversible execution: `{"forward_history_rows":2,"inverse_history_rows":2,"source_restored":true,"source_word":["held"]}`; complete reversible-computation custody: 18 records; all rows preserved; complete finite forward and inverse traces supply the

observation; physical energy, heat, timing, noise and hardware performance remain downstream measurements and cannot select the formal law

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-REVERSIBLE-COMPUTATION-OBSERVER, SFT-V1-V2-QUANTUM-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported reversible or quantum machine selects the law; host absence is not a semantic numerical-zero object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, selected branch, stochastic cause or unregistered oracle; no physical energy or hardware value is inferred without its owning measurement branch; no failed route retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:0646b9cc7f0199d5df18f718fad75b58325570e9dfcd12a45f1ea4eb72792353; independent implementation sha256:31f72fafeca75c4ddd14a1c36917d892f89814a1a0ac6b285131d2a4445d12d3; independent certificate
sha256:acbaef32ec774809c3a955f1488b8ad3ee8d22c16618b06bb146c53ec03757d6; measurement sha256:0d77c2189d689648a9df85b85dc4a621c01fbd31696a70d30bcf6c72f09a2c2e; engine receipt sha256:dc23c8b9762fd8b728f1b2ab6f9d631afd7540540c75e30cf5277b4f3c001a5a at receipts/engine/model_admitted/SFT-QUANTUM-REXX-HISTORY-UNCOMPUTATION-008-dc23c8b9762fd8b7.json.

SFT-QUANTUM-OBL-REXX-009 - Ancilla preparation and exact restoration

- **Claim and ownership:** SFT-QUANTUM-REXX-ANCILLA-RESTORATION-009; Quantum Computation family REXX; Ancilla preparation and exact restoration.
- **Forced law:** An ancilla is lawful only when its preparation identity is retained and the complete process restores it to that exact prepared label before the support is released.
- **Unique surviving structure:** complete-canonical-configuration__source-complete-one-to-one-transition__prepared-work-restored-ancilla__complete-forward-and-inverse-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-logical-physical-handoff. Exact result: REXX-009 uniquely retains prepared-work-restored-ancilla, complete inverse custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete reversible grammar, exact forward/inverse execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-COMP-HAND-CLASSICAL-QUANTUM-004 -> SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-REXX-HISTORY-UNCOMPUTATION-008. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** REXX-009: ancilla_restoration; exact reversible execution: {"ancilla_released_after_restoration":true,"prepared_label":"held","restored_label":"held","work_label":"returned"}; complete reversible-computation custody: 18 records; all rows preserved; complete finite forward and inverse traces supply the observation; physical energy, heat, timing, noise and hardware performance remain downstream measurements and cannot select the formal law
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-REVERSIBLE-COMPUTATION-OBSERVER, SFT-V1-V2-QUANTUM-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported reversible or quantum machine selects the law; host absence is not a semantic numerical-zero object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, selected branch, stochastic cause or unregistered oracle; no physical energy or hardware value is inferred without its owning measurement branch; no failed route retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:31d8b0a8207103280bd789d016f7c8729bcc1d2772b90e8351b7e95cb08530bd; independent implementation sha256:31f72fafeca75c4ddd14a1c36917d892f89814a1a0ac6b285131d2a4445d12d3; independent certificate
sha256:acbaef32ec774809c3a955f1488b8ad3ee8d22c16618b06bb146c53ec03757d6; measurement sha256:8ee8f2c7f69389755347b8f0925ac54c7ebcfb6c01a3660c1ed23de4ac288a6e; engine receipt sha256:5cc7d3edb3e0dafb19726ebbb39088ee64c2dbbf966eb8e031b58ead36c563c1 at receipts/engine/model_admitted/SFT-QUANTUM-REVS-ANCILLA-RESTORATION-009-5cc7d3edb3e0dafb.json.

SFT-QUANTUM-OBL-REVS-010 - Garbage record and cleanup boundary

- **Claim and ownership:** SFT-QUANTUM-REVS-GARBAGE-CLEANUP-010; Quantum Computation family REVS; Garbage record and cleanup boundary.
- **Forced law:** Computational garbage is the retained distinction record not present in the declared output; cleanup is reversible only while that record or an equivalent inverse trace remains available.
- **Unique surviving structure:** complete-canonical-configuration__source-complete-one-to-one-transition__garbage-provenance-and-cleanup__complete-forward-and-inverse-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-logical-physical-handoff. Exact result: REVS-010 uniquely retains garbage-provenance-and-cleanup, complete inverse custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete reversible grammar, exact forward/inverse execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-COMP-HAND-CLASSICAL-QUANTUM-004 -> SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-REVS-ANCILLA-RESTORATION-009. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** REVS-010: garbage_cleanup; exact reversible execution: {"garbage_rows":2,"output_width":2,"source_reconstructible":true,"source_width":2}; complete reversible-computation custody: 18 records; all rows preserved; complete finite forward and inverse traces supply the observation; physical energy, heat, timing, noise and hardware performance remain downstream measurements and cannot select the formal law
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-REVERSIBLE-COMPUTATION-OBSERVER, SFT-V1-V2-QUANTUM-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported reversible or quantum machine selects the law; host absence is not a semantic numerical-zero object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, selected branch, stochastic cause or unregistered oracle; no physical energy or hardware value is inferred without its owning measurement branch; no failed route retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:1d5024d10214f7f4d6c3f29b5a9edb128c40adb78c1c4b38a8ffc132858bb05a; independent implementation sha256:31f72fafeca75c4ddd14a1c36917d892f89814a1a0ac6b285131d2a4445d12d3; independent certificate
sha256:acbaef32ec774809c3a955f1488b8ad3ee8d22c16618b06bb146c53ec03757d6; measurement sha256:1a4e8f7b5d97e680811fb1afddcdfa63dbc91f13193fba96a1e2a37d4a706ec4; engine receipt sha256:de29a80e561df8683ed002701bdf8ff9f8993886a28e63d13241e3b66ff7211 at receipts/engine/model_admitted/SFT-QUANTUM-REVS-GARBAGE-CLEANUP-010-de29a80e561df868.json.

SFT-QUANTUM-OBL-REVS-011 - Logical reversibility and physical erasure handoff

- **Claim and ownership:** SFT-QUANTUM-REVS-ERASURE-HANDOFF-011; Quantum Computation family REVS; Logical reversibility and physical erasure handoff.

- **Forced law:** Logical reversibility is decided by exact predecessor recovery; energy, heat and device erasure belong to a separately measured physical handoff and cannot select the logical law.
- **Unique surviving structure:** complete-canonical-configuration__source-complete-one-to-one-transition__logical-record-to-physical-erasure-boundary__complete-forward-and-inverse-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-logical-physical-handoff. Exact result: REVX-011 uniquely retains logical-record-to-physical-erasure-boundary, complete inverse custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete reversible grammar, exact forward/inverse execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-COMP-HAND-CLASSICAL-QUANTUM-004 -> SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-REXX-GARBAGE-CLEANUP-010. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** REVX-011: logical_physical_erasure_handoff; exact reversible execution: {"logical_predecessor_recoverable":true,"owner":"physics-or-engineering-measurement","physical_energy_value_present":false}; complete reversible-computation custody: 18 records; all rows preserved; complete finite forward and inverse traces supply the observation; physical energy, heat, timing, noise and hardware performance remain downstream measurements and cannot select the formal law
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-REVERSIBLE-COMPUTATION-OBSERVER, SFT-V1-V2-QUANTUM-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported reversible or quantum machine selects the law; host absence is not a semantic numerical-zero object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, selected branch, stochastic cause or unregistered oracle; no physical energy or hardware value is inferred without its owning measurement branch; no failed route retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:edb9406e87e095f629b3d0785cb5c064e3aca41f40d115a52d39d70831418a89; independent implementation sha256:31f72fafeca75c4ddd14a1c36917d892f89814a1a0ac6b285131d2a4445d12d3; independent certificate
sha256:acbaef32ec774809c3a955f1488b8ad3ee8d22c16618b06bb146c53ec03757d6; measurement sha256:620cdfc0963aff4d2cc11ecf638c0a6552a5861390570c71b382b73f4c8d062; engine receipt sha256:18e94b65d4a134c94b03b17c2b2ed92df203f3ad878bd5dec3287fc0a2b51fe4 at receipts/engine/model_admitted/SFT-QUANTUM-REXX-ERASURE-HANDOFF-011-18e94b65d4a134c9.json.

SFT-QUANTUM-OBL-REXX-012 - Reversible simulation of irreversible computation

- **Claim and ownership:** SFT-QUANTUM-REXX-IRREVERSIBLE-SIMULATION-012; Quantum Computation family REXX; Reversible simulation of irreversible computation.
- **Forced law:** Every finite irreversible step is reversibly simulated by pairing its image with the exact predecessor distinction needed to make the extended transition one-to-one.
- **Unique surviving structure:** complete-canonical-configuration__source-complete-one-to-one-transition__irreversible-image-plus-predecessor-ledger__complete-forward-and-inverse-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-logical-physical-handoff. Exact result: REVX-012 uniquely retains irreversible-image-plus-predecessor-ledger, complete inverse custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete reversible grammar, exact forward/inverse execution, source identity, candidate survivor, seal, custody record, control or independent

reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-COMP-HAND-CLASSICAL-QUANTUM-004 -> SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-REXX-ERASURE-HANDOFF-011. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** REXX-012: reversible_irreversible_simulation; exact reversible execution: {"irreversible_image_width":2,"predecessor_record_width":3,"source_restored":true,"source_width":3}; complete reversible-computation custody: 18 records; all rows preserved; complete finite forward and inverse traces supply the observation; physical energy, heat, timing, noise and hardware performance remain downstream measurements and cannot select the formal law
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-REVERSIBLE-COMPUTATION-OBSERVER, SFT-V1-V2-QUANTUM-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported reversible or quantum machine selects the law; host absence is not a semantic numerical-zero object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, selected branch, stochastic cause or unregistered oracle; no physical energy or hardware value is inferred without its owning measurement branch; no failed route retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:5b4b673db0522b42ae1b758caba103416c2c885bb13397a9145397f9df6fca0c; independent implementation sha256:31f72fafeca75c4ddd14a1c36917d892f89814a1a0ac6b285131d2a4445d12d3; independent certificate sha256:acbaef32ec774809c3a955f1488b8ad3ee8d22c16618b06bb146c53ec03757d6; measurement sha256:4b4d7fe0f84a07665791fb8b095e40f2dc838e5c7ce53a48b55f8f4df24f8b5a; engine receipt sha256:4af6000bc8cc6ac4b302c79deab9545af71a5e4c622a5fa1a99b05ee753a456a at receipts/engine/model_admitted/SFT-QUANTUM-REXX-IRREVERSIBLE-SIMULATION-012-4af6000bc8cc6ac4.json.

SFT-QUANTUM-OBL-REXX-013 - Irreversible simulation of reversible computation

- **Claim and ownership:** SFT-QUANTUM-REXX-OBSERVATION-SIMULATION-013; Quantum Computation family REXX; Irreversible simulation of reversible computation.
- **Forced law:** An irreversible process simulates a reversible process only relative to an observation that may close inverse records; the underlying reversible trace remains separately reconstructible.
- **Unique surviving structure:** complete-canonical-configuration__source-complete-one-to-one-transition__observation-closes-reversible-records__complete-forward-and-inverse-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-logical-physical-handoff. Exact result: REXX-013 uniquely retains observation-closes-reversible-records, complete inverse custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete reversible grammar, exact forward/inverse execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-COMP-HAND-CLASSICAL-QUANTUM-004 -> SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-REXX-IRREVERSIBLE-SIMULATION-012. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** REXX-013: irreversible_observation_of_reversible_trace; exact reversible execution: {"closed_inverse_rows_retained_separately":true,"observed_terminal_labels":1,"reversible_trace_width":3}; complete reversible-computation custody: 18 records; all rows preserved; complete finite forward and inverse traces supply the observation; physical energy, heat, timing, noise and hardware performance remain downstream measurements and cannot select the formal law

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-REVERSIBLE-COMPUTATION-OBSERVER, SFT-V1-V2-QUANTUM-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported reversible or quantum machine selects the law; host absence is not a semantic numerical-zero object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, selected branch, stochastic cause or unregistered oracle; no physical energy or hardware value is inferred without its owning measurement branch; no failed route retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:32ddaba4ac841b4042687af1e48125ae80195913aaf2dec141fda5944359dbec; independent implementation sha256:31f72fafeca75c4ddd14a1c36917d892f89814a1a0ac6b285131d2a4445d12d3; independent certificate
sha256:acbaef32ec774809c3a955f1488b8ad3ee8d22c16618b06bb146c53ec03757d6; measurement sha256:f2b79c11155530ae6dd5461b55baef1fb765dd5ad09c1e4257f8645e49c3f772; engine receipt sha256:a228793a7a7417741da1eeccacda28672dbddf562b38bf7f906f3566fd075399 at receipts/engine/model_admitted/SFT-QUANTUM-REXX-OBSERVATION-SIMULATION-013-a228793a7a741774.json.

SFT-QUANTUM-OBL-REXX-014 - Reversible time, space and record tradeoff

- **Claim and ownership:** SFT-QUANTUM-REXX-RESOURCE-TRADEOFF-014; Quantum Computation family REXX; Reversible time, space and record tradeoff.
- **Forced law:** Reversible resource accounting separately retains forward steps, inverse steps, live configurations, history width, ancilla support and released records for the declared representation.
- **Unique surviving structure:** complete-canonical-configuration__source-complete-one-to-one-transition__time-space-history-resource-ledger__complete-forward-and-inverse-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-logical-physical-handoff. Exact result: REXX-014 uniquely retains time-space-history-resource-ledger, complete inverse custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete reversible grammar, exact forward/inverse execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-COMP-HAND-CLASSICAL-QUANTUM-004 -> SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-REXX-OBSERVATION-SIMULATION-013. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** REXX-014: reversible_resource_tradeoff; exact reversible execution: {"all_resources_separate":true,"ancilla_rows":1,"forward_steps":3,"history_rows":3,"inverse_steps":3}; complete reversible-computation custody: 18 records; all rows preserved; complete finite forward and inverse traces supply the observation; physical energy, heat, timing, noise and hardware performance remain downstream measurements and cannot select the formal law
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-REVERSIBLE-COMPUTATION-OBSERVER, SFT-V1-V2-QUANTUM-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported reversible or quantum machine selects the law; host absence is not a semantic numerical-zero object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, selected branch, stochastic cause or unregistered oracle; no physical energy or hardware value is inferred without its owning measurement branch; no failed route retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:509fd8bb4579142019f32fe5efefb7e6dc535dcd2c4df63693cb17cc9b7fa73f; independent implementation sha256:31f72fafeca75c4ddd14a1c36917d892f89814a1a0ac6b285131d2a4445d12d3; independent certificate

sha256:acbaef32ec774809c3a955f1488b8ad3ee8d22c16618b06bb146c53ec03757d6; measurement
 sha256:2d2585748683a5a27e800673751fd96255ba165a2d1c3b4b74d1b9862d7ca41c; engine receipt
 sha256:7dad641a707a1137855e03b850b4ef39b1168855c75e7f5b88dd59bc4a1c43e5 at receipts/
 engine/model_admitted/SFT-QUANTUM-REXX-RESOURCE-TRADEOFF-014-7dad641a707a1137.json.

SFT-QUANTUM-OBL-REXX-015 - Reversible circuit synthesis and decomposition

- **Claim and ownership:** SFT-QUANTUM-REXX-CIRCUIT-SYNTHESIS-015; Quantum Computation family REXX; Reversible circuit synthesis and decomposition.
- **Forced law:** A reversible circuit is a causal composition of finite bijective gates; its inverse is the reverse causal sequence of the exact inverse gates.
- **Unique surviving structure:** complete-canonical-configuration__source-complete-one-to-one-transition__bijective-local-gate-decomposition__complete-forward-and-inverse-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-logical-physical-handoff. Exact result: REXX-015 uniquely retains bijective-local-gate-decomposition, complete inverse custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete reversible grammar, exact forward/inverse execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-COMP-HAND-CLASSICAL-QUANTUM-004 -> SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-REXX-RESOURCE-TRADEOFF-014. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** REXX-015: reversible_circuit; exact reversible execution: {"gate_count":1,"inverse_restores_source":true,"source_word":["held","returned"],"terminal_word":["returned","held"]}; complete reversible-computation custody: 18 records; all rows preserved; complete finite forward and inverse traces supply the observation; physical energy, heat, timing, noise and hardware performance remain downstream measurements and cannot select the formal law
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-REVERSIBLE-COMPUTATION-OBSERVER, SFT-V1-V2-QUANTUM-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported reversible or quantum machine selects the law; host absence is not a semantic numerical-zero object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, selected branch, stochastic cause or unregistered oracle; no physical energy or hardware value is inferred without its owning measurement branch; no failed route retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:746b1f794da4dbbabbba2ab0ab129c4c7748d14ec7f688dc45056638961b6dfd8; independent
 implementation sha256:31f72fafeca75c4ddd14a1c36917d892f89814a1a0ac6b285131d2a4445d12d3;
 independent certificate
 sha256:acbaef32ec774809c3a955f1488b8ad3ee8d22c16618b06bb146c53ec03757d6; measurement
 sha256:c67caf26f4f380f4225ea8a6526770b9b1e3bf6bf9eb49b6dcc9e1c56617837e; engine receipt
 sha256:00b3f31892ee8f8d15dc5cff96ed50443153952bd9b5b75fe5bea34665a491ee at receipts/
 engine/model_admitted/SFT-QUANTUM-REXX-CIRCUIT-SYNTHESIS-015-00b3f31892ee8f8d.json.

SFT-QUANTUM-OBL-REXX-016 - Reversible control and conditional execution

- **Claim and ownership:** SFT-QUANTUM-REXX-CONTROL-016; Quantum Computation family REXX; Reversible control and conditional execution.
- **Forced law:** A reversible conditional retains its control label and applies a bijective target action on exactly the declared control class, preserving one-to-one joint support.
- **Unique surviving structure:** complete-canonical-configuration__source-complete-one-to-one-transition__held-control-bijective-action__complete-forward-and-inverse-trace__lite

ral-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-logical-physical-handoff. Exact result: REVX-016 uniquely retains held-control-bijective-action, complete inverse custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete reversible grammar, exact forward/inverse execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-COMP-HAND-CLASSICAL-QUANTUM-004 -> SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-REXX-CIRCUIT-SYNTHESIS-015. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** REVX-016: reversible_control; exact reversible execution: {"control_classes":["held","returned"],"control_retained":true,"held_target":"held","returned_target":"returned"}; complete reversible-computation custody: 18 records; all rows preserved; complete finite forward and inverse traces supply the observation; physical energy, heat, timing, noise and hardware performance remain downstream measurements and cannot select the formal law
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-REVERSIBLE-COMPUTATION-OBSERVER, SFT-V1-V2-QUANTUM-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported reversible or quantum machine selects the law; host absence is not a semantic numerical-zero object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, selected branch, stochastic cause or unregistered oracle; no physical energy or hardware value is inferred without its owning measurement branch; no failed route retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:eb9a7619abc1b3dd4f3a40404f67deab747e1df3cd472d61506ad06d6d9ce7f5; independent implementation sha256:31f72fafeca75c4ddd14a1c36917d892f89814a1a0ac6b285131d2a4445d12d3; independent certificate
sha256:acbaef32ec774809c3a955f1488b8ad3ee8d22c16618b06bb146c53ec03757d6; measurement sha256:38bc5e5c7f86fe4f7bf76311291be12612ce7a61cf1c024d0536bcdc2a48217d; engine receipt sha256:5ce14dd5b5ee37e3d9fdcfdc00d9d17d9867a5da9cfb8d1e8ed183e9ec1f448d at receipts/engine/model_admitted/SFT-QUANTUM-REXX-CONTROL-016-5ce14dd5b5ee37e3.json.

SFT-QUANTUM-OBL-REXX-017 - Reversible fault and recovery trace

- **Claim and ownership:** SFT-QUANTUM-REXX-FAULT-RECOVERY-017; Quantum Computation family REXX; Reversible fault and recovery trace.
- **Forced law:** A reversible recovery retains the declared fault action, location, prior label and correction trace so the corrected result and faulted predecessor are both reconstructible.
- **Unique surviving structure:** complete-canonical-configuration__source-complete-one-to-one-transition__fault-location-label-and-recovery-record__complete-forward-and-inverse-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-logical-physical-handoff. Exact result: REVX-017 uniquely retains fault-location-label-and-recovery-record, complete inverse custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete reversible grammar, exact forward/inverse execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-COMP-HAND-CLASSICAL-QUANTUM-004 -> SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-REXX-CONTROL-016. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** REVX-017: reversible_fault_recovery; exact reversible execution: {"fault_position":1,"fault_record_retained":true,"faulted_word":["held","returned","held"],"recovered_word":["held","held","held"]}; complete reversible-computation custody: 18 records; all rows preserved; complete finite forward and inverse traces supply the observation; physical energy, heat, timing, noise and hardware performance remain downstream measurements and cannot select the formal law
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-REVERSIBLE-COMPUTATION-OBSERVER, SFT-V1-V2-QUANTUM-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported reversible or quantum machine selects the law; host absence is not a semantic numerical-zero object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, selected branch, stochastic cause or unregistered oracle; no physical energy or hardware value is inferred without its owning measurement branch; no failed route retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:4f6cd4fbc1e384e169f4cdc9835860d65618061569d4a1c65652e847effba1ce; independent implementation sha256:31f72fafeca75c4ddd14a1c36917d892f89814a1a0ac6b285131d2a4445d12d3; independent certificate
sha256:acbaef32ec774809c3a955f1488b8ad3ee8d22c16618b06bb146c53ec03757d6; measurement sha256:f2c20d653e98842bf97817464cbeaa6dab05c48eda146b95c8b686452c1d9877; engine receipt sha256:4d43437bc6ce9aa6ebd6fb865848bc023d82f3c0bf086c78d7e26edf58397cc8 at receipts/engine/model_admitted/SFT-QUANTUM-REVX-FAULT-RECOVERY-017-4d43437bc6ce9aa6.json.

SFT-QUANTUM-OBL-REVX-018 - Reversible-computation completeness certificate

- **Claim and ownership:** SFT-QUANTUM-REVX-COMPLETENESS-018; Quantum Computation family REVX; Reversible-computation completeness certificate.
- **Forced law:** The reversible-computation family is complete exactly when all eighteen frozen obligations have one owner, one unique survivor, complete controls, exact observations, independent reconstructions and current receipts.
- **Unique surviving structure:** complete-canonical-configuration__source-complete-one-to-one-transition__eighteen-obligation-no-omission-ledger__complete-forward-and-inverse-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-logical-physical-handoff. Exact result: REVX-018 uniquely retains eighteen-obligation-no-omission-ledger, complete inverse custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete reversible grammar, exact forward/inverse execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-COMP-HAND-CLASSICAL-QUANTUM-004 -> SFT-QUANTUM-REVERSIBLE-MODEL-001 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-REVX-FAULT-RECOVERY-017. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** REVX-018: reversible_computation_completeness; exact reversible execution: {"duplicate_owners":0,"execution_rows":18,"omitted_owners":0,"registered_obligations":18}; complete reversible-computation custody: 18 records; all rows preserved; complete finite forward and inverse traces supply the observation; physical energy, heat, timing, noise and hardware performance remain downstream measurements and cannot select the formal law
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-REVERSIBLE-COMPUTATION-OBSERVER, SFT-V1-V2-QUANTUM-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported reversible or quantum machine selects the law; host absence is not a semantic numerical-zero object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, selected branch, stochastic cause or unregistered oracle; no physical energy or hardware value is inferred without its owning measurement branch; no failed route retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:ad6b52cae6615d2747042929ea7c79001060f29c11ca940b3bc67b9974a1f6cb; independent implementation sha256:31f72fafeca75c4ddd14a1c36917d892f89814a1a0ac6b285131d2a4445d12d3; independent certificate
sha256:acbaef32ec774809c3a955f1488b8ad3ee8d22c16618b06bb146c53ec03757d6; measurement sha256:141fd744e8ed8de24b7a2d75f43efef497a0b276701aed5d1b5cd537dea7312d; engine receipt sha256:286e0e1e0213794cb1715babb4f3d239b86df8eff7275082ba352b84529e66df at receipts/engine/model_admitted/SFT-QUANTUM-REVS-COMPLETENESS-018-286e0e1e0213794c.json.

36.5 Family QSTATEX - 28/28 complete

Scope. Quantum information units, states, support and composition. This family decides 7,168 candidates, retains 28 unique survivors and passes 112 adverse controls.

SFT-QUANTUM-OBL-QSTATEX-001 - Quantum information-unit identity

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-INFORMATION-UNIT-001; Quantum Computation family QSTATEX; Quantum information-unit identity.
- **Forced law:** One quantum information unit is one generated Fold distinction with exactly the two forced fibre labels and a retained source identity; it is not imported as a binary digit.
- **Unique surviving structure:** complete-canonical-word-support__one-fold-distinction-two-fibre-labels__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-001 uniquely retains one-fold-distinction-two-fibre-labels, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-001: information_unit; exact quantum-state execution: {"distinction_count":1,"fibre_labels":["held","returned"],"labels_distinct":true}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation
sha256:4a9062456fc4b4f33e3de32828fd300fd6971e66d38825e3a9b263779a872c87; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate
sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement sha256:80de0ef7a8c88e2411ccbcf839e98da93fec0e2624239e7877c7f72493684335; engine receipt sha256:4509e5c5939f5156b1060685cd4b7d1dde127e01f8f7a31dff98ac1e69268c98 at receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-INFORMATION-UNIT-001-4509e5c5939f5156.js

on.

SFT-QUANTUM-OBL-QSTATEX-002 - Finite register and word-support construction

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-REGISTER-SUPPORT-002; Quantum Computation family QSTATEX; Finite register and word-support construction.
- **Forced law:** A positive finite register of width k is the complete generated set of length- k Fold words, with every word retained once and no completed-infinite carrier.
- **Unique surviving structure:** complete-canonical-word-support__positive-finite-cartesian-word-support__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-002 uniquely retains positive-finite-cartesian-word-support, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-INFORMATION-UNIT-001. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-002: finite_register; exact quantum-state execution: {"complete_support":true,"width":3,"word_count":8}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation
sha256:6c7711c891c459e4a16cefb593c0703b290f6b4ed403add3376b6feda9dd3b22; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate
sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement sha256:ff59adcc87ecbe0d1d30edd9e1fe2b3d11f19ed813861b5188cf2a1022bca9e8; engine receipt sha256:9ca570a51adb715efd5f6f8ab7cb77051f83767b7a94c9f7e03cd04f31c40f4c at receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-REGISTER-SUPPORT-002-9ca570a51adb715e.js on.

SFT-QUANTUM-OBL-QSTATEX-003 - Canonical state-description identity

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-STATE-IDENTITY-003; Quantum Computation family QSTATEX; Canonical state-description identity.
- **Forced law:** A state description is the canonical ledger of unique supported Fold words, held phase labels, provenance and observation records; order of presentation cannot create a new state.
- **Unique surviving structure:** complete-canonical-word-support__canonical-word-phase-record-ledger__held-relative-phase-ledger__complete-source-and-observation-record__literal

-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-003 uniquely retains canonical-word-phase-record-ledger, complete state custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-REGISTER-SUPPORT-002. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-003: canonical_state; exact quantum-state execution: {"presentation_order_closed":true,"support_rows":2,"unique_words":2}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation sha256:080238e8cd7fb4e00996214bf2efc6e8571fe426ab21f280631ef413b55bfbe9; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement sha256:6a3750dd10c713dba8ac4d3f6acf52b48d6bd1db1aeb82b3bdc5033112c2dbd0; engine receipt sha256:2402952ebc75d9763eb55653db0c6d375dbc5314072973eabebdc9d0125061da at receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-STATE-IDENTITY-003-2402952ebc75d976.json.

SFT-QUANTUM-OBL-QSTATEX-004 - State preparation and provenance

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-PREPARATION-004; Quantum Computation family QSTATEX; State preparation and provenance.
- **Forced law:** Preparation is a registered reversible process from a named source state to canonical support, retaining every transition and phase-label provenance row.
- **Unique surviving structure:** complete-canonical-word-support__source-bound-preparation-trace__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-004 uniquely retains source-bound-preparation-trace, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 ->

SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 ->

SFT-QUANTUM-QSTATEX-STATE-IDENTITY-003. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING.

Imported axioms: 0; free parameters: 0.

- **Observation:** QSTATEX-004: state_preparation; exact quantum-state execution: {"provenance_retained":true,"source":"registered","terminal":"prepared"}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation
sha256:1e345e2eec23bd42a97aa796ba03d96944b6f75c39048848607fada8d033f32b; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate
sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement sha256:33221e3162aebbb069d67cf6fcfa5414ef9e039fd4d8761a7f9db67f1adb664f; engine receipt sha256:70979d98c9cbdaadc5b4cce292ddf8255d0653cfd0582dad870731a779b1c45 at receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-PREPARATION-004-70979d98c9cbdaad.json.

SFT-QUANTUM-OBL-QSTATEX-005 - Product state and component reconstruction

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-PRODUCT-STATE-005; Quantum Computation family QSTATEX; Product state and component reconstruction.
- **Forced law:** A product state is the complete Cartesian concatenation of component supports and permits exact recovery of the declared component partition.
- **Unique surviving structure:** complete-canonical-word-support__cartesian-product-with-exact-factor-recovery__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-005 uniquely retains cartesian-product-with-exact-factor-recovery, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REXX-COMPLETENESS-018 ->
SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 ->
SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 ->
SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 ->
SFT-QUANTUM-QSTATEX-PREPARATION-004. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-005: product_state; exact quantum-state execution: {"components_reconstructible":true,"joint_width":2,"left_width":2,"right_width":1}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status:

empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation
`sha256:ee007cfad80d2c964a55cf648ba62b16f33fb3da2c2cbddec1c586349777c383`; independent
 implementation `sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d`;
 independent certificate
`sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd`; measurement
`sha256:9d153075f400d1218629a34c89788883cb63ef7fa9f23c14d8dfa687d2048e7d`; engine receipt
`sha256:2d85c3e3e84bd556997ecf69a106bc4ff554f17a6751a120e5fd8a5dc4cbcc75` at receipts/
 engine/model_admitted/SFT-QUANTUM-QSTATEX-PRODUCT-STATE-005-2d85c3e3e84bd556.json.

SFT-QUANTUM-OBL-QSTATEX-006 - Joint state and marginal support

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-JOINT-MARGINAL-006; Quantum Computation family QSTATEX; Joint state and marginal support.
- **Forced law:** A joint state is canonical support over concatenated words; a marginal support is the exact distinct projection at a declared partition while the complete joint record remains retained.
- **Unique surviving structure:** complete-canonical-word-support__joint-words-and-projected-sup
 port__held-relative-phase-ledger__complete-source-and-observation-record__literal-
 complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__expl
 icit-formal-physical-handoff. Exact result: QSTATEX-006 uniquely retains
 joint-words-and-projected-support, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REXX-COMPLETENESS-018 ->
 SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 ->
 SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 ->
 SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 ->
 SFT-QUANTUM-QSTATEX-PRODUCT-STATE-005. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING.
 Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-006: joint_marginal; exact quantum-state execution:
 {"joint_record_retained":true,"joint_rows":2,"left_marginal_rows":2,"right_marginal_rows":2}; complete quantum-state
 custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the
 observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER,
 SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status:
 empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation
`sha256:68a583aec296a85ce36b1b7aaa12f442284f65e64bdb946c29bb832dfe3967c9`; independent
 implementation `sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d`;
 independent certificate

sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement
 sha256:d4fe5a95c94a6f146c7ab3e976680cdfa368408bd699122f03a4dc2ce8308b44; engine receipt
 sha256:2b3c0b6607af9ccc4509b1d83a3c33d5b36cad8271ceeb01abb85175e1540f02 at receipts/
 engine/model_admitted/SFT-QUANTUM-QSTATEX-JOINT-MARGINAL-006-2b3c0b6607af9ccc.json.

SFT-QUANTUM-OBL-QSTATEX-007 - Pure-support and mixed-record correspondence

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-PURE-MIXED-007; Quantum Computation family QSTATEX; Pure-support and mixed-record correspondence.
- **Forced law:** Pure support names one complete source-bound preparation class; mixed correspondence is a retained collection of distinct preparation records and cannot be introduced as ontic randomness.
- **Unique surviving structure:** complete-canonical-word-support__support-versus-preparation-record-class__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-007 uniquely retains support-versus-preparation-record-class, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-JOINT-MARGINAL-006. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-007: pure_mixed_correspondence; exact quantum-state execution: {"ontic_randomness_imported":false,"preparation_record_count":2,"support_identity_count":1}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation
 sha256:46053d7008b243b7d2ef5c076b00d52817250acb0f56ed8cff2c1915d84ce973; independent
 implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d;
 independent certificate
 sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement
 sha256:98a25bb2cf0b14a6794d3444018680f3e5191aaf6cf867d8e759a2beab7285c6; engine receipt
 sha256:28e296f15f825dc2c6925f7429132601eb8be317655293c597fcd88dbf2a6755 at receipts/
 engine/model_admitted/SFT-QUANTUM-QSTATEX-PURE-MIXED-007-28e296f15f825dc2.json.

SFT-QUANTUM-OBL-QSTATEX-008 - Superposition-equivalent complete support

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-SUPERPOSITION-SUPPORT-008; Quantum Computation family QSTATEX; Superposition-equivalent complete support.
- **Forced law:** Superposition-equivalent structure is complete generated word support with held relative-phase labels; no irrational or imaginary proof scalar is required.

- **Unique surviving structure:** complete-canonical-word-support__complete-word-support-with-held-phases__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-008 uniquely retains complete-word-support-with-held-phases, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-PURE-MIXED-007. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-008: complete_superposition_support; exact quantum-state execution: {"all_words_present_once":true,"register_width":4,"support_rows":16}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation sha256:b9f580a0cfe62ef32d717505a2ca4eb018c9818558e344f8a54faca40363a96d; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement sha256:da28df6f9c15be09c81d85b36753bd1e40c247d2e3812aa38cc54e8a2f52d6a0; engine receipt sha256:4a42e08c5efe05880736cd8ce0198fe6db4dbfd82c3480e3de048240df13bc30 at receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-SUPERPOSITION-SUPPORT-008-4a42e08c5efe0588.json.

SFT-QUANTUM-OBL-QSTATEX-009 - Relative phase-label identity

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-RELATIVE-PHASE-009; Quantum Computation family QSTATEX; Relative phase-label identity.
- **Forced law:** Relative phase is the comparison class between held phase labels on supported words; only relative classes can alter a later predecessor merge.
- **Unique surviving structure:** complete-canonical-word-support__period-two-held-relative-phase-relation__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-009 uniquely retains period-two-held-relative-phase-relation, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-SUPERPOSITION-SUPPORT-008. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-009: relative_phase; exact quantum-state execution: {"phase_labels":["phase-held","phase-returned"],"relative_classes":["same","distinct"]}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation sha256:5632841741b8f3e32b1bd5b13fffc84986471e7b399749d3f224de864311a784; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement sha256:66b78cc99e92a7cbe0d94076aedd3ffca0e9ea666ed1e1701da2e4a00f207ae5; engine receipt sha256:60dbdbeaa6e2e2942b88084b2468aa50d46427d845004f91f90f4eb8e94e3ee6 at receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-RELATIVE-PHASE-009-60dbdbeaa6e2e294.json.

SFT-QUANTUM-OBL-QSTATEX-010 - Global and relative phase distinction

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-GLOBAL-PHASE-010; Quantum Computation family QSTATEX; Global and relative phase distinction.
- **Forced law:** A common phase action on every supported word changes the global record while preserving all pairwise relative-phase classes.
- **Unique surviving structure:** complete-canonical-word-support__common-phase-action-preserves-relations__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-010 uniquely retains common-phase-action-preserves-relations, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-RELATIVE-PHASE-009. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-010: global_relative_phase; exact quantum-state execution: {"common_action_applied":true,"relative_classes_preserved":true}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation
sha256:e0326e9e302ce238c3f787b973b36b24cd2811ef938a7e1c0f717d5fed150180; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate
sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement sha256:9fb96cb522b025d492c7ddff2044b867fd29da16ee307ab4fdb545c805d2e2de; engine receipt sha256:d2a8f2002a3c22e4db71a7022f70f045177edceee2c17db6ef8286cc4ab24181 at receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-GLOBAL-PHASE-010-d2a8f2002a3c22e4.json.

SFT-QUANTUM-OBL-QSTATEX-011 - Phase composition and inversion

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-PHASE-COMPOSITION-011; Quantum Computation family QSTATEX; Phase composition and inversion.
- **Forced law:** Phase actions compose as exact held-label transformations; the inverse is the action that restores every original phase label on complete support.
- **Unique surviving structure:** complete-canonical-word-support__period-two-phase-action-and-inverse__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-011 uniquely retains period-two-phase-action-and-inverse, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REXX-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-GLOBAL-PHASE-010. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-011: phase_inverse; exact quantum-state execution: {"period":2,"restored_phase":"phase-held","source_phase":"phase-held"}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation
sha256:ca48f07f1e842328c8488fdca4c73b531b519d9cc6cc15196e2c1f2a249de5e1; independent

```
implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d;
independent certificate
sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement
sha256:b0d2819f58cf3459c4b63e15bd6b7c832c698685b6d0323e711f3ed2abf51015; engine receipt
sha256:e4e489d082ea1614dc7bec1fc93b7b502168bd7f955afd564619c4c51c8dc610 at receipts/
engine/model_admitted/SFT-QUANTUM-QSTATEX-PHASE-COMPOSITION-011-e4e489d082ea1614.j
son.
```

SFT-QUANTUM-OBL-QSTATEX-012 - Constructive and destructive interference classes

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-INTERFERENCE-CLASSES-012; Quantum Computation family QSTATEX; Constructive and destructive interference classes.
- **Forced law:** Interference classes are forced when multiple predecessor paths share an image: retained equal relative labels compose constructively while distinct period labels compose destructively under the registered observation.
- **Unique surviving structure:** complete-canonical-word-support__same-and-distinct-phase-predecessor-merge__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-012 uniquely retains same-and-distinct-phase-predecessor-merge, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-PHASE-COMPOSITION-011. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-012: interference_classes; exact quantum-state execution: {"predecessors":2,"same_and_distinct_phase_classes_retained":true,"shared_image":true}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation


```
sha256:2dfe297777c0c3fd563e59a29e9becc447d9e98e70d9ce8b22d9e3ca6582c63; independent
implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d;
independent certificate
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sha256:5b06b7b00a905e065dcddc466202522ffd032ff62a158467621b8d817a8db1a6; engine receipt
sha256:419507ed46703949d71cabbf3cf5b234ef96968a1ca353f22002910ef697ca27 at receipts/
engine/model_admitted/SFT-QUANTUM-QSTATEX-INTERFERENCE-CLASSES-012-419507ed4670394
9.json.
```

SFT-QUANTUM-OBL-QSTATEX-013 - Path composition and predecessor merging

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-PATH-MERGING-013; Quantum Computation family QSTATEX; Path composition and predecessor merging.
- **Forced law:** Path composition retains every predecessor, transition and phase label even when several paths share one observed image; the merged record determines the operational class.
- **Unique surviving structure:** complete-canonical-word-support__many-predecessor-one-image-phase-ledger__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-013 uniquely retains many-predecessor-one-image-phase-ledger, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-INTERFERENCE-CLASSES-012. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-013: path_merging; exact quantum-state execution: {"observed_images":1,"predecessor_ledger_width":2,"predecessor_paths":2}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation sha256:9e1e3d6ff58c3b7d0e2a7558cf94ab28f22b1dce77666991c55118fa0fbde92c; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement sha256:ebca5631c3c0edb4f6eb20edbe9029a1f41dd5d0f6b6328f9b2c2270918d2dd2; engine receipt sha256:9d64a42b48be4ca9f5a3f705acfeb91e0b00c7d94a972d098c9e98fd1a62f5a6 at receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-PATH-MERGING-013-9d64a42b48be4ca9.json.

SFT-QUANTUM-OBL-QSTATEX-014 - Which-path record and interference boundary

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-WHICH-PATH-014; Quantum Computation family QSTATEX; Which-path record and interference boundary.
- **Forced law:** A retained which-path record keeps predecessor classes distinguishable and therefore prevents their treatment as one closed image class; interference requires closure of that distinction at the declared observation.
- **Unique surviving structure:** complete-canonical-word-support__path-record-retains-distinguishability__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-014 uniquely retains

path-record-retains-distinguishability, complete state custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-PATH-MERGING-013. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-014: which_path_boundary; exact quantum-state execution: {"closed_merge_allowed":false,"distinguishable_classes":2,"path_record_retained":true}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation
sha256:2b6f35f03845296a08591e7bca94a904e7def879dee6eed87f89096e653c01ac; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate
sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement sha256:6e06be48a30948e7aded15a17f7cbe311e09b876ecf6485882feab66019495f8; engine receipt sha256:9e88fa83b62ba6010900f67dbaade3367cbda6ec56b649da014a64d1ed35d41 at receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-WHICH-PATH-014-9e88fa83b62ba601.json.

SFT-QUANTUM-OBL-QSTATEX-015 - Bipartite nonfactorable support

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-BIPARTITE-ENTANGLEMENT-015; Quantum Computation family QSTATEX; Bipartite nonfactorable support.
- **Forced law:** Bipartite entanglement is exact joint support that is not the Cartesian product of its two marginal supports, proven by complete product comparison.
- **Unique surviving structure:** complete-canonical-word-support__noncartesian-bipartite-joint-support__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-015 uniquely retains noncartesian-bipartite-joint-support, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-WHICH-PATH-014. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported

axioms: 0; free parameters: 0.

- **Observation:** QSTATEX-015: bipartite_nonfactorability; exact quantum-state execution: {"factorable":false,"joint_rows":2,"marginal_product_rows":4}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation
sha256:6e05057d5f09295414ae8d7c3cb44a3dc239a0b025f0db72103263a8c0a4df49; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate
sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement sha256:6eb721a08bc77a938eff668b6073f591e81dfe48f34422b38c7b51ad08f2d13c; engine receipt sha256:00feac58eb639f6b1ee6b52635ef198ffb08ed4ed536cde5b8c0023026a1cda4 at receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-BIPARTITE-ENTANGLEMENT-015-00feac58eb639f6b.json.

SFT-QUANTUM-OBL-QSTATEX-016 - Multipartite nonfactorable support

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-MULTIPARTITE-ENTANGLEMENT-016; Quantum Computation family QSTATEX; Multipartite nonfactorable support.
- **Forced law:** Multipartite entanglement is joint word support that fails every declared component factorization while retaining the full partition ledger.
- **Unique surviving structure:** complete-canonical-word-support__nonfactorable-multipartite-joint-support__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-016 uniquely retains nonfactorable-multipartite-joint-support, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REXX-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-BIPARTITE-ENTANGLEMENT-015. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-016: multipartite_nonfactorability; exact quantum-state execution: {"factorable_partitions":0,"joint_rows":2,"partitions_checked":2}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation
`sha256:9781912c679689cbee040439239f0adb67007a3549c5f4457a478692b6f6b550`; independent implementation `sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d`; independent certificate
`sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd`; measurement `sha256:263e588349dba478dc22e3ded0650dfa12819ca14d36b74bb6c47c95948cef12`; engine receipt `sha256:73551624cfce10acd3164061ed9d02b2ccd8c201b173984cbcd86e7492a810c0` at `receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-MULTIPARTITE-ENTANGLEMENT-016-73551624cfce10ac.json`.

SFT-QUANTUM-OBL-QSTATEX-017 - Entanglement partition and cut structure

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-ENTANGLEMENT-CUT-017; Quantum Computation family QSTATEX; Entanglement partition and cut structure.
- **Forced law:** Entanglement across a cut is decided by enumerating every joint word and the full product of the two projected supports at that cut.
- **Unique surviving structure:** `complete-canonical-word-support__complete-partition-factorability-census__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff`. Exact result: QSTATEX-017 uniquely retains `complete-partition-factorability-census`, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-MULTIPARTITE-ENTANGLEMENT-016. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-017: `partition_cut`; exact quantum-state execution: `{"complete_census":true,"declared_cuts":[1,2],"nonfactorable_cuts":[1,2]}`; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation
`sha256:9de444b16fe15bda55f47b826ff320d2a79cb4d65acf51d79525f57e5bbdb5e6`; independent implementation `sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d`; independent certificate

sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement
 sha256:92eaa9ffe32b9ae1123612dd36099678e9d9a4e7caa424aab9fb9fc460d418af; engine receipt
 sha256:11f278f78be095cf6e69ad062885f2f1768d30ab562db5c08d028a01d4ae95e3 at receipts/
 engine/model_admitted/SFT-QUANTUM-QSTATEX-ENTANGLEMENT-CUT-017-11f278f78be095cf.js
 on.

SFT-QUANTUM-OBL-QSTATEX-018 - Entanglement swapping correspondence

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-ENTANGLEMENT-SWAP-018; Quantum Computation family QSTATEX; Entanglement swapping correspondence.
- **Forced law:** Entanglement-swapping correspondence is the reversible composition of two joint supports followed by a registered middle observation whose retained record indexes a nonfactorable outer support.
- **Unique surviving structure:** complete-canonical-word-support__joint-observation-repartition s-outer-support__held-relative-phase-ledger__complete-source-and-observation-recor d__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-exec ution__explicit-formal-physical-handoff. Exact result: QSTATEX-018 uniquely retains joint-observation-repartitions-outer-support, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-ENTANGLEMENT-CUT-017. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-018: entanglement_swapping; exact quantum-state execution: {"input_joint_supports":2,"middle_observation_retained":true,"outer_support_repartitioned":true}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation
 sha256:c317e126c6bbe2fc5f0c3080a824adceeb5d02a8735fcac3ee1e898bdad4f801; independent
 implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d;
 independent certificate
 sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement
 sha256:64f87debeb61ea84492003167da9289270ebc2739059c64401b85c09857d309f; engine receipt
 sha256:5e0753f708ad7d648ea6abae55c2f7f642a3ff7e5191ea4d472f8690d1650ad at receipts/
 engine/model_admitted/SFT-QUANTUM-QSTATEX-ENTANGLEMENT-SWAP-018-5e0753f708ad7d64.j
 son.

SFT-QUANTUM-OBL-QSTATEX-019 - Monogamy and shareability boundary

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-MONOGAMY-019; Quantum Computation family QSTATEX; Monogamy and shareability boundary.

- **Forced law:** Shareability is decided by whether one complete joint support can satisfy every requested marginal product relation simultaneously; missing cross-words force the exact monogamy boundary.
- **Unique surviving structure:** complete-canonical-word-support__joint-support-shareability-census__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-019 uniquely retains joint-support-shareability-census, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-RE VX-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-ENTANGLEMENT-SWAP-018. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-019: monogamy_shareability; exact quantum-state execution: {"missing_cross_words":2,"present_joint_rows":2,"requested_product_rows":4}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation sha256:97050d514b11589c1f6d8bb198271dcc2bac9bee01c9d386d855ad6d22ebb178; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement sha256:dfd8b94272cd012887baea3dcf88197f61af7d13904f4bc551d90c4fb5f804c5; engine receipt sha256:a2d4e6c7faeaa6ac9de03d76fe857bd7498a4b1f1c79f1e1acb63009e477a7dc at receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-MONOGAMY-019-a2d4e6c7faeaa6ac.json.

SFT-QUANTUM-OBL-QSTATEX-020 - State purification correspondence

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-PURIFICATION-020; Quantum Computation family QSTATEX; State purification correspondence.
- **Forced law:** A mixed preparation record is purified by adjoining one distinct retained record word per preparation class, forming one larger source-bound joint support.
- **Unique surviving structure:** complete-canonical-word-support__preparation-record-extension__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-020 uniquely retains preparation-record-extension, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or

independent reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-MONOGAMY-019. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-020: purification; exact quantum-state execution: {"extended_support_source_bound":true,"preparation_classes":2,"record_words_added":2}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation sha256:0df6af2e8184d2dc4bbf61d6320ffff0adad7f6479a4504e750122d730946f29; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement sha256:f81a01ef2f4c6bfb7b1d581171842afa85d1f6be97a81c6259c3203bbf2715a9; engine receipt sha256:34ff5cdf590a29ac0dcc8ef08d29aba42c5ceb495a80c893c5a1c20adebdaa83 at receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-PURIFICATION-020-34ff5cdf590a29ac.json.

SFT-QUANTUM-OBL-QSTATEX-021 - Reduced observation and retained environment record

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-REDUCED-OBSERVATION-021; Quantum Computation family QSTATEX; Reduced observation and retained environment record.
- **Forced law:** Reduced observation projects the declared subsystem while retaining the complementary environment and full joint source record needed for reconstruction.
- **Unique surviving structure:** complete-canonical-word-support__projected-observation-with-joint-custody__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-021 uniquely retains projected-observation-with-joint-custody, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-PURIFICATION-020. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-021: reduced_observation; exact quantum-state execution: {"environment_rows_retained":2,"joint_rows_retained":2,"projected_rows":2}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation;

physical frequencies, amplitudes and device outcomes remain downstream measurements

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation
sha256:259cbc062081f82d5b386a2892fdb7b6930bdeb9a7ac568bfb38842284a504ac; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate
sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement sha256:327cf9b87ffef41b652bcc9d13559d9fa69c73d27b7a08df4dfc09d90b6bfb8; engine receipt sha256:0b3a739b0445173ab3a2276746cc431103a49279235c514f8fa2a1294d1c6633 at receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-REDUCED-OBSERVATION-021-0b3a739b0445173a.json.

SFT-QUANTUM-OBL-QSTATEX-022 - Measurement question and outcome-class identity

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-MEASUREMENT-CLASS-022; Quantum Computation family QSTATEX; Measurement question and outcome-class identity.
- **Forced law:** A measurement is a registered exhaustive partition of state support; its outcome is the unique class containing the presented word and its record identifies the question and source state.
- **Unique surviving structure:** complete-canonical-word-support__complete-partition-observation__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-022 uniquely retains complete-partition-observation, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-REDUCED-OBSERVATION-021. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-022: measurement_classes; exact quantum-state execution: {"matched_classes":1,"question_classes":2,"question_identity_retained":true}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.

- **Evidence identities:** derivation

sha256:efb583a7624648f8104ae459e2cce42ff36c7865daf02f3bc771d473fd046c86; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate
sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement sha256:aa6c00d02866870eff52e04ada0bd958a99ae4413a3e1a6a61296f60195c457f; engine receipt sha256:cfa379963b05bf0e9994669b8486a01235b077ea5ed9fb1f6ec81160ae0ecee2 at receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-MEASUREMENT-CLASS-022-cfa379963b05bf0e.json.

SFT-QUANTUM-OBL-QSTATEX-023 - Measurement composition and repeatability

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-MEASUREMENT-REPEATABILITY-023; Quantum Computation family QSTATEX; Measurement composition and repeatability.
- **Forced law:** Repeating the same complete observation on its retained outcome class returns that class unless an intervening registered transformation changes support.
- **Unique surviving structure:** complete-canonical-word-support__retained-outcome-class-repeatability__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-023 uniquely retains retained-outcome-class-repeatability, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-MEASUREMENT-CLASS-022. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-023: measurement_repeatability; exact quantum-state execution: {"first_outcome":"class-held","intervening_transform":false,"repeated_outcome":"class-held"}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation
sha256:650309b4b56dc9d678efd07371c95dceffdcdb3aeba17e1cdfe233da005a8bae; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate
sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement sha256:8820a22924d3c9162d9aebd9e20538db06ea4e03155f51db694dca46849dd12c; engine receipt sha256:eb3cddc2b5a75d3984308d8fcb37ba1714e194d646da05e6959ac0e01d8ff7ca at receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-MEASUREMENT-REPEATABILITY-023-eb3cddc2b5a75d39.json.

SFT-QUANTUM-OBL-QSTATEX-024 - Compatible and incompatible observation relations

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-OBSERVATION-COMPATIBILITY-024; Quantum Computation family QSTATEX; Compatible and incompatible observation relations.
- **Forced law:** Observations are compatible exactly when both causal orders produce the same outcome and retained-record partition; otherwise their order is an operational distinction.
- **Unique surviving structure:** complete-canonical-word-support__observation-order-record-comparison__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-024 uniquely retains observation-order-record-comparison, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REVS-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-MEASUREMENT-REPEATABILITY-023. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-024: observation_compatibility; exact quantum-state execution: {"causal_orders_compared":2,"same_record_required_for_compatibility":true}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation sha256:9821d8f3467c4c951faeb092b2865c43aa996994abaab6e15d0e41e3e87bdef3; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement sha256:ed74781bb71ecb26ff6efe6341242471c459e1c7186a0086c627111c413d5af0; engine receipt sha256:f3b117b3af9056e91d0dbee0fe8539ad1307eb559fae5250a5e438bda6604656 at receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-OBSERVATION-COMPATIBILITY-024-f3b117b3af9056e9.json.

SFT-QUANTUM-OBL-QSTATEX-025 - Deferred-measurement correspondence

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-DEFERRED-MEASUREMENT-025; Quantum Computation family QSTATEX; Deferred-measurement correspondence.
- **Forced law:** A measurement can be deferred when a reversible interaction retains the exact question-class record and later reading it yields the same outcome and downstream support.
- **Unique surviving structure:** complete-canonical-word-support__reversible-question-record-then-read__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__e

explicit-formal-physical-handoff. Exact result: QSTATEX-025 uniquely retains reversible-question-record-then-read, complete state custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-RE VX-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-OBSERVATION-COMPATIBILITY-024. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-025: deferred_measurement; exact quantum-state execution: {"deferred_outcome": "held", "immediate_outcome": "held", "question_record_retained": true}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation sha256:a9be6084ab2d9b8d6864d6ce21280a6f95b25c278f79ccc69d7281b16c8e6125; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement sha256:2c835177b5a18032c4882fe33e8de9c3cb9c96e50c0991f3fbbc837f6e6e60fa; engine receipt sha256:93e35d078d24df28131ffa88edbfb9abb96a1de1bd0a83141dc2e366f660a412 at receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-DEFERRED-MEASUREMENT-025-93e35d078d24df28.json.

SFT-QUANTUM-OBL-QSTATEX-026 - No-cloning and exact copying boundary

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-NO-CLONING-026; Quantum Computation family QSTATEX; No-cloning and exact copying boundary.
- **Forced law:** An unknown nonfactorable or relative-phase support cannot be copied by a source-independent local reversible map; exact copying remains lawful only for an already distinguished orthogonal Fold-label class with its source record.
- **Unique surviving structure:** complete-canonical-word-support__unknown-joint-state-copy-impossibility__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-026 uniquely retains unknown-joint-state-copy-impossibility, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-RE VX-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 ->

SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 ->
 SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 ->
 SFT-QUANTUM-QSTATEX-DEFERRED-MEASUREMENT-025. Registered root:
 SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** QSTATEX-026: no_cloning_boundary; exact quantum-state execution: {"distinguished_label_copy_allowed":true,"local_source_independent_copy":false,"unknown_joint_support_factorable":false}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation
 sha256:084bb4a7cc53b44050b7adda3a8d45506423e7aee4b9b65c553d4e8b4c441793; independent
 implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d;
 independent certificate
 sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement
 sha256:e4d912d7f86bb06add90638e91dfcfd8a067a85418e06f52bb9f94608055d7c8; engine receipt
 sha256:a70d2584130a0489c6d1442e137f8cd77dc8a5bfda38d0cc316ba458b6eab556 at receipts/
 engine/model_admitted/SFT-QUANTUM-QSTATEX-NO-CLONING-026-a70d2584130a0489.json.

SFT-QUANTUM-OBL-QSTATEX-027 - No-deleting and retained-record boundary

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-NO-DELETING-027; Quantum Computation family QSTATEX; No-deleting and retained-record boundary.
- **Forced law:** A reversible process cannot map two distinct supported states to one identical state unless the deleted distinction survives in an explicit environment or history record.
- **Unique surviving structure:** complete-canonical-word-support__reversible-deletion-requires-record__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-027 uniquely retains reversible-deletion-requires-record, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REXX-COMPLETENESS-018 ->
 SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 ->
 SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 ->
 SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 ->
 SFT-QUANTUM-QSTATEX-NO-CLONING-026. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported
 axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-027: no_deleting_boundary; exact quantum-state execution:
 {"distinct_sources":2,"retained_environment_records":2,"terminal_labels":1}; complete quantum-state custody: 28
 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation;
 physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status:

empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation
sha256:10665669683c29e18db83dade4042156aedcea3810145e09dc9228a9d160000b; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate
sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement sha256:42dfd95b61466bdac98266446ff81ba72848fe3e4f259c632fff4c8b80c8a17b; engine receipt sha256:b6891e68c3fb693be79ac5332d00165c11aa974b310b84d63c48f87f4ed4ea17 at receipts/engine/model_admitted/SFT-QUANTUM-QSTATEX-NO-DELETING-027-b6891e68c3fb693b.json.

SFT-QUANTUM-OBL-QSTATEX-028 - Quantum-state structure completeness certificate

- **Claim and ownership:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028; Quantum Computation family QSTATEX; Quantum-state structure completeness certificate.
- **Forced law:** The quantum-state family is complete exactly when all twenty-eight frozen obligations have one owner, one unique survivor, complete controls, exact observations, independent reconstructions and current receipts.
- **Unique surviving structure:** complete-canonical-word-support__twenty-eight-obligation-no-omission-ledger__held-relative-phase-ledger__complete-source-and-observation-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: QSTATEX-028 uniquely retains twenty-eight-obligation-no-omission-ledger, complete state custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete state grammar, exact support/phase/partition execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-REXX-COMPLETENESS-018 -> SFT-QUANTUM-INFORMATION-UNIT-001 -> SFT-QUANTUM-STATE-COMPOSITION-001 -> SFT-QUANTUM-SUPERPOSITION-001 -> SFT-QUANTUM-PHASE-INTERFERENCE-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-MEASUREMENT-001 -> SFT-QUANTUM-QSTATEX-NO-DELETING-027. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSTATEX-028: state_family_completeness; exact quantum-state execution: {"duplicate_owners":0,"execution_rows":28,"omitted_owners":0,"registered_obligations":28}; complete quantum-state custody: 28 records; all rows preserved; complete finite support, phase, path, partition and record execution supplies the observation; physical frequencies, amplitudes and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-STATE-OBSERVER, SFT-V1-V2-QUANTUM-STATE-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space, wavefunction, stochastic-collapse or qubit axiom selects the law; host 0 denotes absence only and is not a numerical-zero state object; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden predecessor, chosen support row, ontic randomness or unregistered oracle; no physical probability, device result or measured constant is inferred inside the formal state family; no first failure retires an obligation or alters the protected authority.
- **Evidence identities:** derivation
sha256:5bad2ef816e75c84b42f491b8a47748829a9077ff33a1726fbc6bc79728e1386; independent implementation sha256:d3176c8bd387bb5eda168956b7735674c8c21cf5ee5375bc3d210be89313e24d; independent certificate

```
sha256:7a3464f10b97d34667b8fc599024208c2f8f44cc9e480c4827860a0b4c6750bd; measurement
sha256:e444adf564c9e80b73979d8a86e7d3c712b34afc9956a261d7ccb668e401eed8; engine receipt
sha256:dba29903af1c70b92bf9bb278fce2ddd10f820809bef4b4a5f534223e130bd2d at receipts/
engine/model_admitted/SFT-QUANTUM-QSTATEX-COMPLETENESS-028-dba29903af1c70b9.json.
```

36.6 Family GATEX - 22/22 complete

Scope. Transformations, gates, circuits, compilation and universality. This family decides 5,632 candidates, retains 22 unique survivors and passes 88 adverse controls.

SFT-QUANTUM-OBL-GATEX-001 - Reversible transformation identity

- **Claim and ownership:** SFT-QUANTUM-GATEX-TRANSFORMATION-001; Quantum Computation family GATEX; Reversible transformation identity.
- **Forced law:** A reversible quantum transformation is a source-complete permutation of canonical Fold support with one exact inverse row and retained phase and provenance records.
- **Unique surviving structure:** complete-canonical-gate-support__complete-support-permutation-with-inverse__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: GATEX-001 uniquely retains complete-support-permutation-with-inverse, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-001: reversible_transformation; exact gate/circuit execution: {"image_rows":2,"inverse_rows":2,"round_trip":true,"source_rows":2}; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation
 sha256:ec5e0de5ff51244830f4d81186aa9c28a5fb46f6e71cfcee28ed27cdab0b6dab; independent implementation sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b; independent certificate
 sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c; measurement sha256:e5ec437652ad0c86d77d219c7818bf8d4edce5ece235e484e447391cc2edfcb9; engine receipt sha256:8b5001ec7f6cb3acb0b6b310de22de61b4eb5083ca6baa4f838d8b97ef273dbd at receipts/engine/model_admitted/SFT-QUANTUM-GATEX-TRANSFORMATION-001-8b5001ec7f6cb3ac.json.

SFT-QUANTUM-OBL-GATEX-002 - Single-unit permutation and phase actions

- **Claim and ownership:** SFT-QUANTUM-GATEX-SINGLE-UNIT-002; Quantum Computation family GATEX; Single-unit permutation and phase actions.

- **Forced law:** The primitive single-unit actions are the complete permutations of the two forced fibre labels together with the independently retained period-phase action.
- **Unique surviving structure:** complete-canonical-gate-support__single-distinction-label-and-phase-actions__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: GATEX-002 uniquely retains single-distinction-label-and-phase-actions, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-TRANSFORMATION-001. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-002: single_unit_actions; exact gate/circuit execution: {"permutations":2,"phase_actions":2,"support_complete":true}; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation
sha256:6f6f29fa8dd36e69bc852b4dfc38869ca5eb60039a11e4eba5e188925c83dcbd; independent implementation sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b; independent certificate
sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c; measurement sha256:d9eb85aab6bc8914483676742d3cd4cdc481a768be13303f90bef3dcbf1fa578; engine receipt sha256:ea3b9e967ed4dbbdf3bf60e2c3d4b9df6a45d2a4c51b19b8b1fda2f5673409ba at receipts/engine/model_admitted/SFT-QUANTUM-GATEX-SINGLE-UNIT-002-ea3b9e967ed4dbbd.json.

SFT-QUANTUM-OBL-GATEX-003 - Controlled transformation law

- **Claim and ownership:** SFT-QUANTUM-GATEX-CONTROLLED-003; Quantum Computation family GATEX; Controlled transformation law.
- **Forced law:** A controlled transformation retains the control label and applies a bijective target action on exactly the registered control class, preserving one-to-one joint support.
- **Unique surviving structure:** complete-canonical-gate-support__retained-control-bijective-target-action__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: GATEX-003 uniquely retains retained-control-bijective-target-action, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-SINGLE-UNIT-002. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-003: controlled_transformation; exact gate/circuit execution: {"bijection":true,"control_retained":true,"joint_rows":4}; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:081ff722ce382f6b8eef5698992c971ddb4da1dced41cd9b2e8156180495f9c6; independent implementation sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b; independent certificate sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c; measurement sha256:2b963427c9dde8abc79e408f9c5c1f1d7e66e68eb7f18af26450e1a87957bb8f; engine receipt sha256:3e13e2a5631fd926b66b5a7f0ddeb18c9ea742fa984b28aeb29b63f0de71391 at receipts/engine/model_admitted/SFT-QUANTUM-GATEX-CONTROLLED-003-3e13e2a5631fd926.json.

SFT-QUANTUM-OBL-GATEX-004 - Two-unit entangling transformation

- **Claim and ownership:** SFT-QUANTUM-GATEX-ENTANGLING-004; Quantum Computation family GATEX; Two-unit entangling transformation.
- **Forced law:** A two-unit transformation is entangling when a product input support can be mapped to nonfactorable joint support while the complete four-word map remains bijective.
- **Unique surviving structure:** complete-canonical-gate-support__two-unit-controlled-nonlocal-support-action__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomp utation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registr y-exact-execution__explicit-formal-physical-handoff. Exact result: GATEX-004 uniquely retains two-unit-controlled-nonlocal-support-action, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-CONTROLLED-003. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-004: two_unit_entangling_gate; exact gate/circuit execution: {"bijection":true,"joint_rows":4,"product_input_can_become_nonfactorable":true}; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation
`sha256:691ed2cb92cd95bb48bfd3c564b72b909649791ef8e3312d8cc57badc47d8c1`; independent implementation `sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b`; independent certificate
`sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c`; measurement `sha256:6d624880f8ef1b4f0bfcd5ae6c8f3c866c238ce18304caa648c4d5f1a44672b`; engine receipt `sha256:2d5cf54185479899eac16361b268746ab889ea24a98db34c68ff7fa91fd0e701` at `receipts/engine/model_admitted/SFT-QUANTUM-GATEX-ENTANGLING-004-2d5cf54185479899.json`.

SFT-QUANTUM-OBL-GATEX-005 - Multi-controlled transformation

- **Claim and ownership:** SFT-QUANTUM-GATEX-MULTICONTROL-005; Quantum Computation family GATEX; Multi-controlled transformation.
- **Forced law:** A multi-controlled transformation applies one target permutation only on its declared complete control word and retains every control label and unaffected row.
- **Unique surviving structure:** `complete-canonical-gate-support__complete-control-word-conditioned-action__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff`. Exact result: GATEX-005 uniquely retains complete-control-word-conditioned-action, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-ENTANGLING-004. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-005: `multi_controlled_gate`; exact gate/circuit execution: `{"control_width":2,"trigger_words":1,"unaffected_words_retained":true}`; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: True.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation
`sha256:7b140765c7edda87916a18f6b1969bb2c35b9ab3c6d95027707a8aa72f7ea7c4`; independent implementation `sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b`; independent certificate
`sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c`; measurement `sha256:b619700fb368f1b17f250903efc84cb851505d40c8db5f06b83521b7c9fa22c2`; engine receipt `sha256:66beb7c02e47e1d52ae04e98a0f05df366c44074f999999f98177e7d12c5ea8b` at `receipts/`

engine/model_admitted/SFT-QUANTUM-GATEX-MULTICONTROL-005-66beb7c02e47e1d5.json.

SFT-QUANTUM-OBL-GATEX-006 - Gate composition and inverse

- **Claim and ownership:** SFT-QUANTUM-GATEX-COMPOSITION-INVERSE-006; Quantum Computation family GATEX; Gate composition and inverse.
- **Forced law:** Gate composition is causal application on complete support; its inverse is the reverse causal sequence of each exact gate inverse.
- **Unique surviving structure:** complete-canonical-gate-support__causal-gate-composition-reverse-inverse__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: GATEX-006 uniquely retains causal-gate-composition-reverse-inverse, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-MULTICONTROL-005. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-006: composition_inverse; exact gate/circuit execution: {"forward_gates":2,"inverse_gates":2,"source_restored":true}; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:d740ac7d11e5a544860c635e8d7b1119e82717d3c5bce8cc277925554e24053f; independent implementation sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b; independent certificate sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c; measurement sha256:d1541fba74a0a90c24df5a7f835833279ac9bdc2c42f48a32c199d67cc7241da; engine receipt sha256:b723ce59cdd03a86c6e71f0dde9dcf301bb35aeb96b6002cd417a19f21702bfc at receipts/engine/model_admitted/SFT-QUANTUM-GATEX-COMPOSITION-INVERSE-006-b723ce59cdd03a86.json.

SFT-QUANTUM-OBL-GATEX-007 - Gate commutation and causal reordering

- **Claim and ownership:** SFT-QUANTUM-GATEX-COMMUTATION-007; Quantum Computation family GATEX; Gate commutation and causal reordering.
- **Forced law:** Two gates commute exactly when both causal orders produce the same word, phase and retained record on every generated input support row.
- **Unique surviving structure:** complete-canonical-gate-support__complete-support-order-equivalence__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: GATEX-007 uniquely retains

complete-support-order-equivalence, complete circuit custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-COMPOSITION-INVERSE-006. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-007: commutation; exact gate/circuit execution: `{"causal_orders":2,"complete_support_compared":true,"same_map":true}`; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation
`sha256:626138f9cf767f8472e68c1fcd5099c1cafff04393ec34949e20ace4bcb35f844`; independent implementation `sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b`; independent certificate
`sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c`; measurement `sha256:7cf863761da4cd603716df99d6f8c479fda8d4aa76e21c98ec1f18b78d3b4d23`; engine receipt `sha256:e03cfd16be48d36d84bca744439b11df8d28babfd801df29e48ec206b6825f7a` at `receipts/engine/model_admitted/SFT-QUANTUM-GATEX-COMMUTATION-007-e03cfd16be48d36d.json`.

SFT-QUANTUM-OBL-GATEX-008 - Universal finite gate-description grammar

- **Claim and ownership:** SFT-QUANTUM-GATEX-DESCRIPTION-GRAMMAR-008; Quantum Computation family GATEX; Universal finite gate-description grammar.
- **Forced law:** A universal finite gate-description grammar generates wire identities, primitive reversible actions, controls, compositions and observations with exact arity and source binding.
- **Unique surviving structure:** `complete-canonical-gate-support__finite-registered-gate-word-grammar__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff`. Exact result: GATEX-008 uniquely retains `finite-registered-gate-word-grammar`, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-COMMUTATION-007. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-008: `gate_grammar`; exact gate/circuit execution: `{"finite_descriptions":true,"primitive_forms":["wire","swap","control","observe"]}`; complete gate/circuit custody: 22

records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation
sha256:bca4d67c87f52ffce5a5a16c99cd178078ed2bddd0ea0190e44675cb5715230c; independent implementation sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b; independent certificate
sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c; measurement sha256:5f081fa19cdcb49e21078987f955c8de690ab89b17e35aa46e1f9a8a7e8811c2; engine receipt sha256:b49926c4a573df6af7dfbf6245d2d626bd7c6db330d4848244bacce2ada6ef0c at receipts/engine/model_admitted/SFT-QUANTUM-GATEX-DESCRIPTION-GRAMMAR-008-b49926c4a573df6a.json.

SFT-QUANTUM-OBL-GATEX-009 - Exact gate synthesis

- **Claim and ownership:** SFT-QUANTUM-GATEX-EXACT-SYNTHESIS-009; Quantum Computation family GATEX; Exact gate synthesis.
- **Forced law:** Every permutation of a declared positive finite support is exactly synthesized by a generated finite sequence of support-row transpositions whose execution reproduces the target map.
- **Unique surviving structure:** complete-canonical-gate-support__finite-permutation-transposition-synthesis__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: GATEX-009 uniquely retains finite-permutation-transposition-synthesis, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-DESCRIPTION-GRAMMAR-008. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-009: exact_synthesis; exact gate/circuit execution: {"support_rows":3,"target_permutation_reproduced":true,"transpositions":2}; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.

- **Evidence identities:** derivation

sha256:612b244d7c0406710a30668c0c0a8243f85fb9fdfeff309c224d7c4b2eaf77e7; independent implementation sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b; independent certificate sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c; measurement sha256:c18b6e380283f9f94ec3df059a90cea56cb2530159edf9c3f4955e99f825c78b; engine receipt sha256:91f18eef9502551589f2f1af93744b35ae326f7f6a075246c547ee221a75271a at receipts/engine/model_admitted/SFT-QUANTUM-GATEX-EXACT-SYNTHESIS-009-91f18eef95025515.json.

SFT-QUANTUM-OBL-GATEX-010 - Approximate gate synthesis with enclosure custody

- **Claim and ownership:** SFT-QUANTUM-GATEX-APPROXIMATE-SYNTHESIS-010; Quantum Computation family GATEX; Approximate gate synthesis with enclosure custody.
- **Forced law:** Approximate synthesis retains exact rational lower and upper enclosures for every unresolved target relation and admits a circuit only when its full action remains inside the registered enclosure.
- **Unique surviving structure:** complete-canonical-gate-support__nested-rational-enclosure-synthesis__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: GATEX-010 uniquely retains nested-rational-enclosure-synthesis, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-EXACT-SYNTHESIS-009. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-010: approximate_synthesis; exact gate/circuit execution: {"bounds_exact_rational":true,"enclosures":2,"target_outside_enclosure_rejected":true}; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:9ff428f2525e95586b2ff47972ef9e9986ab61624c83f963b183c3a159f29512; independent implementation sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b; independent certificate sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c; measurement sha256:03ba79298cce08871e93f7ef7066d1509b28479f8ee1255f63e3da25400b75f7; engine receipt sha256:666608332674b971e5212e4c8cdf54729da6f85e64fd5383e50ab1ca4bcda3c4 at receipts/engine/model_admitted/SFT-QUANTUM-GATEX-APPROXIMATE-SYNTHESIS-010-666608332674b971.json.

SFT-QUANTUM-OBL-GATEX-011 - Circuit wire, register and gate syntax

- **Claim and ownership:** SFT-QUANTUM-GATEX-CIRCUIT-SYNTAX-011; Quantum Computation family GATEX; Circuit wire, register and gate syntax.

- **Forced law:** Circuit syntax is a finite causal word of declared wires, registers, gate instances, controls, observations and terminal outputs with every arity and identity checked.
- **Unique surviving structure:** complete-canonical-gate-support__typed-wire-register-gate-causal-word__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: GATEX-011 uniquely retains typed-wire-register-gate-causal-word, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-APPROXIMATE-SYNTHESIS-010. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-011: circuit_syntax; exact gate/circuit execution: {"arity_checked":true,"gate_instances":1,"terminal_outputs":1,"wires":1}; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation
sha256:ed767442387a38f7a9384a16945167ae561dd05e401b8e77568c2fd199f869ca; independent implementation sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b; independent certificate
sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c; measurement sha256:d75742f5df34a39aa4a9f918ab31a2c9f419c3286d3c4bc4c9966d915755f80a; engine receipt sha256:9cd4df151f0ed87920f6c24f83df29aa963f37525019fc54be651407abc3ce88 at receipts/engine/model_admitted/SFT-QUANTUM-GATEX-CIRCUIT-SYNTHESIS-011-9cd4df151f0ed879.json.

SFT-QUANTUM-OBL-GATEX-012 - Circuit branchwise operational semantics

- **Claim and ownership:** SFT-QUANTUM-GATEX-CIRCUIT-SEMANTICS-012; Quantum Computation family GATEX; Circuit branchwise operational semantics.
- **Forced law:** Circuit semantics executes every generated support word through every gate in causal order while retaining phase, predecessor and resource traces branch by branch.
- **Unique surviving structure:** complete-canonical-gate-support__complete-support-branchwise-execution__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: GATEX-012 uniquely retains complete-support-branchwise-execution, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-CIRCUIT-SYNTAX-011. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-012: branchwise_semantics; exact gate/circuit execution: {"complete_trace":true,"input_branches":2,"terminal_branches":2}; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation
sha256:600834f4c9fc25857b9995531166300e1645adfa5011137e5aa3292f418011c0; independent implementation sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b; independent certificate
sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c; measurement sha256:3ac528d27c655a442b1db90fd98cf43c6b6ad0545fba2a1705bda09b01a7eec; engine receipt sha256:83794eb58a4c8b04026d2a321375704fff221cee9ee8264a16c78e5ec41dde19 at receipts/engine/model_admitted/SFT-QUANTUM-GATEX-CIRCUIT-SEMANTICS-012-83794eb58a4c8b04.json.

SFT-QUANTUM-OBL-GATEX-013 - Circuit observation semantics

- **Claim and ownership:** SFT-QUANTUM-GATEX-CIRCUIT-OBSERVATION-013; Quantum Computation family GATEX; Circuit observation semantics.
- **Forced law:** Circuit observation applies a registered exhaustive partition to terminal support and retains the question, outcome class, predecessor support and downstream record.
- **Unique surviving structure:** complete-canonical-gate-support__registered-partition-and-outcome-record__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: GATEX-013 uniquely retains registered-partition-and-outcome-record, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-CIRCUIT-SEMANTICS-012. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-013: observation_semantics; exact gate/circuit execution: {"outcome_classes":1,"predecessor_support_retained":true,"question_classes":2}; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation
`sha256:c99042d3f8cc479f2a173b995ddb4acf921bd8b321064d43dabcc83b925f055a`; independent implementation `sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b`; independent certificate
`sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c`; measurement `sha256:83b899accb5af7bf6f6f3f3e07f5823b5b95731bd0972269e7da9c327e367229`; engine receipt `sha256:8e6b9386b3c4e1a6984bbc0b5e56215a5775c2f2fd9f99bf6190e81a4d301e5c` at receipts/engine/model_admitted/SFT-QUANTUM-GATEX-CIRCUIT-OBSERVATION-013-8e6b9386b3c4e1a6.json.

SFT-QUANTUM-OBL-GATEX-014 - Circuit inversion and uncomputation

- **Claim and ownership:** SFT-QUANTUM-GATEX-CIRCUIT-INVERSE-014; Quantum Computation family GATEX; Circuit inversion and uncomputation.
- **Forced law:** A circuit is uncomputed by applying exact inverse gates in reverse causal order and restoring every ancilla and retained support record to its declared source state.
- **Unique surviving structure:** `complete-canonical-gate-support__reverse-causal-inverse-circuit__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff`. Exact result: GATEX-014 uniquely retains `reverse-causal-inverse-circuit`, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-CIRCUIT-OBSERVATION-013. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-014: `circuit_inverse`; exact gate/circuit execution: `{"forward_depth":3,"inverse_depth":3,"source_restored":true}`; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: True.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation
`sha256:fd93f799e1a38a95b64190f9f80f954e61a45f65e45057d04c4cd8d9abe1faee`; independent implementation `sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b`; independent certificate
`sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c`; measurement `sha256:7345f9cb6eb1802a0b03363b36beaf4c30303eec5eeeb993dfb9fd99dbf3321`; engine receipt

sha256:345c266b83686cf0eba59615abd04386331e054d61397110e1fe5c20946ff714 at receipts/engine/model_admitted/SFT-QUANTUM-GATEX-CIRCUIT-INVERSE-014-345c266b83686cf0.json.

SFT-QUANTUM-OBL-GATEX-015 - Circuit equivalence and normal form

- **Claim and ownership:** SFT-QUANTUM-GATEX-CIRCUIT-EQUIVALENCE-015; Quantum Computation family GATEX; Circuit equivalence and normal form.
- **Forced law:** Two circuits are equivalent exactly when their word, relative-phase and retained-record maps agree on complete declared support; a normal form is the canonical representative of that map.
- **Unique surviving structure:** complete-canonical-gate-support__complete-support-semantic-equivalence__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: GATEX-015 uniquely retains complete-support-semantic-equivalence, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-CIRCUIT-INVERSE-014. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-015: circuit_equivalence; exact gate/circuit execution: {"circuits_compared":2,"same_canonical_map":true,"support_rows":2}; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:55eceb7a9a48551e4c26437068f810e3a891d533ef88d1b343b378a5461a32f7; independent implementation sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b; independent certificate sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c; measurement sha256:d733f410797a78f868194abee203dee838a4201c3039fde43e3ff7f8784695f9; engine receipt sha256:3fd607c9b9b332cd1a6c7c3dbb213bfecfb4a9b114b1d19d691ca7c9bed2f522 at receipts/engine/model_admitted/SFT-QUANTUM-GATEX-CIRCUIT-EQUIVALENCE-015-3fd607c9b9b332cd.json.

SFT-QUANTUM-OBL-GATEX-016 - Circuit decomposition into local transformations

- **Claim and ownership:** SFT-QUANTUM-GATEX-LOCAL-DECOMPOSITION-016; Quantum Computation family GATEX; Circuit decomposition into local transformations.
- **Forced law:** A circuit decomposition replaces a finite-support transformation with local reversible and controlled transformations whose complete composed map is identical.
- **Unique surviving structure:** complete-canonical-gate-support__finite-local-gate-causal-decomposition__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exa

ct-execution__explicit-formal-physical-handoff. Exact result: GATEX-016 uniquely retains finite-local-gate-causal-decomposition, complete circuit custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-CIRCUIT-EQUIVALENCE-015. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-016: local_decomposition; exact gate/circuit execution: {"composed_map_exact":true,"joint_support_rows":4,"local_gate_count":2}; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation
sha256:448be864f3193d9320c23a4f2b0637a69812c0826f581b1adb8a4b7c8e1e3064; independent implementation sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b; independent certificate
sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c; measurement sha256:c703a54d81295ab8ba670603e2e9fca0fdde263822607077eff8ddf27fc03ba6; engine receipt sha256:86aaf62f90fd19f92b19f67fbbfc374a9bb1a4109ccd190bac12888930ba2c9e at receipts/engine/model_admitted/SFT-QUANTUM-GATEX-LOCAL-DECOMPOSITION-016-86aaf62f90fd19f9.j
son.

SFT-QUANTUM-OBL-GATEX-017 - Circuit size, depth, width and live-support resources

- **Claim and ownership:** SFT-QUANTUM-GATEX-CIRCUIT-RESOURCES-017; Quantum Computation family GATEX; Circuit size, depth, width and live-support resources.
- **Forced law:** Circuit resources separately count gate size, causal depth, register width, complete live support, ancilla records, observations and retained inverse information.
- **Unique surviving structure:** complete-canonical-gate-support__separate-exact-circuit-resource-ledger__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: GATEX-017 uniquely retains separate-exact-circuit-resource-ledger, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-LOCAL-DECOMPOSITION-016. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** GATEX-017: circuit_resources; exact gate/circuit execution: {"depth":3,"live_support":4,"size":3,"width":2}; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation
sha256:69a520f7f6d900b3aced5d51b9e50d6582a03542e0cdc14c7644243030f618cd; independent implementation sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b; independent certificate
sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c; measurement sha256:6609669f4590d21adb5d3c27845ca01d1481a4d2b01f325139966df4ef8538cd; engine receipt sha256:05ca55eae472f5265d17cb719681ede3ab8baa7611c8df02eb7198e40a83e7e9 at receipts/engine/model_admitted/SFT-QUANTUM-GATEX-CIRCUIT-RESOURCES-017-05ca55eae472f526.json.

SFT-QUANTUM-OBL-GATEX-018 - Circuit compilation and semantic preservation

- **Claim and ownership:** SFT-QUANTUM-GATEX-COMPILATION-018; Quantum Computation family GATEX; Circuit compilation and semantic preservation.
- **Forced law:** Compilation is valid only when the generated target circuit reproduces the source circuit's complete support, phase, observation and retained-record semantics.
- **Unique surviving structure:** complete-canonical-gate-support__source-target-circuit-map-equivalence__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: GATEX-018 uniquely retains source-target-circuit-map-equivalence, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-CIRCUIT-RESOURCES-017. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-018: compilation; exact gate/circuit execution: {"semantics_preserved":true,"source_circuit_rows":2,"target_circuit_rows":2}; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an

obligation or changes protected authority.

- **Evidence identities:** derivation

sha256:f1b298769e6f4efbd7ac6844abcb95f8154f80306b48e59533b2a6063cdaf496; independent implementation sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b; independent certificate sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c; measurement sha256:19eb807ab038d5e1584a623bd1083015f85b351b6c84266d26e42a82aede1500; engine receipt sha256:35e44849027ef13fda2c95f5474fb6d77f83c305219ea4ed5fe1e9415f1ef19e at receipts/engine/model_admitted/SFT-QUANTUM-GATEX-COMPILATION-018-35e44849027ef13f.json.

SFT-QUANTUM-OBL-GATEX-019 - Measurement-based computation correspondence

- **Claim and ownership:** SFT-QUANTUM-GATEX-MEASUREMENT-BASED-019; Quantum Computation family GATEX; Measurement-based computation correspondence.
- **Forced law:** Measurement-based computation corresponds to circuit execution when joint preparation, ordered observations and record-controlled corrections reproduce the same terminal support and records.
- **Unique surviving structure:** complete-canonical-gate-support__joint-preparation-observation-correction-correspondence__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: GATEX-019 uniquely retains joint-preparation-observation-correction-correspondence, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-COMPILATION-018. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-019: measurement_based_correspondence; exact gate/circuit execution: {"observations":1,"preparation":1,"record_controlled_corrections":1,"terminal_map_preserved":true}; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation
sha256:172465cda5764be0aec858dc87a3006f3ae56d39e43f1667d59fad64abda7899; independent implementation sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b; independent certificate sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c; measurement sha256:df467d94b557855e381db563010dabf5620ca7b1d61893fb7a71cb4b7cf2235a; engine receipt sha256:471565a1baece55943aa9b8f82ba9a1f79180ee216e84635c681ee9cde1a8e25 at receipts/engine/model_admitted/SFT-QUANTUM-GATEX-MEASUREMENT-BASED-019-471565a1baece559.json.

SFT-QUANTUM-OBL-GATEX-020 - Adiabatic computation correspondence boundary

- **Claim and ownership:** SFT-QUANTUM-GATEX-ADIABATIC-BOUNDARY-020; Quantum Computation family GATEX; Adiabatic computation correspondence boundary.
- **Forced law:** An adiabatic description corresponds to a circuit only through a registered positive-finite state path and terminal map; physical gap, timing and hardware success remain measured handoffs.
- **Unique surviving structure:** complete-canonical-gate-support__finite-state-path-terminal-map-correspondence__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: GATEX-020 uniquely retains finite-state-path-terminal-map-correspondence, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-MEASUREMENT-BASED-019. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-020: adiabatic_boundary; exact gate/circuit execution: {"finite_path_rows":3,"physical_gap_measured_here":false,"terminal_map_retained":true}; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:7cb31ed9adbac7f9947c5d3e7bbda4e018562ce3bfb272d4b369226affef24e; independent implementation sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b; independent certificate sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c; measurement sha256:13374b607e3e76b59abeba14d878e94f6527feb0d8dbdd42e42cabebc49c4f05; engine receipt sha256:75a421b6ff07ed1a747a991322b59212b6bf2d6b507ba562512b298fd06497a8 at receipts/engine/model_admitted/SFT-QUANTUM-GATEX-ADIABATIC-BOUNDARY-020-75a421b6ff07ed1a.js on.

SFT-QUANTUM-OBL-GATEX-021 - Topological computation correspondence boundary

- **Claim and ownership:** SFT-QUANTUM-GATEX-TOPOLOGICAL-BOUNDARY-021; Quantum Computation family GATEX; Topological computation correspondence boundary.
- **Forced law:** A topological description corresponds to a circuit through a finite registered braid or deformation word whose induced support map and observation record are exactly reconstructed; physical realization remains downstream.
- **Unique surviving structure:** complete-canonical-gate-support__finite-braid-word-transformation-correspondence__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff. Exact result: GATEX-021 uniquely

retains finite-braid-word-transformation-correspondence, complete circuit custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-ADIABATIC-BOUNDARY-020. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** GATEX-021: `topological_boundary`; exact gate/circuit execution: `{"crossing_records":1,"finite_braid_words":1,"physical_realization_measured_here":false}`; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation
`sha256:2972b6cfb73f8ab13e57f7ce70a8094ba7364e4d58ee1dd0d8d5875c2f45352f`; independent implementation `sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b`; independent certificate
`sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c`; measurement `sha256:612dc48fe922bb37471af518e35facff1a58331e8eeee32b82bbaa536c9a990a`; engine receipt `sha256:2a5819d1404b1d485a21f9d1268905df7677faea70e4be6bd3cbe43b7312fa97` at `receipts/engine/model_admitted/SFT-QUANTUM-GATEX-TOPOLOGICAL-BOUNDARY-021-2a5819d1404b1d48.json`.

SFT-QUANTUM-OBL-GATEX-022 - Gate-and-circuit completeness certificate

- **Claim and ownership:** SFT-QUANTUM-GATEX-COMPLETENESS-022; Quantum Computation family GATEX; Gate-and-circuit completeness certificate.
- **Forced law:** The gate-and-circuit family is complete exactly when all twenty-two frozen obligations have one owner, one unique survivor, controls, observations, independent reconstructions and current receipts.
- **Unique surviving structure:** `complete-canonical-gate-support__twenty-two-obligation-no-omission-ledger__complete-branchwise-phase-and-record-map__exact-inverse-and-uncomputation-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-handoff`. Exact result: GATEX-022 uniquely retains twenty-two-obligation-no-omission-ledger, complete circuit custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete gate/circuit grammar, exact branchwise execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSTATEX-COMPLETENESS-028 -> SFT-QUANTUM-GATE-001 -> SFT-QUANTUM-CIRCUIT-001 -> SFT-QUANTUM-UNIVERSALITY-001 -> SFT-QUANTUM-GATEX-TOPOLOGICAL-BOUNDARY-021. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** GATEX-022: gate_circuit_completeness; exact gate/circuit execution: {"duplicate_owners":0,"execution_rows":22,"omitted_owners":0,"registered_obligations":22}; complete gate/circuit custody: 22 records; all rows preserved; complete finite support, gate, inverse, circuit, observation and resource execution supplies the observation; physical timing, gaps, error rates and device outcomes remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-GATE-CIRCUIT-OBSERVER, SFT-V1-V2-QUANTUM-CIRCUIT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported unitary matrix, complex amplitude, circuit basis or universality theorem selects the law; host 0 denotes absence only and is not a numerical-zero gate or state; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no sampled support, hidden branch, selected target or unregistered oracle; no physical gap, timing, error rate or hardware result is inferred inside the formal circuit family; no first failure retires an obligation or changes protected authority.
- **Evidence identities:** derivation
sha256:47a77288175c7b1cec0b99614e6718b6ebeca41e26e6183e21c470c5ebc28799; independent implementation sha256:cfbab36a1ed3c5571bcfe64bb901fbc3e8a4c3df148ee9e538dd093ddde1af9b; independent certificate
sha256:08339e2cb625320b21f22b0f4dfb5e1b44658831abd784909c89a9f9e859556c; measurement sha256:e1f9b272e8921154ef791530eba4931d717b626763bc94158aee184d737dedfb; engine receipt sha256:529a40311f05d60d99080f86ba2bbcaa8392a15a8e006c8c7e2c7bf89a49ae8f at receipts/engine/model_admitted/SFT-QUANTUM-GATEX-COMPLETENESS-022-529a40311f05d60d.json.

36.7 Family QALGX - 30/30 complete

Scope. Quantum algorithms and correctness. This family decides 7,680 candidates, retains 30 unique survivors and passes 120 adverse controls.

SFT-QUANTUM-OBL-QALGX-001 - Quantum algorithm specification and correctness trace

- **Claim and ownership:** SFT-QUANTUM-QALGX-SPECIFICATION-001; Quantum Computation family QALGX; Quantum algorithm specification and correctness trace.
- **Forced law:** A quantum algorithm is a registered input and promise grammar, a complete reversible process, an observation rule and an exact correctness trace covering every declared input class.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__input-promise-process-output-proof-trace__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-001 uniquely retains input-promise-process-output-proof-trace, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-001: algorithm_specification; exact quantum-algorithm execution: {"correctness_trace":true,"input_classes":1,"output_classes":1,"process_steps":1}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:9ea5dcdbd37b47e28c79669e612affe81f91be56453efdd0cb579099f21b0a126`; independent implementation `sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a`; independent certificate
`sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3`; measurement `sha256:d06bb935e23c6bc7fe25971bf591adfb576cb44c089c03cb933f54bb2d1ba609`; engine receipt `sha256:99f791836aa124c14c32cbeb4ee8ae4063f52f0fe0df9353ebe43a593ec643e7` at `receipts/engine/model_admitted/SFT-QUANTUM-QALGX-SPECIFICATION-001-99f791836aa124c1.json`.

SFT-QUANTUM-OBL-QALGX-002 - Reversible oracle and query interface

- **Claim and ownership:** SFT-QUANTUM-QALGX-ORACLE-002; Quantum Computation family QALGX; Reversible oracle and query interface.
- **Forced law:** An oracle query is a source-retaining reversible extension of a registered finite relation; its description, query count and target boundary remain explicit.
- **Unique surviving structure:** `complete-registered-input-and-promise-grammar__source-retaining-reversible-query-extension__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff`. Exact result: QALGX-002 uniquely retains source-retaining-reversible-query-extension, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-SPECIFICATION-001. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-002: `reversible_oracle`; exact quantum-algorithm execution: `{"bijection":true,"image_rows":4,"queries":1,"source_rows":4}`; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:2dd97b928e079d7839a941b534102d20dd3e0fcca165d3b38e7afabea4511726`; independent implementation `sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a`; independent certificate
`sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3`; measurement `sha256:b215a2acaa8755d266cef47d7bc5c3fe0efec68afd044573f05a3157280d1300`; engine receipt

sha256:c4dcd4b669adf954991cd5fa669eb73bfe980fca4db19f5256c3e0f65feddfdf at
 receipts/engine/model_admitted/SFT-QUANTUM-QALGX-ORACLE-002-c4dcd4b669adf954.json.

SFT-QUANTUM-OBL-QALGX-003 - Phase kickback correspondence

- **Claim and ownership:** SFT-QUANTUM-QALGX-PHASE-KICKBACK-003; Quantum Computation family QALGX; Phase kickback correspondence.
- **Forced law:** Phase kickback is the retained relative-phase change on a control support induced by a reversible controlled query while the target support returns to its source class.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__control-conditioned-relative-phase-return__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-003 uniquely retains control-conditioned-relative-phase-return, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-ORACLE-002. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-003: phase_kickback; exact quantum-algorithm execution: {"control_retained":true,"relative_phase_recorded":true,"target_restored":true}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:59fcfebe093c2114226aef30aeb2efe182fd8f33f725cdc015b858b85a5454b; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate
 sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:08bbc12f89a7bef2e3d3ccc31c95a5a5b3135d683fe2beea7f1fc8e59f12f47; engine receipt sha256:8e53cd4b8504ed63af45bd97dceaab40cd60c89858b0f9d421db5b0ed965fc0f at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-PHASE-KICKBACK-003-8e53cd4b8504ed63.json.

SFT-QUANTUM-OBL-QALGX-004 - Quantum Fourier transform correspondence

- **Claim and ownership:** SFT-QUANTUM-QALGX-FOURIER-004; Quantum Computation family QALGX; Quantum Fourier transform correspondence.
- **Forced law:** Fourier correspondence is the exact reindexing of positive-finite periodic word support into held phase-relation classes, without importing complex or irrational scalars.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__finite-period-support-phase-reindexing__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-re

gistry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-004 uniquely retains finite-period-support-phase-reindexing, complete algorithm custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-PHASE-KICKBACK-003. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-004: fourier_correspondence; exact quantum-algorithm execution: {"complex_scalars_imported":false,"period_classes":2,"support_rows":4}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:6e2e76f5c83f6c7395c050bf0d6e83ae1135346a0c7c8c6e6e2179dd66f07433; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate
sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:8e818425448d562ce60ecb779549e682aeb9b2bdf1e4c4fd5b7b0e2a4621cd5e; engine receipt sha256:a350b7f7c0159d1aa7361166777e966c06fc38f32cd979b6833b2c6f9bb77c23 at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-FOURIER-004-a350b7f7c0159d1a.json.

SFT-QUANTUM-OBL-QALGX-005 - Phase estimation correspondence

- **Claim and ownership:** SFT-QUANTUM-QALGX-PHASE-ESTIMATION-005; Quantum Computation family QALGX; Phase estimation correspondence.
- **Forced law:** Phase estimation recovers the least repeated held phase-label period over complete finite support with an exact enclosure when the declared resolution is insufficient.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__finite-period-phase-label-recovery__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-005 uniquely retains finite-period-phase-label-recovery, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-FOURIER-004. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** QALGX-005: phase_estimation; exact quantum-algorithm execution: {"exact_enclosure":true,"least_period":2,"phase_rows":4}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:46c2b8ff53859887b3fa7d7cf99f8aa3adb63b700a2557b0286d153ce432c167; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate
sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:02a761422387dd9d39ec50c7468f2f0094d6d3fd8804b232746997faf1cb298e; engine receipt sha256:62e453962fba7cf89d6471b6520fafea927970b3b759623bf50cf1b66a315d79 at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-PHASE-ESTIMATION-005-62e453962fba7cf8.json.

SFT-QUANTUM-OBL-QALGX-006 - Period finding and order finding

- **Claim and ownership:** SFT-QUANTUM-QALGX-PERIOD-ORDER-006; Quantum Computation family QALGX; Period finding and order finding.
- **Forced law:** Period or order is the least positive recurrence width that reproduces every registered function or action row across its complete finite domain.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__least-positive-support-recurrence__complete-branchwise-reversible-trace__same-problem-complete-re-source-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-006 uniquely retains least-positive-support-recurrence, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-PHASE-ESTIMATION-005. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-006: period_order; exact quantum-algorithm execution: {"all_rows_checked":true,"domain_rows":6,"least_period":3}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its

owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:fffea2dd82e714bbb79ab6849cfae499b401cf154e29085cea3af80f6a7b1a85; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate
sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:541126008d820aea4abf3e68ef245a4013ade3a2bcba44d7alefa52f51f04b27; engine receipt sha256:a27386bfddeb7ccf0ce82c1bb6618f0e04f37625a5080cf9f87ea4a261e20bfb at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-PERIOD-ORDER-006-a27386bfddeb7ccf.json.

SFT-QUANTUM-OBL-QALGX-007 - Deutsch-style promise distinction

- **Claim and ownership:** SFT-QUANTUM-QALGX-DEUTSCH-PROMISE-007; Quantum Computation family QALGX; Deutsch-style promise distinction.
- **Forced law:** A Deutsch-style distinction enumerates the complete promised function grammar and uses one reversible phase-sensitive query trace to separate constant from balanced classes.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__constant-versus-balanced-complete-function-census__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-007 uniquely retains constant-versus-balanced-complete-function-census, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-PERIOD-ORDER-006. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-007: deutsch_promise; exact quantum-algorithm execution: {"distinguished":true,"function_rows":2,"promise_classes":2}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:f9f81436fa4f52046b9e7b11d68b7c2b55e654835a5bba5020d621f7d3170853; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate
sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:21a3bdc0aa97aca05ff9c0f416597600c4bf8fea94865b3158e5d3a5504e4120; engine receipt sha256:23c735cd44ecd083f63373633c3ac08bc0b7629502fdf994a52cb5bdeb9d570 at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-DEUTSCH-PROMISE-007-23c735cd44ecd08.json.

SFT-QUANTUM-OBL-QALGX-008 - Simon-style hidden-structure distinction

- **Claim and ownership:** SFT-QUANTUM-QALGX-SIMON-HIDDEN-008; Quantum Computation family QALGX; Simon-style hidden-structure distinction.
- **Forced law:** A Simon-style process derives a hidden Fold-word translation from the complete equal-image pair relation, with every candidate translation enumerated and checked.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__paired-input-hidden-translation-census__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-008 uniquely retains paired-input-hidden-translation-census, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-DEUTSCH-PROMISE-007. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-008: simon_hidden_structure; exact quantum-algorithm execution: {"candidate_translations":4,"unique_consistent_translation":true,"word_width":2}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:b31cf6e5f5c6010f2528c4fd144468b0b8317697db44e038818e4275f3fe3724; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:25bfe5ad15433201e822e2061774a20c8d98c27115872d3274ecb83f2a3e3f3; engine receipt sha256:363db564f687e848d11636c013b2143ad1ba7dcca1af2d216b7a501ab9b22e64 at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-SIMON-HIDDEN-008-363db564f687e848.json.

SFT-QUANTUM-OBL-QALGX-009 - Factorization reduction and resource custody

- **Claim and ownership:** SFT-QUANTUM-QALGX-FACTORIZATION-009; Quantum Computation family QALGX; Factorization reduction and resource custody.
- **Forced law:** Factorization correspondence reduces a positive finite composite to a registered period/order process and validates every resulting divisor by exact arithmetic while retaining all resource rows.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__period-to-positive-divisor-reduction__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-009 uniquely retains period-to-positive-divisor-reduction, complete algorithm custody, root forcing, post-registry

execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-SIMON-HIDDEN-008. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-009: factorization; exact quantum-algorithm execution: {"composite":15,"nontrivial_divisors":[3,5],"positive_divisors":[1,3,5,15]}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:5b422c78fdf31597692fbe67b3172a4a371c27150b2f9ce95ebb89ddf2cf940c; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate
sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:53f6ebbb1454bd6d6b45f3d31ace96bba36d300addc8d9b0bc77131702fefc9a; engine receipt sha256:57a2dac780fbab05c787ffea3bb8edb902c42f906751602e898e294523d9063b at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-FACTORIZATION-009-57a2dac780fbab05.json.

SFT-QUANTUM-OBL-QALGX-010 - Unstructured search and amplitude-amplification correspondence

- **Claim and ownership:** SFT-QUANTUM-QALGX-SEARCH-010; Quantum Computation family QALGX; Unstructured search and amplitude-amplification correspondence.
- **Forced law:** Search correspondence applies a registered marked-support phase action and reversible support reflection; exact path multiplicities, queries and final observation are retained without stochastic causation.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__marked-support-phase-reflection-path-amplification__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-010 uniquely retains marked-support-phase-reflection-path-amplification, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-FACTORIZATION-009. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** QALGX-010: unstructured_search; exact quantum-algorithm execution: {"marked_rows":1,"query_trace_retained":true,"support_rows":4}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:5f12814cfa3ccbee752da83a07e8b630f5e6213ebd8da2baf8f2c768eeee4090; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate
sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:a0bdb7b6860ffaa33958e7815a1f49b4c947a10548588230d6626cdcccca0416; engine receipt sha256:58330f107cb92f7460edcbd01f36e92580181d0995720eaf0aba244de49d766a at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-SEARCH-010-58330f107cb92f74.json.

SFT-QUANTUM-OBL-QALGX-011 - Quantum counting correspondence

- **Claim and ownership:** SFT-QUANTUM-QALGX-COUNTING-011; Quantum Computation family QALGX; Quantum counting correspondence.
- **Forced law:** Quantum counting corresponds to recovering the exact marked-support cardinality from the registered search action's finite period and complete observation record.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__marked-support-cardinality-from-phase-period__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-011 uniquely retains marked-support-cardinality-from-phase-period, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-SEARCH-010. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-011: quantum_counting; exact quantum-algorithm execution: {"cardinality_exact":true,"marked_rows":2,"support_rows":4}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its

owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:fcba7749981718139c246673333f2b816a098e92386f1097e69c52157e1103fd; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate
sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:9b942823a32052b4fe89d0fbfad8c874b5fab6b43fd606e5d550b22c8c2a0aab; engine receipt sha256:4c131477b1ac328d4426a2bf4f0d34c6d8e0754765c5dbd8345c09499d4edb8b at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-COUNTING-011-4c131477b1ac328d.json.

SFT-QUANTUM-OBL-QALGX-012 - Amplitude-estimation correspondence with exact enclosures

- **Claim and ownership:** SFT-QUANTUM-QALGX-AMPLITUDE-ESTIMATION-012; Quantum Computation family QALGX; Amplitude-estimation correspondence with exact enclosures.
- **Forced law:** Amplitude-estimation correspondence reports an exact support ratio when resolved and otherwise retains nested rational lower and upper enclosures, never a fitted floating value.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__support-ratio-rational-enclosure__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-012 uniquely retains support-ratio-rational-enclosure, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-COUNTING-011. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-012: amplitude_estimation; exact quantum-algorithm execution: {"exact_ratio":"1/3","lower_enclosure":"1/4","upper_enclosure":"1/2"}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:bd622de7f8f27002792084275e3935a61b549cfd4c60fbc54e36f9eddf385b45; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate
sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:da1b89d8d8e33717c13c461c4602279274889ece3fbddc14d8b7206db873d2ff; engine receipt sha256:ed56af082d66d5075507a46a54ecc5d6a1f420a4a97f47e437aec0307a6599a5 at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-AMPLITUDE-ESTIMATION-012-ed56af082d66d507.json.

SFT-QUANTUM-OBL-QALGX-013 - Quantum walk state and transition law

- **Claim and ownership:** SFT-QUANTUM-QALGX-WALK-013; Quantum Computation family QALGX; Quantum walk state and transition law.
- **Forced law:** A quantum walk is complete vertex-support with reversible edge transitions, held path-phase labels and one branchwise trace per generated path.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__vertex-edge-phase-reversible-walk__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-013 uniquely retains vertex-edge-phase-reversible-walk, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-AMPLITUDE-ESTIMATION-012. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-013: quantum_walk; exact quantum-algorithm execution: {"source_restored":true,"steps":2,"vertices":2}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:e22db13fcaa39013b48f04055fbbdb0ed7c4876f8b66f8205d478c14dea4d5bd; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate
sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:d4331603eb5b5c1dd3a0463abbfa71fd9ad40cf5f05d7c413b427c8057340a92; engine receipt sha256:f7810258c596ea6d594fd4dbddfc6f81a049dbaba23aeed8d675404ec1fc82b at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-WALK-013-f7810258c596ea6d.json.

SFT-QUANTUM-OBL-QALGX-014 - Quantum walk search boundary

- **Claim and ownership:** SFT-QUANTUM-QALGX-WALK-SEARCH-014; Quantum Computation family QALGX; Quantum walk search boundary.
- **Forced law:** Walk search composes a registered marked-vertex action with the reversible walk and states the exact graph, path, query and observation boundary for any advantage claim.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__marked-vertex-walk-observation__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-014 uniquely retains marked-vertex-walk-observation, complete algorithm custody, root forcing, post-registry execution and no

extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-WALK-013. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-014: `walk_search`; exact quantum-algorithm execution: `{"marked_vertices":1,"observation_retained":true,"path_rows":3}`; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:96ea8d5d4c46e6d632c562bd220b4203165b5493c3e2ca294890fd2cdf525395`; independent implementation `sha256:7aa5c7de4dc2e2404bfb53591970611400ebc3094760e0278e99852dc52d5f37a`; independent certificate
`sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3`; measurement `sha256:a95eb703d83ed330b1ab3240709f0aa3addf9d787a84f509303392fc2df2c54e`; engine receipt `sha256:aa9edc20c205fcde5dfd05debf41c7143947c3470b0fb5ee23e2c093f6444ba8` at `receipts/engine/model_admitted/SFT-QUANTUM-QALGX-WALK-SEARCH-014-aa9edc20c205fcde.json`.

SFT-QUANTUM-OBL-QALGX-015 - Linear-system algorithm correspondence

- **Claim and ownership:** SFT-QUANTUM-QALGX-LINEAR-SYSTEM-015; Quantum Computation family QALGX; Linear-system algorithm correspondence.
- **Forced law:** A linear-system correspondence encodes a finite exact relation and returns supported solution words only when inverse, conditioning enclosure and observation custody are explicit.
- **Unique surviving structure:** `complete-registered-input-and-promise-grammar__finite-exact-relation-solution-support__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff`. Exact result: QALGX-015 uniquely retains `finite-exact-relation-solution-support`, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-WALK-SEARCH-014. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** QALGX-015: linear_system; exact quantum-algorithm execution: {"inverse_explicit":true,"relation_rows":2,"solution_rows":2}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:13d1c082e177e6da5f6f19e2e32613e4ef38d0aa6efe15ad57cb953cdcd9b5ab; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate
sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:f018880e18fd0800f677c137f229a35dd410208a14f00764e011f80e199b1ca5; engine receipt sha256:c77327afa3678f6c95665c5d3aa98d3c95563a71b848d56467eec659fbf9c00c at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-LINEAR-SYSTEM-015-c77327afa3678f6c.json.

SFT-QUANTUM-OBL-QALGX-016 - Eigenvalue and mode-estimation algorithm

- **Claim and ownership:** SFT-QUANTUM-QALGX-EIGENMODE-016; Quantum Computation family QALGX; Eigenvalue and mode-estimation algorithm.
- **Forced law:** Mode estimation recovers invariant support cycles and their held phase periods under a registered reversible transformation, using exact recurrence or rational enclosure.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__recurrent-mode-phase-period-recovery__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-016 uniquely retains recurrent-mode-phase-period-recovery, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-LINEAR-SYSTEM-015. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-016: eigenmode; exact quantum-algorithm execution: {"least_period":2,"mode_rows":4,"phase_record_retained":true}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its

owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:049df7f3ce18f9d9d934c9c5f8b135395c2e364c220f21457967d3627fb47e79; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:d3dcbe262315502d7538fccb199cb196f6fc73da1770e4fc8e695928a82c97f0; engine receipt sha256:7881703e0b15e360d809276d83265d1bc3ab18b2165ebecd76010eecdcb33f6 at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-EIGENMODE-016-7881703e0b15e360.json.

SFT-QUANTUM-OBL-QALGX-017 - Hamiltonian-simulation algorithm interface

- **Claim and ownership:** SFT-QUANTUM-QALGX-HAMILTONIAN-INTERFACE-017; Quantum Computation family QALGX; Hamiltonian-simulation algorithm interface.
- **Forced law:** Simulation consumes a registered finite local-generator word and duration count; the physical generator values remain owned measurements and cannot select the computational law.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__registered-local-generator-word-interface__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-017 uniquely retains registered-local-generator-word-interface, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-EIGENMODE-016. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-017: hamiltonian_interface; exact quantum-algorithm execution: {"duration_count":1,"local_generators":2,"physical_values_present":false}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:8f7d0f48fa569098086c5c71731f4c9bb20d76202a70abc73da7bf73f621d1aa; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:621bfd6fffc44665b4c9c57bf996f009e4d1dccf3d8b29a474c8a3f95dbf1a18c; engine receipt sha256:fa1aeab8485737f751127afb14d71b1a26e963daff7c869e73ea142ead6de278 at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-HAMILTONIAN-INTERFACE-017-fa1aeab8485737f7.json.

SFT-QUANTUM-OBL-QALGX-018 - Product-formula simulation and error custody

- **Claim and ownership:** SFT-QUANTUM-QALGX-PRODUCT-FORMULA-018; Quantum Computation family QALGX; Product-formula simulation and error custody.
- **Forced law:** Product-formula simulation composes exact local generator actions in registered order and retains a rational error enclosure, causal trace and omitted-term boundary.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__ordered-local-generator-product-with-enclosure__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-018 uniquely retains ordered-local-generator-product-with-enclosure, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-HAMILTONIAN-INTERFACE-017. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-018: product_formula; exact quantum-algorithm execution: {"error_enclosure_exact":true,"local_actions":4,"ordering_retained":true}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:b672bc88bd8c583914613771e374c4b230d47e73d236cfb8f89510cb4db40c74; independent implementation sha256:7aa5c7de4dc2e2404bfb53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:a32f933281ebf2093a38866e8e30aa83e222420d0c81fea56d4704c0664e9d87; engine receipt sha256:edff0ad0eaa1280f08480f78331c5be991ee4e6b91e49b13db1f31c152833de2 at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-PRODUCT-FORMULA-018-edff0ad0eaa1280f.json.

SFT-QUANTUM-OBL-QALGX-019 - Combinatorial optimization algorithm boundary

- **Claim and ownership:** SFT-QUANTUM-QALGX-COMBINATORIAL-OPTIMIZATION-019; Quantum Computation family QALGX; Combinatorial optimization algorithm boundary.
- **Forced law:** Quantum optimization must enumerate the declared feasible grammar, evaluate its exact ordered objective and preserve the full survivor and resource comparison; the target cannot select the search law.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__complete-candidate-order-and-minimum-custody__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-019 uniquely retains complete-candidate-order-and-minimum-custody, complete algorithm custody, root

forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-PRODUCT-FORMULA-018. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-019: `combinatorial_optimization`; exact quantum-algorithm execution: `{"candidate_rows":3,"full_order_retained":true,"unique_minimum":"b"}`; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:d9b0486338452b636b11bc65055dce2eafd5ebe339855d13a252ca0c3baf917c`; independent implementation `sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a`; independent certificate
`sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3`; measurement `sha256:27a04b26daf793860ff1ec00455b4c45e1145891f50d6f33538ace91af3fea90`; engine receipt `sha256:ecb3da65494f8e3e97a64631d914f7f6bac906efea433bf9fc48030750e98003` at `receipts/engine/model_admitted/SFT-QUANTUM-QALGX-COMBINATORIAL-OPTIMIZATION-019-ecb3da65494f8e3e.json`.

SFT-QUANTUM-OBL-QALGX-020 - Variational algorithm parameter-prohibition boundary

- **Claim and ownership:** SFT-QUANTUM-QALGX-VARIATIONAL-BOUNDARY-020; Quantum Computation family QALGX; Variational algorithm parameter-prohibition boundary.
- **Forced law:** A variational correspondence is admissible only when every control is generated by a finite Fold grammar and exhaustively enumerated; fitted continuous parameters cannot enter an SFT derivation.
- **Unique surviving structure:** `complete-registered-input-and-promise-grammar__generated-control-grammar-without-fitting__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff`. Exact result: QALGX-020 uniquely retains `generated-control-grammar-without-fitting`, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-COMBINATORIAL-OPTIMIZATION-019. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** QALGX-020: variational_boundary; exact quantum-algorithm execution: {"fitted_controls":0,"generated_controls":4,"unique_survivors":1}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:34b19c8108a22b8f9583750a77726b0e4f4fb3997ad2015c89214a7f24d8b1e6; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate
sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:73bccf8090563300dee6ea7ac790c6f97f64bf501508bbca4132d06b6973a33b; engine receipt sha256:155051cedf924f8b67569617fa99546cfb91c54a9d9c5fb59722dcdd12cced54 at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-VARIATIONAL-BOUNDARY-020-155051cedf924f8b.json.

SFT-QUANTUM-OBL-QALGX-021 - Quantum annealing correspondence boundary

- **Claim and ownership:** SFT-QUANTUM-QALGX-ANNEALING-BOUNDARY-021; Quantum Computation family QALGX; Quantum annealing correspondence boundary.
- **Forced law:** Annealing correspondence is a registered finite path of reversible state transformations and observations; physical schedule, gap and device behavior remain measured handoffs.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__finite-generate d-state-path-correspondence__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-021 uniquely retains finite-generated-state-path-correspondence, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-VARIATIONAL-BOUNDARY-020. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-021: annealing_boundary; exact quantum-algorithm execution: {"finite_path_rows":3,"physical_schedule_present":false,"terminal_map_retained":true}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational,

imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:517375c4aedd9dbd55c2df4e2fc24e817f46c198ef0a07a9bb8ab972f4eb30d; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:e555812bd169c7f4534078589bce054a72946c52d2bcd4ce37a6597573cb401; engine receipt sha256:246ec51cdcc2fc1ae0962f9cab40abf60edce266a71a521820b330f5bdb53683 at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-ANNEALING-BOUNDARY-021-246ec51cdcc2fc1a.json.

SFT-QUANTUM-OBL-QALGX-022 - Quantum sampling and output-support custody

- **Claim and ownership:** SFT-QUANTUM-QALGX-SAMPLING-022; Quantum Computation family QALGX; Quantum sampling and output-support custody.

- **Forced law:** Sampling claims retain complete generated output support, exact multiplicities, the observation selection protocol and every favorable or adverse row; ontic randomness is not imported.

- **Unique surviving structure:** complete-registered-input-and-promise-grammar__deterministic-complete-output-support-and-counts__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-022 uniquely retains deterministic-complete-output-support-and-counts, complete algorithm custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-ANNEALING-BOUNDARY-021. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** QALGX-022: sampling_custody; exact quantum-algorithm execution: {"exact_multiplicities":{"held":2,"returned":1},"ontic_randomness_imported":false,"output_rows":3}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:e94ecad1d0ddc5a881b31932c6406c1ff2e20040633116d2a89f8278fe48a9d9; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:b2c830fddd5cc842ddecf221967ee3febab834d0f1cc4713143dbc73017ca0c2; engine receipt

sha256:6ef91e7464c4f4369c2564a180da11dc5e444b75e6b18583755157deec3cc382 at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-SAMPLING-022-6ef91e7464c4f436.json.

SFT-QUANTUM-OBL-QALGX-023 - Bosonic-sampling correspondence boundary

- **Claim and ownership:** SFT-QUANTUM-QALGX-BOSONIC-SAMPLING-023; Quantum Computation family QALGX; Bosonic-sampling correspondence boundary.
- **Forced law:** Bosonic-sampling correspondence enumerates finite occupation words and indistinguishable permutation paths with held phases; device probability and scale remain external measurements.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__occupation-word-permutation-path-census__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-023 uniquely retains occupation-word-permutation-path-census, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-SAMPLING-022. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-023: bosonic_sampling_boundary; exact quantum-algorithm execution: {"device_probability_present":false,"occupation_words":2,"permutation_classes":2}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:2dff47a55af40af305913d941dfe6584355e85fa094010d5bf1556bcd8825b83; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate
sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:a3f140f2ae21abb2a9b8d325ca8131f7e178851085fb7ab1e48bc5398c1d7b5b; engine receipt sha256:0ec830399029b17ae876e01bf52d27acc07e9a14c5362487626389a21a190243 at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-BOSONIC-SAMPLING-023-0ec830399029b17a.json.

SFT-QUANTUM-OBL-QALGX-024 - Hidden-subgroup algorithm family

- **Claim and ownership:** SFT-QUANTUM-QALGX-HIDDEN-SUBGROUP-024; Quantum Computation family QALGX; Hidden-subgroup algorithm family.
- **Forced law:** A hidden-subgroup family derives candidate cosets from the complete equal-image relation and eliminates every candidate inconsistent with the finite phase and observation trace.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__complete-coset-equality-and-phase-census__complete-branchwise-reversible-trace__same-problem-comp

lete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-024 uniquely retains complete-coset-equality-and-phase-census, complete algorithm custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-BOSONIC-SAMPLING-023. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-024: hidden_subgroup; exact quantum-algorithm execution: {"candidate_partitions_exhausted":true,"cosets":2,"support_rows":4}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:ad8a5e2d25e08727d677f4198d697d3c47a0327c1b862767b02ae02bca14942f; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate
sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:aa102648532bea28afc25b4e1a0708b565638e6a25bad3c56bbc4adb4ee50063; engine receipt sha256:3bb52cf39d5ccd8e08410e32bd8b33164a43b5bad73cac2890cf8905093dd435 at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-HIDDEN-SUBGROUP-024-3bb52cf39d5ccd8e.json.

SFT-QUANTUM-OBL-QALGX-025 - Quantum dynamic programming correspondence

- **Claim and ownership:** SFT-QUANTUM-QALGX-DYNAMIC-PROGRAMMING-025; Quantum Computation family QALGX; Quantum dynamic programming correspondence.
- **Forced law:** Quantum dynamic-programming correspondence retains every subproblem state, recurrence dependency, reversible update and observation while comparing resources with the corresponding classical recurrence.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__reversible-subproblem-recurrence-ledger__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-025 uniquely retains reversible-subproblem-recurrence-ledger, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-HIDDEN-SUBGROUP-024. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING.

Imported axioms: 0; free parameters: 0.

- **Observation:** QALGX-025: dynamic_programming; exact quantum-algorithm execution: {"dependency_rows_retained":true,"subproblems":4,"terminal_optimum":2}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:4875a42779da9d6f66b624eec6c12e79fd474c031dc0b8ce92f6cace596e64f5; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate
sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:1a6143a7053f0d13b3ad99012b830922c6326c946120a9f6eca4b5fc3f911c18; engine receipt sha256:e6657124994f2f4af35f0984d4555a2a08b920aa1f0768d8b1d9ca93dc06afa9 at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-DYNAMIC-PROGRAMMING-025-e6657124994f2f4a.json.

SFT-QUANTUM-OBL-QALGX-026 - Quantum parallelism and output-access boundary

- **Claim and ownership:** SFT-QUANTUM-QALGX-PARALLELISM-BOUNDARY-026; Quantum Computation family QALGX; Quantum parallelism and output-access boundary.
- **Forced law:** Complete support permits branchwise parallel transformation, but accessible output is restricted by the registered observation; unobserved branch values cannot be claimed as simultaneously returned.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__complete-branch-execution-single-observation-boundary__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-026 uniquely retains complete-branch-execution-single-observation-boundary, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-DYNAMIC-PROGRAMMING-025. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-026: quantum_parallelism; exact quantum-algorithm execution: {"all_branch_values_claimed_returned":false,"branch_rows":8,"registered_output_observations":1}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:95e7e83173b6ed4c0544347d3e11c7f09aed564662df2648a81eaf14ec5311ca`; independent implementation `sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a`; independent certificate
`sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3`; measurement `sha256:7daaa81d46c022d6b812b765860e45e155f06ed208de5e743dcf20e7c83f5031`; engine receipt `sha256:c9f957aa2df8a9c632991d04bf4f5ad1dfe4f915c15400081fd542229cfd7069` at `receipts/engine/model_admitted/SFT-QUANTUM-QALGX-PARALLELISM-BOUNDARY-026-c9f957aa2df8a9c6.json`.

SFT-QUANTUM-OBL-QALGX-027 - Classical preprocessing and postprocessing custody

- **Claim and ownership:** SFT-QUANTUM-QALGX-CLASSICAL-CUSTODY-027; Quantum Computation family QALGX; Classical preprocessing and postprocessing custody.
- **Forced law:** Every algorithm separates classical preparation and decoding from the reversible quantum core and charges their time, space, query and retained-information resources.
- **Unique surviving structure:** `complete-registered-input-and-promise-grammar__separate-classical-quantum-resource-ledger__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff`. Exact result: QALGX-027 uniquely retains `separate-classical-quantum-resource-ledger`, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-PARALLELISM-BOUNDARY-026. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-027: `classical_custody`; exact quantum-algorithm execution: `{"classical_post_steps":1,"classical_pre_steps":1,"quantum_core_steps":1,"resources_separate":true}`; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:a270db884628a4ea3b47f5b47975861667aea30ce7401ccd365fc67d896d019e`; independent implementation `sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a`; independent certificate

sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:6564a4ecbf59b0a954fe01829bbba5a982374da61e6ddbf7835cc03de107d42; engine receipt sha256:09ca9d6e5a3e5efa4ed3d02f7fd63605a8b625e79b9c7fb64ed264e2a937b9dd at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-CLASSICAL-CUSTODY-027-09ca9d6e5a3e5efa.json.

SFT-QUANTUM-OBL-QALGX-028 - Speedup definition and comparison grammar

- **Claim and ownership:** SFT-QUANTUM-QALGX-SPEEDUP-028; Quantum Computation family QALGX; Speedup definition and comparison grammar.
- **Forced law:** A speedup is a proven comparison of complete resource vectors for the same input grammar, success condition and observation boundary; asymptotic, exact and finite-census claims remain distinct.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__same-problem-resource-vector-comparison__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-028 uniquely retains same-problem-resource-vector-comparison, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-CLASSICAL-CUSTODY-027. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-028: speedup_grammar; exact quantum-algorithm execution: {"observation_boundaries_equal":true,"resource_vectors":2,"same_problem":true}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:c5ed884e9e5d88136573825737a4422b877d81b14d5a7ad4fd7aea83d24e9c13; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:a67a38ac0120f4e444249b87db73595c3a358b4d3101a4bfc72a448fb4a78eed; engine receipt sha256:6fb381c39102f62d4e9ac871ba972d2d74e1ac52c4acc43c8226fe343440d92b at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-SPEEDUP-028-6fb381c39102f62d.json.

SFT-QUANTUM-OBL-QALGX-029 - Algorithm lower and upper resource witnesses

- **Claim and ownership:** SFT-QUANTUM-QALGX-RESOURCE-BOUNDS-029; Quantum Computation family QALGX; Algorithm lower and upper resource witnesses.
- **Forced law:** An upper bound requires an executing algorithm and a lower bound requires an exhaustive or adversary witness over the same grammar; both retain exact query, time, space, depth and record coordinates.

- **Unique surviving structure:** complete-registered-input-and-promise-grammar__constructive-upper-and-adversary-lower-witnesses__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-029 uniquely retains constructive-upper-and-adversary-lower-witnesses, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-SPEEDUP-028. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-029: resource_bounds; exact quantum-algorithm execution: {"all_exact_values_enclosed":true,"lower_witnesses":2,"upper_witnesses":2}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:ef13d9ef2d10ff21e607fbb231f9c585b0bee5308b34eb2ae2cd4cb8229f94ad; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:0ac4c7bc0419944640652112852176a52e4c09ae9a471cc6eea415a11c35122f; engine receipt sha256:adf146bef541f68b5015d0838d5d652173d444744fbd347e36ec9d8d7fb4a244 at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-RESOURCE-BOUNDS-029-adf146bef541f68b.json.

SFT-QUANTUM-OBL-QALGX-030 - Quantum-algorithm completeness certificate

- **Claim and ownership:** SFT-QUANTUM-QALGX-COMPLETENESS-030; Quantum Computation family QALGX; Quantum-algorithm completeness certificate.
- **Forced law:** The quantum-algorithm family is complete exactly when all thirty frozen obligations have one owner, one unique survivor, controls, observations, independent reconstructions and current receipts.
- **Unique surviving structure:** complete-registered-input-and-promise-grammar__thirty-obligation-no-omission-ledger__complete-branchwise-reversible-trace__same-problem-complete-resource-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-and-comparison-handoff. Exact result: QALGX-030 uniquely retains thirty-obligation-no-omission-ledger, complete algorithm custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete algorithm grammar, exact execution/resource trace, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-GATEX-COMPLETENESS-022 -> SFT-QUANTUM-ALGORITHMS-001 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-QUANTUM-QALGX-RESOURCE-BOUNDS-029. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QALGX-030: algorithm_completeness; exact quantum-algorithm execution: {"duplicate_owners":0,"execution_rows":30,"omitted_owners":0,"registered_obligations":30}; complete quantum-algorithm custody: 30 records; all rows preserved; complete finite input, path, query, result and resource execution supplies the observation; physical hardware performance and unrestricted advantage remain downstream measurements or explicit boundaries
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-ALGORITHM-OBSERVER, SFT-V1-V2-QUANTUM-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported quantum algorithm, oracle, complex transform or speedup theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, resource or parameter; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled candidate grammar, hidden branch, target-selected survivor or ontic randomness; no physical Hamiltonian, gap, schedule, device rate or probability is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:6a709620fe6c53c4e99752096a09d3ba63448818be54430f00a2e953174dbdc3; independent implementation sha256:7aa5c7de4dc2e2404bf53591970611400ebc3094760e0278e99852dc52d5f37a; independent certificate
sha256:a80466c0a56106a6e60187b52dc54c7d39a59e8851dba549b3f82243c5b1bbd3; measurement sha256:1089e9298c9a62a75d8937cc4348f4cb6b2f0bbf53b60090f9a679bd134509f3; engine receipt sha256:be183aeb9c83cd98d8d6a59b0df147aa0f8f334a2cae8c2ff521fba02c33eb0d at receipts/engine/model_admitted/SFT-QUANTUM-QALGX-COMPLETENESS-030-be183aeb9c83cd98.json.

36.8 Family QCPLXX - 26/26 complete

Scope. Quantum resources, complexity and bounds. This family decides 6,656 candidates, retains 26 unique survivors and passes 104 adverse controls.

SFT-QUANTUM-OBL-QCPLXX-001 - Canonical quantum input-size carrier

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-INPUT-SIZE-001; Quantum Computation family QCPLXX; Canonical quantum input-size carrier.
- **Forced law:** Quantum input size is the exact length of the canonical finite Fold description, including registered promise and precision records; representation changes require an explicit translation cost.
- **Unique surviving structure:** canonical-input-and-promise-size__canonical-positive-finite-input-description-width__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-001 uniquely retains canonical-positive-finite-input-description-width, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-001: input_size; exact quantum-complexity execution: {"description_width":3,"promise_rows":1,"representation_canonical":true}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the

observation; physical device performance and unrestricted asymptotic claims remain downstream

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:aed1c829f467e2de863676da1656792c3afdee93d6c40395ae115cf38aace81d; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf; independent certificate
 sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:777548a2b19988c08748141a915ac69cd2fe0bcc26c7202bcf1583a2b3b440da; engine receipt sha256:9ae4fd9294b5e8ab57800c754a037ea8ea0b805366df78c8fdd00b4627326e78 at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-INPUT-SIZE-001-9ae4fd9294b5e8ab.json.

SFT-QUANTUM-OBL-QCPLXX-002 - Gate count and circuit-depth resource

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-GATE-DEPTH-002; Quantum Computation family QCPLXX; Gate count and circuit-depth resource.
- **Forced law:** Gate count enumerates executed gate instances while depth counts the longest causal layer chain; neither may silently substitute for the other.
- **Unique surviving structure:** canonical-input-and-promise-size__separate-gate-count-and-causal-depth__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-002 uniquely retains separate-gate-count-and-causal-depth, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-INPUT-SIZE-001. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-002: gate_depth; exact quantum-complexity execution: {"coordinates_separate":true,"depth":2,"gate_count":3}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:ef0e07bfcdb7dca679feb0f89e99f154ebc48e499b164fecbe27ec4fadc626c39; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf; independent certificate
sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:6b9727598759eebdb110681d3e5b048aeb18bdd715f7d22db79309f5ab0f4009; engine receipt sha256:9c3a3b0862561a294c83710b159ad630f0183658c957b3d8601171ac4ad1621b at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-GATE-DEPTH-002-9c3a3b0862561a29.json.

SFT-QUANTUM-OBL-QCPLXX-003 - Live branch-support and state-description resource

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-BRANCH-SUPPORT-003; Quantum Computation family QCPLXX; Live branch-support and state-description resource.
- **Forced law:** Quantum space accounting includes complete live word support, held phase rows and exact state-description width at every circuit stage.
- **Unique surviving structure:** canonical-input-and-promise-size__complete-live-support-and-description-ledger__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-003 uniquely retains complete-live-support-and-description-ledger, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-GATE-DEPTH-002. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-003: live_support; exact quantum-complexity execution: {"description_rows":4,"live_support_rows":4,"register_width":2}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:91d75fc83c518fdaa9d7054b8f94b0d4cc99d75c53069acf594d63faaea24690; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf; independent certificate
sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:2c947ea228e2913f5e6f2de07736f8380d26d9318d987a1c686900316040b1b3; engine receipt sha256:312b3f9ec032529c5caba6067369ba55fc83709380a6c13d5831956ef80638fe at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-BRANCH-SUPPORT-003-312b3f9ec032529c.json.

SFT-QUANTUM-OBL-QCPLXX-004 - Ancilla, measurement and retained-record resource

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-RECORD-RESOURCE-004; Quantum Computation family QCPLXX; Ancilla, measurement and retained-record resource.

- **Forced law:** Ancilla support, observations, classical outcome records and inverse-history custody are separately counted resources and cannot be discarded from a comparison.
- **Unique surviving structure:** canonical-input-and-promise-size__separate-ancilla-observation-record-costs__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-004 uniquely retains separate-ancilla-observation-record-costs, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-BRANCH-SUPPORT-003. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-004: record_resources; exact quantum-complexity execution: {"ancilla":2,"measurements":1,"retained_records":3}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:9a312f69ec6d91ad95051071db6d79e8a53a8ccf59a7f810e50b31bb4ba03901; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf; independent certificate
sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:6b091059a0b9a8b47342e9dfb2007e9535b29e959024bcf42396ab5905e49e50; engine receipt sha256:bbf945918ab9ed9da16b1bf8761208494543a0704e29cc564cf72a0a71c9f6d7 at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-RECORD-RESOURCE-004-bbf945918ab9ed9d.json.

SFT-QUANTUM-OBL-QCPLXX-005 - Quantum query complexity

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-QUERY-005; Quantum Computation family QCPLXX; Quantum query complexity.
- **Forced law:** Quantum query complexity counts each invocation of the registered reversible oracle interface, including controlled and inverse invocations, over the same problem grammar.
- **Unique surviving structure:** canonical-input-and-promise-size__registered-reversible-query-count__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-005 uniquely retains registered-reversible-query-count, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent

reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-RECORD-RESOURCE-004. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-005: query_complexity; exact quantum-complexity execution: {"inverse_queries_included":true,"reversible_queries":3}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:ccd00586cd1914f48ce218b9c7c203be5c4ddbc3d2b590c2f50c767a5e4410ce; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf; independent certificate
sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:512eac005d1080e63d24b23c4094464de907b12c891ef9d5d52bebd098a263df; engine receipt sha256:9917d13cf77dfd378f79bf7f2e9a5d38c416e94733667a71288671200fc045d1 at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-QUERY-005-9917d13cf77dfd37.json.

SFT-QUANTUM-OBL-QCPLXX-006 - Quantum communication complexity

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-COMMUNICATION-006; Quantum Computation family QCPLXX; Quantum communication complexity.
- **Forced law:** Quantum communication complexity counts transmitted Fold distinctions, rounds, shared supports and retained correction records under a declared party partition.
- **Unique surviving structure:** canonical-input-and-promise-size__transmitted-distinction-and-shared-record-ledger__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-006 uniquely retains transmitted-distinction-and-shared-record-ledger, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-QUERY-005. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-006: communication_complexity; exact quantum-complexity execution: {"rounds":3,"shared_records_retained":true,"transmitted_distinctions":5}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:e2ece624b13c60b7ffd9212049fd2a16d75dff9ff5b3992d785cab5272ad246d; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf; independent certificate
sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:7111994c00def88ba26f66991e2fed13dd52d051dece134e0625e9dbe9bfb472; engine receipt sha256:553aabdl1eb1ef19db3570bf61927723037a9219fa63acb87c202d472f9bfce30 at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-COMMUNICATION-006-553aabdl1eb1ef19d.json.

SFT-QUANTUM-OBL-QCPLXX-007 - Quantum decision and recognition class

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-DECISION-CLASS-007; Quantum Computation family QCPLXX; Quantum decision and recognition class.
- **Forced law:** A quantum decision or recognition class is a set of canonical inputs whose complete executions meet a registered accept/reject support condition inside an exact resource bound.
- **Unique surviving structure:** canonical-input-and-promise-size__resource-bounded-accept-reject-support__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-007 uniquely retains resource-bounded-accept-reject-support, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-COMMUNICATION-006. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-007: decision_class; exact quantum-complexity execution: {"accept_rows":2,"input_classes":3,"reject_rows":1}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:01e9c6b668174e584f17301682fbe0d1f8cb20ecb90b2808b60295963aadd83b; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf;

independent certificate

sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement
sha256:5605246068929d0c9eb2366673da68361a2d533d50a392dac8507919c9f79fad; engine receipt
sha256:c4620b96478b9638b904831a3198b63473ad396f90517378f3687a9cdef77fbb at receipts/
engine/model_admitted/SFT-QUANTUM-QCPLXX-DECISION-CLASS-007-c4620b96478b9638.json.

SFT-QUANTUM-OBL-QCPLXX-008 - Bounded-error deterministic-support correspondence

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-BOUNDED-ERROR-008; Quantum Computation family QCPLXX; Bounded-error deterministic-support correspondence.
- **Forced law:** Bounded-error correspondence is an exact favorable-to-complete support ratio under deterministic generated paths and a registered observation; it does not introduce ontic randomness.
- **Unique surviving structure:** canonical-input-and-promise-size__exact-favorable-support-ratio-boundary__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-008 uniquely retains exact-favorable-support-ratio-boundary, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-DECISION-CLASS-007. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-008: bounded_error; exact quantum-complexity execution: {"complete_support":4,"exact_ratio":"3/4","favorable_support":3,"ontic_randomness":false}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:8dbe827b1a60e60f9495f8a15098639c9cf318e25091e5570f13d35d6e8d05c5; independent
implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf;
independent certificate
sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement
sha256:3636491a507f3b2491b9eb1d50bacc1aa0e43354bcd4c566f8508e19a4428b65; engine receipt
sha256:a6a049352785a4bb608475615a833a54f1315371cc67a98108e6ce3dec04573b at receipts/
engine/model_admitted/SFT-QUANTUM-QCPLXX-BOUNDED-ERROR-008-a6a049352785a4bb.json.

SFT-QUANTUM-OBL-QCPLXX-009 - Exact quantum decision class

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-EXACT-DECISION-009; Quantum Computation family QCPLXX; Exact quantum decision class.

- **Forced law:** Exact quantum decision requires every generated support row to yield the correct registered decision class with no adverse row.
- **Unique surviving structure:** canonical-input-and-promise-size__complete-support-correct-decision__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-009 uniquely retains complete-support-correct-decision, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-BOUNDED-ERROR-008. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-009: exact_decision; exact quantum-complexity execution: {"complete_support":2,"correct_support":2,"exact_ratio":"1/1"}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:d2bc9bb31ac70c15be54b45245118c0a18513ebf8a6658256ec0e7c64199f890; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf; independent certificate
sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:7f1880742ea0af30fdc88a44236074a94a430870edba85be49b421ad491be8db; engine receipt sha256:bb305d629813ac758b15e228becacbfba5501e36d0d1409e7c4211513b6a7579 at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-EXACT-DECISION-009-bb305d629813ac75.json.

SFT-QUANTUM-OBL-QCPLXX-010 - One-sided-error correspondence boundary

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-ONE-SIDED-ERROR-010; Quantum Computation family QCPLXX; One-sided-error correspondence boundary.
- **Forced law:** One-sided-error correspondence requires complete correctness for one decision class and a stated exact support-ratio bound for the other.
- **Unique surviving structure:** canonical-input-and-promise-size__one-class-complete-other-class-bounded-support__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-010 uniquely retains one-class-complete-other-class-bounded-support, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent

reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-EXACT-DECISION-009. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-010: one_sided_error; exact quantum-complexity execution: {"adverse_side_error_ratio": "0/1", "complete_side_ratio": "1/1"}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:eeff030236001f4a64b204d1e35e533ab8ccfe0a74d6662ad0997cc0c2bdc64c; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcb09d292acf; independent certificate
sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:ab766b6cdaed131e71928ab45b03276db4b89baf83ff8ab86ed37c12e2fc3eea; engine receipt sha256:dc3dc960cf5ffcd638c2d09c0bb4eb26592ab2914e0b5f0b7219ae5a639d4eba at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-ONE-SIDED-ERROR-010-dc3dc960cf5ffcd6.json.

SFT-QUANTUM-OBL-QCPLXX-011 - Quantum nondeterminism correspondence boundary

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-NONDETERMINISM-011; Quantum Computation family QCPLXX; Quantum nondeterminism correspondence boundary.
- **Forced law:** Quantum nondeterminism corresponds to existence of an accepting branch in the complete deterministically generated support; no uncaused choice or ontic randomness is admitted.
- **Unique surviving structure:** canonical-input-and-promise-size__existential-accepting-branch-on-deterministic-support__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-011 uniquely retains existential-accepting-branch-on-deterministic-support, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-ONE-SIDED-ERROR-010. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-011: nondeterminism_correspondence; exact quantum-complexity execution: {"accepting_branches": 1, "generated_branches": 3, "ontic_choice_imported": false}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:309e38d5945c4d2fbf532dbd0c7e456a5caf686da8f6218953ea7df3fc6f9de7; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf; independent certificate
sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:b526027a2bfc2aa6af904ccfe7b198b0450e668d541b7f764bac4c37108e6ce1; engine receipt sha256:9ddbfa0268eed7e12174b1e5491df76b3c11c93c92839df0b77d4b965f664eee at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-NONDETERMINISM-011-9ddbfa0268eed7e1.json.

SFT-QUANTUM-OBL-QCPLXX-012 - Quantum polynomial-time correspondence

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-POLYNOMIAL-TIME-012; Quantum Computation family QCPLXX; Quantum polynomial-time correspondence.
- **Forced law:** Polynomial-time correspondence requires an exact positive-finite polynomial resource envelope over canonical input size, with uniform circuit generation and observation costs included.
- **Unique surviving structure:** canonical-input-and-promise-size__positive-finite-polynomial-resource-envelope__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-012 uniquely retains positive-finite-polynomial-resource-envelope, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-NONDETERMINISM-011. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-012: polynomial_time; exact quantum-complexity execution: {"input_sizes":[1,2,3],"resource_costs":[1,4,9],"uniform_envelope":true}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:5723a871277fc762b7dd834448dae65b6fe0e9d223a7abd703e2228cc2081430; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf;

independent certificate

sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement
sha256:42d3896deaa2672fe6c903beb2b431233b8927268ed172785c15a2efee00a731; engine receipt
sha256:0cfdda54a20502f672fac23627d8f7f07af6e13526b93a6bacdf6a7a627e3319 at receipts/
engine/model_admitted/SFT-QUANTUM-QCPLXX-POLYNOMIAL-TIME-012-0cfdda54a20502f6.json.

SFT-QUANTUM-OBL-QCPLXX-013 - Quantum witness-verification class

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-WITNESS-VERIFY-013; Quantum Computation family QCPLXX; Quantum witness-verification class.
- **Forced law:** A quantum witness class enumerates the declared witness grammar and accepts exactly when at least one witness has a complete resource-bounded verification trace meeting the observation condition.
- **Unique surviving structure:** canonical-input-and-promise-size__existential-witness-complete-verification-census__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff.
Exact result: QCPLXX-013 uniquely retains existential-witness-complete-verification-census, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-POLYNOMIAL-TIME-012. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-013: witness_verification; exact quantum-complexity execution: {"accepting_witnesses":1,"complete_census":true,"witnesses":2}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:6238c4a11d6820f7be6ae41d0c1cac6d7c4dcab84f839b9e172cfa507cb48d4d; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf; independent certificate
sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:5334759d2bab3c266b6d815c0fa0370ff2d699f39248e580e50e04c2df32bc01; engine receipt sha256:d380def03ff20463a7c5eceb523208bd0de9c14ae81818c8cb4dcd8bcdef439a at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-WITNESS-VERIFY-013-d380def03ff20463.json.

SFT-QUANTUM-OBL-QCPLXX-014 - Interactive quantum proof correspondence

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-INTERACTIVE-PROOF-014; Quantum Computation family QCPLXX; Interactive quantum proof correspondence.
- **Forced law:** Interactive quantum proof correspondence retains each party, message, challenge, response, round, private support and final verification record under an exact resource bound.

- **Unique surviving structure:** canonical-input-and-promise-size__round-challenge-response-decision-ledger__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-014 uniquely retains round-challenge-response-decision-ledger, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-WITNESS-VERIFY-013. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-014: interactive_proof; exact quantum-complexity execution: {"final_decisions":1,"messages":2,"private_support_retained":true,"rounds":1}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:74c08a49ed5895668c0f7196928a3713a191e3b9194d2a0bbb5e2d273df1de91; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf; independent certificate
sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:ad0c3c3c7c850b4f35f80565ce9f3f0726767c2d63831a3397b41002e74dfe71; engine receipt sha256:4d91ba16b6fde9099733e3584629d28e981b72b94658aac4db74a25a8e7809af at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-INTERACTIVE-PROOF-014-4d91ba16b6fde909.json.

SFT-QUANTUM-OBL-QCPLXX-015 - Quantum space complexity

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-SPACE-015; Quantum Computation family QCPLXX; Quantum space complexity.
- **Forced law:** Quantum space is the maximum simultaneous canonical register, branch-support, phase, ancilla and retained-record width across the execution.
- **Unique surviving structure:** canonical-input-and-promise-size__maximum-live-support-record-width__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-015 uniquely retains maximum-live-support-record-width, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-INTERACTIVE-PROOF-014. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-015: space_complexity; exact quantum-complexity execution: {"maximum_live_width":3,"stage_widths":[2,3,2,1]}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:8c102a8bdaa1236926126b0fcb24f5495a200e6a3e545011cc5573bdadb932b3; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcb09d292acf; independent certificate sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:007e1699bbf41bf573988bb2f0316a8f8cc4f4619a85dd93cded759766370c29; engine receipt sha256:587dc8e1da18658d8b74e093bcac722ba3e4253784697cff4b5893c96388d279 at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-SPACE-015-587dc8e1da18658d.json.

SFT-QUANTUM-OBL-QCPLXX-016 - Quantum parallel complexity

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-PARALLEL-016; Quantum Computation family QCPLXX; Quantum parallel complexity.
- **Forced law:** Quantum parallel complexity retains total work, causal depth, available processors, register width, live support and communication as separate coordinates.
- **Unique surviving structure:** canonical-input-and-promise-size__work-depth-width-support-vector__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-016 uniquely retains work-depth-width-support-vector, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-SPACE-015. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-016: parallel_complexity; exact quantum-complexity execution: {"coordinates_separate":true,"depth":2,"processors":3,"work":6}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:ea6b47619fc0503907ef9d448c7b367abd45030d1f43ea5a00892638bfc22ba3`; independent implementation `sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf`; independent certificate
`sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919`; measurement `sha256:d1e30be8d93cb2c5b1afc0f568955bd51173d9311db302bcbde454c9e75d3f61`; engine receipt `sha256:f7a0404fc737283bba59728d9b5424ffab40ac748a06ddf5ac746ec074e3413d` at `receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-PARALLEL-016-f7a0404fc737283b.json`.

SFT-QUANTUM-OBL-QCPLXX-017 - Quantum circuit-family uniformity

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-UNIFORMITY-017; Quantum Computation family QCPLXX; Quantum circuit-family uniformity.
- **Forced law:** A circuit family is uniform only when one registered finite generator produces and verifies the circuit for every positive finite input size within its own declared resource bound.
- **Unique surviving structure:** `canonical-input-and-promise-size__single-generator-for-every-input-size__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff`. Exact result: QCPLXX-017 uniquely retains `single-generator-for-every-input-size`, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-PARALLEL-016. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-017: uniformity; exact quantum-complexity execution: `{"generated_circuits":3,"input_sizes":3,"single_generator":true}`; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:456a7380b470496ea2a75e4660c9fe642e0ece95bc1c806bc6d236f58dfef4f1`; independent implementation `sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf`; independent certificate
`sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919`; measurement `sha256:0c1d70fe8ac62dcd5697dfb3ee6c244a42e0c642e3cda3f05c05528e53f39fb`; engine receipt

sha256:c55fae1e1863fc5820515fb493be82c11bece3a827995f11b9be688693e3f312 at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-UNIFORMITY-017-c55fae1e1863fc58.json.

SFT-QUANTUM-OBL-QCPLXX-018 - Quantum reduction and completeness

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-REDUCTION-018; Quantum Computation family QCPLXX; Quantum reduction and completeness.
- **Forced law:** A quantum reduction maps every canonical source instance to a target instance while preserving decisions and charging the complete transformation and observation resources; completeness requires all owned reductions.
- **Unique surviving structure:** canonical-input-and-promise-size__semantics-resource-preserving-problem-map__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-018 uniquely retains semantics-resource-preserving-problem-map, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-UNIFORMITY-017. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-018: reduction_completeness; exact quantum-complexity execution: {"decision_preserved":true,"mapping_rows":1,"source_classes":1,"target_classes":1}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:bd8f9ee0330ea5f011c1c230a57807b6962f8fcec45dfd0204c9111078230681; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf; independent certificate
sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:64c8dda4cb8c3b075bcc1d9ed764b92b3f049cd39bfc4d274aed7bb6acd8b730; engine receipt sha256:c56d004bfc36401e94937879b9c9824273d90bc18915c897562268fa948ad413 at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-REDUCTION-018-c56d004bfc36401e.json.

SFT-QUANTUM-OBL-QCPLXX-019 - Quantum lower-bound adversary certificate

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-ADVERSARY-LOWER-019; Quantum Computation family QCPLXX; Quantum lower-bound adversary certificate.
- **Forced law:** An adversary lower bound supplies a complete family of indistinguishable input pairs and proves the minimum queries or transitions required to create the registered distinguishing observation.
- **Unique surviving structure:** canonical-input-and-promise-size__complete-distinguishability-adversary-ledger__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__po

st-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-019 uniquely retains complete-distinguishability-adversary-ledger, complete resource custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-REDUCTION-018. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-019: adversary_lower_bound; exact quantum-complexity execution: {"input_pairs":2,"lower_witnesses":[2,3],"observed_costs":[3,4]}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:6051fef6fe7036b4003556d0a672c2089d4aade57995063d6133f65d34d347a2; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf; independent certificate
sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:e960196d1282b7984984bf09d365ea8ea6ed89ffa8f04fda80f9ae0ce05e0243; engine receipt sha256:bd32fcf800d0653fc4ffd8b443ea1d1394a3a0e9617f7fd96187e3a990029eca at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-ADVERSARY-LOWER-019-bd32fcf800d0653f.json.

SFT-QUANTUM-OBL-QCPLXX-020 - Polynomial method correspondence boundary

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-POLYNOMIAL-METHOD-020; Quantum Computation family QCPLXX; Polynomial method correspondence boundary.
- **Forced law:** Polynomial-method correspondence translates a finite query trace into exact integer or rational degree constraints; it cannot import continuous coefficients or an unrestricted bound without proof.
- **Unique surviving structure:** canonical-input-and-promise-size__exact-degree-query-trace-correspondence__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-020 uniquely retains exact-degree-query-trace-correspondence, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-ADVERSARY-LOWER-019. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** QCPLXX-020: polynomial_method; exact quantum-complexity execution: {"constraint_rows":3,"continuous_coefficients_imported":false,"query_degrees":[1,2,3]}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:9f76d319028662270649827fa38c4abbbbeba54da722ed8c48d870f672726277d; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf; independent certificate sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:40002265421b8bf56b1db2c89d21c429a8f00bbf32259ced77375a4e69809ca8; engine receipt sha256:49b2fee86b80771f99488941875b603c3442ba63bff63cbcb1912fa6d39c8e4f at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-POLYNOMIAL-METHOD-020-49b2fee86b80771f.js on.

SFT-QUANTUM-OBL-QCPLXX-021 - Classical simulation resource boundary

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-CLASSICAL-SIMULATION-021; Quantum Computation family QCPLXX; Classical simulation resource boundary.
- **Forced law:** Classical simulation reproduces the complete quantum support, phase and observation trace under an explicit classical resource vector; finite and asymptotic boundaries remain distinct.
- **Unique surviving structure:** canonical-input-and-promise-size__same-trace-classical-simulation-vector__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-021 uniquely retains same-trace-classical-simulation-vector, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-POLYNOMIAL-METHOD-020. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-021: classical_simulation; exact quantum-complexity execution: {"classical_vector":[5,8,4],"complete_trace_reproduced":true,"quantum_vector":[3,4,2]}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost,

sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:18cceb7bcfbef0a77120b9958e6d7c088ef1e5da5f6e0b23d52c36805da4b6f; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf; independent certificate sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:30407e8bef6e2fa9d1743c62eb9bfe67ee68f00091482818efc06abd1af0d7bd; engine receipt sha256:c2779f9eda27d0f0e8ecf5305ae16d04db236548913ae73ac2fc867e5c670571 at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-CLASSICAL-SIMULATION-021-c2779f9eda27d0f0.json.

SFT-QUANTUM-OBL-QCPLXX-022 - Quantum advantage and separation certificate

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-ADVANTAGE-022; Quantum Computation family QCPLXX; Quantum advantage and separation certificate.
- **Forced law:** Quantum advantage requires a strict resource-vector separation for the same problem grammar, correctness condition, representation and observation boundary, with both algorithms executed or bounded.
- **Unique surviving structure:** canonical-input-and-promise-size__same-problem-strict-resource-separation__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-022 uniquely retains same-problem-strict-resource-separation, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-CLASSICAL-SIMULATION-021. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-022: advantage_separation; exact quantum-complexity execution: {"comparison_vector":[5,8,4],"quantum_vector":[3,4,2],"same_problem":true,"strict_coordinate":true}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:3f229e3d6b45c4db9dd177d40a2b6f75abdb0cdaaeaf17b99c2040ae8d080f7f; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf; independent certificate sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:b8373d3a1752b7662cd39642a0fc638672742a57e29348b0e4c9cf7432258a20; engine receipt sha256:2acc6d3fec6f694907d190d3c82c0eee707fe40bae87ac2f0e74e107cf92b147 at receipts/

engine/model_admitted/SFT-QUANTUM-QCPLXX-ADVANTAGE-022-2acc6d3fec6f6949.json.

SFT-QUANTUM-OBL-QCPLXX-023 - Average-case and distribution custody

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-AVERAGE-CASE-023; Quantum Computation family QCPLXX; Average-case and distribution custody.
- **Forced law:** Average-case complexity uses a complete registered finite case-support and exact rational weights generated independently of outcomes; every favorable and adverse case remains present.
- **Unique surviving structure:** canonical-input-and-promise-size__exact-generated-case-weight-ledger__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-023 uniquely retains exact-generated-case-weight-ledger, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-ADVANTAGE-022. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-023: average_case; exact quantum-complexity execution: {"all_cases_retained":true,"case_rows":3,"exact_average":"10/3","weight_denominator":6}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:239e1c8d08d05e95526e1f8675d0f1adaf1e004f8e6e748b3f912ba83ca858ca; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf; independent certificate
sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:4b042ab32764329fb121b1728f988e6ec68fe872adab9dc2b27db700718c4b2a; engine receipt sha256:691f996178c6f59f55ef921c6812d758d49e607cc9d2eedb48498b803449a3be at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-AVERAGE-CASE-023-691f996178c6f59f.json.

SFT-QUANTUM-OBL-QCPLXX-024 - Parameterized quantum complexity

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-PARAMETERIZED-024; Quantum Computation family QCPLXX; Parameterized quantum complexity.
- **Forced law:** Parameterized quantum complexity records canonical input size, a separately generated finite parameter and a complete resource bound for every parameter slice.
- **Unique surviving structure:** canonical-input-and-promise-size__input-size-parameter-resource-slices__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-regis

try-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-024 uniquely retains input-size-parameter-resource-slices, complete resource custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-AVERAGE-CASE-023. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-024: parameterized_complexity; exact quantum-complexity execution: {"input_size_separate":true,"parameter_slices":2,"resource_rows":3}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:c8e0e6ca221dded101ec7798fe3f33b0d5af7908451a32a80cd264822bd31870; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcbb09d292acf; independent certificate
sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:fa238644fd35fa76a6e2d9828c8617f93429d887577ea2290e7b2f32d2a9eab2; engine receipt sha256:3d4c9d10a5fcea9713c3f225538e9959cac1e73112a5a60408936f4abb4e75c6 at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-PARAMETERIZED-024-3d4c9d10a5fcea97.json.

SFT-QUANTUM-OBL-QCPLXX-025 - Quantum descriptive complexity

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-DESCRIPTIVE-025; Quantum Computation family QCPLXX; Quantum descriptive complexity.
- **Forced law:** Quantum descriptive complexity is the least canonical Fold-machine description that generates the target support and observation trace under the frozen description grammar.
- **Unique surviving structure:** canonical-input-and-promise-size__least-complete-machine-description__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-025 uniquely retains least-complete-machine-description, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-PARAMETERIZED-024. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** QCPLXX-025: descriptive_complexity; exact quantum-complexity execution: {"grammar_frozen":true,"machine_description_tokens":3,"minimum_checked":true}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost, sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:a8f5ebbc0be3cf878993079f36c14f0f21309f03bba73e48a5f419e0a47071ba; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcb09d292acf; independent certificate
sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:79e974784a9f174b532278a028ea9790a7029e6109a4f71078ed085b5263308c; engine receipt sha256:ff4c459d85411a61b053743f19306a30f58c7dc01f0c2153e6dc1f61c3ac4799 at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-DESCRIPTIVE-025-ff4c459d85411a61.json.

SFT-QUANTUM-OBL-QCPLXX-026 - Quantum-complexity completeness certificate

- **Claim and ownership:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026; Quantum Computation family QCPLXX; Quantum-complexity completeness certificate.
- **Forced law:** The quantum-complexity family is complete exactly when all twenty-six frozen obligations have one owner, one unique survivor, controls, observations, independent reconstructions and current receipts.
- **Unique surviving structure:** canonical-input-and-promise-size__twenty-six-obligation-no-omission-ledger__complete-time-space-query-depth-record-vector__same-problem-same-observation-comparison__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-finite-asymptotic-and-physical-handoff. Exact result: QCPLXX-026 uniquely retains twenty-six-obligation-no-omission-ledger, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete complexity grammar, exact resource execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QALGX-COMPLETENESS-030 -> SFT-QUANTUM-COMPLEXITY-001 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-QUANTUM-QCPLXX-DESCRIPTIVE-025. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCPLXX-026: complexity_completeness; exact quantum-complexity execution: {"duplicate_owners":0,"execution_rows":26,"omitted_owners":0,"registered_obligations":26}; complete quantum-complexity custody: 26 records; all rows preserved; complete finite input, execution, decision, resource and comparison records supply the observation; physical device performance and unrestricted asymptotic claims remain downstream
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMPLEXITY-OBSERVER, SFT-V1-V2-QUANTUM-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported complexity class, stochastic error premise, asymptotic theorem or lower bound selects the law; host 0 denotes absence only and is not a numerical-zero input, resource or probability; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no mismatched problem grammar, hidden cost,

sampled case or target-selected comparison; no physical device performance is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:4158292e6504b5aa6bb211c8206bbf60320afa725935f9d3f4f1d52c5cf2d7d1; independent implementation sha256:5b552f57d938ba0a865127c15cb7984165a22c9a9dca8718f40bcb09d292acf; independent certificate sha256:8ac2228b13c88950798db591cd3c0458bb96ae3ee7a20c1e7026d28f9731c919; measurement sha256:c114f00822164c7f5737b41a757ab3da88c36b88bdc5095c9959e0d91e94dd88; engine receipt sha256:d84ae8e7486c5a85ec27e57471c7f9e5a675f5821f0903ef767c5530f68b30a8 at receipts/engine/model_admitted/SFT-QUANTUM-QCPLXX-COMPLETENESS-026-d84ae8e7486c5a85.json.

36.9 Family QCOMM - 24/24 complete

Scope. Quantum channels, communication and cryptography. This family decides 6,144 candidates, retains 24 unique survivors and passes 96 adverse controls.

SFT-QUANTUM-OBL-QCOMM-001 - Quantum channel as exact relation

- **Claim and ownership:** SFT-QUANTUM-QCOMM-CHANNEL-001; Quantum Computation family QCOMM; Quantum channel as exact relation.
- **Forced law:** A quantum channel is the complete exact relation from canonical source support to image support, phase records and any retained environment distinctions.
- **Unique surviving structure:** complete-source-image-environment-relation__complete-source-image-environment-relation__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-not-hing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMM-001 uniquely retains complete-source-image-environment-relation, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMM-001: channel_relation; exact communication execution: {"complete":true,"environment_rows":2,"image_rows":2,"source_rows":2}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:68d226914de40905a21eaaecfccc072ceca61d42ac89ccd3e60c4bc3ca4739b; independent implementation sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e; independent certificate sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45; measurement

sha256:550763659c952461be8a9e8eeaf70e7ad00df7594f608c6d5519522e8bf4811a; engine receipt
 sha256:59b7627bcc3cad71c2500ada3f272b4e86f4fc0bc3e8bc6cb3433690a3da7b55 at receipts/
 engine/model_admitted/SFT-QUANTUM-QCOMM-X-CHANNEL-001-59b7627bcc3cad71.json.

SFT-QUANTUM-OBL-QCOMM-X-002 - Channel composition and memory boundary

- **Claim and ownership:** SFT-QUANTUM-QCOMM-X-COMPOSITION-002; Quantum Computation family QCOMM-X; Channel composition and memory boundary.
- **Forced law:** Channel composition joins exact image/source rows in causal order; memory is present precisely when an earlier source or environment record changes a later channel row.
- **Unique surviving structure:** complete-source-image-environment-relation__causal-relation-composition-with-memory-record__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMM-X-002 uniquely retains causal-relation-composition-with-memory-record, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMM-X-CHANNEL-001. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMM-X-002: channel_composition; exact communication execution: {"component_channels":2,"composed_rows":2,"memory_record_explicit":true}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:f3b9899b97b0a5feac2cd0c8fa3673558e99ac523255ff380dadb870b4511235; independent implementation sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e; independent certificate
 sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45; measurement sha256:7c7a6a26175f1b7e69e905c76c1235d9019764b2c3049e1baa76a149d3e534f6; engine receipt sha256:589c4daf64d0143ba453051b5f1e2c9afad93dcf06e578607d2952b99bbf4a42 at receipts/engine/model_admitted/SFT-QUANTUM-QCOMM-X-COMPOSITION-002-589c4daf64d0143b.json.

SFT-QUANTUM-OBL-QCOMM-X-003 - Classical information over a quantum channel

- **Claim and ownership:** SFT-QUANTUM-QCOMM-X-CLASSICAL-OVER-QUANTUM-003; Quantum Computation family QCOMM-X; Classical information over a quantum channel.
- **Forced law:** Classical information over a quantum channel is a generated set of distinguishable message classes, channel encodings and an exhaustive decoding observation.
- **Unique surviving structure:** complete-source-image-environment-relation__distinguished-message-class-channel-coding__complete-causal-message-and-record-transcript__complete-

link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMMXX-003 uniquely retains distinguished-message-class-channel-coding, complete transcript custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMMXX-COMPOSITION-002. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMMXX-003: classical_over_quantum; exact communication execution: {"decoder_complete":true,"distinguishable_images":2,"message_classes":2}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:f08d0bc03851b67d4284d224ac6a0486f257c6325bc615026a4d6fdf968e9c6a; independent implementation sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e; independent certificate
sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45; measurement sha256:5c2dda934a2dae84d6b5be25bb79a1a3e183ea3970faccd03ce4597e9aad4c14; engine receipt sha256:ca1b04a7ddcac9e59815f58896a1e47f4af6f9f655b5aadf246f449973afb9f1 at receipts/engine/model_admitted/SFT-QUANTUM-QCOMMXX-CLASSICAL-OVER-QUANTUM-003-ca1b04a7ddcac9e5.json.

SFT-QUANTUM-OBL-QCOMMXX-004 - Quantum information transfer

- **Claim and ownership:** SFT-QUANTUM-QCOMMXX-QUANTUM-TRANSFER-004; Quantum Computation family QCOMMXX; Quantum information transfer.
- **Forced law:** Quantum information transfer preserves a canonical word-support, relative-phase and joint-record description through a registered channel, not merely its terminal label.
- **Unique surviving structure:** complete-source-image-environment-relation__word-phase-joint-record-transfer__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMMXX-004 uniquely retains word-phase-joint-record-transfer, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 ->

SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMM-CLASSICAL-OVER-QUANTUM-003.

Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** QCOMM-004: quantum_transfer; exact communication execution:
{"all_preserved":true,"phase_rows":1,"source_records":1,"word_rows":1}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:53024577b45d28473d89728ff17406b844f3acdec32e86627729f97a0ecea757; independent implementation sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e; independent certificate
sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45; measurement sha256:7a61983e9bada54b394778f9e6bb80c5b1c4cdab865d51d632b493124242cf41; engine receipt sha256:8116bc92126a510b919b089914971c78f3f0ecfeffce88b7fbf2c4db43e76d38 at receipts/engine/model_admitted/SFT-QUANTUM-QCOMM-QUANTUM-TRANSFER-004-8116bc92126a510b.json.

SFT-QUANTUM-OBL-QCOMM-005 - Entanglement-assisted communication

- **Claim and ownership:** SFT-QUANTUM-QCOMM-ENTANGLEMENT-ASSISTED-005; Quantum Computation family QCOMM; Entanglement-assisted communication.
- **Forced law:** Entanglement-assisted communication explicitly accounts for pre-shared nonfactorable support, transmitted distinctions, local transformations and observation records.
- **Unique surviving structure:** complete-source-image-environment-relation__shared-nonfactorable-support-assisted-channel__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMM-005 uniquely retains shared-nonfactorable-support-assisted-channel, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMM-QUANTUM-TRANSFER-004. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMM-005: entanglement_assistance; exact communication execution:
{"left_marginal_rows":2,"right_marginal_rows":2,"shared_joint_rows":2}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:8d23de208b9c601e473b07e86bcc465d33f95ed720417d87ca752fa469fb16b9`; independent implementation `sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e`; independent certificate
`sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45`; measurement `sha256:171566c7638b1080179bb7997c1fc3b56d146191e4bd71c51f9cb2bb7efebc6b`; engine receipt `sha256:9cf3bbe8093792283b6ba3719388ac066eaf273ec4bf336f6c317b6faeb5c7a8` at receipts/engine/model_admitted/SFT-QUANTUM-QCOMMXX-ENTANGLEMENT-ASSISTED-005-9cf3bbe809379228.json.

SFT-QUANTUM-OBL-QCOMMXX-006 - Teleportation-equivalent state transfer

- **Claim and ownership:** SFT-QUANTUM-QCOMMXX-TELEPORTATION-006; Quantum Computation family QCOMMXX; Teleportation-equivalent state transfer.
- **Forced law:** Teleportation correspondence transfers an unknown state description by joint observation, two retained classical fibre records and a record-controlled reversible reconstruction on shared joint support.
- **Unique surviving structure:** `complete-source-image-environment-relation__joint-observation-record-controlled-reconstruction__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff`. Exact result: QCOMMXX-006 uniquely retains joint-observation-record-controlled-reconstruction, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMMXX-ENTANGLEMENT-ASSISTED-005. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMMXX-006: `teleportation_correspondence`; exact communication execution: `{"classical_records":2,"controlled_corrections":1,"joint_observations":1,"state_reconstructed":true}`; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:6c2fc42652ba128e84208a95736f04db8baea30c00f0ed754b03f5030d78d17c`; independent implementation `sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e`; independent certificate
`sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45`; measurement

sha256:0b294afde50ca0f10c60dc4e3a99806705c47890e197b913a2bf1fab5376e98d; engine receipt
 sha256:a98394422b828be384aaa52f7f173faaf5e3b24339ae66ed4090830ff36737e4 at receipts/
 engine/model_admitted/SFT-QUANTUM-QCOMMXX-TELEPORTATION-006-a98394422b828be3.json.

SFT-QUANTUM-OBL-QCOMMXX-007 - Dense-coding-equivalent distinction transfer

- **Claim and ownership:** SFT-QUANTUM-QCOMMXX-DENSE-CODING-007; Quantum Computation family QCOMMXX; Dense-coding-equivalent distinction transfer.
- **Forced law:** Dense-coding correspondence maps four generated message distinctions to local actions on pre-shared joint support and one transmitted unit, with an exhaustive four-class joint observation.
- **Unique surviving structure:** complete-source-image-environment-relation__four-message-joint-support-encoding__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMMXX-007 uniquely retains four-message-joint-support-encoding, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMMXX-TELEPORTATION-006. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMMXX-007: dense_coding_correspondence; exact communication execution: {"message_classes":4,"shared_joint_support":true,"transmitted_units":1}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:aa60f3f14997f1b310890b4eddf5912b864018badc2a5dc837b67625f52d262d; independent implementation sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e; independent certificate
 sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45; measurement sha256:236020428ddee2ba484dc468ee93aa4ee2fa8d68cda0bb3632c82f66cf7a4123; engine receipt sha256:bd8e3adb57c298bad71c94498d786f41d417d70ad485f129287dea6cc5d86f06 at receipts/engine/model_admitted/SFT-QUANTUM-QCOMMXX-DENSE-CODING-007-bd8e3adb57c298ba.json.

SFT-QUANTUM-OBL-QCOMMXX-008 - No-signalling operational boundary

- **Claim and ownership:** SFT-QUANTUM-QCOMMXX-NO-SIGNALLING-008; Quantum Computation family QCOMMXX; No-signalling operational boundary.
- **Forced law:** No-signalling is the exact invariance of a remote marginal observation under local reversible actions when no communication or shared outcome record crosses the partition.
- **Unique surviving structure:** complete-source-image-environment-relation__remote-marginal-invariance-without-record-transfer__complete-causal-message-and-record-transcript__c

complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMMXX-008 uniquely retains remote-marginal-invariance-without-record-transfer, complete transcript custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMMXX-DENSE-CODING-007. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMMXX-008: no_signalling; exact communication execution: {"record_transferred":false,"remote_marginal_after":["held","returned"],"remote_marginal_before":["held","returned"]}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:9a0a1aa38b119670cf1e4fe084e7a869efe259fbc9a605814ca4371b281837ae; independent implementation sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e; independent certificate sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45; measurement sha256:030ce7f20a8c77276182ae6da8a2d0ff2c5d7f9f48e7d3e75f06ff5da312d5dc; engine receipt sha256:b7ff278c0ace85e20357d456d100ef48970e3d4ea0cf867929c8e723e79d0bec at receipts/engine/model_admitted/SFT-QUANTUM-QCOMMXX-NO-SIGNALLING-008-b7ff278c0ace85e2.json.

SFT-QUANTUM-OBL-QCOMMXX-009 - Channel capacity as generated support rate

- **Claim and ownership:** SFT-QUANTUM-QCOMMXX-CAPACITY-009; Quantum Computation family QCOMMXX; Channel capacity as generated support rate.
- **Forced law:** Channel capacity is the maximum exact distinguishable message-support rate over the complete generated code grammar at a stated positive finite use count and error condition.
- **Unique surviving structure:** complete-source-image-environment-relation__maximum-distinguishable-support-per-use__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMMXX-009 uniquely retains maximum-distinguishable-support-per-use, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 ->

SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMM-X-NO-SIGNALING-008. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** QCOMM-X-009: channel_capacity; exact communication execution: {"channel_uses":2,"distinguishable_messages":4,"exact_rate":"2/1"}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:397d9ae07cb2f796a6b6f569cfea2709911a9746e6923c89235be2b3c47d3382; independent implementation sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e; independent certificate
sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45; measurement sha256:914acb56dc93fd4855018ff11c328cabd02ab8b866ecd2d93c7e9d1d77d2e82c; engine receipt sha256:9877068a5d841f06bded5bd2f51ae9610016490f02238e3455a9588167cad58b at receipts/engine/model_admitted/SFT-QUANTUM-QCOMM-X-CAPACITY-009-9877068a5d841f06.json.

SFT-QUANTUM-OBL-QCOMM-X-010 - Private and coherent information correspondence

- **Claim and ownership:** SFT-QUANTUM-QCOMM-X-PRIVATE-COHERENT-010; Quantum Computation family QCOMM-X; Private and coherent information correspondence.
- **Forced law:** Private and coherent information correspondences compare exact distinctions retained by receiver and environment after one source-bound channel execution.
- **Unique surviving structure:** complete-source-image-environment-relation__receiver-environment-distinction-ledger__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMM-X-010 uniquely retains receiver-environment-distinction-ledger, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMM-X-CAPACITY-009. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMM-X-010: private_coherent_information; exact communication execution: {"comparison_exact":true,"environment_distinctions":2,"receiver_distinctions":4}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:8877cd9ba0047d6ae22d279ba23a3e09b419f8367c682855eddfb6126828db2a`; independent implementation `sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e`; independent certificate
`sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45`; measurement `sha256:7ca43b290f34a98ccaa3130562fa0fcd3f02d02294f4840fe498809c17b7b63`; engine receipt `sha256:99193247bc475c647196d2ac7c4a3145777ebaaf5694bef3142f77cff778bb03` at receipts/engine/model_admitted/SFT-QUANTUM-QCOMM-X-PRIVATE-COHERENT-010-99193247bc475c64.json.

SFT-QUANTUM-OBL-QCOMM-X-011 - Quantum data-processing relation

- **Claim and ownership:** SFT-QUANTUM-QCOMM-X-DATA-PROCESSING-011; Quantum Computation family QCOMM-X; Quantum data-processing relation.
- **Forced law:** A downstream channel cannot reopen a distinction closed by an upstream observation unless an independent retained record supplies it; exact distinguishability therefore cannot increase by postprocessing alone.
- **Unique surviving structure:** complete-source-image-environment-relation__postprocessing-can-not-reopen-closed-distinction__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMM-X-011 uniquely retains postprocessing-cannot-reopen-closed-distinction, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMM-X-PRIVATE-COHERENT-010. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMM-X-011: data_processing; exact communication execution: {"distinction_counts":[4,3,2,1],"increases":0}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:582c87b3e5f0987858d67c5cdab60788fa1bdc8610d88d774961308d1723a52b`; independent implementation `sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e`; independent certificate
`sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45`; measurement

sha256:2dfe84a60df594b83376aa5efede7e745ad515f6aa68f6ecc0617ca950543be0; engine receipt
 sha256:22ba20947483cd6a836efbde48b11bf8a5c124ff5967d834287c01939c444415 at receipts/
 engine/model_admitted/SFT-QUANTUM-QCOMM-X-DATA-PROCESSING-011-22ba20947483cd6a.json.

SFT-QUANTUM-OBL-QCOMM-X-012 - Channel noise and environment record

- **Claim and ownership:** SFT-QUANTUM-QCOMM-X-NOISE-ENVIRONMENT-012; Quantum Computation family QCOMM-X; Channel noise and environment record.
- **Forced law:** Channel noise is an exact source/image mismatch class accompanied by the environment, fault or missing-record ledger required to reconstruct its origin.
- **Unique surviving structure:** complete-source-image-environment-relation__source-image-error-environment-custody__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMM-X-012 uniquely retains source-image-error-environment-custody, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMM-X-DATA-PROCESSING-011. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMM-X-012: noise_environment; exact communication execution: {"environment_records":1,"mismatched_images":1,"source_rows":1}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:aa6cee66395c18e1a206e72d6e9a5468714fc37dec3854a3e9a62e97913ea0db; independent implementation sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e; independent certificate
 sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45; measurement sha256:8f6ecbcbca80c92abfb4098291126d60f305b66dd45e38dc2a62cafb2d078aac; engine receipt sha256:4cf3a01e853a41a74645ee334655711a972807106122d45bb55eb507640d90d4 at receipts/engine/model_admitted/SFT-QUANTUM-QCOMM-X-NOISE-ENVIRONMENT-012-4cf3a01e853a41a7.json.

SFT-QUANTUM-OBL-QCOMM-X-013 - Entanglement distribution and swapping

- **Claim and ownership:** SFT-QUANTUM-QCOMM-X-ENTANGLEMENT-DISTRIBUTION-013; Quantum Computation family QCOMM-X; Entanglement distribution and swapping.
- **Forced law:** Entanglement distribution and swapping compose elementary joint links and a retained middle observation to produce an exact outer nonfactorable support.

- **Unique surviving structure:** complete-source-image-environment-relation__link-support-joint-observation-outer-repartition__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMMXX-013 uniquely retains link-support-joint-observation-outer-repartition, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMMXX-NOISE-ENVIRONMENT-012. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMMXX-013: entanglement_swapping; exact communication execution: {"elementary_links":2,"middle_observations":1,"outer_links":1}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:6db0bd744749125c9d79faeeba88eb263db8d1a349556c7807ce29470f60a57; independent implementation sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e; independent certificate
sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45; measurement sha256:8b1e3eed4360a9d483b8b77db1dd4d75b01cb13a325c1ceca40a2ab4df429128; engine receipt sha256:71a463ad79adbfe6f9937120beeb8c3938b2213905c0f931f949fbfb1c96f7f at receipts/engine/model_admitted/SFT-QUANTUM-QCOMMXX-ENTANGLEMENT-DISTRIBUTION-013-71a463ad79adbfe6.json.

SFT-QUANTUM-OBL-QCOMMXX-014 - Quantum repeater correspondence

- **Claim and ownership:** SFT-QUANTUM-QCOMMXX-REPEATER-014; Quantum Computation family QCOMMXX; Quantum repeater correspondence.
- **Forced law:** A repeater correspondence is a finite causal chain of link generation, verified filtering or correction, swapping and classical-record propagation with complete resource custody.
- **Unique surviving structure:** complete-source-image-environment-relation__link-generation-filtering-swapping-resource-chain__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMMXX-014 uniquely retains link-generation-filtering-swapping-resource-chain, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMMXX-ENTANGLEMENT-DISTRIBUTION-013. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMMXX-014: repeater_correspondence; exact communication execution: {"end_links":1,"link_generations":1,"swaps":1,"verified_filters":1}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:09f12a45472742b3e1048bf2e8982e7c2bac566234ef3830fd771b66c7169705; independent implementation sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e; independent certificate sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45; measurement sha256:1dcc6be3763330dd1f95bf51235e36af626c83c82fdb550bab9e311d1cf015f; engine receipt sha256:b09de71a3b5c42d89e8ce4353418f5fe5d5e9719fb357fccfe94dee295d62f58 at receipts/engine/model_admitted/SFT-QUANTUM-QCOMMXX-REPEATER-014-b09de71a3b5c42d8.json.

SFT-QUANTUM-OBL-QCOMMXX-015 - Quantum network node and link identity

- **Claim and ownership:** SFT-QUANTUM-QCOMMXX-NETWORK-IDENTITY-015; Quantum Computation family QCOMMXX; Quantum network node and link identity.
- **Forced law:** A quantum network is an exact finite graph of typed nodes, quantum links, classical links, local supports and interface records, with every ownership boundary explicit.
- **Unique surviving structure:** complete-source-image-environment-relation__typed-node-link-support-graph__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMMXX-015 uniquely retains typed-node-link-support-graph, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMMXX-REPEATER-014. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMMXX-015: network_identity; exact communication execution: {"links":2,"nodes":3,"ownership_boundaries_explicit":true}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:03ddd06e73af5f258fac87a5f465903c18a662800b17fec9d60edd404189b002`; independent implementation `sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e`; independent certificate
`sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45`; measurement `sha256:645e9065723c13068136017a29ee7f484a94ed219524152a556b06fb77758dc7`; engine receipt `sha256:f23149520ad72273eb34621de6e3d5c21279ab9f8b43d640d5e25668b4efb548` at `receipts/engine/model_admitted/SFT-QUANTUM-QCOMM-X-NETWORK-IDENTITY-015-f23149520ad72273.json`.

SFT-QUANTUM-OBL-QCOMM-X-016 - Distributed quantum process causality

- **Claim and ownership:** SFT-QUANTUM-QCOMM-X-CAUSALITY-016; Quantum Computation family QCOMM-X; Distributed quantum process causality.
- **Forced law:** Distributed quantum causality is the partial order generated by local transitions, transmissions, receipts and shared observations; spacelike or unordered events remain explicitly incomparable.
- **Unique surviving structure:** `complete-source-image-environment-relation__partial-order-event-and-message-ledger__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff`. Exact result: QCOMM-X-016 uniquely retains partial-order-event-and-message-ledger, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMM-X-NETWORK-IDENTITY-015. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMM-X-016: distributed `causality`; exact communication execution: `{"cycles":0,"events":3,"ordered_pairs":2}`; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:2d3b732a1497721b9e4d5f53fc2e77a74893621540b8f08a270745145c2a4290`; independent implementation `sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e`; independent certificate
`sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45`; measurement

sha256:ed92fc3cd6e8d1c19a6e085abcf00cef189940a97949e2a0d74c2a836d5eb468; engine receipt
 sha256:078d540c771aa19eb0b6a43d1150b5a197b1e9f298078a05bbe0e31aa06caf84 at receipts/
 engine/model_admitted/SFT-QUANTUM-QCOMMXX-CAUSALITY-016-078d540c771aa19e.json.

SFT-QUANTUM-OBL-QCOMMXX-017 - Network entanglement routing

- **Claim and ownership:** SFT-QUANTUM-QCOMMXX-ROUTING-017; Quantum Computation family QCOMMXX; Network entanglement routing.
- **Forced law:** Entanglement routing enumerates all lawful network paths and selects only through a generated exact link-resource order, retaining every rejected path and correction record.
- **Unique surviving structure:** complete-source-image-environment-relation__complete-path-link-resource-selection__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-hand off. Exact result: QCOMMXX-017 uniquely retains complete-path-link-resource-selection, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMMXX-CAUSALITY-016. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMMXX-017: routing; exact communication execution: {"all_link_records_retained":true,"candidate_path_length":4,"source":"a","target":"d"}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:4d18d6e8423aa6388291c61e91964bc662a3c3113133b9a9cef752d936c5408f; independent
 implementation sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e;
 independent certificate
 sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45; measurement
 sha256:182e3481ce41750fd4d84df39ab43a0e145721259e0fff62eae3937392ec4dca; engine receipt
 sha256:ff79382ba10ee289ef5ab35e94d4c873a0295304b8fbbf025df0acb36eb3db88 at receipts/
 engine/model_admitted/SFT-QUANTUM-QCOMMXX-ROUTING-017-ff79382ba10ee289.json.

SFT-QUANTUM-OBL-QCOMMXX-018 - Quantum key-distribution correctness boundary

- **Claim and ownership:** SFT-QUANTUM-QCOMMXX-QKD-018; Quantum Computation family QCOMMXX; Quantum key-distribution correctness boundary.
- **Forced law:** Quantum key-distribution correctness requires exact preparation, basis, observation, sifting, disclosed-test and retained-key transcripts; secrecy additionally requires a separately defined adversary and bound.
- **Unique surviving structure:** complete-source-image-environment-relation__prepare-observe-sift-test-key-transcript__complete-causal-message-and-record-transcript__complete-li

nk-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMMXX-018 uniquely retains prepare-observe-sift-test-key-transcript, complete transcript custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMMXX-ROUTING-017. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMMXX-018: qkd_correctness; exact communication execution: {"disclosed_test":1,"matched_bases":2,"prepared":4,"retained_key":1}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:de1d832a2cf774aaf68f726eeae85212970b9d396f648565019f127ba3e8dc17; independent implementation sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e; independent certificate
sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45; measurement sha256:7afd269cb89f0930e3c7f8297416ade6ce563ce358b720bc640c6ab85c70f128; engine receipt sha256:97ca19f650b83a7a40e8951a9f56fefae9a8abae12407193b48ed5b1d75e1ce2 at receipts/engine/model_admitted/SFT-QUANTUM-QCOMMXX-QKD-018-97ca19f650b83a7a.json.

SFT-QUANTUM-OBL-QCOMMXX-019 - Quantum authentication interface

- **Claim and ownership:** SFT-QUANTUM-QCOMMXX-AUTHENTICATION-019; Quantum Computation family QCOMMXX; Quantum authentication interface.
- **Forced law:** A quantum authentication interface binds message support, key description, tag generation, channel transcript and accept/reject verification while preserving every tamper control.
- **Unique surviving structure:** complete-source-image-environment-relation__message-tag-channel-verification-ledger__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMMXX-019 uniquely retains message-tag-channel-verification-ledger, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMMXX-QKD-018. Registered root:

SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** QCOMM-019: authentication; exact communication execution: {"message_rows":1,"tag_rows":1,"tamper_controls_retained":true,"verified_rows":1}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:079171feb2314d7e231129b773e4e2fd3c08dc659cc72fe01e6219d854d6f732; independent implementation sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e; independent certificate
sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45; measurement sha256:e60b657d74a98d93a02128c7ba335c6a49eb7933d13a7c8e032120962fd077f8; engine receipt sha256:a031faa6c525e870cc7fa104545e67893ebe5f011d806e8501fe89fdf2ba4bf8 at receipts/engine/model_admitted/SFT-QUANTUM-QCOMM-AUTHENTICATION-019-a031faa6c525e870.json.

SFT-QUANTUM-OBL-QCOMM-020 - Quantum secret-sharing correspondence

- **Claim and ownership:** SFT-QUANTUM-QCOMM-SECRET-SHARING-020; Quantum Computation family QCOMM; Quantum secret-sharing correspondence.
- **Forced law:** Quantum secret-sharing correspondence enumerates every participant subset, proves exact reconstruction for authorized subsets and exact missing-distinction closure for forbidden subsets.
- **Unique surviving structure:** complete-source-image-environment-relation__authorized-subset-reconstruction-forbidden-subset-closure__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMM-020 uniquely retains authorized-subset-reconstruction-forbidden-subset-closure, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMM-AUTHENTICATION-019. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMM-020: secret_sharing; exact communication execution: {"authorized_pairs":3,"forbidden_singletons":3,"participants":3}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:48280032b120ce56dbe52ee4beebac2d3c31962b4628be10d67352c6c674654c`; independent implementation `sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e`; independent certificate
`sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45`; measurement `sha256:570aa7fed977d99ff2b38daebb2dc6745029c56b5b3630db611eb44d843d7773`; engine receipt `sha256:be797c98156fcb0498b7a7b3b84668299f469c0e32de24425e5a1d820ee2023e` at receipts/engine/model_admitted/SFT-QUANTUM-QCOMM-X-SECRET-SHARING-020-be797c98156fcb04.json.

SFT-QUANTUM-OBL-QCOMM-X-021 - Device-independent security handoff

- **Claim and ownership:** SFT-QUANTUM-QCOMM-X-DEVICE-INDEPENDENT-021; Quantum Computation family QCOMM-X; Device-independent security handoff.
- **Forced law:** Device-independent security retains the formal input/output transcript, causal assumptions and adversary model while handing physical isolation, loopholes and measured violation values to experiment.
- **Unique surviving structure:** complete-source-image-environment-relation__formal-transcript-to-physical-test-handoff__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-not-hing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMM-X-021 uniquely retains formal-transcript-to-physical-test-handoff, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMM-X-SECRET-SHARING-020. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMM-X-021: device_independent_handoff; exact communication execution: {"formal_transcript_complete":true,"physical_loophole_values_present":false}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:7812339d69d740eb1478dc0bd7d8db01d7a2778a86cda6b556c2470febaac39e`; independent implementation `sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e`; independent certificate
`sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45`; measurement `sha256:527dabc15ff0555e3d9fd288547da7e40a921a02fca2fe5d2e358f0c537135b7`; engine receipt

sha256:c2ec2267046c481b786d01c73946c6aa1708fcbafbccf41a8c973c11da081d8b at receipts/engine/model_admitted/SFT-QUANTUM-QCOMM-X-DEVICE-INDEPENDENT-021-c2ec2267046c481b.json.

SFT-QUANTUM-OBL-QCOMM-X-022 - Quantum adversary and transcript custody

- **Claim and ownership:** SFT-QUANTUM-QCOMM-X-ADVERSARY-TRANSCRIPT-022; Quantum Computation family QCOMM-X; Quantum adversary and transcript custody.
- **Forced law:** A quantum adversary is an exact generated action and resource grammar; its complete queries, states, observations, side information and final view remain in the transcript.
- **Unique surviving structure:** complete-source-image-environment-relation__adversary-action-resource-view-ledger__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMM-X-022 uniquely retains adversary-action-resource-view-ledger, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMM-X-DEVICE-INDEPENDENT-021. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMM-X-022: adversary_transcript; exact communication execution: {"guesses":1,"measurements":1,"queries":1,"responses":1}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:5623d47c38d78cb065b21831513bf7c1e501cb3d2b9eed23c9d6ee2c6319557f; independent implementation sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e; independent certificate
sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45; measurement sha256:06da82386b6239c2db4e8b09e54f2122f9ae66482575ef452b996733ef3d71e5; engine receipt sha256:fe11c0c71438066d6618a813dedd5fb6dc41ff8c14b4940d80cee9db69f43421 at receipts/engine/model_admitted/SFT-QUANTUM-QCOMM-X-ADVERSARY-TRANSCRIPT-022-fe11c0c71438066d.json.

SFT-QUANTUM-OBL-QCOMM-X-023 - Post-quantum classical-security handoff

- **Claim and ownership:** SFT-QUANTUM-QCOMM-X-POST-QUANTUM-HANDOFF-023; Quantum Computation family QCOMM-X; Post-quantum classical-security handoff.
- **Forced law:** Post-quantum security is a handoff from a classical scheme to an exact quantum-adversary grammar and reduction; no hardness or security claim is inherited without its own proof.

- **Unique surviving structure:** complete-source-image-environment-relation__classical-scheme-quantum-adversary-reduction-interface__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMMXX-023 uniquely retains classical-scheme-quantum-adversary-reduction-interface, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMMXX-ADVERSARY-TRANSCRIPT-022. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMMXX-023: post_quantum_handoff; exact communication execution: {"classical_scheme_registered":true,"quantum_adversary_bounded":true,"reduction_required":true}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:385be0480fbb2f59edbc93ae86b1ec4ec9e3753bb93bd66d6602e2d0177bd2be; independent implementation sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e; independent certificate
sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45; measurement sha256:eb8d61f698ea07c5539d6ec812522be6ba6981c156056240688b0d6ec239b08f; engine receipt sha256:9647836ad4a693e5e1a7aeafa3168dc04c2f87f7471490157f3b205dc8f46af7 at receipts/engine/model_admitted/SFT-QUANTUM-QCOMMXX-POST-QUANTUM-HANDOFF-023-9647836ad4a693e5.json.

SFT-QUANTUM-OBL-QCOMMXX-024 - Quantum communication and security completeness certificate

- **Claim and ownership:** SFT-QUANTUM-QCOMMXX-COMPLETENESS-024; Quantum Computation family QCOMMXX; Quantum communication and security completeness certificate.
- **Forced law:** The quantum communication family is complete exactly when all twenty-four frozen obligations have one owner, one unique survivor, controls, observations, independent reconstructions and current receipts.
- **Unique surviving structure:** complete-source-image-environment-relation__twenty-four-obligation-no-omission-ledger__complete-causal-message-and-record-transcript__complete-link-round-support-and-adversary-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-security-and-physical-handoff. Exact result: QCOMMXX-024 uniquely retains twenty-four-obligation-no-omission-ledger, complete transcript custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete channel/protocol grammar, exact transcript, source identity, candidate survivor, seal, custody record, control or independent

reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-QCPLXX-COMPLETENESS-026 -> SFT-QUANTUM-COMMUNICATION-001 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-INFO-CHAN-COMPLETENESS-018 -> SFT-QUANTUM-QCOMMXX-POST-QUANTUM-HANDOFF-023. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCOMMXX-024: communication_completeness; exact communication execution: {"duplicate_owners":0,"execution_rows":24,"omitted_owners":0,"registered_obligations":24}; complete communication custody: 24 records; all rows preserved; complete finite channel, transcript, adversary and resource records supply the observation; physical links, loopholes and device security remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-COMMUNICATION-OBSERVER, SFT-V1-V2-QUANTUM-COMMUNICATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported channel matrix, entropy formula, security theorem or stochastic message premise selects the law; host 0 denotes absence only and is not a numerical-zero rate, error or key object; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden transcript, adversary action, selected code, omitted adverse row or ontic randomness; no physical distance, rate, loophole or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:261bc46c9fca2d3204d94470e20c8cf2e03b0a8dd2abf41e347e1f05fa7648a6; independent implementation sha256:c30fb851de5c582d7b4e3facc636e98d28938d05cb24a8e02c35fe273f07705e; independent certificate
sha256:bc0228764ae5a44091121528fa3f3d0e301f641495983779600979c1ff510a45; measurement sha256:6a63f6de38744c1b614bbb59675821c9dac306dce9a5bcc7e3a3a521a74505d6; engine receipt sha256:549cdb70cffd760ae9952d5bef571058e2e583ed5e17cf5a0be52f1a608e00ce at receipts/engine/model_admitted/SFT-QUANTUM-QCOMMXX-COMPLETENESS-024-549cdb70cffd760a.json.

36.10 Family QCODEX - 32/32 complete

Scope. Quantum coding, error correction and fault tolerance. This family decides 8,192 candidates, retains 32 unique survivors and passes 128 adverse controls.

SFT-QUANTUM-OBL-QCODEX-001 - Logical information and physical carrier distinction

- **Claim and ownership:** SFT-QUANTUM-QCODEX-LOGICAL-PHYSICAL-001; Quantum Computation family QCODEX; Logical information and physical carrier distinction.
- **Forced law:** A logical information class is distinct from each physical carrier word; encoding is the registered relation between one logical label and a complete physical support class.
- **Unique surviving structure:** complete-logical-physical-code-support__logical-class-physical-word-encoding-relation__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-001 uniquely retains logical-class-physical-word-encoding-relation, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMMXX-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-001: logical_physical; exact coding/fault execution: {"carrier_distinction_retained":true,"logical_classes":2,"physical_width":3}; complete coding/fault custody: 32 records; all

rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:37e804fd0b3786db014ad3df04c11d021a2933b2f18a6926057c725c8fb36f2b; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
 sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:6848ced18cc82ef5b0f4316eba101a2840b7b1eb0cb4c69965d83580d6bb4527; engine receipt sha256:8ca0045d3a6c9bf4a3e49d5401c9425e067efd91accc4cde2ab5b35ef6ff947e at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-LOGICAL-PHYSICAL-001-8ca0045d3a6c9bf4.json.

SFT-QUANTUM-OBL-QCODEX-002 - Quantum encoder and reversible decoder

- **Claim and ownership:** SFT-QUANTUM-QCODEX-ENCODER-DECODER-002; Quantum Computation family QCODEX; Quantum encoder and reversible decoder.
- **Forced law:** A quantum encoder maps logical support reversibly into physical code support and a decoder returns the logical class while retaining syndrome, fault and discarded-carrier records.
- **Unique surviving structure:** complete-logical-physical-code-support__source-bound-reversible-encoding-decoding__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-002 uniquely retains source-bound-reversible-encoding-decoding, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-LOGICAL-PHYSICAL-001. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-002: encoder_decoder; exact coding/fault execution: {"encoded_words":2,"logical_inputs":2,"round_trip_complete":true}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred

without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:17a8ef5b5a17eee371381228a9c47bc5685312f3583101666287a009318ea225; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:019250e1fefeda5975c9e66ff0fcc1882653d248a156ea9e0ae803ff503c05e9; engine receipt sha256:8ce63850b94782800e2110a923b7f251498401cf2f6a8e82ac926aa29a228152 at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-ENCODER-DECODER-002-8ce63850b9478280.json.

SFT-QUANTUM-OBL-QCODEX-003 - Error action and error-family registration

- **Claim and ownership:** SFT-QUANTUM-QCODEX-ERROR-FAMILY-003; Quantum Computation family QCODEX; Error action and error-family registration.
- **Forced law:** An error family is frozen as a complete generated grammar of locations, label actions, phase actions, erasures, correlations and environment records before recovery outcomes are opened.
- **Unique surviving structure:** complete-logical-physical-code-support__complete-generated-error-action-grammar__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-003 uniquely retains complete-generated-error-action-grammar, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-ENCODER-DECODER-002. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-003: error_family; exact coding/fault execution: {"frozen_before_outcomes":true,"masks_through_order_one":4,"width":3}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:8c827f131f085e8dc251e5a49ba9ce4d370ad48e5481b92e384c1b3f768ab515; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:c0a834a24c5dc05128ffc6630a3aa12ddcb80efd3cc6a09d642aef409fa17a95; engine receipt sha256:821de04dfd308d93ad60a96f3764dc9fe3f320549e4d6940e366dc68716a079c at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-ERROR-FAMILY-003-821de04dfd308d93.json.

SFT-QUANTUM-OBL-QCODEX-004 - Error detection and syndrome distinction

- **Claim and ownership:** SFT-QUANTUM-QCODEX-SYNDROME-004; Quantum Computation family QCODEX; Error detection and syndrome distinction.
- **Forced law:** A syndrome is the complete exact distinction pattern between a received word and the code's generated constraint relations, without revealing or altering its logical class.
- **Unique surviving structure:** complete-logical-physical-code-support__code-constraint-comparison-syndrome__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-004 uniquely retains code-constraint-comparison-syndrome, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-ERROR-FAMILY-003. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-004: syndrome; exact coding/fault execution: {"constraint_rows":2,"distinct_rows":2,"logical_class_hidden":true}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:da6c4f44ced0265ac3fc51f3be0a4fc98081dfdc7362a8173861c101227f0e0e; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
 sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:c49866655b15f975f27e6e04c41a0c59fb77ec71ba9402853572bb8dacf9af4e; engine receipt sha256:51c83c233ac90d7f55a3986eee36eec6e40a3b19d615f90812eac18ca61c81fe at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-SYNDROME-004-51c83c233ac90d7f.json.

SFT-QUANTUM-OBL-QCODEX-005 - Correctable-error condition correspondence

- **Claim and ownership:** SFT-QUANTUM-QCODEX-CORRECTABLE-CONDITION-005; Quantum Computation family QCODEX; Correctable-error condition correspondence.
- **Forced law:** A registered error family is correctable exactly when its code-space images are disjoint or accompanied by distinct retained syndrome records that select one reversible recovery.
- **Unique surviving structure:** complete-logical-physical-code-support__distinct-error-images-or-syndrome-records__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result:

QCODEX-005 uniquely retains distinct-error-images-or-syndrome-records, complete fault custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-Completeness-024 -> SFT-QUANTUM-Coding-001 -> SFT-QUANTUM-Error-Correction-001 -> SFT-QUANTUM-Fault-Tolerance-001 -> SFT-QUANTUM-Unbounded-Finite-Fault-Tolerance-002 -> SFT-QUANTUM-QCODEX-Syndrome-004. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-005: `correctable_condition`; exact coding/fault execution: `{"distinct_syndromes":3,"recoveries":3,"single_faults":3}`; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:aeb7153ec27b87daf550e7ca22e7c74f368f2d289d0f9841c157734a37c91ffb`; independent implementation `sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75`; independent certificate
`sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465`; measurement `sha256:d7f5946ff3a6cfff68ba1b900e9889bc9c50419da93065b7b8bc4565b292698a`; engine receipt `sha256:1daa05a01c33f5fd18bcd8c745915e7b0a4847314fbb9e2eb2d59130ea1db923` at `receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-CORRECTABLE-CONDITION-005-1daa05a01c33f5fd.json`.

SFT-QUANTUM-OBL-QCODEX-006 - Bit-label error repetition code

- **Claim and ownership:** SFT-QUANTUM-QCODEX-BIT-REPETITION-006; Quantum Computation family QCODEX; Bit-label error repetition code.
- **Forced law:** The bit-label repetition code uses width $2t+1$ and recovers the logical fibre label after every generated set of at most t label faults by exact majority distinction.
- **Unique surviving structure:** `complete-logical-physical-code-support__odd-width-label-majority-code__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff`. Exact result: QCODEX-006 uniquely retains odd-width-label-majority-code, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-Completeness-024 -> SFT-QUANTUM-Coding-001 -> SFT-QUANTUM-Error-Correction-001 -> SFT-QUANTUM-Fault-Tolerance-001 -> SFT-QUANTUM-Unbounded-Finite-Fault-Tolerance-002 ->

SFT-QUANTUM-QCODEX-CORRECTABLE-CONDITION-005. Registered root:

SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** QCODEX-006: bit_repetition; exact coding/fault execution: {"all_recovered":true,"fault_order":1,"mask_rows":4,"width":3}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:c6cd711691ea9abb6e79240228abacf119725305f8302e2f2a913e5ccafe3ae3; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:0a49061dd0ec3b454583e14962b82d136032b7636e4caf0d00a279feb584a635; engine receipt sha256:2c82e56699ed69e38036bf91e65ba8a7237d39f14a3eccc12c653b4718c9dcb3 at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-BIT-REPETITION-006-2c82e56699ed69e3.json.

SFT-QUANTUM-OBL-QCODEX-007 - Phase-label error repetition code

- **Claim and ownership:** SFT-QUANTUM-QCODEX-PHASE-REPETITION-007; Quantum Computation family QCODEX; Phase-label error repetition code.
- **Forced law:** The phase-label repetition code applies the same forced $2t+1$ construction to held period-phase labels and recovers every generated set of at most t phase faults.
- **Unique surviving structure:** complete-logical-physical-code-support__odd-width-phase-majority-code__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-007 uniquely retains odd-width-phase-majority-code, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-BIT-REPETITION-006. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-007: phase_repetition; exact coding/fault execution: {"all_recovered":true,"fault_order":1,"mask_rows":4,"width":3}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:a9b82c64ecf39c4d03988a30d579548305cf3d67a2dbe1c956526576cb1c6fa9`; independent implementation `sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75`; independent certificate
`sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465`; measurement `sha256:c3456dbc4399fbd7926877a2085c9a90e0ce866f11d40201758d375f07acc797`; engine receipt `sha256:e4819c0f9bd9086c5bceb58bb0d4cbc3b0411d5de688e407f0c9cfcdb5f142d` at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-PHASE-REPETITION-007-e4819c0f9bd9086c.json.

SFT-QUANTUM-OBL-QCODEX-008 - Joint bit-phase error composition

- **Claim and ownership:** SFT-QUANTUM-QCODEX-JOINT-ERROR-008; Quantum Computation family QCODEX; Joint bit-phase error composition.
- **Forced law:** Joint error composition retains label and phase actions as separate coordinates plus their common location and causal record so neither component is hidden.
- **Unique surviving structure:** complete-logical-physical-code-support__separate-label-phase-and-joint-fault-ledger__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-008 uniquely retains separate-label-phase-and-joint-fault-ledger, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-PHASE-REPETITION-007. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-008: joint_error; exact coding/fault execution: {"joint_records":1,"label_faults":1,"phase_faults":1}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:3e04dfbcd9c3bd511b0f2a232b25f9d4bb50a64c4835852441376b17de51d2a9`; independent implementation `sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75`; independent certificate

```
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement
sha256:a8f4374294e46db25b61205b4269e8f14d60e5f844844c6611e056671c06b0f8; engine receipt
sha256:0f5399fe917d08f0c68691ac90af1c49aabffa7ee7ac58efbe048090af698a87 at receipts/
engine/model_admitted/SFT-QUANTUM-QCODEX-JOINT-ERROR-008-0f5399fe917d08f0.json.
```

SFT-QUANTUM-OBL-QCODEX-009 - Single-error exhaustive recovery

- **Claim and ownership:** SFT-QUANTUM-QCODEX-ONE-ERROR-009; Quantum Computation family QCODEX; Single-error exhaustive recovery.
- **Forced law:** Width three is the unique least odd repetition width correcting every location mask of at most one registered label or phase fault; all four masks are executed.
- **Unique surviving structure:** complete-logical-physical-code-support__width-three-all-one-fault-census__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-009 uniquely retains width-three-all-one-fault-census, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-JOINT-ERROR-008. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-009: single_error; exact coding/fault execution: {"exhaustive_masks":4,"failures":0,"fault_order":1,"width":3}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation

```
sha256:abfe5bd1a3718c0b743c3a78747a658f0f5b63adeabb74a699b53e60f0d5cf0; independent
implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75;
independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement
sha256:8f8a121a1434c5e2305598c04b8f0c02e385e655cb89cec05eccd389d76b04b2; engine receipt
sha256:e035f2b790fb4cf453501440e863a18df5c04e67a267e50029c75266f77716b8 at receipts/
engine/model_admitted/SFT-QUANTUM-QCODEX-ONE-ERROR-009-e035f2b790fb4cf4.json.
```

SFT-QUANTUM-OBL-QCODEX-010 - Two-error exhaustive recovery

- **Claim and ownership:** SFT-QUANTUM-QCODEX-TWO-ERROR-010; Quantum Computation family QCODEX; Two-error exhaustive recovery.
- **Forced law:** Width five is the unique least odd repetition width correcting every location mask of at most two registered faults; all sixteen masks are executed.

- **Unique surviving structure:** complete-logical-physical-code-support__width-five-all-two-fault-census__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-010 uniquely retains width-five-all-two-fault-census, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-ONE-ERROR-009. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-010: two_error; exact coding/fault execution: {"exhaustive_masks":16,"failures":0,"fault_order":2,"width":5}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:d41df48ee2328bb4ccd7dd5df5c70be5753b0f717754aa987feccff30ab00808; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:f90b65f46182afceaa07286ad221e677ecd199e8b3388189f631e2d767a50c5f; engine receipt sha256:3f3a6c8b07f87d0646a6cd3e6e2a59a81152ad6305c63841a41fc4af2e7b1e63 at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-TWO-ERROR-010-3f3a6c8b07f87d06.json.

SFT-QUANTUM-OBL-QCODEX-011 - Three-error exhaustive recovery

- **Claim and ownership:** SFT-QUANTUM-QCODEX-THREE-ERROR-011; Quantum Computation family QCODEX; Three-error exhaustive recovery.
- **Forced law:** Width seven is the unique least odd repetition width correcting every location mask of at most three registered faults; all sixty-four masks are executed.
- **Unique surviving structure:** complete-logical-physical-code-support__width-seven-all-three-fault-census__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-011 uniquely retains width-seven-all-three-fault-census, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-TWO-ERROR-010. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-011: three_error; exact coding/fault execution: {"exhaustive_masks":64,"failures":0,"fault_order":3,"width":7}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:7bd21ec2affc3067322582472a9274b186910e7ce4309c22409be56e2948366a; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:fed103a258251d7bdc2f6c96cb27bee64e1f18cd33272411a9b0e8e09bf86bb8; engine receipt sha256:11c4d8cbf4530006b9a0d1c79d5fc006dba818518b19c9c4516c0a85a0fcd17b at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-THREE-ERROR-011-11c4d8cbf4530006.json.

SFT-QUANTUM-OBL-QCODEX-012 - Positive-finite multi-error successor law

- **Claim and ownership:** SFT-QUANTUM-QCODEX-MULTI-ERROR-SUCCESSOR-012; Quantum Computation family QCODEX; Positive-finite multi-error successor law.
- **Forced law:** For every supplied positive finite fault order t , width $2t+1$ is uniquely least because t faults leave $t+1$ unchanged labels; induction appends two carriers and one admissible fault.
- **Unique surviving structure:** complete-logical-physical-code-support__two-t-plus-one-unbounded-finite-successor__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-012 uniquely retains two-t-plus-one-unbounded-finite-successor, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-THREE-ERROR-011. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-012: multi_error_successor; exact coding/fault execution: {"induction_depth_independent":true,"law":" $2t+1$ ","tested_orders":[1,2,3,4,5]}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:7b2c0c1364df41f84a8195466b8e4d22dea64b50765a3199d077729e50134263; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
 sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:9d4e73ec203892adbf9b92fb7534b9a2fa71ea36ec3c9ad5b3b3823f963a7b7a; engine receipt sha256:0cf0fc7fdcd1e25a550860a42df007fa1ca84f6c0ab5b561998a53a23c27d44 at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-MULTI-ERROR-SUCCESSOR-012-0cf0fc7fdcd1e25.json.

SFT-QUANTUM-OBL-QCODEX-013 - Erasure error and located recovery

- **Claim and ownership:** SFT-QUANTUM-QCODEX-ERASURE-013; Quantum Computation family QCODEX; Erasure error and located recovery.
- **Forced law:** An erasure supplies its exact missing-carrier location; recovery reconstructs that carrier from retained code constraints and preserves the erasure record.
- **Unique surviving structure:** complete-logical-physical-code-support__located-missing-carrier-reconstruction__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-013 uniquely retains located-missing-carrier-reconstruction, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-MULTI-ERROR-SUCCESSOR-012. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-013: erasure; exact coding/fault execution: {"located_position":2,"reconstructed":true,"source_width":3}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:03e61b7d3ed02c5a6adf3ae82ed9c94c53842260faeeb6582457b5241ebfe9af; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:54f052ac72a1589061365b78b6ee6d816061b1d82cbde107974bf5467211e2bf; engine receipt sha256:97dea103cb489fbbaed2e02e16cfdcae13f6fc2826c1aebc0de4637d341dd0ee at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-ERASURE-013-97dea103cb489fbb.json.

SFT-QUANTUM-OBL-QCODEX-014 - Amplitude-loss correspondence boundary

- **Claim and ownership:** SFT-QUANTUM-QCODEX-AMPLITUDE-LOSS-014; Quantum Computation family QCODEX; Amplitude-loss correspondence boundary.
- **Forced law:** Amplitude-loss correspondence records support transferred to an environment and any located loss; physical decay rates and device dynamics remain measured handoffs.
- **Unique surviving structure:** complete-logical-physical-code-support__located-support-loss-environment-handoff__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-014 uniquely retains located-support-loss-environment-handoff, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-ERASURE-013. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-014: amplitude_loss; exact coding/fault execution: {"located_loss_record":true,"physical_decay_rate_present":false}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:918a6cc7c22f438a98e072f8c40669c9bc1cf746cfb1aa4b7f93858d34f90458; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:d964101de70b50d706a2874dfaa08415ec747c18e3a142986d554e25434d9d64; engine receipt sha256:48827434f3ce4b62a448e87aa5d438e27fdbb3ae1580e0ee01d17c68702873 at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-AMPLITUDE-LOSS-014-48827434f3ce4b62.json.

SFT-QUANTUM-OBL-QCODEX-015 - Dephasing correspondence boundary

- **Claim and ownership:** SFT-QUANTUM-QCODEX-DEPHASING-015; Quantum Computation family QCODEX; Dephasing correspondence boundary.

- **Forced law:** Dephasing correspondence is a registered change or closure of relative-phase labels with an exact environment or missing-record boundary; physical rates remain downstream.
- **Unique surviving structure:** complete-logical-physical-code-support__relative-phase-fault-and-environment-record__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-015 uniquely retains relative-phase-fault-and-environment-record, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-AMPLITUDE-LOSS-014. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-015: dephasing; exact coding/fault execution: {"environment_records":1,"phase_fault_rows":1,"physical_rate_present":false}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:476c962124ef916fc0f5a17eed0cb958d78edb4f8071efca6236292831e3b7fb; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:a76af0ea8bf3d10893dfb0d28e753bdfdf8b1d21f63102b13848bb23455565995; engine receipt sha256:a3d92c7192cd0cd8abecbefaa0d4a025cf2d72e60082e00453908dc2547124c4 at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-DEPHASING-015-a3d92c7192cd0cd8.json.

SFT-QUANTUM-OBL-QCODEX-016 - Depolarizing-support correspondence boundary

- **Claim and ownership:** SFT-QUANTUM-QCODEX-DEPOLARIZING-016; Quantum Computation family QCODEX; Depolarizing-support correspondence boundary.
- **Forced law:** Depolarizing correspondence enumerates label, phase and joint error support and exact weights or counts only when supplied by the owning physical measurement.
- **Unique surviving structure:** complete-logical-physical-code-support__complete-label-phase-joint-error-support__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-016 uniquely retains complete-label-phase-joint-error-support, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault

grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-DEPHASING-015. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-016: depolarizing_support; exact coding/fault execution: {"error_classes":["label","phase","joint"],"physical_weights_present":false}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:80a94c0352c28ae99193e25c065dc384aa3c999c4cd0e6c2a96710ec790ec84d; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:568685fbda5c82fc2ca048634a14197548b6fdb609fc8fb14d900c19ffc585fe; engine receipt sha256:ba048dd0d7e964be0acbea310c29e06215ead43bfb0a51ff7b307a0be2d7c150 at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-DEPOLARIZING-016-ba048dd0d7e964be.json.

SFT-QUANTUM-OBL-QCODEX-017 - Stabilizer-code correspondence

- **Claim and ownership:** SFT-QUANTUM-QCODEX-STABILIZER-017; Quantum Computation family QCODEX; Stabilizer-code correspondence.
- **Forced law:** Stabilizer correspondence is a generated family of compatible reversible code constraints whose unchanged class defines code support and whose changed classes form syndromes.
- **Unique surviving structure:** complete-logical-physical-code-support__commuting-code-constraint-syndrome-correspondence__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-017 uniquely retains commuting-code-constraint-syndrome-correspondence, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-DEPOLARIZING-016. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-017: stabilizer_correspondence; exact coding/fault execution: {"code_syndrome":["same","same"],"constraint_rows":2,"imported_algebra":false}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the

observation; physical thresholds and device values remain downstream measurements

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:9ab654154c27bc753fd1269ac7e48698063b66de9bb0e144787d6113dc994795; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
 sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:1d8e8d02b1a17100348f29f82591cac928a64054e2f03b04424100d3aef507d4; engine receipt sha256:e57a4d8fc09f8f601080d354bb32ef3f1339e9d7b4cd091c4a388f373632f198 at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-STABILIZER-017-e57a4d8fc09f8f60.json.

SFT-QUANTUM-OBL-QCODEX-018 - CSS-code correspondence

- **Claim and ownership:** SFT-QUANTUM-QCODEX-CSS-018; Quantum Computation family QCODEX; CSS-code correspondence.
- **Forced law:** CSS correspondence separates commuting label-error and phase-error constraint families, then composes their complete syndrome and recovery records.
- **Unique surviving structure:** complete-logical-physical-code-support__separate-label-and-phase-constraint-families__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-018 uniquely retains separate-label-and-phase-constraint-families, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-Completeness-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-STABILIZER-017. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-018: css_correspondence; exact coding/fault execution: {"label_constraint_families":1,"phase_constraint_families":1,"records_composed":true}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:071392b163b34f1650686f08e30f093c5c699080ba72bcdef8e7c48d4cab8a6c; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:cd17afabb5b68b9048e6981a9b5c006083ebd16e7cb75e69ffafceeb8ec2e3d8; engine receipt sha256:67cd6ea19e6be988e13351fe43785fecb6fa354457bb877016cc2bdd910d047c at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-CSS-018-67cd6ea19e6be988.json.

SFT-QUANTUM-OBL-QCODEX-019 - Subsystem-code correspondence

- **Claim and ownership:** SFT-QUANTUM-QCODEX-SUBSYSTEM-019; Quantum Computation family QCODEX; Subsystem-code correspondence.
- **Forced law:** Subsystem-code correspondence partitions encoded support into logical, gauge and syndrome factors; gauge changes cannot alter the retained logical class.
- **Unique surviving structure:** complete-logical-physical-code-support__logical-gauge-syndrome-factor-ledger__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-019 uniquely retains logical-gauge-syndrome-factor-ledger, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-CSS-018. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-019: subsystem_correspondence; exact coding/fault execution: {"gauge_factors":1,"logical_factors":1,"syndrome_factors":1}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:cd309f80941b5e0b8d989b2e7ec1e700693e6209cd5066fb8ae1045a56067f24; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:a8d48007e27db802a675338bb39bb76c93b4aa83cf4d57e6120d1ac2b56f60c5; engine receipt sha256:81d92c24a4c93d689f835e2e8e320a604e243c6acc377b644e090d76891ec27c at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-SUBSYSTEM-019-81d92c24a4c93d68.json.

SFT-QUANTUM-OBL-QCODEX-020 - Topological-code correspondence boundary

- **Claim and ownership:** SFT-QUANTUM-QCODEX-TOPOLOGICAL-020; Quantum Computation family QCODEX; Topological-code correspondence boundary.

- **Forced law:** Topological-code correspondence derives code constraints and syndrome chains from a finite generated cell complex; physical locality and thresholds remain downstream measurements.
- **Unique surviving structure:** complete-logical-physical-code-support__cell-complex-chain-syndrome-correspondence__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-020 uniquely retains cell-complex-chain-syndrome-correspondence, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-Completeness-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-SUBSYSTEM-019. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-020: topological_boundary; exact coding/fault execution: {"cell_edges":3,"physical_locality_present":false,"syndrome_chains":1}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:0c7b6ffab679bc01a17249d19a24b4848691b6b2678442a7666e92b2a49752df; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:c7b1cb46cd87b17a30a54898a14481f36ba6d4612a82f21b359311900be4d1c5; engine receipt sha256:a5d177e8ee3e84961d62932036e533027ca8ff582971a4ca72be1d27b9ebc4f7 at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-TOPOLOGICAL-020-a5d177e8ee3e8496.json.

SFT-QUANTUM-OBL-QCODEX-021 - Surface-code correspondence boundary

- **Claim and ownership:** SFT-QUANTUM-QCODEX-SURFACE-021; Quantum Computation family QCODEX; Surface-code correspondence boundary.
- **Forced law:** Surface-code correspondence is a finite surface cellulation with generated vertex/face constraints, syndrome endpoints and correction chains; no physical threshold is imported.
- **Unique surviving structure:** complete-logical-physical-code-support__finite-surface-cell-check-and-chain-ledger__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-021 uniquely retains finite-surface-cell-check-and-chain-ledger, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault

grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-TOPOLOGICAL-020. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-021: surface_boundary; exact coding/fault execution: {"edges":4,"faces":1,"physical_threshold_present":false,"vertices":4}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:e4be59890f831da0de67774924bd070ae899b51b4ead575d796154ddf974d4ea; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:aa01c2f876db6746c30704faaf04529f579c831fbba6c2bda8c0bd3a939f01b2; engine receipt sha256:6cb978aae152056c782701e1f9b375c640f819a73fc1b8b767933a6b2040c8b7 at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-SURFACE-021-6cb978aae152056c.json.

SFT-QUANTUM-OBL-QCODEX-022 - Concatenated-code composition

- **Claim and ownership:** SFT-QUANTUM-QCODEX-CONCATENATION-022; Quantum Computation family QCODEX; Concatenated-code composition.
- **Forced law:** Concatenation replaces every outer physical carrier with a complete inner codeword and composes their encoders, syndromes, decoders and resource ledgers.
- **Unique surviving structure:** complete-logical-physical-code-support__outer-inner-codeword-s ubstitution__complete-registered-fault-family__complete-syndrome-environment-and-r ecovery-record__literal-complete-product__there-is-no-nothing-lineage__post-regist ry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-022 uniquely retains outer-inner-codeword-substitution, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-SURFACE-021. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-022: concatenation; exact coding/fault execution: {"inner_width":3,"outer_width":3,"physical_carriers":9}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:60139faebd1d13677ff39004d9404634263a3f0ad829db8977e48520bd3e2137; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:5476fe8a970c28ed37615d4992e7301b82d5f6c5c7eae3b7c7ce0957d91dd9a5; engine receipt sha256:564c2c3e139937f1de6add70e781a91091a37a14c7884604e205d41df577d172 at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-CONCATENATION-022-564c2c3e139937f1.json.

SFT-QUANTUM-OBL-QCODEX-023 - Logical gate and encoded transformation

- **Claim and ownership:** SFT-QUANTUM-QCODEX-LOGICAL-GATE-023; Quantum Computation family QCODEX; Logical gate and encoded transformation.
- **Forced law:** A logical gate is a physical-support permutation that maps every codeword to the codeword of the transformed logical class and retains all syndrome action.
- **Unique surviving structure:** complete-logical-physical-code-support__code-space-preserving-physical-transformation__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-023 uniquely retains code-space-preserving-physical-transformation, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-CONCATENATION-022. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-023: logical_gate; exact coding/fault execution: {"code_space_preserved":true,"encoded_images":2,"logical_classes":2}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:6ebaa8b9d514fba916e6baf78baf67c58cea675c84a54279cca90188068f9c5d; independent

implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75;
 independent certificate
 sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement
 sha256:b6a454c6ce315e444f76ba2a41025aa6a693cb6de975e8007ff25f161071fd9e; engine receipt
 sha256:7a209b712ac6cb872b7b0e76b39f2fd70bed7e992b097d626b3bca6ccbe9c437 at receipts/
 engine/model_admitted/SFT-QUANTUM-QCODEX-LOGICAL-GATE-023-7a209b712ac6cb87.json.

SFT-QUANTUM-OBL-QCODEX-024 - Transversal-gate containment

- **Claim and ownership:** SFT-QUANTUM-QCODEX-TRANSVERSAL-024; Quantum Computation family QCODEX; Transversal-gate containment.
- **Forced law:** A transversal gate acts carrierwise across code blocks so one physical fault cannot spread to multiple carriers in the same block under the registered fault grammar.
- **Unique surviving structure:** complete-logical-physical-code-support__carrierwise-gate-within-block-fault-containment__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-024 uniquely retains carrierwise-gate-within-block-fault-containment, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-LOGICAL-GATE-023. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-024: transversal_containment; exact coding/fault execution: {"carrierwise_actions":3,"contained":true,"within_block_spread":1}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:af840a44499ce3acdf083fe467b84e62ea5d80c6c68b16c5540b6163ddc8d2c3; independent
 implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75;
 independent certificate
 sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement
 sha256:022f39f654c3b66b6aaedd48b7c217b004847ba500cbfb4b43d6024876900388; engine receipt
 sha256:756cb6fe6ccd6d93519f6bad52ca7ec393678323fd80c587eee4259956b0d61b at receipts/
 engine/model_admitted/SFT-QUANTUM-QCODEX-TRANSVERSAL-024-756cb6fe6ccd6d93.json.

SFT-QUANTUM-OBL-QCODEX-025 - Syndrome-extraction fault custody

- **Claim and ownership:** SFT-QUANTUM-QCODEX-SYNDROME-FAULT-025; Quantum Computation family QCODEX; Syndrome-extraction fault custody.

- **Forced law:** Syndrome extraction retains every data/ancilla interaction, observation and extraction fault so a faulty syndrome cannot silently select recovery.
- **Unique surviving structure:** complete-logical-physical-code-support__data-ancilla-measurement-fault-transcript__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-025 uniquely retains data-ancilla-measurement-fault-transcript, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-TRANSVERSAL-024. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-025: syndrome_extraction_fault; exact coding/fault execution: {"data_rows":1,"extraction_fault_rows":1,"syndrome_rows":1}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:0323d1f2e317381477b42cccaad280e72ba9eeaca77cd960f282c7215365ff30; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:e30cd32eeb18052d61eac6af3ad33e01edcc8b5b7a03cbdfd9e36718344c1062; engine receipt sha256:38c39f21129fe64fd1f24a01f6668e89a4453c94cb5d2b333faf7617519930cb at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-SYNDROME-FAULT-025-38c39f21129fe64f.json.

SFT-QUANTUM-OBL-QCODEX-026 - Fault-tolerant location composition

- **Claim and ownership:** SFT-QUANTUM-QCODEX-FAULT-LOCATION-026; Quantum Computation family QCODEX; Fault-tolerant location composition.
- **Forced law:** Fault-tolerant composition partitions a process into registered locations and proves that every admissible location-fault set remains correctable after causal composition.
- **Unique surviving structure:** complete-logical-physical-code-support__location-fault-order-composable-containment__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-026 uniquely retains location-fault-order-composable-containment, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault

grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-SYNDROME-FAULT-025. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-026: fault_tolerant_locations; exact coding/fault execution: {"composable_containment":true,"location_orders":[1,2,3]}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:1e0c7633f9a5ee201c384011a0a955cba79bc7de4446c5e6b7ff7cc4085ece06; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:321250bfe8c89259e2e4f1346acb6157b508315398d0dd81086e4b59db7000dc; engine receipt sha256:c6e3f3fd0f7a2f8de9e95ff7cbb32213e0830bda9c25be614f3d6414f17c7481 at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-FAULT-LOCATION-026-c6e3f3fd0f7a2f8d.json.

SFT-QUANTUM-OBL-QCODEX-027 - Malignant-fault-set boundary

- **Claim and ownership:** SFT-QUANTUM-QCODEX-MALIGNANT-SET-027; Quantum Computation family QCODEX; Malignant-fault-set boundary.
- **Forced law:** A malignant set is a generated location subset whose exact execution causes logical failure; every subset through the declared order must be retained, including benign and adverse rows.
- **Unique surviving structure:** complete-logical-physical-code-support__complete-fault-subset-logical-failure-census__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-027 uniquely retains complete-fault-subset-logical-failure-census, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-FAULT-LOCATION-026. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-027: malignant_fault_sets; exact coding/fault execution: {"all_retained":true,"benign_rows":1,"fault_subsets":2,"logical_failure_rows":1}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the

observation; physical thresholds and device values remain downstream measurements

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:366f8c0c59a6220499676fc090cb0a4062ec5f938b4d3b64b625b61dee56a8c2; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
 sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:35034678922f2cc3046060db0a8c114fa145f8653b120a5ac5306b029fb761f2; engine receipt sha256:1d8459ef5bc47efe43799e28591cb59060d0a05a99af9a0d0f50a71349f683ea at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-MALIGNANT-SET-027-1d8459ef5bc47efe.json.

SFT-QUANTUM-OBL-QCODEX-028 - Correlated and nonlocal fault boundary

- **Claim and ownership:** SFT-QUANTUM-QCODEX-CORRELATED-FAULT-028; Quantum Computation family QCODEX; Correlated and nonlocal fault boundary.
- **Forced law:** Correlated and nonlocal faults require a separately registered joint-cause grammar; independent-fault conclusions cannot be extended to them without a new exhaustive census.
- **Unique surviving structure:** complete-logical-physical-code-support__joint-cause-multilocation-error-grammar__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-028 uniquely retains joint-cause-multilocation-error-grammar, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-MALIGNANT-SET-027. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-028: correlated_fault; exact coding/fault execution: {"fault_locations":2,"independence_assumed":false,"joint_cause_records":1}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:7ac0e1bd64bbd015efcf693966246ee29b18858a3e287962e7c30f435657a385; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:a38bee914fc5f2acac43dd713b144dcbfb5b9856a07e1d4e613296e40c3cd994; engine receipt sha256:654b4c4a3bf993eda8b2fe3635e84644bb1c57d999542407d591a48fc05337fa at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-CORRELATED-FAULT-028-654b4c4a3bf993ed.json.

SFT-QUANTUM-OBL-QCODEX-029 - Leakage and loss fault boundary

- **Claim and ownership:** SFT-QUANTUM-QCODEX-LEAKAGE-LOSS-029; Quantum Computation family QCODEX; Leakage and loss fault boundary.
- **Forced law:** Leakage and loss are explicit departures from declared code support, with location, environment and return-or-replacement records; ordinary in-support correction does not silently cover them.
- **Unique surviving structure:** complete-logical-physical-code-support__outside-code-support-and-located-loss-ledger__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-029 uniquely retains outside-code-support-and-located-loss-ledger, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-CORRELATED-FAULT-028. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-029: leakage_loss; exact coding/fault execution: {"code_support_classes":2,"located_loss_classes":1,"outside_support_classes":1}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:b413d5ce16b9c55fd9db42340c0076d4721e6a99dd9c4c0177037b805272cd74; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:5bb4bbfa2ff01ac26b98c073b3779837d7acfd8b8f842d13a026fc88f8263e3; engine receipt sha256:2f3608afbc751872ca474a3f9d663417aba27959b2a82cd691123b69fbc4e56 at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-LEAKAGE-LOSS-029-2f3608afbc751872.json.

SFT-QUANTUM-OBL-QCODEX-030 - State and magic-resource distillation correspondence

- **Claim and ownership:** SFT-QUANTUM-QCODEX-DISTILLATION-030; Quantum Computation family QCODEX; State and magic-resource distillation correspondence.
- **Forced law:** Distillation correspondence consumes a generated finite resource-support, applies exact checks and retains accepted, rejected and output records; physical fidelity remains measured.
- **Unique surviving structure:** complete-logical-physical-code-support__multi-input-check-select-output-transcript__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff. Exact result: QCODEX-030 uniquely retains multi-input-check-select-output-transcript, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-X-COMPLETENESS-024 -> SFT-QUANTUM-CODING-001 -> SFT-QUANTUM-ERROR-CORRECTION-001 -> SFT-QUANTUM-FAULT-TOLERANCE-001 -> SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002 -> SFT-QUANTUM-QCODEX-LEAKAGE-LOSS-029. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-030: distillation; exact coding/fault execution: {"accepted":1,"input_resources":3,"physical_fidelity_present":false,"rejected":2}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:cb30631ccbfc6df6cdc41d4f802bd9098ddcd9eb88b8ca5acb9171fd2774794f; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:82d4e9804721a772dbbe0d855485c087c171025ec876918405593b1404ebe676; engine receipt sha256:3b3e32f94925d00e48b57c7d04bba7aefe09bbe5d9b530d69b2cfb4f8d693988 at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-DISTILLATION-030-3b3e32f94925d00e.json.

SFT-QUANTUM-OBL-QCODEX-031 - Physical threshold constant measurement handoff

- **Claim and ownership:** SFT-QUANTUM-QCODEX-PHYSICAL-THRESHOLD-031; Quantum Computation family QCODEX; Physical threshold constant measurement handoff.
- **Forced law:** The formal $2t+1$ successor theorem is not a physical hardware threshold; a threshold constant requires measured device error processes, correlations, leakage, geometry and decoding costs in Physics or Engineering.
- **Unique surviving structure:** complete-logical-physical-code-support__formal-fault-census-to-measured-threshold-handoff__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff.

Exact result: QCODEX-031 uniquely retains formal-fault-census-to-measured-threshold-handoff, complete fault custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-Completeness-024 -> SFT-QUANTUM-Coding-001 -> SFT-QUANTUM-Error-Correction-001 -> SFT-QUANTUM-Fault-Tolerance-001 -> SFT-QUANTUM-Unbounded-Finite-Fault-Tolerance-002 -> SFT-QUANTUM-QCODEX-Distillation-030. Registered root: SFT-ROOT-THERE-IS-NO-Nothing. Imported axioms: 0; free parameters: 0.
- **Observation:** QCODEX-031: `threshold_handoff`; exact coding/fault execution: `{"formal_fault_grammar_complete":true,"owner":"physics-engineering-measurement","physical_threshold_value_present":false}`; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-Coding-OBSERVER, SFT-V1-V2-QUANTUM-Coding-Fault-Observation-CORPUS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:47c9a1974c97ec2424c7a4e8ad216f62394555d6e7a4d90e53359485b84d2592`; independent implementation `sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75`; independent certificate
`sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465`; measurement `sha256:df07c3f25f764b9ea01e406cc9d608bdacaa90877271ac84fd4a2b05b37fcaa5`; engine receipt `sha256:2b894a90da74f8447dd997e8013d68513a7ca520fd471ab3fb34cb904b27cb01` at `receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-PHYSICAL-THRESHOLD-031-2b894a90da74f844.json`.

SFT-QUANTUM-OBL-QCODEX-032 - Quantum coding and fault-tolerance completeness certificate

- **Claim and ownership:** SFT-QUANTUM-QCODEX-Completeness-032; Quantum Computation family QCODEX; Quantum coding and fault-tolerance completeness certificate.
- **Forced law:** The quantum coding family is complete exactly when all thirty-two frozen obligations have one owner, one unique survivor, controls, observations, independent reconstructions and current receipts.
- **Unique surviving structure:** `complete-logical-physical-code-support__thirty-two-obligation-no-omission-ledger__complete-registered-fault-family__complete-syndrome-environment-and-recovery-record__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-physical-threshold-handoff`. Exact result: QCODEX-032 uniquely retains thirty-two-obligation-no-omission-ledger, complete fault custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete code/fault grammar, exhaustive recovery execution, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCOMM-Completeness-024 -> SFT-QUANTUM-Coding-001 -> SFT-QUANTUM-Error-Correction-001 -> SFT-QUANTUM-Fault-Tolerance-001 -> SFT-QUANTUM-Unbounded-Finite-Fault-Tolerance-002 ->

SFT-QUANTUM-QCODEX-PHYSICAL-THRESHOLD-031. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** QCODEX-032: coding_completeness; exact coding/fault execution: {"duplicate_owners":0,"execution_rows":32,"omitted_owners":0,"registered_obligations":32}; complete coding/fault custody: 32 records; all rows preserved; complete finite code, error-mask, syndrome, recovery and adverse-fault records supply the observation; physical thresholds and device values remain downstream measurements
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-QUANTUM-CODING-OBSERVER, SFT-V1-V2-QUANTUM-CODING-FAULT-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported code, stabilizer algebra, channel model, threshold or fault-rate premise selects the law; host 0 denotes absence only and is not a numerical-zero carrier, error or rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled fault masks, omitted malignant sets, hidden correlations or target-selected recovery; no physical error rate, threshold, fidelity, geometry or device value is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:a373c98950c6d99bf30407b91725b65417e424cd6ffa57df7e74eade7092ff7b; independent implementation sha256:4b4f6a419a50381ca76e4c74e00ed9469233cb20edfcd106a68151febb4edc75; independent certificate
sha256:1690adf45bfc5cb4e8b3c9912ab451e317a49c39e09d85eecd47d891a0666465; measurement sha256:7950b2ebcf25f539c841800319a41fc3dc198b3fdf7c277948b4a09c2bd23ac4; engine receipt sha256:9b130b5c4fd4046768d3a237915c0ce918376574e692cb6f5743aaf49836bb0c at receipts/engine/model_admitted/SFT-QUANTUM-QCODEX-COMPLETENESS-032-9b130b5c4fd40467.json.

36.11 Family QSIMX - 24/24 complete

Scope. Quantum simulation and verification. This family decides 6,144 candidates, retains 24 unique survivors and passes 96 adverse controls.

SFT-QUANTUM-OBL-QSIMX-001 - Quantum model and simulator identity

- **Claim and ownership:** SFT-QUANTUM-QSIMX-MODEL-SIMULATOR-IDENTITY-001; Quantum Computation family QSIMX; Quantum model and simulator identity.
- **Forced law:** A quantum model is its complete generated support and update relation; a simulator is identical to that model only when every source, image, phase, observation and resource row corresponds in both directions.
- **Unique surviving structure:** complete-source-bound-Fold-model__source-bound-model-simulator-bisimulation__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-001 uniquely retains source-bound-model-simulator-bisimulation, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-001: model_simulator_identity; exact simulation/verification execution: {"forward_reverse_correspondence":true,"model_rows":4,"simulator_rows":4}; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs

- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:e8590b490d83c9aea3803c056d5413ce54a61ad9704005976a40611559a2e114; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate
sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:005802cc9b2556a4fe947fff4de169a6a4e1737ba81ade7438ce097716e4b3ce; engine receipt sha256:3da80c2fdf954b7f41a48d6ddfe329469d9eb131eb1d2f10fa90996509dcf89e at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-MODEL-SIMULATOR-IDENTITY-001-3da80c2fdf954b7f.json.

SFT-QUANTUM-OBL-QSIMX-002 - Finite target support encoding

- **Claim and ownership:** SFT-QUANTUM-QSIMX-TARGET-SUPPORT-ENCODING-002; Quantum Computation family QSIMX; Finite target support encoding.
- **Forced law:** Finite target support encoding assigns every generated target distinction one canonical Fold word and preserves the inverse decoding relation without omitted or duplicated support.
- **Unique surviving structure:** complete-source-bound-Fold-model__complete-finite-target-word-encoding__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-002 uniquely retains complete-finite-target-word-encoding, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-MODEL-SIMULATOR-IDENTITY-001. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-002: finite_target_support; exact simulation/verification execution: {"duplicate_words":0,"generated_words":8,"word_width":3}; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy,

probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:d776ebf74e8f9acb999a35a2c3bd76039a291038060e4a63b80e893035752598; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:148759ad3744d1fb280346caa4e947f2f1ae7951154fa46dc5d25f06378810ef; engine receipt sha256:9b40db1a1d66ddffef51c2d4a8ed1bcbe4126c44ff57e621958e4cccc72d381 at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-TARGET-SUPPORT-ENCODING-002-9b40db1a1d66ddff.json.

SFT-QUANTUM-OBL-QSIMX-003 - Digital quantum simulation

- **Claim and ownership:** SFT-QUANTUM-QSIMX-DIGITAL-SIMULATION-003; Quantum Computation family QSIMX; Digital quantum simulation.
- **Forced law:** Digital quantum simulation is a finite source-bound sequence of reversible local transformations with complete phase, interference, observation and resource traces.
- **Unique surviving structure:** complete-source-bound-Fold-model__finite-reversible-update-sequence-simulation__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-003 uniquely retains finite-reversible-update-sequence-simulation, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-TARGET-SUPPORT-ENCODING-002. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-003: digital_simulation; exact simulation/verification execution: {"image":["returned","held"],"schedule":[1,2],"source":["held","returned"],"trace_rows":2}; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:ba4b581a46c394899b5f5dd6a0dbcfea8a1a5f61f7c13f22e29ac6c3e99c0425; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f;

independent certificate

sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement
sha256:ed7f2db079c79edb5a3f25d8650ceff485c6d8a4dcfcd11716dbef9cd130bce; engine receipt
sha256:c839b40be7727d9073a5f3d67fb1df6b613d1e1a6cdc78454b089417277cd362 at receipts/
engine/model_admitted/SFT-QUANTUM-QSIMX-DIGITAL-SIMULATION-003-c839b40be7727d90.json.

SFT-QUANTUM-OBL-QSIMX-004 - Analog quantum simulation correspondence boundary

- **Claim and ownership:** SFT-QUANTUM-QSIMX-ANALOG-BOUNDARY-004; Quantum Computation family QSIMX; Analog quantum simulation correspondence boundary.
- **Forced law:** Analog-simulation correspondence identifies a formal Fold support/update relation with a physical system only through separately measured preparation, dynamics, observables, scale and uncertainty.
- **Unique surviving structure:** complete-source-bound-Fold-model__formal-support-to-physical-dynamics-handoff__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-004 uniquely retains formal-support-to-physical-dynamics-handoff, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-DIGITAL-SIMULATION-003. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-004: analog_boundary; exact simulation/verification execution: {"formal_support_complete":true,"measurement_handoff_required":true,"physical_mapping_present":false}; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:d7aacbdf39feba004a6f4b921bcb6ad085ccce1d2533fa81f2ebc28ff9c71995; independent
implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f;
independent certificate
sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement
sha256:4417b7f64678d8292048d0c3bf3bfa8d465872ad1f7b268fb146bf99f296c84b; engine receipt
sha256:feb6969efd7e3b075ccbf68941eab9eef5ccc15f2bca51d5107892c481b6e283 at receipts/
engine/model_admitted/SFT-QUANTUM-QSIMX-ANALOG-BOUNDARY-004-feb6969efd7e3b07.json.

SFT-QUANTUM-OBL-QSIMX-005 - Local interaction and update composition

- **Claim and ownership:** SFT-QUANTUM-QSIMX-LOCAL-UPDATE-005; Quantum Computation family QSIMX; Local interaction and update composition.
- **Forced law:** Local interactions compose by their exact support overlap and causal order; disjoint updates commute while overlapping updates retain their order and predecessor records.
- **Unique surviving structure:** complete-source-bound-Fold-model__causally-ordered-local-update-e-composition__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-005 uniquely retains causally-ordered-local-update-composition, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-ANALOG-BOUNDARY-004. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-005: local_update_composition; exact simulation/verification execution: {"causal_trace_retained":true,"local_updates":2,"support_places":2}; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:fd056c549a23bd3e3b4f508bc8bb52f96c1536e9cfd09bec4c224f122c15e61; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate
 sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:16161edf4783e1e5dfabc5754624461dd118d2deaf112714214cca58982d1ac0; engine receipt sha256:d61405660a791dd4190b44c3be267d5c370e1c9850b2ec093c7ca864cb447b7e at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-LOCAL-UPDATE-005-d61405660a791dd4.json.

SFT-QUANTUM-OBL-QSIMX-006 - Hamiltonian correspondence without imported continuum proof values

- **Claim and ownership:** SFT-QUANTUM-QSIMX-HAMILTONIAN-CORRESPONDENCE-006; Quantum Computation family QSIMX; Hamiltonian correspondence without imported continuum proof values.
- **Forced law:** Hamiltonian correspondence is the downstream name for a complete finite reversible generator relation and its phase/action ledger; no continuum operator, negative energy or imaginary proof scalar is imported.
- **Unique surviving structure:** complete-source-bound-Fold-model__finite-reversible-generator-relation__complete-update-phase-observation-trace__complete-support-depth-round-and

d-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-006 uniquely retains finite-reversible-generator-relation, complete trace custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-LOCAL-UPDATE-005. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-006: hamiltonian_correspondence; exact simulation/verification execution: {"generator_rows":4,"imaginary_proof_scalar":false,"imported_continuum_operator":false,"inverse_rows":4}; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:64b98ac830c28fb68271d81cf7c41ec1cd791aa43a9f8c6c20cc2df854a38713; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate
sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:629825a3a6900df26b8d86b3c334ca3cfce8b11a549a42344a34161c136e3a1b; engine receipt sha256:28a3e3e662b3773c4a140c1b84da25906492432d0b5d7e2eb4aecf29c151120d at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-HAMILTONIAN-CORRESPONDENCE-006-28a3e3e662b3773c.json.

SFT-QUANTUM-OBL-QSIMX-007 - Time-evolution approximation and enclosure custody

- **Claim and ownership:** SFT-QUANTUM-QSIMX-TIME-EVOLUTION-ENCLOSURE-007; Quantum Computation family QSIMX; Time-evolution approximation and enclosure custody.
- **Forced law:** A time-evolution approximation is a registered finite refinement whose exact trace and enclosure relation are retained; convergence is claimed only under a separately proved positive-finite successor certificate.
- **Unique surviving structure:** complete-source-bound-Fold-model__nested-finite-update-enclosure-ledger__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-007 uniquely retains nested-finite-update-enclosure-ledger, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record,

control or independent reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-HAMILTONIAN-CORRESPONDENCE-006. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-007: evolution_enclosure; exact simulation/verification execution: {"complete_trace_each_depth":true,"finite_refinement_depths":[1,2,3],"unregistered_limit_claimed":false}; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:570035bb597fa6e2641bd2978e0da7199e3f50473efea6c4d3d94d5d9c95edc1; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:906b939799562254f36f83d33a4b6502d972bfd4782ad95b39f7417fb015cb65; engine receipt sha256:a7816fb5812f8ed1da1a7cc70f6143809f3746c6e975dc0acbab31822e71c9d8 at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-TIME-EVOLUTION-ENCLOSURE-007-a7816fb5812f8ed1.json.

SFT-QUANTUM-OBL-QSIMX-008 - Many-body support organization

- **Claim and ownership:** SFT-QUANTUM-QSIMX-MANY-BODY-SUPPORT-008; Quantum Computation family QSIMX; Many-body support organization.
- **Forced law:** Many-body support is the complete joint Fold-word product with every body/place incidence, interaction edge, phase relation and observation partition explicitly retained.
- **Unique surviving structure:** complete-source-bound-Fold-model__complete-joint-word-incidence-support__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-008 uniquely retains complete-joint-word-incidence-support, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-TIME-EVOLUTION-ENCLOSURE-007. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-008: many_body_support; exact simulation/verification execution: {"body_places":3,"incidence_rows_per_word":3,"joint_words":8}; complete simulation/verification custody: 24 records;

all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs

- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:ee60677b565cffd3ed11010a7ba4aef19ea83dbbeeb4a8e506aa2563fe5ff53f; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate
 sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:5d78626c907609bf1c57c535449c664ab93d04de3de03904578835bd29fa5eb5; engine receipt sha256:01ea3b4501528d0fc6145f2497fa68a65a64cdd8d16a208dalacd9402cb4a063 at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-MANY-BODY-SUPPORT-008-01ea3b4501528d0f.json.

SFT-QUANTUM-OBL-QSIMX-009 - Fermionic and bosonic encoding correspondence

- **Claim and ownership:** SFT-QUANTUM-QSIMX-FERMION-BOSON-ENCODING-009; Quantum Computation family QSIMX; Fermionic and bosonic encoding correspondence.
- **Forced law:** Particle-statistics correspondence is encoded by exact exchange classes, occupancy distinctions and retained period-phase labels; conventional fermionic and bosonic names enter only after this structure is derived.
- **Unique surviving structure:** complete-source-bound-Fold-model__exchange-class-and-phase-record-encoding__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-009 uniquely retains exchange-class-and-phase-record-encoding, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-MANY-BODY-SUPPORT-008. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-009: fermion_boson_correspondence; exact simulation/verification execution: {"conventional_statistics_imported":false,"distinct_exchange_phase":"phase-returned","same-label_exchange_phase":"phase-held"}; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:ce72ea0dc2fc676f16ba24623b16a1554d243ee497316962985a3f3d88910f3c; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:7180cb0a69bbdc6c84330e3d6102a780804b69631871bcd64bb3c280eb4aeb6a; engine receipt sha256:bc581e6f325eeaf7407cf3f67d7be08f872c86ecb70dbce74c88edbfcc9a7cc at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-FERMION-BOSON-ENCODING-009-bc581e6f325eeaf7.json.`

SFT-QUANTUM-OBL-QSIMX-010 - Lattice and field discretization handoff

- **Claim and ownership:** SFT-QUANTUM-QSIMX-LATTICE-FIELD-HANDOFF-010; Quantum Computation family QSIMX; Lattice and field discretization handoff.
- **Forced law:** A lattice/field discretization is a finite generated cell-incidence support and update ledger; physical spacing, field scale, boundary data and continuum interpretation remain owning-domain measurements.
- **Unique surviving structure:** `complete-source-bound-Fold-model__finite-cell-incidence-to-physical-field-handoff__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff.` Exact result: QSIMX-010 uniquely retains finite-cell-incidence-to-physical-field-handoff, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-FERMION-BOSON-ENCODING-009. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-010: `lattice_field_handoff`; exact simulation/verification execution: `{"incidence_complete":true,"links":2,"physical_spacing_present":false,"sites":3}`; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:94ca7fbbe59abf14a733f4cea0c7df9313a0a5c2395c153bf12343b1fbd26d77; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate
sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:6eead11375e42d1c88dab730f5a095c0f50ea3ec627d0e7d422904453f730608; engine receipt sha256:bbd24be374b9b9a9b27c39210944aa85fecc73631dd8f784fd95958acddcadd at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-LATTICE-FIELD-HANDOFF-010-bbd24be374b9b9a9.json.

SFT-QUANTUM-OBL-QSIMX-011 - Open-system simulation and environment record

- **Claim and ownership:** SFT-QUANTUM-QSIMX-OPEN-SYSTEM-011; Quantum Computation family QSIMX; Open-system simulation and environment record.
- **Forced law:** Open-system simulation embeds each apparent system-only loss in a reversible joint system/environment update and retains the environment record required for reconstruction.
- **Unique surviving structure:** complete-source-bound-Fold-model__system-environment-reversible-extension__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-011 uniquely retains system-environment-reversible-extension, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-LATTICE-FIELD-HANDOFF-010. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-011: open_system_record; exact simulation/verification execution: {"cause_records":1,"environment_rows":1,"joint_reconstruction":true,"system_rows":1}; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:be395daf81f1bbb5c19535a51f65daf9cdedd33268ac0971413f06d77c0abf7e; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate
sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:2848a20ad745a94fdc9f33e0f3f691d0265d62d1eca876deb6dc59dab9a09eb9; engine receipt sha256:9cb61672aa4c64f08de9a372c44c21dae38e644815f642a4130e1b9d1dc60b32 at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-OPEN-SYSTEM-011-9cb61672aa4c64f0.json.

SFT-QUANTUM-OBL-QSIMX-012 - Noise simulation and source custody

- **Claim and ownership:** SFT-QUANTUM-QSIMX-NOISE-CUSTODY-012; Quantum Computation family QSIMX; Noise simulation and source custody.
- **Forced law:** Noise simulation enumerates the complete registered source/image/cause grammar and never replaces deterministic support by an ontic random premise or an unregistered sampled error.
- **Unique surviving structure:** complete-source-bound-Fold-model__registered-source-image-cause-noise-ledger__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-012 uniquely retains registered-source-image-cause-noise-ledger, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-OPEN-SYSTEM-011. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-012: noise_source_custody; exact simulation/verification execution: {"cause_rows":1,"image_rows":1,"ontic_randomness_imported":false,"source_rows":1}; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:606ed468c56544668eae4b02c0dde950e90bac75f4410a4796a636a5ebba67f7; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:ecfb86b73ae85a03c9879dc17343a9a06dda9a6857970e6cefb62811727bffbcb; engine receipt sha256:3726582b1fc5fb2242ddf579f5551098c55c7564aca774c2bab5b2b26241932b at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-NOISE-CUSTODY-012-3726582b1fc5fb22.json.

SFT-QUANTUM-OBL-QSIMX-013 - Quantum chemistry simulation handoff

- **Claim and ownership:** SFT-QUANTUM-QSIMX-CHEMISTRY-HANDOFF-013; Quantum Computation family QSIMX; Quantum chemistry simulation handoff.
- **Forced law:** Quantum chemistry simulation consumes a sealed formal simulator and hands molecular identity, geometry, scales, observables and uncertainty to Chemistry; chemistry data cannot select the quantum-computational law.
- **Unique surviving structure:** complete-source-bound-Fold-model__formal-simulator-to-chemistry-measurement-interface__complete-update-phase-observation-trace__complete-support

-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-line age__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-013 uniquely retains formal-simulator-to-chemistry-measurement-interface, complete trace custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-NOISE-CUSTODY-012. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-013: chemistry_handoff; exact simulation/verification execution: {"chemistry_target_owner": "chemistry", "formal_simulator_sealed": true, "target_value_present": false}; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:38c39767321d35f2b7111f3d27b0c8bf546e3cafa1a1dedc099493ace0cca9d5; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:b5355ee6ec3cba4274f362308c984a497747ce21b6f77254f93eb4e6856ac218; engine receipt sha256:e575b052b5482a607eba363a0633f57293ddc89d03299470fc904590dbd9208c at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-CHEMISTRY-HANDOFF-013-e575b052b5482a60.json.

SFT-QUANTUM-OBL-QSIMX-014 - Materials simulation handoff

- **Claim and ownership:** SFT-QUANTUM-QSIMX-MATERIALS-HANDOFF-014; Quantum Computation family QSIMX; Materials simulation handoff.
- **Forced law:** Materials simulation consumes a sealed formal simulator and hands composition, structure, boundary conditions, observables and uncertainty to Materials Science without importing them into the formal law.
- **Unique surviving structure:** complete-source-bound-Fold-model__formal-simulator-to-material-s-measurement-interface__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-line age__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-014 uniquely retains formal-simulator-to-materials-measurement-interface, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record,

control or independent reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-CHEMISTRY-HANDOFF-013. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-014: materials_handoff; exact simulation/verification execution: {"formal_simulator_sealed":true,"materials_target_owner":"materials","target_value_present":false}; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:0c4f0f6be05ae67f455544616965b44966292932644ad7c062b148e903e0d4ec; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:411bc91221adf6b9a83fb1c3ec2a271707915517acf056e023c9f55618399a26; engine receipt sha256:3d0fc0ab83e09458a9c9c95cbe7fec4b07457bd8b96ee0e1be3a1d4617206f5a at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-MATERIALS-HANDOFF-014-3d0fc0ab83e09458.json.

SFT-QUANTUM-OBL-QSIMX-015 - Verification of a quantum computation

- **Claim and ownership:** SFT-QUANTUM-QSIMX-COMPUTATION-VERIFICATION-015; Quantum Computation family QSIMX; Verification of a quantum computation.
- **Forced law:** Quantum computation verification replays the registered source, transformation schedule, branch/phase trace and observation record and accepts exactly when the claimed image equals the independently reconstructed image.
- **Unique surviving structure:** complete-source-bound-Fold-model__source-schedule-trace-image-verification__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-015 uniquely retains source-schedule-trace-image-verification, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-MATERIALS-HANDOFF-014. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-015: computation_verification; exact simulation/verification execution: {"claimed_image_verified":true,"schedule_rows":2,"source_rows":1,"trace_rows":2}; complete simulation/verification

custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs

- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:ea10b54046bb2b158df3de143b4a06ce29287c58274b1d43eef38b709771af1d; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate
sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:2f0344889516ae3f23d4aa403b7c710041926a68e1123236f1d5e0b572f2ba4d; engine receipt sha256:99e6893148d1365275022f7600678f0bfff99d43c6bb9a5239ec634b377a361c7 at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-COMPUTATION-VERIFICATION-015-99e6893148d13652.json.

SFT-QUANTUM-OBL-QSIMX-016 - Interactive quantum verification

- **Claim and ownership:** SFT-QUANTUM-QSIMX-INTERACTIVE-VERIFICATION-016; Quantum Computation family QSIMX; Interactive quantum verification.
- **Forced law:** Interactive verification is a finite causal challenge/response process whose complete round support, verifier choices, prover responses, observations, acceptance rule and resource bounds are registered.
- **Unique surviving structure:** complete-source-bound-Fold-model__challenge-response-round-transcript-verification__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-016 uniquely retains challenge-response-round-transcript-verification, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-COMPUTATION-VERIFICATION-015. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-016: interactive_verification; exact simulation/verification execution: {"challenge_rounds":3,"complete_transcript":true,"response_rows":3}; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:fe340e5025f18af3d61b39b8214d027ea937165e6722421b66e15de2484ca058; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate`
`sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:c50f433e6623f809bf440f031f4223e2a6e370620c9e7b3c57decbb0e757f8f7; engine receipt sha256:25ca3df8eaeceb2418d39a33e03aba3183c924377153274ed1d8e26dd25ba3c1e at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-INTERACTIVE-VERIFICATION-016-25ca3df8eaeceb241.json.`

SFT-QUANTUM-OBL-QSIMX-017 - Blind delegated-computation correspondence

- **Claim and ownership:** SFT-QUANTUM-QSIMX-BLIND-DELEGATION-017; Quantum Computation family QSIMX; Blind delegated-computation correspondence.
- **Forced law:** Blind delegation correspondence separates a sealed task commitment from the executor's declared view while preserving the complete execution and verification transcript needed to test correctness and blindness.
- **Unique surviving structure:** `complete-source-bound-Fold-model__sealed-task-view-bounded-delegation-transcript__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff`. Exact result: QSIMX-017 uniquely retains sealed-task-view-bounded-delegation-transcript, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-INTERACTIVE-VERIFICATION-016. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-017: `blind_delegation`; exact simulation/verification execution: `{"complete_verification_trace":true,"executor_target_access":false,"task_commitment_present":true}`; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:5b107e54ba3ab239711b43b6be719f9c2219b77aabe1bbf6f03c8dc6b882d867; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate
sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:be0ffd0fb8b1cf1436f269ce379750cc3c01d1cd663db260dd7b9f69aa9a716f; engine receipt sha256:3c4869bb0c0ee070e7f57a80f650175f88e00c6f076324f45da920e75e6a7005 at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-BLIND-DELEGATION-017-3c4869bb0c0ee070.json.

SFT-QUANTUM-OBL-QSIMX-018 - Self-testing correspondence boundary

- **Claim and ownership:** SFT-QUANTUM-QSIMX-SELF-TESTING-018; Quantum Computation family QSIMX; Self-testing correspondence boundary.
- **Forced law:** Self-testing correspondence may identify a formal behavior relation inside a registered grammar; physical device identity, isolation and closeness require separately measured assumptions and bounds.
- **Unique surviving structure:** complete-source-bound-Fold-model__behavior-relation-to-device-identity-boundary__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-018 uniquely retains behavior-relation-to-device-identity-boundary, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-BLIND-DELEGATION-017. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-018: self_testing_boundary; exact simulation/verification execution: {"formal_behavior_relation_id_identified":true,"measurement_handoff_required":true,"physical_device_identity_claimed":false}; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:e70ebe685fb2da485c2ef0cc962b520d16cb68e24a335cf1fbbc7595bc3e20bd; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate
sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:3b4d9e88fbfc1e0d92947ce092f6eab762f8493f5f6bc9c92575a164de9a04f8; engine receipt sha256:d0dc6ab0e5fdf134a7b9cb9e1333b4cdc45bbbd724289d26f06c1bc6e5c7545b at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-SELF-TESTING-018-d0dc6ab0e5fdf134.json.

SFT-QUANTUM-OBL-QSIMX-019 - Tomography and state-reconstruction boundary

- **Claim and ownership:** SFT-QUANTUM-QSIMX-TOMOGRAPHY-019; Quantum Computation family QSIMX; Tomography and state-reconstruction boundary.
- **Forced law:** Tomography reconstructs a finite declared state-support exactly only when the observation grammar distinguishes every registered class; finite or noisy physical records require explicit enclosures and cannot imply an ungenerated continuum.
- **Unique surviving structure:** complete-source-bound-Fold-model__complete-observation-class-support-reconstruction__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-019 uniquely retains complete-observation-class-support-reconstruction, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-SELF-TESTING-018. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-019: tomography_boundary; exact simulation/verification execution: {"continuum_claimed":false,"declared_support_rows":4,"observation_rows":4,"reconstructed_rows":4}; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:fa034a331a1aebd110add8ee1b568f99f4e65c82e7287f292c90e8de9ab535a; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate
 sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:f7c19e5cc6a1510b99c96993bb0dcf52697e341d5b3366c438282af8a265bf76; engine receipt sha256:eb9109ef49265eabccd261b69bb0a18fdee0cb8ea3f62e7ef15a5ed0f49a390d at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-TOMOGRAPHY-019-eb9109ef49265eab.json.

SFT-QUANTUM-OBL-QSIMX-020 - Process reconstruction and channel verification

- **Claim and ownership:** SFT-QUANTUM-QSIMX-PROCESS-CHANNEL-VERIFICATION-020; Quantum Computation family QSIMX; Process reconstruction and channel verification.
- **Forced law:** Process reconstruction enumerates every declared source support and its exact image, phase, environment and observation records; a channel is verified only across this complete registered table.

- **Unique surviving structure:** complete-source-bound-Fold-model__complete-source-image-process-table-reconstruction__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-020 uniquely retains complete-source-image-process-table-reconstruction, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-TOMOGRAPHY-019. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-020: process_channel_verification; exact simulation/verification execution: {"complete_table":true,"duplicate_images":0,"image_rows":4,"source_rows":4}; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:98baa917b15580c7aee7303b99b1f9e8cc8b8a784ee21db86415a678c3ea88bf; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:471c781b7da73c9ea8e2fa3917f4b39733b3ab4c26a02710cb252a50ca200c9b; engine receipt sha256:3f85d6c82346e7fae0bca236969c49bc41402fa60b6d5cb430a3205406a46b1 at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-PROCESS-CHANNEL-VERIFICATION-020-3f85d6c823346e7f.json.

SFT-QUANTUM-OBL-QSIMX-021 - Randomized benchmarking deterministic-support boundary

- **Claim and ownership:** SFT-QUANTUM-QSIMX-DETERMINISTIC-BENCHMARKING-021; Quantum Computation family QSIMX; Randomized benchmarking deterministic-support boundary.
- **Forced law:** Randomized benchmarking correspondence is a deterministic complete support of benchmark schedules plus a registered observation/coarsening rule; randomness is epistemic selection over that support, not an ontic cause.
- **Unique surviving structure:** complete-source-bound-Fold-model__complete-generated-benchmark-schedule-support__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-021 uniquely retains complete-generated-benchmark-schedule-support, complete trace custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-PROCESS-CHANNEL-VERIFICATION-020. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-021: `deterministic_benchmarking`; exact simulation/verification execution: `{"complete_schedules":8,"ontic_randomness":false,"sampled_schedules":0,"schedule_depth":3}`; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:783ab7ad5a72d360d230beecb0e5cfd13300813a5038bbfd74c728f029b3e9c0`; independent implementation `sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f`; independent certificate
`sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9`; measurement `sha256:32cc7835cb15adff1cb7f28470482d35bd6787b9ce086551c2514a565952f6e9`; engine receipt `sha256:1089b279ab1a0170cd091d632796b64da1cd7b678b52c0439f36a06e05da731f` at `receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-DETERMINISTIC-BENCHMARKING-021-1089b279ab1a0170.json`.

SFT-QUANTUM-OBL-QSIMX-022 - Simulation validation against owning-domain data

- **Claim and ownership:** SFT-QUANTUM-QSIMX-OWNING-DOMAIN-VALIDATION-022; Quantum Computation family QSIMX; Simulation validation against owning-domain data.
- **Forced law:** A simulation is empirically validated only after its formal result and target identity are sealed, then compared against the owning domain's authoritative data with values, units, uncertainty, adverse rows and provenance preserved.
- **Unique surviving structure:** `complete-source-bound-Fold-model__sealed-formal-result-to-blind-owning-data-comparison__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff`. Exact result: QSIMX-022 uniquely retains sealed-formal-result-to-blind-owning-data-comparison, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-DETERMINISTIC-BENCHMARKING-021.

Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** QSIMX-022: owning_domain_validation; exact simulation/verification execution: {"adverse_rows_preserved":true,"data_selects_law":false,"formal_result_sealed":true,"target_identity_sealed":true}; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032, SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS, OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:b390d8b0dca60ade388c3871e5bb747474b63f53d9207fa3e2c1ab17085f7234; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:40aec0c116d5129b45ebe069af01609117c6e62081f7ed0fd8c04f055c9fdf82; engine receipt sha256:b0309a07cd6d8b0a25f54b31835c4a9bad1dca840373c307e89ca0c43f1d128e at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-OWNING-DOMAIN-VALIDATION-022-b0309a07cd6d8b0a.json.

SFT-QUANTUM-OBL-QSIMX-023 - Reproducible quantum-workflow provenance

- **Claim and ownership:** SFT-QUANTUM-QSIMX-WORKFLOW-PROVENANCE-023; Quantum Computation family QSIMX; Reproducible quantum-workflow provenance.
- **Forced law:** A reproducible quantum workflow binds its registry, source identities, exact inputs, complete trace, result, controls, independent reconstruction, measurement record and engine receipt in one replayable chain.
- **Unique surviving structure:** complete-source-bound-Fold-model__registry-source-trace-result-control-receipt-chain__complete-update-phase-observation-trace__complete-support-depth-round-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QSIMX-023 uniquely retains registry-source-trace-result-control-receipt-chain, complete trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 -> SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-OWNING-DOMAIN-VALIDATION-022. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-023: workflow_provenance; exact simulation/verification execution: {"chain_fields":6,"independent_reconstruction_required":true,"source_and_trace_hashes_required":true}; complete simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations; physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001, SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032,

SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS,
 OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status:
 empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy, probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity, chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:66038fb17337a0385f5f70083a57c96191b060dc6d714cb6903288d74f690ef6; independent
 implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f;
 independent certificate
 sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement
 sha256:e58a0dd288269f11c4986b0ba96da49a9d0bd94854bdb546a1f68852ef89be23; engine receipt
 sha256:956fd986d7484e5804c1bef19f21766c9e203106b0d89ff84bad64f1ff5d3de7 at receipts/
 engine/model_admitted/SFT-QUANTUM-QSIMX-WORKFLOW-PROVENANCE-023-956fd986d7484e58.j
 son.

SFT-QUANTUM-OBL-QSIMX-024 - Quantum simulation and verification completeness certificate

- **Claim and ownership:** SFT-QUANTUM-QSIMX-COMPLETENESS-024; Quantum Computation family QSIMX;
 Quantum simulation and verification completeness certificate.
- **Forced law:** The QSIMX family is complete exactly when all twenty-four frozen obligations have one owner, one unique survivor, controls, post-registry observations, independent reconstructions and current receipts.
- **Unique surviving structure:** complete-source-bound-Fold-model__twenty-four-obligation-no-om
 ission-ledger__complete-update-phase-observation-trace__complete-support-depth-rou
 nd-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-
 registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result:
 QSIMX-024 uniquely retains twenty-four-obligation-no-omission-ledger, complete trace custody, root forcing,
 post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls
 passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete
 simulator/verifier grammar, exact execution or explicit handoff, source identity, candidate survivor, seal, custody record,
 control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QCODEX-COMPLETENESS-032 ->
 SFT-QUANTUM-SIMULATION-001 -> SFT-QUANTUM-VERIFICATION-001 ->
 SFT-COMP-SCI-SIMULATION-001 -> SFT-QUANTUM-QSIMX-WORKFLOW-PROVENANCE-023. Registered root:
 SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QSIMX-024: simulation_verification_completeness; exact simulation/verification execution:
 {"duplicate_owners":0,"execution_rows":24,"omitted_owners":0,"registered_obligations":24}; complete
 simulation/verification custody: 24 records; all rows preserved; exact finite operational rows supply formal observations;
 physical dynamics, scales, fidelities and owning-domain values remain explicit measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QUANTUM-SIMULATION-001,
 SFT-V3-ADMITTED-QUANTUM-VERIFICATION-001, SFT-V3-ADMITTED-QCODEX-COMPLETENESS-032,
 SFT-V1-V2-QUANTUM-SIMULATION-VERIFICATION-COMPARISON-CORPUS,
 OWNING-DOMAIN-PHYSICAL-MEASUREMENT-HANDOFFS. External status:
 empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hamiltonian, continuum dynamics, amplitude calculus, tomography
 formula or verification theorem selects the law; host 0 denotes absence only and is not a numerical-zero state, energy,
 probability or error rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no sampled
 schedule, selected state, hidden environment, omitted adverse row or ontic randomness; no physical scale, fidelity,

chemistry, materials or device result is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:c9d151ec8a75990a698315b7c4cfcf8c672be79e8b861f797969175dffacd030; independent implementation sha256:a4227680778da04359aa807877d08ea908406f29cc09aa78146695820f6b0a2f; independent certificate sha256:988e49456ce8a31a7dbf2a3987f997964f68267bcc94b1b150114056d94221c9; measurement sha256:d2c447be424940ae879e0494254661cc7b25282cd0064046b1cb5fe9d82087de; engine receipt sha256:eeb3915656fe41fb894b3b70cf49b9b1b64bdd75cbb2621ff2f65a1fea9c3ad5 at receipts/engine/model_admitted/SFT-QUANTUM-QSIMX-COMPLETENESS-024-eeb3915656fe41fb.json.

36.12 Family QLEARNX - 22/22 complete

Scope. Quantum learning and intelligence. This family decides 5,632 candidates, retains 22 unique survivors and passes 88 adverse controls.

SFT-QUANTUM-OBL-QLEARNX-001 - Quantum learning problem and example identity

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-PROBLEM-EXAMPLE-IDENTITY-001; Quantum Computation family QLEARNX; Quantum learning problem and example identity.
- **Forced law:** A quantum learning problem registers its source domain, example identities, target relation, admissible observations, loss/order relation and resources before any outcome is opened.
- **Unique surviving structure:** registered-source-target-resource-problem__typed-problem-example-target-resource-registry__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-001 uniquely retains typed-problem-example-target-resource-registry, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-001: problem_example_identity; exact quantum-learning execution: {"domain_examples":2,"duplicate_example_ids":0,"target_rows":2}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:df4ffcf0489bfd6442327d1ed5b0ff7520e8a0f1056b9a02f5296bd705cb0495; independent

```
implementation sha256:eb111ec3fb38565b0fa56f6ea369da57abf71f6602836cd8f086b259a29bdbb8;
independent certificate
sha256:48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement
sha256:3d7dfb1e49efd1d84b3035009108df5114293762fcc90fe84202694c057661c0; engine receipt
sha256:65f97c528556510cd79d279c31548e9e08796bb44d0082b6729a91d99bd08e7b at receipts/
engine/model_admitted/SFT-QUANTUM-QLEARNX-PROBLEM-EXAMPLE-IDENTITY-001-65f97c52855
6510c.json.
```

SFT-QUANTUM-OBL-QLEARNX-002 - Classical-data quantum-process boundary

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-CLASSICAL-DATA-BOUNDARY-002; Quantum Computation family QLEARNX; Classical-data quantum-process boundary.
- **Forced law:** Classical data enters a quantum process only through a canonical reversible encoding with complete source identity and decoding records; encoding does not create information.
- **Unique surviving structure:** registered-source-target-resource-problem__canonical-classical-data-to-Fold-support-encoding__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-002 uniquely retains canonical-classical-data-to-Fold-support-encoding, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-PROBLEM-EXAMPLE-IDENTITY-001. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-002: classical_data_boundary; exact quantum-learning execution: {"encoded_rows":2,"inverse_records":2,"source_rows":2}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation


```
sha256:94f969bfb4c1b6444db98c0e49c5082cecf9ad8ea491b7f3ecb15300b7821652; independent
implementation sha256:eb111ec3fb38565b0fa56f6ea369da57abf71f6602836cd8f086b259a29bdbb8;
independent certificate
sha256:48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement
sha256:6d9e0f155724dc8eeb05aecf366fab51461201ac4025f95867dc186ae7b2844d; engine receipt
sha256:66466af4b2f4a7312fc81d4d6d29159a8e3c1db2dcd6d989650637d72d8206cb at receipts/
engine/model_admitted/SFT-QUANTUM-QLEARNX-CLASSICAL-DATA-BOUNDARY-002-66466af4b2f4a731.json.
```

SFT-QUANTUM-OBL-QLEARNX-003 - Quantum-data support and observation custody

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-QUANTUM-DATA-CUSTODY-003; Quantum Computation family QLEARNX; Quantum-data support and observation custody.
- **Forced law:** Quantum data is a complete finite Fold support with phase and joint records; each observation retains its class, closed distinctions and source custody.
- **Unique surviving structure:** registered-source-target-resource-problem__complete-quantum-example-support-observation-ledger__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__the-re-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-003 uniquely retains complete-quantum-example-support-observation-ledger, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-CLASSICAL-DATA-BOUNDARY-002. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-003: quantum_data_custody; exact quantum-learning execution: {"source_records":4,"support_rows":4,"support_width":2}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:3266de75ba63db5a582a33037190c701c604910926c3f6337ea7844ee5ad3e58; independent implementation sha256:eb111ec3fb38565b0fa56feea369da57abf71f6602836cd8f086b259a29bdbb8; independent certificate
sha256:48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement sha256:48722dc3c0b65629fba54e76f0bf9eb33282615ba1b5ae9af582325c2a05d030; engine receipt sha256:aaf22c744004c2430b18272dcaab6a0b2f4909d6bf81c0fc61225b7e8374a40e at receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-QUANTUM-DATA-CUSTODY-003-aaf22c744004c243.json.

SFT-QUANTUM-OBL-QLEARNX-004 - Quantum hypothesis family generation

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-HYPOTHESIS-FAMILY-004; Quantum Computation family QLEARNX; Quantum hypothesis family generation.
- **Forced law:** A quantum hypothesis family is the complete generated set of source-bound processes admitted before training targets are opened, with no pretrained or silently selected member.

- **Unique surviving structure:** registered-source-target-resource-problem__complete-generated-quantum-hypothesis-family__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-004 uniquely retains complete-generated-quantum-hypothesis-family, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-QUANTUM-DATA-CUSTODY-003. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-004: hypothesis_family; exact quantum-learning execution: {"domain_rows":2,"generated_hypotheses":4,"pretrained_hypotheses":0}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:e3f20dfcb077d8cb6b87c27e480ffbca39e5fa4fc7f01b19785285b9dbb28ef8; independent implementation sha256:eb111ec3fb38565b0fa56feea369da57abf71f6602836cd8f086b259a29bdbb8; independent certificate
sha256:48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement sha256:36fb5d7292119adc5fdc8c36d7cf21d576e55d3fdd68d8184af10caccab07bdf; engine receipt sha256:4054b109c0d66e370772229353c625ccc26b579f278bf988a9f19bdfbfa00139 at receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-HYPOTHESIS-FAMILY-004-4054b109c0d66e37.json.

SFT-QUANTUM-OBL-QLEARNX-005 - Quantum feature-map correspondence

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-FEATURE-MAP-005; Quantum Computation family QLEARNX; Quantum feature-map correspondence.
- **Forced law:** Feature-map correspondence is a reversible source-to-Fold-support relation whose collisions, phase actions, resource cost and inverse records are explicit.
- **Unique surviving structure:** registered-source-target-resource-problem__reversible-source-feature-support-map__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-005 uniquely retains reversible-source-feature-support-map, complete learning custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-HYPOTHESIS-FAMILY-004. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-005: `feature_map`; exact quantum-learning execution: `{"collisions":0,"feature_rows":4,"source_rows":4}`; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: `empirically_tested_and_independently_replicated`; `all rows preserved`: True.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:548c7f8ac6eee3c8ffa208c287c31c3d17ab0cd032213a2566fb8e6b3e48894a`; independent implementation `sha256:eb111ec3fb38565b0fa56feea369da57abf71f6602836cd8f086b259a29bdbb8`; independent certificate
`sha256:48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2`; measurement `sha256:568d255a1fda826551567aff959f3dd7a53934206322ab123d4eb8d1ca55dcf3`; engine receipt `sha256:ead50c2898ab96c41f3902a7920db272982b56fa1015876a181879ea330b0ae3` at `receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-FEATURE-MAP-005-ead50c2898ab96c4.json`.

SFT-QUANTUM-OBL-QLEARNX-006 - Quantum kernel correspondence boundary

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-KERNEL-BOUNDARY-006; Quantum Computation family QLEARNX; Quantum kernel correspondence boundary.
- **Forced law:** Quantum-kernel correspondence is an exact pair-comparison observation relation over registered feature supports; no continuum inner product or physical advantage is imported.
- **Unique surviving structure:** `registered-source-target-resource-problem__exact-pair-comparison-observation-relation__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff`. Exact result: QLEARNX-006 uniquely retains exact-pair-comparison-observation-relation, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-FEATURE-MAP-005. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** QLEARNX-006: kernel_boundary; exact quantum-learning execution: {"continuum_inner_product_imported":false,"pair_width":2,"retained_matches":1}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:2b1cbc2ef55fef38470a17982f71b81c861790498778462e062a9c4c947bda7e; independent implementation sha256:eb111ec3fb38565b0fa56feea369da57abf71f6602836cd8f086b259a29bdbb8; independent certificate
sha256:48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement sha256:34d2853d138ce0d311bdea0c91cfale9d47f6a30be9b4ffe38047f31bf898857; engine receipt sha256:04ef3e26e8db92331ab1f9d2a2faccd904c0cce51384928ca391a8106c4fa269 at receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-KERNEL-BOUNDARY-006-04ef3e26e8db9233.json.

SFT-QUANTUM-OBL-QLEARNX-007 - Quantum classification process

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-CLASSIFICATION-007; Quantum Computation family QLEARNX; Quantum classification process.
- **Forced law:** Classification enumerates every registered hypothesis and example, computes exact class observations, preserves ties and adverse rows, and selects only a uniquely forced survivor.
- **Unique surviving structure:** registered-source-target-resource-problem__complete-hypothesis__class-observation-selection__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-007 uniquely retains complete-hypothesis-class-observation-selection, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-KERNEL-BOUNDARY-006. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-007: classification; exact quantum-learning execution: {"hypotheses_evaluated":4,"training_rows":2,"unique_survivors":1}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026,

SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS,
HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status:
empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:9b99ee26de76a6411e112c8524f9489938129fa26ebd7a931ddfed70469455fa; independent implementation sha256:eb111ec3fb38565b0fa56feea369da57abf71f6602836cd8f086b259a29bdbb8; independent certificate
sha256:48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement sha256:e3c9b197c1010c6bae66cf26fae2afedca2734819ab22d0ba7443a5372464e7d; engine receipt sha256:3033bc11ffc2918c0c3be978de28cb36b2cc2cbeca51493a7e13932427a0b1d6 at receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-CLASSIFICATION-007-3033bc11ffc2918c.json.

SFT-QUANTUM-OBL-QLEARNX-008 - Quantum regression process

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-REGRESSION-008; Quantum Computation family QLEARNX; Quantum regression process.
- **Forced law:** Regression correspondence maps examples to exact finite ordered output support and reports an enclosure when observation cannot distinguish a unique output; no irrational continuum target is presumed.
- **Unique surviving structure:** registered-source-target-resource-problem__exact-ordered-output-support-relation__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-008 uniquely retains exact-ordered-output-support-relation, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-CLASSIFICATION-007. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-008: regression_correspondence; exact quantum-learning execution: {"exact_support_only":true,"irrational_target_imported":false,"ordered_output_classes":3}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024,
SFT-V3-ADMITTED-QUANTUM-LEARNING-001,
SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026,
SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS,
HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status:
empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or

changes the protected authority.

- **Evidence identities:** derivation

sha256: fec93d149daa31e59cee5247d6927efdadaa6e10fada6bd7c871f98593a7c82; independent implementation sha256: eb111ec3fb38565b0fa56feea369da57abf71f6602836cd8f086b259a29bdbb8; independent certificate
sha256: 48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement sha256: c0ff0ad7a99af9c610cc97c191306bf936b863936c20d224d73fa0f9bbcb4e0c; engine receipt sha256: 51c7e80569552b9bfc0a4862d7b4d372afe0827368c3524872ea3dc9385060f6 at receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-REGRESSION-008-51c7e80569552b9b.json.

SFT-QUANTUM-OBL-QLEARNX-009 - Quantum generative-support reconstruction

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-GENERATIVE-SUPPORT-009; Quantum Computation family QLEARNX; Quantum generative-support reconstruction.
- **Forced law:** Generative learning reconstructs the complete registered finite support and exact multiplicity/order records; a sampled subset cannot certify missing support.
- **Unique surviving structure:** registered-source-target-resource-problem__complete-source-support-reconstruction__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-009 uniquely retains complete-source-support-reconstruction, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-REGRESSION-008. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-009: generative_support; exact quantum-learning execution: {"declared_rows":4,"missing_rows":0,"reconstructed_rows":4}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256: 206c7f65d5912a308b9d7a261607c23c667b11131c3d6ca001dc7a93afd062f8; independent implementation sha256: eb111ec3fb38565b0fa56feea369da57abf71f6602836cd8f086b259a29bdbb8; independent certificate
sha256: 48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement sha256: 72fc33b916c2efb5f75cb9c94665ba027279fc4139d61b2394dbc5027cc79c7a; engine receipt sha256: 698359a3853280aa3f6964cf450acff2f7def7b1ab614b671fb515e65dbb6b0d at receipts/

engine/model_admitted/SFT-QUANTUM-QLEARNX-GENERATIVE-SUPPORT-009-698359a3853280aa.json.

SFT-QUANTUM-OBL-QLEARNX-010 - Quantum clustering correspondence

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-CLUSTERING-010; Quantum Computation family QLEARNX; Quantum clustering correspondence.
- **Forced law:** Clustering correspondence generates an exact distinguishability relation and its complete equivalence-class partition, preserving boundary and tied assignments.
- **Unique surviving structure:** registered-source-target-resource-problem__generated-equivalence-class-partition__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-010 uniquely retains generated-equivalence-class-partition, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-GENERATIVE-SUPPORT-009. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-010: clustering; exact quantum-learning execution: {"clusters":2,"rows_per_cluster":2,"support_rows":4}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:cd42e2ca8d9238e50a1d2187865ffca705e5a2cfaf9a59b012a7d982b84c309e; independent implementation sha256:eb111ec3fb38565b0fa56f6ea369da57abf71f6602836cd8f086b259a29bdbb8; independent certificate sha256:48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement sha256:37133511159d85f23570b5057e74bf97b134a7eca71d3e0e586fd793cea97786; engine receipt sha256:059b43ed158be4743b233a277ce5268a617fd9f96c718039d2d0b619b95d5918 at receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-CLUSTERING-010-059b43ed158be474.json.

SFT-QUANTUM-OBL-QLEARNX-011 - Quantum principal-structure correspondence

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-PRINCIPAL-STRUCTURE-011; Quantum Computation family QLEARNX; Quantum principal-structure correspondence.
- **Forced law:** Principal-structure correspondence orders generated finite incidence patterns by exact retained distinction counts while preserving every tie and discarded-coordinate record.

- **Unique surviving structure:** registered-source-target-resource-problem__exact-incidence-rank-and-tie-ledger__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-011 uniquely retains exact-incidence-rank-and-tie-ledger, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-CLUSTERING-010. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-011: principal_structure; exact quantum-learning execution: {"incidence_patterns":2,"ranked_patterns":2,"ties_preserved":true}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:301d7e796d4b6ef3d601f8f2ca19a63ebe754e06bfd80e0b8b1f9c40bd54602f; independent implementation sha256:eb111ec3fb38565b0fa56f6ea369da57abf71f6602836cd8f086b259a29bdbb8; independent certificate sha256:48cdecbb24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement sha256:5537011ef7ddfce6da22e31bb8e9f5165c7080ae27d6f3da502d884c071a1037; engine receipt sha256:ef0e0427f11710f30504f8083d70982cc2dc1e0b99439d3a850092faed9f39c7 at receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-PRINCIPAL-STRUCTURE-011-ef0e0427f11710f3.json.

SFT-QUANTUM-OBL-QLEARNX-012 - Quantum optimization in learning

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-OPTIMIZATION-012; Quantum Computation family QLEARNX; Quantum optimization in learning.
- **Forced law:** Learning optimization enumerates the complete hypothesis/resource grammar, evaluates its exact order relation, retains ties and selects only an independently reconstructed unique optimum.
- **Unique surviving structure:** registered-source-target-resource-problem__exhaustive-hypothesis-is-score-order-selection__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-012 uniquely retains exhaustive-hypothesis-score-order-selection, complete learning custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-PRINCIPAL-STRUCTURE-011. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-012: `learning_optimization`; exact quantum-learning execution: `{"candidates":4,"exact_scores":4,"unique_optimum":true}`; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: True.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:b3a24aa62755b05a42773ae152ac5e156c7440a1245d5be407d1341eaed627d9`; independent implementation `sha256:eb111ec3fb38565b0fa56feea369da57abf71f6602836cd8f086b259a29bdbb8`; independent certificate
`sha256:48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2`; measurement `sha256:e629bde4e73981ac168326809e48437da00660c4f288189ad6eafa6c6f70220c`; engine receipt `sha256:7a3a839aed9cbd6a6717036844844ba79dcadc50972366a3130d3aed53c459e4` at `receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-OPTIMIZATION-012-7a3a839aed9cbd6a.json`.

SFT-QUANTUM-OBL-QLEARNX-013 - Variational quantum learning parameter boundary

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-VARIATIONAL-BOUNDARY-013; Quantum Computation family QLEARNX; Variational quantum learning parameter boundary.
- **Forced law:** A variational setting is a generated finite control word, not a free or fitted proof parameter; target data may compare sealed settings but cannot alter their grammar or law.
- **Unique surviving structure:** `registered-source-target-resource-problem__generated-setting-not-fitted-parameter-boundary__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff`. Exact result: QLEARNX-013 uniquely retains generated-setting-not-fitted-parameter-boundary, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-OPTIMIZATION-012. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** QLEARNX-013: variational_boundary; exact quantum-learning execution: {"free_parameters":0,"generated_settings":4,"target_fitted":false}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:49330443515c4af21815c914474414ded3f733c3a7f755e89fa09ab308dd05ae; independent implementation sha256:eb111ec3fb38565b0fa56feea369da57abf71f6602836cd8f086b259a29bdbb8; independent certificate
sha256:48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement sha256:7c0ed878546456d18122cb8cfa8b1d5a1ed1b611abaf8ab6644f400757d21112; engine receipt sha256:e9981b411ea9c81555ca809e1f97215cdc6f8eab2e74d60b6cb92aefa3befe4b at receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-VARIATIONAL-BOUNDARY-013-e9981b411ea9c815.json.

SFT-QUANTUM-OBL-QLEARNX-014 - Quantum reinforcement-process correspondence

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-REINFORCEMENT-014; Quantum Computation family QLEARNX; Quantum reinforcement-process correspondence.
- **Forced law:** Reinforcement correspondence is a complete state/action/transition/observation transcript with exact return ordering and causal custody; stochastic policy is represented by deterministic support plus observation.
- **Unique surviving structure:** registered-source-target-resource-problem__state-action-transition-observation-transcript__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-014 uniquely retains state-action-transition-observation-transcript, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-VARIATIONAL-BOUNDARY-013. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-014: reinforcement_process; exact quantum-learning execution: {"actions":2,"ontic_randomness":false,"states":4,"transition_rows":8}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026,

SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS,
HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status:
empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:7b3d3cf61ce9e1d76e3ba540869e4af36637e7ecfb69bf859caafc30a896233f; independent implementation sha256:eb111ec3fb38565b0fa56feea369da57abf71f6602836cd8f086b259a29bdbb8; independent certificate
sha256:48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement sha256:fea5926507f22edaf3a81d9d84cc45441960403cf31d15188d29d79792e47242; engine receipt sha256:32629b84aa53b7b4c407cfd8f46fb1eac517b525fe6c58fc8a46dc0fa370efb7 at receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-REINFORCEMENT-014-32629b84aa53b7b4.json.

SFT-QUANTUM-OBL-QLEARNX-015 - Quantum online-learning correspondence

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-ONLINE-015; Quantum Computation family QLEARNX; Quantum online-learning correspondence.
- **Forced law:** Online learning retains every causal example, prediction, observation and hypothesis update; future targets remain inaccessible before their prediction seal.
- **Unique surviving structure:** registered-source-target-resource-problem__causal-example-update-prediction-transcript__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-015 uniquely retains causal-example-update-prediction-transcript, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-REINFORCEMENT-014. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-015: online_learning; exact quantum-learning execution: {"causal_examples":2,"future_target_access":false,"updates":2}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or

changes the protected authority.

- **Evidence identities:** derivation

sha256:0b11eb64276a3ca7ef07caf37319c15f7a817a0ffc869f67540b2ebcecc312c6; independent implementation sha256:eb111ec3fb38565b0fa56f6ea369da57abf71f6602836cd8f086b259a29bdbb8; independent certificate
sha256:48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement sha256:bb6d784f698e93f07d0169bda9ebff0edc50e102b931a415fde020634fa943a2; engine receipt sha256:58d7f9aeb24dd4337260b52e802ce0cdce9c995cfd6328295a044d37a2f4c8e0 at receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-ONLINE-015-58d7f9aeb24dd433.json.

SFT-QUANTUM-OBL-QLEARNX-016 - Quantum sample complexity

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-SAMPLE-COMPLEXITY-016; Quantum Computation family QLEARNX; Quantum sample complexity.

- **Forced law:** Sample complexity is the least positive finite example support forcing the registered learning condition across the complete hypothesis family, with every smaller support counterexample retained.

- **Unique surviving structure:** registered-source-target-resource-problem__least-example-support-for-declared-learning-condition__complete-example-hypothesis-branch-observation__trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-016 uniquely retains least-example-support-for-declared-learning-condition, complete learning custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-ONLINE-015. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** QLEARNX-016: sample_complexity; exact quantum-learning execution: {"hypotheses":4,"least_forcing_examples":2,"smaller_support_counterexample_retained":true}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs

- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:c68f1686bb96cc67e1811dbc029edc51c6f8cd578b8306682ca2e791f6df3320; independent implementation sha256:eb111ec3fb38565b0fa56f6ea369da57abf71f6602836cd8f086b259a29bdbb8; independent certificate
sha256:48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement sha256:a2aeae304057982bc64442147f42145562981db95cf652fd61e1cebad1a80309; engine receipt

sha256:eeaa42743d74ca8285fee800856ce254fc11b3396e36d904929cc347166e8c03 at receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-SAMPLE-COMPLEXITY-016-eeaa42743d74ca82.json.

SFT-QUANTUM-OBL-QLEARNX-017 - Quantum query complexity of learning

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-QUERY-COMPLEXITY-017; Quantum Computation family QLEARNX; Quantum query complexity of learning.
- **Forced law:** Quantum learning query complexity is the least positive finite source-bound query transcript forcing the registered condition, counting preparation, oracle, observation and retained records.
- **Unique surviving structure:** registered-source-target-resource-problem__least-query-transcript-for-declared-learning-condition__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-017 uniquely retains least-query-transcript-for-declared-learning-condition, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-SAMPLE-COMPLEXITY-016. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-017: query_complexity; exact quantum-learning execution: {"domain_rows":2,"preparation_and_observation_counted":true,"queries":2}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:71bf23b61dec43a01701fdf815a626aa49a3082877440a71b36deeba0a01408d; independent implementation sha256:eb111ec3fb38565b0fa56f6ea369da57abf71f6602836cd8f086b259a29bdbb8; independent certificate
 sha256:48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement sha256:1e014bdd9cc4ad45858929ad07fe2a0ab86980c00c2c2f9f438737bbdfef437b8; engine receipt sha256:16decc3feb3ceeb9f811959705a7856a395b6fffb03047b24fafaadc837b92889 at receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-QUERY-COMPLEXITY-017-16decc3feb3ceeb9.json.

SFT-QUANTUM-OBL-QLEARNX-018 - Generalization and held-out target custody

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-GENERALIZATION-CUSTODY-018; Quantum Computation family QLEARNX; Generalization and held-out target custody.

- **Forced law:** Generalization requires a frozen hypothesis and target identity before held-out outcomes are released; all favorable, adverse, absent and unresolved rows remain in the comparison.
- **Unique surviving structure:** registered-source-target-resource-problem__sealed-hypothesis-blind-held-out-comparison__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-018 uniquely retains sealed-hypothesis-blind-held-out-comparison, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-QUERY-COMPLEXITY-017.
Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-018: generalization_custody; exact quantum-learning execution: {"held_out_rows":1,"target_opened_after_hypothesis_seal":true,"training_rows":1}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:77a9ecffbec9027adb6d3e75b1f20c046e39e77b5dad2ee87a50a066ef1b0949; independent implementation sha256:eb111ec3fb38565b0fa56feea369da57abf71f6602836cd8f086b259a29bdbb8; independent certificate
sha256:48cdecbb24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement sha256:f5ab38cbb5859dee0c524822bf9783b35ba52b0645c15a2b44d0afab8de4307e; engine receipt sha256:5e6b50d613bf5064b9ad756aef50209728f6d72e553fa187da770e5766270973 at receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-GENERALIZATION-CUSTODY-018-5e6b50d613bf5064.json.

SFT-QUANTUM-OBL-QLEARNX-019 - Quantum advantage in learning certificate

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-ADVANTAGE-CERTIFICATE-019; Quantum Computation family QLEARNX; Quantum advantage in learning certificate.
- **Forced law:** A quantum-learning advantage certificate requires the same task and error condition, complete classical and quantum resource ledgers, lower and upper bounds, and separately measured physical timing where claimed.
- **Unique surviving structure:** registered-source-target-resource-problem__same-task-complete-resource-ledger-separation__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-019 uniquely retains same-task-complete-resource-ledger-separation, complete

learning custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-GENERALIZATION-CUSTODY-018. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-019: advantage_certificate; exact quantum-learning execution: `{"classical_resource_ledger":true,"physical_speedup_claimed":false,"quantum_resource_ledger":true,"same_task":true};` complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:1a12b663f41f01ca7d74c9c54ca7668f84dbaba5b052756712395163e4b576e1`; independent implementation `sha256:eb111ec3fb38565b0fa56feea369da57abf71f6602836cd8f086b259a29bdbb8`; independent certificate
`sha256:48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2`; measurement `sha256:66914e6fecad42f548c95d48bbd6c34f635a6a1d6f9411687ce51f4d20be976c`; engine receipt `sha256:bdd0de5541769817f63620be42a4d26cd9e7243a1a1f4f5f90f2f0a7d261d948` at `receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-ADVANTAGE-CERTIFICATE-019-bdd0de5541769817.json`.

SFT-QUANTUM-OBL-QLEARNX-020 - Interpretability and branch-trace reconstruction

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-INTERPRETABILITY-020; Quantum Computation family QLEARNX; Interpretability and branch-trace reconstruction.
- **Forced law:** Interpretability is exact reconstruction of the input, feature support, hypothesis action, phase/interference branches, observation and output from the retained trace.
- **Unique surviving structure:** `registered-source-target-resource-problem__input-feature-hypothesis-branch-output-reconstruction__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff`. Exact result: QLEARNX-020 uniquely retains `input-feature-hypothesis-branch-output-reconstruction`, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-ADVANTAGE-CERTIFICATE-019. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-020: interpretability; exact quantum-learning execution: {"branch_reconstruction_complete":true,"opaque_predictor":false,"trace_stages":5}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:28e423019528bd0a7acc87f78977030900249e1db88d3253e250e769bb11d83e; independent implementation sha256:eb111ec3fb38565b0fa56feea369da57abf71f6602836cd8f086b259a29bdbb8; independent certificate
sha256:48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement sha256:41cf2d4a783d113af7f639ad2a4fc27a4964640a5661eba755662d46c7e0b14a; engine receipt sha256:8085c9eed22079f9054685025fa0be89b2d63f8a25646a92229b8f356df18389 at receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-INTERPRETABILITY-020-8085c9eed22079f9.js on.

SFT-QUANTUM-OBL-QLEARNX-021 - Verification and robustness of quantum learners

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-ROBUSTNESS-021; Quantum Computation family QLEARNX; Verification and robustness of quantum learners.
- **Forced law:** Learner robustness is verified only over a frozen complete perturbation and adversary grammar with every changed output, invariant output, rejection and unresolved boundary preserved.
- **Unique surviving structure:** registered-source-target-resource-problem__complete-perturbation-tamper-verification-census__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-021 uniquely retains complete-perturbation-tamper-verification-census, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-INTERPRETABILITY-020. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-021: robustness; exact quantum-learning execution: {"adverse_rows_preserved":true,"registered_perturbations":2,"support_rows":4}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs

- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:f167893927b4d15821a039b452512d3b998ae00e08891b46ea163e32b1dc5897; independent implementation sha256:eb111ec3fb38565b0fa56feea369da57abf71f6602836cd8f086b259a29bdbb8; independent certificate
 sha256:48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement sha256:483d12da8ea03dad80c7773ef3ae3337a36705f80347d1e90f498c1dedb9113; engine receipt sha256:84b3219b563e427ed92c822984b710d224a6d626fa956a17f0c2387adf435629 at receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-ROBUSTNESS-021-84b3219b563e427e.json.

SFT-QUANTUM-OBL-QLEARNX-022 - Quantum-learning completeness certificate

- **Claim and ownership:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022; Quantum Computation family QLEARNX; Quantum-learning completeness certificate.
- **Forced law:** QLEARNX is complete exactly when all twenty-two frozen obligations have one owner, one unique survivor, controls, post-registry observations, independent reconstructions and current receipts.
- **Unique surviving structure:** registered-source-target-resource-problem__twenty-two-obligation-no-omission-ledger__complete-example-hypothesis-branch-observation-trace__complete-sample-query-depth-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLEARNX-022 uniquely retains twenty-two-obligation-no-omission-ledger, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete learning grammar, held-out custody, exact execution or explicit handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QSIMX-COMPLETENESS-024 -> SFT-QUANTUM-LEARNING-001 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-QUANTUM-QLEARNX-ROBUSTNESS-021. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLEARNX-022: quantum_learning_completeness; exact quantum-learning execution: {"duplicate_owners":0,"execution_rows":22,"omitted_owners":0,"registered_obligations":22}; complete learning and held-out custody: 22 records; all rows preserved; exact finite learning rows supply formal observations; physical advantage, device behavior and application targets remain explicit owning-domain measurement handoffs
- **Sources and custody:** SFT-V3-ADMITTED-QSIMX-COMPLETENESS-024, SFT-V3-ADMITTED-QUANTUM-LEARNING-001, SFT-V3-ADMITTED-CLASSICAL-LEARNING-COMPLETENESS-026, SFT-V1-V2-QUANTUM-LEARNING-COMPARISON-CORPUS, HELD-OUT-OWNING-DOMAIN-TARGET-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no pretrained model, conventional learning theorem, continuum kernel or target-selected hypothesis selects the law; host 0 denotes absence only and is not a numerical-zero probability, loss, feature

or learning rate; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no hidden training row, held-out leakage, sampled-only support, omitted tie/adverse row or ontic randomness; no physical advantage, timing, fidelity or domain prediction is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:60314303d1fff2b6fa28646eae59d4c3af366c3090575d3c0644ec1dac6c1d5f; independent implementation sha256:eb111ec3fb38565b0fa56feea369da57abf71f6602836cd8f086b259a29bdbb8; independent certificate sha256:48cdec24ad6054c2ebb54261028f3301a58746d53d5e936bd2986d656f2d7b2; measurement sha256:82c28fac83bd5d15cf572828bc5b9fac103c07ed665bc7aeb5e29aac11fd882a; engine receipt sha256:eb2d67d692a56175ec70aa9980d6f5e18ee213e36594c1233817cefe87d22c14 at receipts/engine/model_admitted/SFT-QUANTUM-QLEARNX-COMPLETENESS-022-eb2d67d692a56175.json.

36.13 Family QLIMITX - 22/22 complete

Scope. Classical-quantum correspondence and computational limits. This family decides 5,632 candidates, retains 22 unique survivors and passes 88 adverse controls.

SFT-QUANTUM-OBL-QLIMITX-001 - Classical-state embedding in the quantum machine

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-CLASSICAL-STATE-EMBEDDING-001; Quantum Computation family QLIMITX; Classical-state embedding in the quantum machine.
- **Forced law:** A classical machine state embeds as one Fold word with one held phase class and a retained classical record inside the complete quantum-machine state grammar.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__single-word-phase-held-classical-embedding__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLIMITX-001 uniquely retains single-word-phase-held-classical-embedding, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-001: classical_embedding; exact classical-quantum correspondence/limit execution: {"classical_words":1,"phase_class":"phase-held","support_rows":1}; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:28e4843176a58b34a867a73b84b4496d5df8878e8f09f68a9d8caf7689d292f1; independent implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fe9d12580a5a42d25dcde28c; independent certificate
sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement sha256:feaa59de70d281e88a80579542bcefcfb50ee8f9decfc44643feb1e62f5bcda0; engine receipt sha256:97589da6e1d4dad086cc7acdf225734834f51ab8e88533a194a8f668520ece1 at receipts/engine/model_admitted/SFT-QUANTUM-QLIMITX-CLASSICAL-STATE-EMBEDDING-001-97589da6e1d4dad0.json.

SFT-QUANTUM-OBL-QLIMITX-002 - Classical reversible submodel correspondence

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-REVERSIBLE-SUBMODEL-002; Quantum Computation family QLIMITX; Classical reversible submodel correspondence.
- **Forced law:** Classical reversible computation is the phase-invariant word-permutation submodel of the quantum Fold machine, with identical source, image and reversal records.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__phase-invariant-word-permutation-submodel__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLIMITX-002 uniquely retains phase-invariant-word-permutation-submodel, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-CLASSICAL-STATE-EMBEDDING-001. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-002: reversible_submodel; exact classical-quantum correspondence/limit execution: {"image_rows":4,"inverse_rows":4,"source_rows":4}; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:a249011b446c1f91a041130b909b34adb49e2f3bee8c19c9742e999b0b6c6bb2; independent implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fe9d12580a5a42d25dcde28c; independent certificate
sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement sha256:fde67e450a44d796311c1d054011200aec71507a8ebba72adda2e57d152a4320; engine receipt sha256:860500bec17e76ea67cf6824b7458f8309c27869821cb44a45b16644c8389494 at receipts/

engine/model_admitted/SFT-QUANTUM-QLIMITX-REVERSIBLE-SUBMODEL-002-860500bec17e76ea.json.

SFT-QUANTUM-OBL-QLIMITX-003 - Classical probabilistic support correspondence

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-PROBABILISTIC-SUPPORT-003; Quantum Computation family QLIMITX; Classical probabilistic support correspondence.
- **Forced law:** Classical probabilistic computation corresponds to deterministic complete generated support plus an observation/coarsening relation and exact branch multiplicities; randomness is not an ontic premise.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__deterministic-complete-support-plus-observation-correspondence__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLIMITX-003 uniquely retains deterministic-complete-support-plus-observation-correspondence, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-REVERSIBLE-SUBMODEL-002. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-003: probabilistic_support; exact classical-quantum correspondence/limit execution: {"deterministic_support_rows":4,"observation_relation_present":true,"ontic_randomness":false}; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation sha256:82b423ef3a5db52e04b81c77e12d4ebdbf331de3bcdadaf11dc0b7c5afda3d1f; independent implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fef12580a5a42d25dcde28c; independent certificate sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement sha256:a204fd9d580fac2d83a286b600a9800eb2b8f1e6ce25c4c39e1c599993f4c98d; engine receipt sha256:7692326b8a9aa1afc54d31347d3e93018d021cc66369e9c8042372de304add83 at receipts/engine/model_admitted/SFT-QUANTUM-QLIMITX-PROBABILISTIC-SUPPORT-003-7692326b8a9aa1af.json.

SFT-QUANTUM-OBL-QLIMITX-004 - Quantum-to-classical measurement decoder

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-MEASUREMENT-DECODER-004; Quantum Computation family QLIMITX; Quantum-to-classical measurement decoder.

- **Forced law:** The quantum-to-classical decoder emits the selected observation class while retaining source support, phase, closed distinctions and the record needed to reconstruct the measurement relation.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__selected-class-and-closed-distinction-decoder__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLIMITX-004 uniquely retains selected-class-and-closed-distinction-decoder, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-PROBABILISTIC-SUPPORT-003. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-004: measurement_decoder; exact classical-quantum correspondence/limit execution: {"closed_classes":1,"selected_classes":1,"source_record_retained":true}; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:4e81760c111a814b3221ca126e025a1f182faf0e4f2925f1b15af5e1a24cd1b6; independent implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fefbd12580a5a42d25dcde28c; independent certificate
sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement sha256:8b41451e9441ca0ca956c3d4d5a15f0cef063bcfca56d13d8ad99d2d8f97c4dc; engine receipt sha256:30be3ef2a45fce2ef7154469bc053fff098f74b6c0f2bf9ae2c51062d8ed060a at receipts/engine/model_admitted/SFT-QUANTUM-QLIMITX-MEASUREMENT-DECODER-004-30be3ef2a45fce2e.json.

SFT-QUANTUM-OBL-QLIMITX-005 - Operational simulation in both directions

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-BIDIRECTIONAL-SIMULATION-005; Quantum Computation family QLIMITX; Operational simulation in both directions.
- **Forced law:** Classical and quantum processes simulate one another only where a source/image/trace correspondence exists in both directions with the same declared resource ledger.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__source-image-trace-bisimulation__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result:

QLIMITX-005 uniquely retains source-image-trace-bisimulation, complete boundary custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-MEASUREMENT-DECODER-004. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-005: `bidirectional_simulation`; exact classical-quantum correspondence/limit execution: `{"forward_rows":4,"resource_ledgers_compared":true,"reverse_rows":4}`; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:4efd86850b515a43479c806613198844504dae6e785ea71f515e50c39811ce79`; independent implementation `sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fefbd12580a5a42d25dcde28c`; independent certificate
`sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1`; measurement `sha256:653806c5615fd67420196091d505121b34f608130a8d27b3e94e61fe7954d230`; engine receipt `sha256:13724cc428ecbb46dbbd0bf0f86e0993b76c8d9e6f4b322b2b6a3f7d7fb20957` at `receipts/engine/model_admitted/SFT-QUANTUM-QLIMITX-BIDIRECTIONAL-SIMULATION-005-13724cc428ecbb46.json`.

SFT-QUANTUM-OBL-QLIMITX-006 - Classical simulation efficient-region certificate

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-EFFICIENT-REGION-006; Quantum Computation family QLIMITX; Classical simulation efficient-region certificate.
- **Forced law:** An efficient classical-simulation region is a frozen positive-finite input family with an explicit simulator, exact error condition and proved resource bound; it is not extrapolated beyond that region.
- **Unique surviving structure:** `complete-classical-quantum-Fold-model__positive-finite-resource-bound-region__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff`. Exact result: QLIMITX-006 uniquely retains positive-finite-resource-bound-region, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.

- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-BIDIRECTIONAL-SIMULATION-005. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-006: efficient_region; exact classical-quantum correspondence/limit execution: {"outside_region_claimed":false,"resource_bound":"two rows per depth","tested_depths":[1,2,3]}; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:bdae81c9a3dae2232caf1b5072f1a00f996bfa9a38f76cd7763c4161284770c0; independent implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fed12580a5a42d25dcde28c; independent certificate
 sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement sha256:6db2910bf7058be660f2aad0dee06dd4e424c5296493b2fe49df81d417dbef20; engine receipt sha256:1b166eda6ee3fb25ff456a662ac72c2ae24d4c2a168e6fb106b86100e9c8296a at receipts/engine/model_admitted/SFT-QUANTUM-QLIMITX-EFFICIENT-REGION-006-1b166eda6ee3fb25.js on.

SFT-QUANTUM-OBL-QLIMITX-007 - Phase-sensitive separation witness

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-PHASE-SEPARATION-007; Quantum Computation family QLIMITX; Phase-sensitive separation witness.
- **Forced law:** A phase-sensitive separation requires two states with identical word support but distinct retained phase whose later exact interference observations differ; deleting phase makes the simulator non-equivalent.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__same-word-distinct-phase-operational-witness__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLIMITX-007 uniquely retains same-word-distinct-phase-operational-witness, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-EFFICIENT-REGION-006. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-007: phase_separation; exact classical-quantum correspondence/limit execution: {"distinct_phase_classes":2,"phase_deletion_equivalent":false,"word_supports":1}; complete limit and handoff custody: 22

records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements

- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:da5fb2f8b73e8b89cb25d733802f810f08b6be9d9c25fedb0415158c90cbcd23; independent implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fefdd12580a5a42d25dcde28c; independent certificate
 sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement sha256:4ce0a2327d5734fce379e6c18e3d5a67d3aa576cc53574a1a70a38f6d9e15acd; engine receipt sha256:36ceb1d4b0a69ccb938467ad9096e76316672fce7594886055de0e454f7f2463 at receipts/engine/model_admitted/SFT-QUANTUM-QLIMITX-PHASE-SEPARATION-007-36ceb1d4b0a69ccb.js on.

SFT-QUANTUM-OBL-QLIMITX-008 - Entanglement-sensitive separation witness

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-ENTANGLEMENT-SEPARATION-008; Quantum Computation family QLIMITX; Entanglement-sensitive separation witness.
- **Forced law:** An entanglement-sensitive separation requires nonfactorable joint support whose complete correlations cannot be reconstructed from independent component marginals without the missing joint record.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__nonfactorable-joint-support-operational-witness__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLIMITX-008 uniquely retains nonfactorable-joint-support-operational-witness, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-PHASE-SEPARATION-007. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-008: entanglement_separation; exact classical-quantum correspondence/limit execution: {"factorable":false,"independent_product_rows":4,"joint_rows":2}; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS,

PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:42796b2eb5640e4f9df1521078e6e7711f25b5742c40f3248be8624aeaa70d07; independent implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fefed12580a5a42d25dcde28c; independent certificate
sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement sha256:232d4566343d8fc2a7cc93e64140465d00f1ff73a3203b020b33888449ecc4e; engine receipt sha256:eb35ea4e46230bfd10c53f9af278ade18e0f375b2f9f31a6356be78975a2c627 at receipts/engine/model_admitted/SFT-QUANTUM-QLIMITX-ENTANGLEMENT-SEPARATION-008-eb35ea4e46230bfd.json.

SFT-QUANTUM-OBL-QLIMITX-009 - No-cloning computational limit

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-NO-CLONING-009; Quantum Computation family QLIMITX; No-cloning computational limit.
- **Forced law:** A universal cloner cannot be a reversible source-bound map because distinct unknown phase/joint descriptions with the same observed word would collapse to one copied terminal-label image unless the supposedly unknown description is supplied.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__unknown-phase-support-copying-noninjectivity__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-not-hing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLIMITX-009 uniquely retains unknown-phase-support-copying-noninjectivity, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-ENTANGLEMENT-SEPARATION-008. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-009: no_cloning; exact classical-quantum correspondence/limit execution: {"distinct_unknown_phase_descriptions":2,"same_observed_word":true,"universal_reversible_cloner":false}; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no

bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:c54255335b57bbb614a32efc7f663b296ed2387ec8caa130c3cacdadf209afef; independent implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fef12580a5a42d25dcde28c; independent certificate sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement sha256:d1f464c88ad57bc3eb8852b6c81dd47ee6c3f6038ee950621b3eb2a4fd6add4b; engine receipt sha256:e24b56950bfc68daa0e65624757b9180cbb50ceb276410acba3e7eb0c4f6527e at receipts/engine/model_admitted/SFT-QUANTUM-QLIMITX-NO-CLONING-009-e24b56950bfc68da.json.

SFT-QUANTUM-OBL-QLIMITX-010 - Measurement disturbance computational boundary

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-MEASUREMENT-DISTURBANCE-010; Quantum Computation family QLIMITX; Measurement disturbance computational boundary.
- **Forced law:** Measurement closes distinctions outside the selected class; reversal therefore requires the retained source-support and phase record, and without it the prior state is computationally unavailable.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__observation-closure-and-record-cost__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLIMITX-010 uniquely retains observation-closure-and-record-cost, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-NO-CLONING-009. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-010: measurement_disturbance; exact classical-quantum correspondence/limit execution: {"closed_classes":1,"reverse_record_required":true,"selected_classes":1}; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:7e1b6a28c16aa61b7da7f78014132cc0efff97d7b60f690a68a36744e053e71b; independent implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fef12580a5a42d25dcde28c; independent certificate

sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement
 sha256:7f328125f2c245a13ba6849568c3bd93dd4eafda350ced9535fdd5ead4e25f23; engine receipt
 sha256:dlee1ea629331f0dccc805e5890f20b1aaf25fe8b6a32266e9ef0e371f47b1de at receipts/
 engine/model_admitted/SFT-QUANTUM-QLIMITX-MEASUREMENT-DISTURBANCE-010-dlee1ea62933
 1f0d.json.

SFT-QUANTUM-OBL-QLIMITX-011 - Quantum halting and self-reference transfer

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-HALTING-SELF-REFERENCE-011; Quantum Computation family QLIMITX; Quantum halting and self-reference transfer.
- **Forced law:** Because the quantum Fold machine contains the classical universal submodel, a purported total internal halting decider can be composed with the same exact self-negating process and cannot remain total and correct.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__quantum-process-self-negating-halting-transfer__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLIMITX-011 uniquely retains quantum-process-self-negating-halting-transfer, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-MEASUREMENT-DISTURBANCE-010. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-011: halting_self_reference; exact classical-quantum correspondence/limit execution: {"embedded_decider_answers":2,"self_negating_rows":2,"total_consistent_decider":false}; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
 sha256:067cc2f7d0d60d286b711b9bb71f5a486bac1d06ef211e22df5ad72a9384665e; independent
 implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fefbd12580a5a42d25dcde28c;
 independent certificate
 sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement
 sha256:14ca1ef2a1ee2bcf4d8b01401c36de3dbb5e6810b0217567637614dfc627c195; engine receipt
 sha256:f47b1c54b9109ad48e8fa13cf7f209bfff60e49cff3d49bc8bd714644c2fe9483 at receipts/
 engine/model_admitted/SFT-QUANTUM-QLIMITX-HALTING-SELF-REFERENCE-011-f47b1c54b9109
 ad4.json.

SFT-QUANTUM-OBL-QLIMITX-012 - Quantum undecidability boundary

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-UNDECIDABILITY-012; Quantum Computation family QLIMITX; Quantum undecidability boundary.
- **Forced law:** Every classical undecidable instance embeds in the quantum machine; a total quantum decider would decide the embedded classical family, so reversible branches and phase cannot remove the boundary.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__classical-undecidable-s
ubmodel-reduction__complete-source-phase-joint-observation-trace__complete-support
-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing
-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff.
Exact result: QLIMITX-012 uniquely retains classical-undecidable-submodel-reduction, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-HALTING-SELF-REFERENCE-011. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-012: undecidability; exact classical-quantum correspondence/limit execution: {"classical_embedding_complete":true,"quantum_total_decider":false}; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:ae19d938382bf31ef8fa739197d4db6d8a12373ebc5fcc456a97f54731fd1c01; independent
implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fef12580a5a42d25dcde28c;
independent certificate
sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement
sha256:1b5c23de514d877afb408ade7d8c7d41a157219b6c465f82fedc712d6b1155ef; engine receipt
sha256:29d7ca0b7be87c7945b4e4b0605e7d29a4535bf2eb8b554495bec5d5892047 at receipts/
engine/model_admitted/SFT-QUANTUM-QLIMITX-UNDECIDABILITY-012-29d7ca0b7be87c79.json.

SFT-QUANTUM-OBL-QLIMITX-013 - Quantum incompleteness boundary

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-INCOMPLETENESS-013; Quantum Computation family QLIMITX; Quantum incompleteness boundary.
- **Forced law:** Any sufficiently expressive finite quantum proof system inherits the self-description boundary: complete internal verification of every true statement would decide its embedded classical self-referential family.

- **Unique surviving structure:** complete-classical-quantum-Fold-model__finite-proof-system-self-description-boundary__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLIMITX-013 uniquely retains finite-proof-system-self-description-boundary, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-UNDECIDABILITY-012. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-013: incompleteness; exact classical-quantum correspondence/limit execution: {"complete_internal_self_truth":false,"finite_proof_system":true}; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:da2385651019681eb0f2f36643c05f479259306dc4fef81c516fa742f0be490b; independent implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fefbd12580a5a42d25dcde28c; independent certificate
sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement sha256:a526158112244e545032c0de37dee2119f7fef27b739f9085224c0a7e98b4473; engine receipt sha256:c158dc4afca40053535c31975bcbca6c0970b4a536cd12e52e54cc0690e08f42 at receipts/engine/model_admitted/SFT-QUANTUM-QLIMITX-INCOMPLETENESS-013-c158dc4afca40053.json.

SFT-QUANTUM-OBL-QLIMITX-014 - No-hypercomputation boundary

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-NO-HYPERCOMPUTATION-014; Quantum Computation family QLIMITX; No-hypercomputation boundary.
- **Forced law:** A finite generated quantum process remains a computable transition system; superposition-equivalent support and interference do not supply an unregistered oracle, completed infinity or noncomputable transition.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__finite-generated-transition-no-oracle-boundary__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLIMITX-014 uniquely retains finite-generated-transition-no-oracle-boundary, complete boundary custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-INCOMPLETENESS-013. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-014: no_hypercomputation; exact classical-quantum correspondence/limit execution: {"finite_generated_transition":true,"hypercomputation_claim":false,"unregistered_oracle":false}; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:44d7fef2af8a9ed5c015523d110bf2085081327d3b347eac7ac6ba0f799a6b83; independent implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fefbd12580a5a42d25dcde28c; independent certificate
sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement sha256:d40861d624071aa68017d082d8322687f91c5eb23b0978b0ff18103ee8b26967; engine receipt sha256:2db3d79829f3ee699197e369c9968d1c4ff376c96be97905d33849b333dbc030 at receipts/engine/model_admitted/SFT-QUANTUM-QLIMITX-NO-HYPERCOMPUTATION-014-2db3d79829f3ee69.json.

SFT-QUANTUM-OBL-QLIMITX-015 - Finite support and completed-infinity prohibition

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-FINITE-SUPPORT-015; Quantum Computation family QLIMITX; Finite support and completed-infinity prohibition.
- **Forced law:** Every admitted quantum support is positive finite and extended by a successor certificate; no completed infinite register, continuum branch set or infinite parallel evaluation is created.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__positive-finite-success-or-support-only__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLIMITX-015 uniquely retains positive-finite-successor-support-only, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 ->

SFT-QUANTUM-QLIMITX-NO-HYPERCOMPUTATION-014. Registered root:

SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** QLIMITX-015: finite_support; exact classical-quantum correspondence/limit execution: {"completed_infinity":false,"support_counts":[2,4,8,16],"tested_widths":[1,2,3,4]}; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:7ac57d2a5c2bb6b141ac7352e563af4eea01bc104d2ecebc1044e72485ab4ca1; independent implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fef12580a5a42d25dcde28c; independent certificate
sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement sha256:5eb40c9a2fb1a3c84edd6a41f185861c3628f4d6b64808c4279d765dafa3e6b7; engine receipt sha256:f0433854b807b74562a3c6dc2d9ea09ccf7e9484cd87c29dc3f4377f62420487 at receipts/engine/model_admitted/SFT-QUANTUM-QLIMITX-FINITE-SUPPORT-015-f0433854b807b745.json.

SFT-QUANTUM-OBL-QLIMITX-016 - No unrestricted advantage from bounded examples

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-NO-UNRESTRICTED-ADVANTAGE-016; Quantum Computation family QLIMITX; No unrestricted advantage from bounded examples.
- **Forced law:** A bounded algorithm or circuit census proves only its frozen family; unrestricted quantum advantage requires a general upper/lower-bound certificate and cannot be inferred from finite favorable examples.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__bounded-separation-scope-custody__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLIMITX-016 uniquely retains bounded-separation-scope-custody, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-FINITE-SUPPORT-015. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-016: bounded_example_limit; exact classical-quantum correspondence/limit execution: {"bounded_depths":[1,2,3],"unrestricted_advantage_claimed":false}; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements

- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:55926712fa37b40a67f36a8609126c2cb6884e03c456a0df59b608404e891b45; independent implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fefbd12580a5a42d25dcde28c; independent certificate
sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement sha256:75a53cfc172b5b7efce1b5bdd43907cf95460381eb045a22421e7e448a132f9; engine receipt sha256:8b69498005cf382b784c448e4c7718a1a6c022d3f1d8c3c332e54899d7cd9b5b at receipts/engine/model_admitted/SFT-QUANTUM-QLIMITX-NO-UNRESTRICTED-ADVANTAGE-016-8b69498005cf382b.json.

SFT-QUANTUM-OBL-QLIMITX-017 - No physical speedup without device measurement

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-NO-UNMEASURED-SPEEDUP-017; Quantum Computation family QLIMITX; No physical speedup without device measurement.
- **Forced law:** Formal query, gate or depth separation is not physical speedup; physical speedup additionally requires sealed implementations and measured timing, error, energy and control overhead on the same task.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__formal-resource-to-measured-device-speedup-handoff__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLIMITX-017 uniquely retains formal-resource-to-measured-device-speedup-handoff, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-NO-UNRESTRICTED-ADVANTAGE-016. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-017: physical_speedup_handoff; exact classical-quantum correspondence/limit execution: {"device_timing_present":false,"formal_resource_separation":true,"measurement_required":true}; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:9e6881f01b50bfcaf91114fd19371e2e8dab131313e13677da440b97548426d7`; independent implementation `sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fe9d12580a5a42d25dcde28c`; independent certificate
`sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1`; measurement `sha256:eb93dc147e19f7065733b027fa65500ad6464d737bb65113682cf8d25573feb4`; engine receipt `sha256:54cf1ce61fef584862d1941b57662cb8fff20527a0eb849033ad6cdbc28987e8` at `receipts/engine/model_admitted/SFT-QUANTUM-QLIMITX-NO-UNMEASURED-SPEEDUP-017-54cf1ce61fef5848.json`.

SFT-QUANTUM-OBL-QLIMITX-018 - No hardware threshold from a formal code theorem

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-NO-FORMAL-HARDWARE-THRESHOLD-018; Quantum Computation family QLIMITX; No hardware threshold from a formal code theorem.
- **Forced law:** The exact width $2t+1$ code theorem supplies no hardware threshold constant; a threshold requires measured correlated faults, leakage, geometry, decoder behavior and device operations.
- **Unique surviving structure:** `complete-classical-quantum-Fold-model__formal-fault-order-to-measured-threshold-handoff__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff`. Exact result: QLIMITX-018 uniquely retains formal-fault-order-to-measured-threshold-handoff, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-NO-UNMEASURED-SPEEDUP-017. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-018: `hardware_threshold_handoff`; exact classical-quantum correspondence/limit execution: `{"formal_code_width":"2t+1","hardware_threshold_present":false,"measurement_required":true}`; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.

- **Evidence identities:** derivation

sha256:6f94f797ed1ed1f32353ef7ec2586a9828b41c60b39f93d3b7bf80debc5598ad; independent implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fe fd12580a5a42d25dcde28c; independent certificate
sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement sha256:79b47657a12cad96833be2904116cae87485158c0e9b08515d94262caa044f8d; engine receipt sha256:8bf2ab0b75e1b7b7febb67b936aca8287f252acf62e711cceb1814cb94c31ddb at receipts/engine/model_admitted/SFT-QUANTUM-QLIMITX-NO-FORMAL-HARDWARE-THRESHOLD-018-8bf2ab0b75e1b7b7.json.

SFT-QUANTUM-OBL-QLIMITX-019 - Energy, timing and implementation handoff

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-IMPLEMENTATION-HANDOFF-019; Quantum Computation family QLIMITX; Energy, timing and implementation handoff.
- **Forced law:** Energy, wall time, control precision, temperature, geometry and fabrication are implementation measurements and cannot be read from abstract transition counts without an owning physical law.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__formal-resource-ledger-to-implementation-measurement__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLIMITX-019 uniquely retains formal-resource-ledger-to-implementation-measurement, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-NO-FORMAL-HARDWARE-THRESHOLD-018. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-019: implementation_handoff; exact classical-quantum correspondence/limit execution: {"measurement_classes":["energy","timing","geometry","control","temperature"],"values_present":false}; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:1670c94644fee6f45a25fd7eb71e2b7743e7f379d698d13b8401ab375d92f3e8; independent implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fe fd12580a5a42d25dcde28c; independent certificate
sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement sha256:884c4422eb424bc5bc01d840db20608a50a4cb12e2d6faa72450d328c2e92258; engine receipt sha256:349bfa13c73b0182eb1454536efa2b1fa582f43ecb30341fbb3bda2bc882799b at receipts/

engine/model_admitted/SFT-QUANTUM-QLIMITX-IMPLEMENTATION-HANDOFF-019-349bfa13c73b0182.json.

SFT-QUANTUM-OBL-QLIMITX-020 - Quantum-to-Physics measurement boundary

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-PHYSICS-MEASUREMENT-BOUNDARY-020; Quantum Computation family QLIMITX; Quantum-to-Physics measurement boundary.
- **Forced law:** Quantum Computation supplies sealed operational predictions and resource identities; Physics owns target preparation, units, uncertainty, apparatus, blind measurement and adverse-result interpretation.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__sealed-formal-prediction-to-blind-physical-measurement__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLIMITX-020 uniquely retains sealed-formal-prediction-to-blind-physical-measurement, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-IMPLEMENTATION-HANDOFF-019. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-020: physics_boundary; exact classical-quantum correspondence/limit execution: {"blind_measurement_owner":"physics","formal_prediction_sealed":true,"physical_target_identity_sealed":true}; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:9a9496d393098c97837c317bf785179eaab32b678ae779b09a55ad2f9cdd1f7b; independent implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4feffd12580a5a42d25dcde28c; independent certificate
sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement sha256:c982b4d6fa9575a7d8aeb40b0f0e285a7bb27dbec08cea7039c15cbcd16775b9; engine receipt sha256:bd7fb477471ca1b208e034d7b05723782ea84110187bf53a88f0281fca4b74a2 at receipts/engine/model_admitted/SFT-QUANTUM-QLIMITX-PHYSICS-MEASUREMENT-BOUNDARY-020-bd7fb477471ca1b2.json.

SFT-QUANTUM-OBL-QLIMITX-021 - Open extension and falsification boundary

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-OPEN-FALSIFICATION-021; Quantum Computation family QLIMITX; Open extension and falsification boundary.

- **Forced law:** Quantum closure is dated to the frozen census and remains open to a new value-free obligation, complete enumeration, adverse controls, independent reconstruction and post-registration falsification.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__dated-closure-open-lawful-extension__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff. Exact result: QLIMITX-021 uniquely retains dated-closure-open-lawful-extension, complete boundary custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-PHYSICS-MEASUREMENT-BOUNDARY-020. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-021: open_falsification; exact classical-quantum correspondence/limit execution: {"adverse_rows_preserved":true,"dated_closure":true,"lawful_extensions_allowed":true}; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
sha256:eea612cc2498c1b700d7bf1ad562da3d5bdd61eed18cc8b139f587d41a868bed; independent implementation sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fefbd12580a5a42d25dcde28c; independent certificate
sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1; measurement sha256:d4660a71a1f0c4b7b59184e1968f15886f9363949bef45abca70e6bf8f3b424c; engine receipt sha256:b2fb7db464e8881aa24e1b22b3d79852ae7ac258a9a042d71cd4287c642a55e5 at receipts/engine/model_admitted/SFT-QUANTUM-QLIMITX-OPEN-FALSIFICATION-021-b2fb7db464e8881a.json.

SFT-QUANTUM-OBL-QLIMITX-022 - Quantum computation limits completeness certificate

- **Claim and ownership:** SFT-QUANTUM-QLIMITX-COMPLETENESS-022; Quantum Computation family QLIMITX; Quantum computation limits completeness certificate.
- **Forced law:** QLIMITX is complete exactly when all twenty-two frozen correspondence and limit obligations have one owner, one unique survivor, controls, post-registry observations, independent reconstructions and current receipts.
- **Unique surviving structure:** complete-classical-quantum-Fold-model__twenty-two-obligation-no-omission-ledger__complete-source-phase-joint-observation-trace__complete-support-time-space-query-and-record-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__explicit-formal-finite-physical-handoff.

Exact result: QLIMITX-022 uniquely retains twenty-two-obligation-no-omission-ledger, complete boundary custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete correspondence/limit grammar, exact witness or explicit physical handoff, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLEARNX-COMPLETENESS-022 -> SFT-QUANTUM-LIMITS-001 -> SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-QUANTUM-QLIMITX-OPEN-FALSIFICATION-021. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** QLIMITX-022: `limits_completeness`; exact classical-quantum correspondence/limit execution: `{"duplicate_owners":0,"execution_rows":22,"omitted_owners":0,"registered_obligations":22}`; complete limit and handoff custody: 22 records; all rows preserved; exact finite limit witnesses supply formal observations; physical speedup, energy, timing, fidelity and thresholds remain explicit owning-domain measurements
- **Sources and custody:** SFT-V3-ADMITTED-QLEARNX-COMPLETENESS-022, SFT-V3-ADMITTED-QUANTUM-LIMITS-001, SFT-V3-ADMITTED-COMPUTABILITY-AND-INCOMPLETENESS-FAMILIES, SFT-V1-V2-CLASSICAL-QUANTUM-LIMITS-COMPARISON-CORPUS, PHYSICS-DEVICE-ENERGY-TIMING-THRESHOLD-MEASUREMENT-HANDOFFS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no imported Hilbert-space limit theorem, oracle, complexity separation or conventional impossibility proof selects the law; host 0 denotes absence only and is not a numerical-zero state, probability, runtime, energy or threshold; no negative, irrational, imaginary, floating, fitted or completed-infinite proof scalar; no bounded-example extrapolation, hidden oracle, discarded measurement record, omitted adverse row or ontic randomness; no physical speedup, energy, timing, fidelity or threshold is inferred without its owning measurement handoff; no first failure retires an obligation or changes the protected authority.
- **Evidence identities:** derivation
`sha256:4b6c7d3195662e1ac82347228c9162c9477a04eca36e3f93c4880dd113a6de7d`; independent implementation `sha256:59e02644e5d0c6b1c0b1c4f19c4cc91640b068b4fef12580a5a42d25dcde28c`; independent certificate
`sha256:83b92d285dd66bb2e53c5b2a04b9d384c054a089f3848f47fa9a4e9d4f4859a1`; measurement `sha256:b4e36f517a5bee11ec96e353a428961736472feae7a64c1b8bf2ef0ed0355019`; engine receipt `sha256:b16d42d36e8e0925408519a05d95c13308a5b24e41ce2f8cfbced8ce800132a0` at `receipts/engine/model_admitted/SFT-QUANTUM-QLIMITX-COMPLETENESS-022-b16d42d36e8e0925.json`.

36.14 Family VALID - 12/12 complete

Scope. Dated full-field validation and grand lock. This family decides 3,072 candidates, retains 12 unique survivors and passes 48 adverse controls.

SFT-QUANTUM-OBL-VALID-001 - Reversible-computation complete validation vector

- **Claim and ownership:** SFT-QUANTUM-VALID-REVERSIBLE-001; Quantum Computation family VALID; Reversible-computation complete validation vector.
- **Forced law:** The reversible-computation validation vector reproduces every REVX receipt, candidate census, control, observation and independent certificate without omission.
- **Unique surviving structure:** `complete-frozen-family-coverage__complete-reversible-family-receipt-vector__current-reproduced-engine-receipt__complete-control-observation-independent-certificate__literal-complete-product__there-is-no-nothing-lineage__all-status-and-ownership-rows-preserved__dated-lock-explicit-handoffs-open-extension`. Exact result: VALID-001 uniquely retains complete-reversible-family-receipt-vector, current receipt replay, full adverse custody and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete current receipt/evidence replay, adverse/ownership vector, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLIMITX-COMPLETENESS-022. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** VALID-001: `revx_vector`; exact Quantum family validation replay: `{"all_rows_replayed":true,"controls_and_certificates_present":true,"family":"RE VX","receipt_rows":18}`; complete formal/empirical/adverse custody: 12 vectors; all rows preserved; the Grand Lock binds current receipts and evidence without inventing downstream measured values or closing lawful extensions
- **Sources and custody:** QUANTUM-RECONCILIATION-V11, ALL-CURRENT-QUANTUM-FAMILY-AUDITS, ALL-CURRENT-QUANTUM-RECEIPTS, ALL-CURRENT-QUANTUM-ADVERSE-AND-HANDOFF-ROWS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no selected receipt, favorable-only vector or handwritten closure status; host 0 denotes absence only and is not a numerical-zero proof object; no negative irrational imaginary floating fitted or completed-infinite proof scalar; no omitted halt adverse absent unavailable unresolved or standing-prediction row; no physical value is invented at a formal handoff; no verifier engine or frozen census mutation.
- **Evidence identities:** derivation
`sha256:0f6b480dd9e3b401196f4af565ce750755ffaeea711e3579534e5179c4fa5c7c`; independent implementation `sha256:954e25b97cce44a272e3aad5929024c3936ab783fdc89b9830f854630e0c774a`; independent certificate
`sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17`; measurement `sha256:4df34b4704bbf976bf41336b0b949055a6b334616120a5b1090bf4d0cb318f57`; engine receipt `sha256:48792da96d5f5b8ce8fa11b5363ce2cf8a3c548bf3465c304f43ad43f699cb47` at `receipts/engine/model_admitted/SFT-QUANTUM-VALID-REVERSIBLE-001-48792da96d5f5b8c.json`.

SFT-QUANTUM-OBL-VALID-002 - Quantum-state-structure complete validation vector

- **Claim and ownership:** SFT-QUANTUM-VALID-STATE-002; Quantum Computation family VALID; Quantum-state-structure complete validation vector.
- **Forced law:** The state-structure validation vector reproduces every QSTATEX support, phase, interference, entanglement, measurement, receipt and adverse row.
- **Unique surviving structure:** `complete-frozen-family-coverage__complete-state-family-receipt-vector__current-reproduced-engine-receipt__complete-control-observation-independent-certificate__literal-complete-product__there-is-no-nothing-lineage__all-status-and-ownership-rows-preserved__dated-lock-explicit-handoffs-open-extension`. Exact result: VALID-002 uniquely retains complete-state-family-receipt-vector, current receipt replay, full adverse custody and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete current receipt/evidence replay, adverse/ownership vector, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLIMITX-COMPLETENESS-022 -> SFT-QUANTUM-VALID-REVERSIBLE-001. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** VALID-002: `qstatex_vector`; exact Quantum family validation replay: `{"all_rows_replayed":true,"controls_and_certificates_present":true,"family":"QSTATEX","receipt_rows":28}`; complete formal/empirical/adverse custody: 12 vectors; all rows preserved; the Grand Lock binds current receipts and evidence without inventing downstream measured values or closing lawful extensions
- **Sources and custody:** QUANTUM-RECONCILIATION-V11, ALL-CURRENT-QUANTUM-FAMILY-AUDITS, ALL-CURRENT-QUANTUM-RECEIPTS, ALL-CURRENT-QUANTUM-ADVERSE-AND-HANDOFF-ROWS.

External status: empirically_tested_and_independently_replicated; all rows preserved: True.

- **Prohibited imports and boundary:** no selected receipt, favorable-only vector or handwritten closure status; host 0 denotes absence only and is not a numerical-zero proof object; no negative irrational imaginary floating fitted or completed-infinite proof scalar; no omitted halt adverse absent unavailable unresolved or standing-prediction row; no physical value is invented at a formal handoff; no verifier engine or frozen census mutation.
- **Evidence identities:** derivation
sha256:b08625dcd2706cd819288b3842eb59d551a8c6567a40553faf7cac56373fcd1a; independent
implementation sha256:954e25b97cce44a272e3aad5929024c3936ab783fdc89b9830f854630e0c774a;
independent certificate
sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17; measurement
sha256:4ca9f7768618bed038bdc83acd933d8d088d2aaf825023600f8bad69d3b8807d; engine receipt
sha256:b51f0b544cad8490da7ab2b3bbf691b2e4cad802ee5f99b6e4497e7261092154 at
receipts/engine/model_admitted/SFT-QUANTUM-VALID-STATE-002-b51f0b544cad8490.json.

SFT-QUANTUM-OBL-VALID-003 - Gate-and-circuit complete validation vector

- **Claim and ownership:** SFT-QUANTUM-VALID-GATE-CIRCUIT-003; Quantum Computation family VALID; Gate-and-circuit complete validation vector.
- **Forced law:** The gate/circuit validation vector reproduces every GATEX transformation, circuit semantic, universality, resource, receipt and control row.
- **Unique surviving structure:** complete-frozen-family-coverage__complete-gate-circuit-family-receipt-vector__current-reproduced-engine-receipt__complete-control-observation-in-dependent-certificate__literal-complete-product__there-is-no-nothing-lineage__all-status-and-ownership-rows-preserved__dated-lock-explicit-handoffs-open-extension. Exact result: VALID-003 uniquely retains complete-gate-circuit-family-receipt-vector, current receipt replay, full adverse custody and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete current receipt/evidence replay, adverse/ownership vector, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLIMITX-COMPLETENESS-022 -> SFT-QUANTUM-VALID-STATE-002. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** VALID-003: gatex_vector; exact Quantum family validation replay:
{ "all_rows_replayed":true, "controls_and_certificates_present":true, "family":"GATEX", "receipt_rows":22 }; complete formal/empirical/adverse custody: 12 vectors; all rows preserved; the Grand Lock binds current receipts and evidence without inventing downstream measured values or closing lawful extensions
- **Sources and custody:** QUANTUM-RECONCILIATION-V11, ALL-CURRENT-QUANTUM-FAMILY-AUDITS, ALL-CURRENT-QUANTUM-RECEIPTS, ALL-CURRENT-QUANTUM-ADVERSE-AND-HANDOFF-ROWS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no selected receipt, favorable-only vector or handwritten closure status; host 0 denotes absence only and is not a numerical-zero proof object; no negative irrational imaginary floating fitted or completed-infinite proof scalar; no omitted halt adverse absent unavailable unresolved or standing-prediction row; no physical value is invented at a formal handoff; no verifier engine or frozen census mutation.
- **Evidence identities:** derivation
sha256:82335e86a9b8d7c9ab1a1634faaaf2f733d5fca64a68982b385ad146adae0cef; independent
implementation sha256:954e25b97cce44a272e3aad5929024c3936ab783fdc89b9830f854630e0c774a;
independent certificate
sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17; measurement
sha256:e20acb7a1387a1351e547910555e39a4888e266c955fabd793446d2644c3001a; engine receipt
sha256:9b72cec0cdf572064df960865cd370007c6b516537971cfac8bf85f62ee97865 at receipts/
engine/model_admitted/SFT-QUANTUM-VALID-GATE-CIRCUIT-003-9b72cec0cdf57206.json.

SFT-QUANTUM-OBL-VALID-004 - Quantum-algorithm complete validation vector

- **Claim and ownership:** SFT-QUANTUM-VALID-ALGORITHM-004; Quantum Computation family VALID; Quantum-algorithm complete validation vector.
- **Forced law:** The algorithm validation vector reproduces all QALGX families, exact traces, resource comparisons, bounds, receipts and adverse cases.
- **Unique surviving structure:** complete-frozen-family-coverage__complete-algorithm-family-receipt-vector__current-reproduced-engine-receipt__complete-control-observation-independent-certificate__literal-complete-product__there-is-no-nothing-lineage__all-status-and-ownership-rows-preserved__dated-lock-explicit-handoffs-open-extension. Exact result: VALID-004 uniquely retains complete-algorithm-family-receipt-vector, current receipt replay, full adverse custody and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete current receipt/evidence replay, adverse/ownership vector, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLIMITX-COMPLETENESS-022 -> SFT-QUANTUM-VALID-GATE-CIRCUIT-003. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** VALID-004: qalgx_vector; exact Quantum family validation replay: {"all_rows_replayed":true,"controls_and_certificates_present":true,"family":"QALGX","receipt_rows":30}; complete formal/empirical/adverse custody: 12 vectors; all rows preserved; the Grand Lock binds current receipts and evidence without inventing downstream measured values or closing lawful extensions
- **Sources and custody:** QUANTUM-RECONCILIATION-V11, ALL-CURRENT-QUANTUM-FAMILY-AUDITS, ALL-CURRENT-QUANTUM-RECEIPTS, ALL-CURRENT-QUANTUM-ADVERSE-AND-HANDOFF-ROWS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no selected receipt, favorable-only vector or handwritten closure status; host 0 denotes absence only and is not a numerical-zero proof object; no negative irrational imaginary floating fitted or completed-infinite proof scalar; no omitted halt adverse absent unavailable unresolved or standing-prediction row; no physical value is invented at a formal handoff; no verifier engine or frozen census mutation.
- **Evidence identities:** derivation
sha256:4929a8ae9040ba593a22898bdfd330b70dca9e561216541d48ed9599a6904179; independent implementation sha256:954e25b97cce44a272e3aad5929024c3936ab783fdc89b9830f854630e0c774a; independent certificate
sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17; measurement sha256:bc6a1ded1bb189edb666e5c27c81cd9bce12fba73b29ce632127cc7c3e40dab9; engine receipt sha256:81c4b69f3fff4896eb168b883d002c69e654db101b805b52e01097608338a6178 at receipts/engine/model_admitted/SFT-QUANTUM-VALID-ALGORITHM-004-81c4b69f3fff4896e.json.

SFT-QUANTUM-OBL-VALID-005 - Quantum-complexity complete validation vector

- **Claim and ownership:** SFT-QUANTUM-VALID-COMPLEXITY-005; Quantum Computation family VALID; Quantum-complexity complete validation vector.
- **Forced law:** The complexity validation vector reproduces all QCPLXX definitions, reductions, upper/lower bounds, scope limits, receipts and controls.
- **Unique surviving structure:** complete-frozen-family-coverage__complete-complexity-family-receipt-vector__current-reproduced-engine-receipt__complete-control-observation-independent-certificate__literal-complete-product__there-is-no-nothing-lineage__all-status-and-ownership-rows-preserved__dated-lock-explicit-handoffs-open-extension. Exact result: VALID-005 uniquely retains complete-complexity-family-receipt-vector, current receipt replay, full adverse custody and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete current receipt/evidence replay, adverse/ownership vector, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLIMITX-COMPLETENESS-022 -> SFT-QUANTUM-VALID-ALGORITHM-004. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** VALID-005: `qcplx_vector`; exact Quantum family validation replay: `{"all_rows_replayed":true,"controls_and_certificates_present":true,"family":"QCPLXX","receipt_rows":26}`; complete formal/empirical/adverse custody: 12 vectors; all rows preserved; the Grand Lock binds current receipts and evidence without inventing downstream measured values or closing lawful extensions
- **Sources and custody:** QUANTUM-RECONCILIATION-V11, ALL-CURRENT-QUANTUM-FAMILY-AUDITS, ALL-CURRENT-QUANTUM-RECEIPTS, ALL-CURRENT-QUANTUM-ADVERSE-AND-HANDOFF-ROWS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no selected receipt, favorable-only vector or handwritten closure status; host 0 denotes absence only and is not a numerical-zero proof object; no negative irrational imaginary floating fitted or completed-infinite proof scalar; no omitted halt adverse absent unavailable unresolved or standing-prediction row; no physical value is invented at a formal handoff; no verifier engine or frozen census mutation.
- **Evidence identities:** derivation
`sha256:1e5e850fe84cb6c871049632fdd909f8644fa6a160c5ef998a1f35b35c326e1b`; independent implementation `sha256:954e25b97cce44a272e3aad5929024c3936ab783fdc89b9830f854630e0c774a`; independent certificate
`sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17`; measurement `sha256:3a1c227c98de8b5cca0df49ed66b7469d9e79ce24f52c312a04006f0d5e56aee`; engine receipt `sha256:165a3b9abd922134c3d6522cccc2a15205078207b1980594f7fd16d5fc964046` at `receipts/engine/model_admitted/SFT-QUANTUM-VALID-COMPLEXITY-005-165a3b9abd922134.json`.

SFT-QUANTUM-OBL-VALID-006 - Quantum-communication and security complete validation vector

- **Claim and ownership:** SFT-QUANTUM-VALID-COMMUNICATION-006; Quantum Computation family VALID; Quantum-communication and security complete validation vector.
- **Forced law:** The communication/security vector reproduces every QCOMM channel, protocol, adversary, network, security boundary, receipt and transcript.
- **Unique surviving structure:** `complete-frozen-family-coverage__complete-communication-family-receipt-vector__current-reproduced-engine-receipt__complete-control-observation-independent-certificate__literal-complete-product__there-is-no-nothing-lineage__all-status-and-ownership-rows-preserved__dated-lock-explicit-handoffs-open-extension`. Exact result: VALID-006 uniquely retains complete-communication-family-receipt-vector, current receipt replay, full adverse custody and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete current receipt/evidence replay, adverse/ownership vector, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLIMITX-COMPLETENESS-022 -> SFT-QUANTUM-VALID-COMPLEXITY-005. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** VALID-006: `qcommx_vector`; exact Quantum family validation replay: `{"all_rows_replayed":true,"controls_and_certificates_present":true,"family":"QCOMM","receipt_rows":24}`; complete formal/empirical/adverse custody: 12 vectors; all rows preserved; the Grand Lock binds current receipts and evidence without inventing downstream measured values or closing lawful extensions

- **Sources and custody:** QUANTUM-RECONCILIATION-V11, ALL-CURRENT-QUANTUM-FAMILY-AUDITS, ALL-CURRENT-QUANTUM-RECEIPTS, ALL-CURRENT-QUANTUM-ADVERSE-AND-HANDOFF-ROWS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no selected receipt, favorable-only vector or handwritten closure status; host 0 denotes absence only and is not a numerical-zero proof object; no negative irrational imaginary floating fitted or completed-infinite proof scalar; no omitted halt adverse absent unavailable unresolved or standing-prediction row; no physical value is invented at a formal handoff; no verifier engine or frozen census mutation.
- **Evidence identities:** derivation
sha256:979d88a9b52c9090c3758646c78c08b331f05283f48fee293e46d2cb2f8069b1; independent implementation sha256:954e25b97cce44a272e3aad5929024c3936ab783fdc89b9830f854630e0c774a; independent certificate
sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17; measurement sha256:f01d489d9ca3955758ecdaf270b1ae241c11adbe8f7291523e98eac872b91ff0; engine receipt sha256:6952f6ac6b8d2cf3cf220baaeaf0e8be0d9bd4179f8927881e2351a22a658bfb at receipts/engine/model_admitted/SFT-QUANTUM-VALID-COMMUNICATION-006-6952f6ac6b8d2cf3.json.

SFT-QUANTUM-OBL-VALID-007 - Quantum-coding and fault-tolerance complete validation vector

- **Claim and ownership:** SFT-QUANTUM-VALID-CODING-007; Quantum Computation family VALID; Quantum-coding and fault-tolerance complete validation vector.
- **Forced law:** The coding vector reproduces all QCODEX codes, exhaustive masks, recoveries, fault boundaries, threshold handoffs, receipts and controls.
- **Unique surviving structure:** complete-frozen-family-coverage__complete-coding-family-receipt-vector__current-reproduced-engine-receipt__complete-control-observation-independent-certificate__literal-complete-product__there-is-no-nothing-lineage__all-status-and-ownership-rows-preserved__dated-lock-explicit-handoffs-open-extension. Exact result: VALID-007 uniquely retains complete-coding-family-receipt-vector, current receipt replay, full adverse custody and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete current receipt/evidence replay, adverse/ownership vector, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLIMITX-COMPLETENESS-022 -> SFT-QUANTUM-VALID-COMMUNICATION-006. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** VALID-007: qcodex_vector; exact Quantum family validation replay:
{ "all_rows_replayed":true, "controls_and_certificates_present":true, "family":"QCODEX", "receipt_rows":32 }; complete formal/empirical/adverse custody: 12 vectors; all rows preserved; the Grand Lock binds current receipts and evidence without inventing downstream measured values or closing lawful extensions
- **Sources and custody:** QUANTUM-RECONCILIATION-V11, ALL-CURRENT-QUANTUM-FAMILY-AUDITS, ALL-CURRENT-QUANTUM-RECEIPTS, ALL-CURRENT-QUANTUM-ADVERSE-AND-HANDOFF-ROWS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no selected receipt, favorable-only vector or handwritten closure status; host 0 denotes absence only and is not a numerical-zero proof object; no negative irrational imaginary floating fitted or completed-infinite proof scalar; no omitted halt adverse absent unavailable unresolved or standing-prediction row; no physical value is invented at a formal handoff; no verifier engine or frozen census mutation.
- **Evidence identities:** derivation
sha256:53c08d4d3be76447e12caa4cc1cdf881f725f17e5e548ef0838dee17dffa7bab; independent implementation sha256:954e25b97cce44a272e3aad5929024c3936ab783fdc89b9830f854630e0c774a; independent certificate
sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17; measurement sha256:eadba8d0721a2fd4dc28dfce88ec1d06d0afbb4cfade97e05449240d0f4e388a; engine receipt

sha256:d57b2f167d86fff531433017aa2233cdf5586539a679277aa80d6fdb69cc1e62 at
receipts/engine/model_admitted/SFT-QUANTUM-VALID-CODING-007-d57b2f167d86fff5.json.

SFT-QUANTUM-OBL-VALID-008 - Quantum-simulation and verification complete validation vector

- **Claim and ownership:** SFT-QUANTUM-VALID-SIMULATION-008; Quantum Computation family VALID; Quantum-simulation and verification complete validation vector.
- **Forced law:** The simulation/verification vector reproduces all QSIMX models, traces, verification transcripts, handoffs, receipts and adverse rows.
- **Unique surviving structure:** complete-frozen-family-coverage__complete-simulation-family-receipt-vector__current-reproduced-engine-receipt__complete-control-observation-independent-certificate__literal-complete-product__there-is-no-nothing-lineage__all-status-and-ownership-rows-preserved__dated-lock-explicit-handoffs-open-extension. Exact result: VALID-008 uniquely retains complete-simulation-family-receipt-vector, current receipt replay, full adverse custody and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete current receipt/evidence replay, adverse/ownership vector, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLIMITX-COMPLETENESS-022 -> SFT-QUANTUM-VALID-CODING-007. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** VALID-008: qsimx_vector; exact Quantum family validation replay: {"all_rows_replayed":true,"controls_and_certificates_present":true,"family":"QSIMX","receipt_rows":24}; complete formal/empirical/adverse custody: 12 vectors; all rows preserved; the Grand Lock binds current receipts and evidence without inventing downstream measured values or closing lawful extensions
- **Sources and custody:** QUANTUM-RECONCILIATION-V11, ALL-CURRENT-QUANTUM-FAMILY-AUDITS, ALL-CURRENT-QUANTUM-RECEIPTS, ALL-CURRENT-QUANTUM-ADVERSE-AND-HANDOFF-ROWS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no selected receipt, favorable-only vector or handwritten closure status; host 0 denotes absence only and is not a numerical-zero proof object; no negative irrational imaginary floating fitted or completed-infinite proof scalar; no omitted halt adverse absent unavailable unresolved or standing-prediction row; no physical value is invented at a formal handoff; no verifier engine or frozen census mutation.
- **Evidence identities:** derivation
sha256:202b4d93e8292427223ac91947c85f12a7ecca5c16551fe58e74ef84fef5934b; independent implementation sha256:954e25b97cce44a272e3aad5929024c3936ab783fdc89b9830f854630e0c774a; independent certificate
sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17; measurement sha256:92b68d1fad9f79256772b8066ca6326f5cd10d4d703a73e1f8ee6f29c9adef56; engine receipt sha256:902817b43c056bcad5a32aa52a5b32311283db1d396ce77ba8b6b801ee01ac09 at receipts/engine/model_admitted/SFT-QUANTUM-VALID-SIMULATION-008-902817b43c056bca.json.

SFT-QUANTUM-OBL-VALID-009 - Quantum-learning complete validation vector

- **Claim and ownership:** SFT-QUANTUM-VALID-LEARNING-009; Quantum Computation family VALID; Quantum-learning complete validation vector.
- **Forced law:** The learning vector reproduces all QLEARNX example, hypothesis, query, held-out, advantage-boundary, receipt and robustness rows.
- **Unique surviving structure:** complete-frozen-family-coverage__complete-learning-family-receipt-vector__current-reproduced-engine-receipt__complete-control-observation-independent-certificate__literal-complete-product__there-is-no-nothing-lineage__all-status-and-ownership-rows-preserved__dated-lock-explicit-handoffs-open-extension. Exact result: VALID-009 uniquely retains complete-learning-family-receipt-vector, current receipt replay, full adverse custody

and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete current receipt/evidence replay, adverse/ownership vector, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLIMITX-COMPLETENESS-022 -> SFT-QUANTUM-VALID-SIMULATION-008. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** VALID-009: `qlearnx_vector`; exact Quantum family validation replay: `{"all_rows_replayed":true,"controls_and_certificates_present":true,"family":"QLEARNX","receipt_rows":22}`; complete formal/empirical/adverse custody: 12 vectors; all rows preserved; the Grand Lock binds current receipts and evidence without inventing downstream measured values or closing lawful extensions
- **Sources and custody:** QUANTUM-RECONCILIATION-V11, ALL-CURRENT-QUANTUM-FAMILY-AUDITS, ALL-CURRENT-QUANTUM-RECEIPTS, ALL-CURRENT-QUANTUM-ADVERSE-AND-HANDOFF-ROWS. External status: `empirically_tested_and_independently_replicated`; all rows preserved: `True`.
- **Prohibited imports and boundary:** no selected receipt, favorable-only vector or handwritten closure status; host 0 denotes absence only and is not a numerical-zero proof object; no negative irrational imaginary floating fitted or completed-infinite proof scalar; no omitted halt adverse absent unavailable unresolved or standing-prediction row; no physical value is invented at a formal handoff; no verifier engine or frozen census mutation.
- **Evidence identities:** derivation
`sha256:849f9660e162ce966602313abe404c43abe3cdb68fc459ef18a471601050af9c`; independent implementation `sha256:954e25b97cce44a272e3aad5929024c3936ab783fdc89b9830f854630e0c774a`; independent certificate
`sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17`; measurement `sha256:92d9136e4add66879f3b4c354978dc9dc221b0bea75c0bfb2d71944bef440d49`; engine receipt `sha256:bfe8847f8595aaef4dac29e4cd82f42c0b1f9511c14bab612b995650a67b331d` at receipts/engine/model_admitted/SFT-QUANTUM-VALID-LEARNING-009-bfe8847f8595aaef.json.

SFT-QUANTUM-OBL-VALID-010 - Classical-quantum correspondence and limits validation vector

- **Claim and ownership:** SFT-QUANTUM-VALID-LIMITS-010; Quantum Computation family VALID; Classical-quantum correspondence and limits validation vector.
- **Forced law:** The limits vector reproduces every QLIMITX correspondence, separation, impossibility, handoff, scope boundary, receipt and control.
- **Unique surviving structure:** `complete-frozen-family-coverage__complete-limits-family-receipt-vector__current-reproduced-engine-receipt__complete-control-observation-independent-certificate__literal-complete-product__there-is-no-nothing-lineage__all-status-and-ownership-rows-preserved__dated-lock-explicit-handoffs-open-extension`. Exact result: VALID-010 uniquely retains complete-limits-family-receipt-vector, current receipt replay, full adverse custody and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is `depth_independent`. Falsification condition: Reject if the value-free registry, complete current receipt/evidence replay, adverse/ownership vector, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLIMITX-COMPLETENESS-022 -> SFT-QUANTUM-VALID-LEARNING-009. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** VALID-010: `qlimitx_vector`; exact Quantum family validation replay: `{"all_rows_replayed":true,"controls_and_certificates_present":true,"family":"QLIMITX","receipt_rows":22}`; complete formal/empirical/adverse custody: 12 vectors; all rows preserved; the Grand Lock binds current receipts and evidence without inventing downstream measured values or closing lawful extensions

- **Sources and custody:** QUANTUM-RECONCILIATION-V11, ALL-CURRENT-QUANTUM-FAMILY-AUDITS, ALL-CURRENT-QUANTUM-RECEIPTS, ALL-CURRENT-QUANTUM-ADVERSE-AND-HANDOFF-ROWS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no selected receipt, favorable-only vector or handwritten closure status; host 0 denotes absence only and is not a numerical-zero proof object; no negative irrational imaginary floating fitted or completed-infinite proof scalar; no omitted halt adverse absent unavailable unresolved or standing-prediction row; no physical value is invented at a formal handoff; no verifier engine or frozen census mutation.
- **Evidence identities:** derivation
sha256:5c5cf97088d2ec8fe3e27bc0a1ac5874e3c4a409299a5b24b66c67306135e3f0; independent implementation sha256:954e25b97cce44a272e3aad5929024c3936ab783fdc89b9830f854630e0c774a; independent certificate
sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17; measurement sha256:2195cc725dc19e6b6a8238fbb504a582ecf6d18f348babc3f9929980892c2ce0; engine receipt sha256:fd44600f7c180ac9e7e816cb8588d49f939c6003db44f911ba0f83b1240a6821 at receipts/engine/model_admitted/SFT-QUANTUM-VALID-LIMITS-010-fd44600f7c180ac9.json.

SFT-QUANTUM-OBL-VALID-011 - Adverse, absent, unresolved and ownership-boundary vector

- **Claim and ownership:** SFT-QUANTUM-VALID-ADVERSE-OWNERSHIP-011; Quantum Computation family VALID; Adverse, absent, unresolved and ownership-boundary vector.
- **Forced law:** The adverse vector preserves every favorable, adverse, absent, unavailable, unresolved, standing-prediction and one-owner boundary row across the branch.
- **Unique surviving structure:** complete-frozen-family-coverage__complete-adverse-absent-unresolved-ownership-ledger__current-reproduced-engine-receipt__complete-control-observation-independent-certificate__literal-complete-product__there-is-no-nothing-lineage__all-status-and-ownership-rows-preserved__dated-lock-explicit-handoffs-open-extension. Exact result: VALID-011 uniquely retains complete-adverse-absent-unresolved-ownership-ledger, current receipt replay, full adverse custody and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete current receipt/evidence replay, adverse/ownership vector, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLIMITX-COMPLETENESS-022 -> SFT-QUANTUM-VALID-LIMITS-010. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** VALID-011: adverse_ownership_vector; exact Quantum family validation replay: {"all_status_classes_preserved":true,"completed_families":11,"one_owner_boundaries_preserved":true}; complete formal/empirical/adverse custody: 12 vectors; all rows preserved; the Grand Lock binds current receipts and evidence without inventing downstream measured values or closing lawful extensions
- **Sources and custody:** QUANTUM-RECONCILIATION-V11, ALL-CURRENT-QUANTUM-FAMILY-AUDITS, ALL-CURRENT-QUANTUM-RECEIPTS, ALL-CURRENT-QUANTUM-ADVERSE-AND-HANDOFF-ROWS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no selected receipt, favorable-only vector or handwritten closure status; host 0 denotes absence only and is not a numerical-zero proof object; no negative irrational imaginary floating fitted or completed-infinite proof scalar; no omitted halt adverse absent unavailable unresolved or standing-prediction row; no physical value is invented at a formal handoff; no verifier engine or frozen census mutation.
- **Evidence identities:** derivation
sha256:52fc8c5e70bc35b2a6b45205b857a4e0176839e168c3311bfbf6e8daef870c81; independent implementation sha256:954e25b97cce44a272e3aad5929024c3936ab783fdc89b9830f854630e0c774a; independent certificate
sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17; measurement sha256:23a75cc112bb02c0b81ad12c94e43e73dbea35213a70cde55a89083a5500d0f7; engine receipt

sha256:add981310a5a650816faf2bb9f22471dece1e8ab95e6ecf575e2ee3504077626 at receipts/engine/model_admitted/SFT-QUANTUM-VALID-ADVERSE-OWNERSHIP-011-add981310a5a6508.json.

SFT-QUANTUM-OBL-VALID-012 - Reversible and Quantum Computation empirical and formal Grand Lock

- **Claim and ownership:** SFT-QUANTUM-VALID-GRAND-LOCK-012; Quantum Computation family VALID; Reversible and Quantum Computation empirical and formal Grand Lock.
- **Forced law:** The Quantum Grand Lock binds the frozen census, root lineage, all pre-lock receipts, family vectors, observations, controls, independent certificates and adverse/ownership rows as one acyclic identity graph.
- **Unique surviving structure:** complete-frozen-family-coverage__complete-quantum-formal-empirical-identity-graph__current-reproduced-engine-receipt__complete-control-observation-independent-certificate__literal-complete-product__there-is-no-nothing-lineage__all-status-and-ownership-rows-preserved__dated-lock-explicit-handoffs-open-extension. Exact result: VALID-012 uniquely retains complete-quantum-formal-empirical-identity-graph, current receipt replay, full adverse custody and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, complete current receipt/evidence replay, adverse/ownership vector, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-QLIMITX-COMPLETENESS-022 -> SFT-QUANTUM-VALID-ADVERSE-OWNERSHIP-011. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** VALID-012: quantum_grand_lock; exact Quantum family validation replay: {"acyclic_root_lineage":true,"frozen_obligations":288,"pre_lock_families":11,"pre_lock_receipt_rows":270,"protected_authority_changed":false}; complete formal/empirical/adverse custody: 12 vectors; all rows preserved; the Grand Lock binds current receipts and evidence without inventing downstream measured values or closing lawful extensions
- **Sources and custody:** QUANTUM-RECONCILIATION-V11, ALL-CURRENT-QUANTUM-FAMILY-AUDITS, ALL-CURRENT-QUANTUM-RECEIPTS, ALL-CURRENT-QUANTUM-ADVERSE-AND-HANDOFF-ROWS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no selected receipt, favorable-only vector or handwritten closure status; host 0 denotes absence only and is not a numerical-zero proof object; no negative irrational imaginary floating fitted or completed-infinite proof scalar; no omitted halt adverse absent unavailable unresolved or standing-prediction row; no physical value is invented at a formal handoff; no verifier engine or frozen census mutation.
- **Evidence identities:** derivation
 sha256:97bb1929bb65e9c0b13ffdd849a5504c14561629771c57ab6b65da74ee74431a; independent implementation sha256:954e25b97cce44a272e3aad5929024c3936ab783fdc89b9830f854630e0c774a; independent certificate
 sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17; measurement sha256:97c3a30cb363e2236ecda461adb2cddc9b8ca9f5ba5d5bcf46e80573a9bb8d98; engine receipt sha256:f6ad2363ac3de7e83a60b977ed24642bc1cc68d5f9b535838f8e6aab2a087f37 at receipts/engine/model_admitted/SFT-QUANTUM-VALID-GRAND-LOCK-012-f6ad2363ac3de7e8.json.

36.15 Family HAND - 6/6 complete

Scope. Single-owner downstream handoffs and open extension. This family decides 1,536 candidates, retains 6 unique survivors and passes 24 adverse controls.

SFT-QUANTUM-OBL-HAND-001 - Quantum Computation one-owner downstream handoff

- **Claim and ownership:** SFT-QUANTUM-HAND-DOWNSTREAM-OWNERSHIP-001; Quantum Computation family HAND; Quantum Computation one-owner downstream handoff.
- **Forced law:** Every Quantum result has one scientific owner and explicit consumer interfaces; downstream use cannot duplicate ownership or retroactively select the law.

- **Unique surviving structure:** single-registered-owner__single-owner-consumer-interface-ledger__complete-formal-result-resource-and-boundary-payload__explicit-owning-domain-measurement__literal-complete-product__there-is-no-nothing-lineage__all-status-rows-preserved__dated-closure-open-lawful-extension. Exact result: HAND-001 uniquely retains single-owner-consumer-interface-ledger, single ownership, complete payload and open extension.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, single-owner handoff grammar, complete interface payload, status custody, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-VALID-GRAND-LOCK-012. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** HAND-001: downstream_ownership; exact Quantum ownership/handoff execution: {"consumer_interfaces":4,"duplicate_owners":0,"scientific_owner":"quantum_computation"}; complete owner, consumer and extension custody: 6 records; all rows preserved; downstream sciences and engineering own their measurements and implementations; no handoff invents a value or selects the source law
- **Sources and custody:** QUANTUM-RECONCILIATION-V12, QUANTUM-GRAND-LOCK-012, CROSS-BRANCH-OWNERSHIP-MAP, PHYSICS-CHEMISTRY-MATERIALS-ENGINEERING-EXTENSION-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no duplicated owner or application-selected law; host 0 denotes absence only; no negative irrational imaginary floating fitted or completed-infinite proof scalar; no invented physical chemistry materials or engineering value; no closed extension gate or suppressed adverse handoff; no protected authority mutation.
- **Evidence identities:** derivation
sha256:f536284663ab416eea3738798807d4fe4cd543b1874317ec86377dc3e6249ef6; independent implementation sha256:4758092639c0a90a8c9e0b79c26af9527ed3632765fb66773bbf15683de5a6a4; independent certificate
sha256:14aa427629e75135c5645f40e2f50246fa9ff6641dc466de204baa168496891f; measurement sha256:99d56307ae45f1e383edc983610dd030a5dc6093a5eb9e5b5a7d14448552a1f5; engine receipt sha256:4cb88c6a40b9b7958701d787327752b3d6bd6b5e11abe04d68ca38108afec28f at receipts/engine/model_admitted/SFT-QUANTUM-HAND-DOWNSTREAM-OWNERSHIP-001-4cb88c6a40b9b795.json.

SFT-QUANTUM-OBL-HAND-002 - Quantum physical-measurement handoff

- **Claim and ownership:** SFT-QUANTUM-HAND-PHYSICS-MEASUREMENT-002; Quantum Computation family HAND; Quantum physical-measurement handoff.
- **Forced law:** Physics receives sealed state, process and resource identities and owns preparation, units, uncertainty, apparatus, timing, energy, fidelity and threshold measurements.
- **Unique surviving structure:** single-registered-owner__sealed-formal-result-to-Physics-measurement-interface__complete-formal-result-resource-and-boundary-payload__explicit-owning-domain-measurement__literal-complete-product__there-is-no-nothing-lineage__all-status-rows-preserved__dated-closure-open-lawful-extension. Exact result: HAND-002 uniquely retains sealed-formal-result-to-Physics-measurement-interface, single ownership, complete payload and open extension.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, single-owner handoff grammar, complete interface payload, status custody, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-VALID-GRAND-LOCK-012 -> SFT-QUANTUM-HAND-DOWNSTREAM-OWNERSHIP-001. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Observation:** HAND-002: physics_measurement; exact Quantum ownership/handoff execution: {"formal_payload_sealed":true,"invented_values":0,"measurement_owner":"physics"}; complete owner, consumer and extension custody: 6 records; all rows preserved; downstream sciences and engineering own their measurements and implementations; no handoff invents a value or selects the source law
- **Sources and custody:** QUANTUM-RECONCILIATION-V12, QUANTUM-GRAND-LOCK-012, CROSS-BRANCH-OWNERSHIP-MAP, PHYSICS-CHEMISTRY-MATERIALS-ENGINEERING-EXTENSION-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no duplicated owner or application-selected law; host 0 denotes absence only; no negative irrational imaginary floating fitted or completed-infinite proof scalar; no invented physical chemistry materials or engineering value; no closed extension gate or suppressed adverse handoff; no protected authority mutation.
- **Evidence identities:** derivation
sha256:75a397a7f26222ee32df9299bb4c17ccaee76c2d7596aafa25a1b33b585294ce; independent implementation sha256:4758092639c0a90a8c9e0b79c26af9527ed3632765fb66773bbf15683de5a6a4; independent certificate
sha256:14aa427629e75135c5645f40e2f50246fa9ff6641dc466de204baa168496891f; measurement sha256:dcc53c0f4ef1416d11d16720257b8e79ccde961ccd64913c36a6cc757f1b54fb; engine receipt sha256:98289f64312bfb3bc2dec26aec199d4766d193723018ff50a0353f76a228ccc2 at receipts/engine/model_admitted/SFT-QUANTUM-HAND-PHYSICS-MEASUREMENT-002-98289f64312bfb3b.json.

SFT-QUANTUM-OBL-HAND-003 - Quantum chemistry and materials handoff

- **Claim and ownership:** SFT-QUANTUM-HAND-CHEMISTRY-MATERIALS-003; Quantum Computation family HAND; Quantum chemistry and materials handoff.
- **Forced law:** Chemistry and Materials consume admitted quantum simulators while owning composition, structure, target observables, measurements and domain-specific falsification.
- **Unique surviving structure:** single-registered-owner__sealed-simulator-to-chemistry-materials-interface__complete-formal-result-resource-and-boundary-payload__explicit-owning-domain-measurement__literal-complete-product__there-is-no-nothing-lineage__all-status-rows-preserved__dated-closure-open-lawful-extension. Exact result: HAND-003 uniquely retains sealed-simulator-to-chemistry-materials-interface, single ownership, complete payload and open extension.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, single-owner handoff grammar, complete interface payload, status custody, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-VALID-GRAND-LOCK-012 -> SFT-QUANTUM-HAND-PHYSICS-MEASUREMENT-002. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** HAND-003: chemistry_materials; exact Quantum ownership/handoff execution: {"application_selects_law":false,"consumers":["chemistry","materials"]}; complete owner, consumer and extension custody: 6 records; all rows preserved; downstream sciences and engineering own their measurements and implementations; no handoff invents a value or selects the source law
- **Sources and custody:** QUANTUM-RECONCILIATION-V12, QUANTUM-GRAND-LOCK-012, CROSS-BRANCH-OWNERSHIP-MAP, PHYSICS-CHEMISTRY-MATERIALS-ENGINEERING-EXTENSION-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no duplicated owner or application-selected law; host 0 denotes absence only; no negative irrational imaginary floating fitted or completed-infinite proof scalar; no invented physical chemistry materials or engineering value; no closed extension gate or suppressed adverse handoff; no protected authority mutation.

- **Evidence identities:** derivation

sha256:ac2576aa75bc63d9092afc6068edcaee6084c94b98bcd522e0a04f57aef59990; independent implementation sha256:4758092639c0a90a8c9e0b79c26af9527ed3632765fb66773bbf15683de5a6a4; independent certificate
sha256:14aa427629e75135c5645f40e2f50246fa9ff6641dc466de204baa168496891f; measurement sha256:03afcd2f15fb52793bb9be37cb2d581cbc6d3218b0703c779874979ddce6124c; engine receipt sha256:10444146c044f15ef45a1cce089151fc0ec6963602b277b8d3d14fd4258c8c38 at receipts/engine/model_admitted/SFT-QUANTUM-HAND-CHEMISTRY-MATERIALS-003-10444146c044f15e.json.

SFT-QUANTUM-OBL-HAND-004 - Quantum software and hardware engineering handoff

- **Claim and ownership:** SFT-QUANTUM-HAND-SOFTWARE-HARDWARE-004; Quantum Computation family HAND; Quantum software and hardware engineering handoff.
- **Forced law:** Engineering consumes formal machines, circuits, codes and resource ledgers while owning implementation architecture, reliability, control, performance and safety.
- **Unique surviving structure:** single-registered-owner__formal-machine-to-engineering-realization-interface__complete-formal-result-resource-and-boundary-payload__explicit-owning-domain-measurement__literal-complete-product__there-is-no-nothing-lineage__all-status-rows-preserved__dated-closure-open-lawful-extension. Exact result: HAND-004 uniquely retains formal-machine-to-engineering-realization-interface, single ownership, complete payload and open extension.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, single-owner handoff grammar, complete interface payload, status custody, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-VALID-GRAND-LOCK-012 -> SFT-QUANTUM-HAND-CHEMISTRY-MATERIALS-003. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** HAND-004: software_hardware; exact Quantum ownership/handoff execution: {"consumers":["software-engineering","hardware-engineering"],"implementation_values_present":false}; complete owner, consumer and extension custody: 6 records; all rows preserved; downstream sciences and engineering own their measurements and implementations; no handoff invents a value or selects the source law
- **Sources and custody:** QUANTUM-RECONCILIATION-V12, QUANTUM-GRAND-LOCK-012, CROSS-BRANCH-OWNERSHIP-MAP, PHYSICS-CHEMISTRY-MATERIALS-ENGINEERING-EXTENSION-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no duplicated owner or application-selected law; host 0 denotes absence only; no negative irrational imaginary floating fitted or completed-infinite proof scalar; no invented physical chemistry materials or engineering value; no closed extension gate or suppressed adverse handoff; no protected authority mutation.
- **Evidence identities:** derivation
sha256:1232254b0e26146ba0eb0da333f0964359de988fdb7a3ef84278f21d13b2ed7a; independent implementation sha256:4758092639c0a90a8c9e0b79c26af9527ed3632765fb66773bbf15683de5a6a4; independent certificate
sha256:14aa427629e75135c5645f40e2f50246fa9ff6641dc466de204baa168496891f; measurement sha256:588d9c3deb5895a2d922721ea75e75af6641a1334bccd045cee6dfe1200750f8; engine receipt sha256:8cf3f413acff88600c32dc5c26639b0e888da2349158a1aead746377d15b0258 at receipts/engine/model_admitted/SFT-QUANTUM-HAND-SOFTWARE-HARDWARE-004-8cf3f413acff8860.json.

SFT-QUANTUM-OBL-HAND-005 - Quantum Computation open-extension handoff

- **Claim and ownership:** SFT-QUANTUM-HAND-OPEN-EXTENSION-005; Quantum Computation family HAND; Quantum Computation open-extension handoff.

- **Forced law:** The branch remains open to new discoveries only through a value-free obligation, dependency ownership, complete enumeration, controls, independent reconstruction and post-registry evidence.
- **Unique surviving structure:** single-registered-owner__dated-closure-new-obligation-admission-interface__complete-formal-result-resource-and-boundary-payload__explicit-owning-domain-measurement__literal-complete-product__there-is-no-nothing-lineage__all-status-rows-preserved__dated-closure-open-lawful-extension. Exact result: HAND-005 uniquely retains dated-closure-new-obligation-admission-interface, single ownership, complete payload and open extension.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, single-owner handoff grammar, complete interface payload, status custody, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-VALID-GRAND-LOCK-012 -> SFT-QUANTUM-HAND-SOFTWARE-HARDWARE-004. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Observation:** HAND-005: open_extension; exact Quantum ownership/handoff execution: {"dated_closure":true,"lawful_extension_gate_open":true,"standards_unchanged":true}; complete owner, consumer and extension custody: 6 records; all rows preserved; downstream sciences and engineering own their measurements and implementations; no handoff invents a value or selects the source law
- **Sources and custody:** QUANTUM-RECONCILIATION-V12, QUANTUM-GRAND-LOCK-012, CROSS-BRANCH-OWNERSHIP-MAP, PHYSICS-CHEMISTRY-MATERIALS-ENGINEERING-EXTENSION-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no duplicated owner or application-selected law; host 0 denotes absence only; no negative irrational imaginary floating fitted or completed-infinite proof scalar; no invented physical chemistry materials or engineering value; no closed extension gate or suppressed adverse handoff; no protected authority mutation.
- **Evidence identities:** derivation
sha256:742d42483fc9ff46b52f77cb9214176b5719a35d547acbb79a5a09852af51731; independent implementation sha256:4758092639c0a90a8c9e0b79c26af9527ed3632765fb66773bbf15683de5a6a4; independent certificate
sha256:14aa427629e75135c5645f40e2f50246fa9ff6641dc466de204baa168496891f; measurement sha256:2e172278731b4c6c8435b081cd92e2dc25a2da28878efab8e2f4aa121b9d9771; engine receipt sha256:7617d7cd2287fbe3eb9e0e724b6a3eb70934e995878357a0a14792524ea52b7c at receipts/engine/model_admitted/SFT-QUANTUM-HAND-OPEN-EXTENSION-005-7617d7cd2287fbe3.json.

SFT-QUANTUM-OBL-HAND-006 - Quantum Computation cross-branch completeness certificate

- **Claim and ownership:** SFT-QUANTUM-HAND-CROSS-BRANCH-COMPLETENESS-006; Quantum Computation family HAND; Quantum Computation cross-branch completeness certificate.
- **Forced law:** Cross-branch completion requires all six frozen handoffs, one owner per result, no invented downstream value, current receipts and explicit open-extension custody.
- **Unique surviving structure:** single-registered-owner__six-handoff-no-omission-ledger__complete-formal-result-resource-and-boundary-payload__explicit-owning-domain-measurement__literal-complete-product__there-is-no-nothing-lineage__all-status-rows-preserved__dated-closure-open-lawful-extension. Exact result: HAND-006 uniquely retains six-handoff-no-omission-ledger, single ownership, complete payload and open extension.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 adverse controls passed; closure is depth_independent. Falsification condition: Reject if the value-free registry, single-owner handoff grammar, complete interface payload, status custody, source identity, candidate survivor, seal, control or independent reconstruction is missing, changed, duplicated or opened out of order.
- **Dependency and root trace:** SFT-QUANTUM-VALID-GRAND-LOCK-012 -> SFT-QUANTUM-HAND-OPEN-EXTENSION-005. Registered root: SFT-ROOT-THERE-IS-NO-NOTHING. Imported

axioms: 0; free parameters: 0.

- **Observation:** HAND-006: cross_branch_completeness; exact Quantum ownership/handoff execution: {"duplicate_owners":0,"execution_rows":6,"omitted_handoffs":0,"registered_handoffs":6}; complete owner, consumer and extension custody: 6 records; all rows preserved; downstream sciences and engineering own their measurements and implementations; no handoff invents a value or selects the source law
- **Sources and custody:** QUANTUM-RECONCILIATION-V12, QUANTUM-GRAND-LOCK-012, CROSS-BRANCH-OWNERSHIP-MAP, PHYSICS-CHEMISTRY-MATERIALS-ENGINEERING-EXTENSION-HANDOFFS. External status: empirically_tested_and_independently_replicated; all rows preserved: True.
- **Prohibited imports and boundary:** no duplicated owner or application-selected law; host 0 denotes absence only; no negative irrational imaginary floating fitted or completed-infinite proof scalar; no invented physical chemistry materials or engineering value; no closed extension gate or suppressed adverse handoff; no protected authority mutation.
- **Evidence identities:** derivation
 sha256:e61eb09b8421fa367b82e13c03f8c99e927b6fd24269c4c5b05df2e031f22350; independent implementation sha256:4758092639c0a90a8c9e0b79c26af9527ed3632765fb66773bbf15683de5a6a4; independent certificate
 sha256:14aa427629e75135c5645f40e2f50246fa9ff6641dc466de204baa168496891f; measurement sha256:85a9e123f4460e091beddbf312fdb7cd94cc249c65e720ce3ade1f320ca1027b; engine receipt sha256:ef8df1b6bb47eea3ca46acd3458598c247270d8628de0d964b43b104af03c8d6 at receipts/engine/model_admitted/SFT-QUANTUM-HAND-CROSS-BRANCH-COMPLETENESS-006-ef8df1b6bb47eea3.json.

37. Empirical constitution: what empirical means in this branch

Empirical means grounded in observation or experience rather than accepted solely by theory or authority. In formal quantum computation the directly observable objects are generated supports, machine states, transformations, phase labels, predecessor merges, joint cells, measurement records, error masks, traces, resources, outputs, tamper responses and independent executions. Each target identity is registered before its outcome is opened; all favourable, adverse, absent and unresolved rows are preserved. Where a claim concerns physical hardware or nature, this branch records the handoff instead of pretending that a formal execution measured a device.

The branch admits no result-only oracle. A result requires its initial support, generated description, full transition and phase trace, dependencies, resource ledger, controls, source identities, independent reconstruction and engine receipt. Apparent randomization is complete deterministic schedule support plus observation-relative uncertainty. This permits probabilistic and quantum correspondence without introducing an uncaused event into a superdeterministic model.

38. Preserved halt, lawful retry and engine integrity

The first official SFT-QUANTUM-VALID-REVERSIBLE-001 submission halted because a sealed prediction label and an already-frozen vector label did not match. The rejected receipt remains preserved as sha256:bc2e0eb42e8650c848de9872a8666fc879d84aec5c21201b378721f00fd8a62e. It was not counted as a success, hidden or converted into an admission. A separately versioned retry corrected only the protocol label binding and then passed the unchanged acceptance route. The candidate grammar, registry, observation vector, scientific law, engine, verifier and acceptance condition were not modified.

Focused Quantum tests pass 40/40 and lightweight repository validation records 2,751 admitted V3 claims at this checkpoint. The canonical engine seal remains sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a and the verification-authority seal remains sha256:bff810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8. The heavy repository-wide verification is deliberately reserved for the final all-branch Grand Lock and is not claimed here.

39. Dated completion, falsification and extension

The branch is complete to frozen census

sha256:a7f855b7e9fc6ffe4ae9485faf26f1be44bd672b843ecbfbad42b0f523726e3c and reconciliation sha256:ac4acee07a49cbde19eee7baacd7be7d8c87c93e4d75a47467160fb31499e650: 288 closed, structural absence of an open frozen row, and no omitted census member. A reviewer may invalidate a claim with an omitted same-boundary coordinate, a second survivor, an induction counterexample, a failed inverse, an incomplete observation record, an uncorrected registered error mask, a source mismatch, an unreproducible reconstruction or a broken receipt. A lawful discovery enters as a new versioned obligation without rewriting prior evidence.

The twelve VALID claims are the dated Quantum Computation validation Grand Lock: a complete reconciliation and adverse-evidence layer. The six HAND claims then assign physical measurement, chemistry, materials, software/hardware engineering and later applications exactly once. Grand Lock does not mean permanent doctrinal closure. Open criticism remains unrestricted; scientific admission requires the public protocol.

40. Complete-field conclusion

Reversible and Quantum Computation is complete to its frozen 29 July 2026 census: **288/288 obligations, 13/13 families, 73,728 candidate decisions, 288 unique survivors, 1,152 passed adverse controls, 288 independent reconstructions and no open frozen obligation.**

The contribution is one unified computational object. Reversible state, quantum support, phase, interference, entanglement, measurement, gate, circuit, algorithm, code, simulator, verifier and learner are successive exact organisations of retained Fold distinctions. Historical formalisms are comparison boundaries after derivation, not premises before it. The branch also states its limits with equal force: no-cloning, disturbance, halting, undecidability, incompleteness and absent hypercomputation survive the quantum extension.

The preserved SFT-QUANTUM-VALID-REVERSIBLE-001 halt remains part of the scientific history; the separately versioned protocol-label correction did not alter the law, target, grammar, engine, verifier or acceptance condition. The validation Grand Lock retains favourable, adverse, absent and unresolved rows without treating formal execution as a physical-device measurement. Six terminal handoffs keep hardware, Chemistry, Materials and later applications with their owning branches. The branch is complete only to this frozen census and remains open to lawful correction, falsification and extension.

The public contribution is the evidence route itself. Maria Smith's lack of credentialed access is not evidence for a theorem; it is evidence of what credential and capital gates can exclude. The open corpus invites unrestricted criticism, reuse and lawful extension while reserving the Ernos Labs designation for work that preserves full derivation chains, adverse evidence, unchanged admission routes, retained authorship and transparent community review.

Data and code availability

The open repository is `'MettaMazza/ernos-labs-sft-platform'`. The live model-admitted index is `census/claims.json`; the Reversible and Quantum Computation field boundary is `census/quantum_computation_discipline_obligations.json`. Every represented claim has its own package under `claims/<claim-id>/` and its immutable model-admitted receipt under `receipts/engine/model_admitted/`.

The publication evidence map is

`publications/successors/quantum_computation/evidence_map_v1_4.json` and the candidate manifest is `publications/successors/quantum_computation/manifest_v1_4.json`. Registered external records and source captures are retained under `experiments/external_sources/quantum_computation/` together with their source identities, transport outcomes and hashes. Files too large for ordinary Git remain checksum-bound and are distributed through the applicable existing Zenodo record rather than omitted or replaced.

Build, render, replay and verification programs are retained under `tools/`, with branch implementations under `sft/`, independent reconstructions under `generated/` and tests under `tests/`. The final approved Markdown, rendered PDF, evidence map, manifest, metadata and checksums must agree before deposit. No successful test is treated as empirical confirmation unless the claim-specific evidence protocol separately authorises that status.

Shared claim-record clauses

The clauses below previously appeared verbatim in multiple claim records. They are preserved once here to reduce mechanical repetition. Each in-text clause reference applies this exact wording at that claim location; no scientific statement, status, condition or qualification has been removed.

QUANT-S001 - applied at 22 claim locations

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

QUANT-S002 - applied at 22 claim locations

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

QUANT-S003 - applied at 21 claim locations

The unique all-preserving member is complete-generated-branch-support__one-Fold-distinction-unit__complete-pair-cell-joint-support__period-held-phase-action__total-reversible-source-bound-map__retained-class-and-complete-record__branch-and-depth-successor-certificat
e__no-extra-quantum-premise.

QUANT-S004 - applied at 22 claim locations

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

QUANT-S005 - applied at 22 claim locations

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

QUANT-S006 - applied at 21 claim locations

Successor: Adding one generated branch, component, fault position or circuit location appends all new joint cells, phase rows, transformations and observation distinctions while retaining prior records.

QUANT-S007 - applied at 22 claim locations

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

References

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Complete current-census successor supplement

The authoritative current scope of this paper is 290 live model-admitted claims (quantum_computation 290). Counts retained inside the preceding-version narrative describe the dated freeze at which those passages were written; this successor table and the machine census control the current total.

The preceding manuscript did not contain 2 later-admitted claim section(s). They are included below in dependency order so the successor covers the complete current branch surface.

SFT reconstruction of distribution-uniform quantum parallel repetition

Claim ID: SFT-QUANTUM-OAI26-PARALLEL-REPETITION-001

Exact statement: There exists one positive exact-real name c such that for every complete finite generated question/answer carrier and every admitted encoded two-player game with nonempty answer carriers and positive exact entangled-value gap, every positive generated repetition count satisfies the complete source exponential inequality under the exact strategy-value and logarithm/exponential enclosure translations.

Dependency route: SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-QCPLXX-PARALLEL-016 -> SFT-QUANTUM-QCPLXX-REDUCTION-018 -> SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006

Candidate generator: Compose, without an outcome axis, the admitted candidate generators of: finite game carriers x complete strategy supports x exact entangled-value enclosures x repetition successor traces x parallel-resource certificates. Generate theorem-specific witnesses, arbitrary-input proof steps, eliminations and certificates.

Grammar boundary: Every SFT-admissible encoding of every source binder and every original hypothesis/conjunct. Universal natural scope requires base-and-successor closure; existential scope requires the complete registered witness grammar; a rejected carrier is not a mathematical negation.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: exact-frozen-source__exact-quantifier-conjunct-order__exact-native-correspondence__preexisting-branch-grammar-composition__complete-prior-receipt-trace__complete-root-to-result-chain__complete-executable-certificates__sft-root-bound-forward-derivation

Exact result: PROVED: There exists one positive exact-real name c such that for every complete finite generated question/answer carrier and every admitted encoded two-player game with nonempty answer carriers and positive exact entangled-value gap, every positive generated repetition count satisfies the complete source exponential inequality under the exact strategy-value and logarithm/exponential enclosure translations.

Closure: depth_independent; minimality True; named-shape uniqueness True

External status: independently_replicated

Engine receipt: sha256:f0dbd3cd8484f63cb2a4a453583003df3a9271257eb6489214ffb0e071ac5dfc at receipts/engine/model_admitted/SFT-QUANTUM-OAI26-PARALLEL-REPETITION-001-f0dbd3cd8484f63c.json

Control	Passed	Expected	Observed
false_promise	True	reject a changed source target	A changed or paraphrased target is not the registered proposition.
tampered_source	True	reject a changed source identity	changed source identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional survivor	the complete route product contains exactly one all-admitted derivation
boundary	True	reject carrier denial finite examples imported proof authority or incomplete chains	exact correspondence, full proof graph, arbitrary-input closure and executable certificates all passed

SFT disproof of source-artifact validity: QuantumParallelRepetition.distributionUniformExponential

Claim ID: SFT-QUANTUM-OAI26-PARALLEL-REPETITION-VALIDITY-001

Exact statement: Not SFTValid(exact frozen artifact QuantumParallelRepetition.distributionUniformExponential): assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector [propext, Classical.choice, Quot.sound] and requires real suprema over complex density-matrix and POVM strategies followed by a completed exponential bound.

Dependency route: SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-FOUNDATION-EXACT-OPERATIONS-001 -> SFT-QUANTUM-ENTANGLEMENT-001 -> SFT-QUANTUM-QCPLXX-PARALLEL-016 -> SFT-QUANTUM-QCPLXX-REDUCTION-018

Candidate generator: Compose exact source custody, exact quotation, exposed axiom/carrier evidence, pre-existing admission laws, a complete contradiction chain, executable checks and a strict native/source nontransfer boundary; no verdict coordinate.

Grammar boundary: SFTValid(exact frozen artifact QuantumParallelRepetition.distributionUniformExponential)

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: exact-frozen-artifact__exact-statement-and-quantifier-quotation__nonempty-source-axiom-vector-exposed__necessary-source-component-exposed__preexisting-admission-and-domain-laws__assume-valid-and-derive-actual-contradiction__complete-executable-evidence__native-reconstruction-kept-distinct

Exact result: DISPROVED: SFTValid(exact frozen artifact QuantumParallelRepetition.distributionUniformExponential). SFT validity forces an empty axiom vector and admitted carriers; the exact source exposes three axioms and the theorem-specific excluded component. The assumption therefore entails a contradiction.

Closure: depth_independent; minimality True; named-shape uniqueness True

External status: independently_replicated

Engine receipt: sha256:141430b51bc2ddc8e83656de2d0c45b0afeb3020aa1c30842b39316856b3ed05 at receipts/engine/model_admitted/SFT-QUANTUM-OAI26-PARALLEL-REPETITION-VALIDITY-001-141430b51bc2ddc8.json

Control	Passed	Expected	Observed
false_promise	True	reject the false premise that the frozen source axiom vector is empty	the exact three-entry source vector is nonempty
tampered_source	True	reject any changed source identity	the changed identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional complete contradiction route	the complete route product has one all-evidence survivor
boundary	True	reject transfer from the SFT-native reconstruction to the external artifact	the contradiction proves only the exact source-validity negation and records the reconstruction as distinct

Preceding-version abstract preserved for chronology

The following abstract is reproduced from version 1.4.0. Its counts and status language describe that dated version and do not override the current abstract or census above.

This paper reports the complete-field Reversible and Quantum Computation branch of the third clean-room reconstruction of Smithian Fold Theory (SFT), complete to its frozen dated census and explicitly open to lawful extension. The census contains **288 obligations in 13 dependency-ordered families**. Every obligation has an untouched-engine model-admission receipt, a complete 256-member candidate census, exactly one survivor, four adverse controls, an explicit depth-independent or registered finite boundary, an implementation-distinct reconstruction and a post-registration observation record with preserved custody. Together the branch decides **73,728 generated candidates**, retains **288 unique survivors**, passes **1,152 adverse controls** and reconciles to sha256:ac4acee07a49cbde19eee7baacd7be7d8c87c93e4d75a47467160fb31499e650.

The reconstruction derives reversible computation; quantum information units; state preparation and composition; complete-word superposition-equivalent support; finite held phase and exact interference; nonfactorable entangling composition; measurement with retained records; transformations, gates, circuits, compilation and universality; thirty quantum-algorithm laws; twenty-six resource and complexity laws; twenty-four channel and communication laws; thirty-two coding, error-correction and fault-tolerance laws; twenty-four simulation and verification laws; twenty-two learning laws; and twenty-two classical-quantum correspondence and limit laws. The quantum and classical modes are two exact regimes of one Fold machine rather than unrelated formalisms.

No conventional quantum postulate, Hilbert-space axiom, complex amplitude, irrational normalization, imaginary proof value, stochastic-collapse premise, ontic randomness, fitted parameter, physical benchmark, pretrained model or authority label selects a law. Displayed 0 denotes structural absence rather than a numerical proof magnitude. Error correction forces width $2\tau+1$ for every supplied positive finite fault order τ ; quantum speedup, device thresholds, timing, energy and natural measurement distributions remain explicitly assigned to their empirical physical or engineering owners.

Version 1.4 preserves the full version 1.3 foundation, public scientific mission and 22 extensive foundational derivations, then documents all 266 executed full-field extensions claim by claim. No protected engine or verification-authority source was modified. A preserved first-attempt halt and separately versioned retry demonstrate that the engine remains allowed to reject the work.

Keywords: Smithian Fold Theory; complete-field quantum computation; reversible computation; quantum information; superposition; interference; entanglement; measurement; quantum circuits; quantum algorithms; quantum complexity; quantum communication; quantum coding; fault tolerance; quantum simulation; quantum learning; open science.

Scope and ownership boundary

This paper owns the `quantum_computation` surface comprising 290 current claims. Ownership identifies responsibility for derivation and falsification; it does not prevent criticism or lawful cross-branch use. Application performance, consensus, credentials and Lean acceptance cannot transfer ownership or select a law.

Registered source surface

Human-readable scholarly references remain in the inherited paper. Machine source IDs, custodians, releases, locators, source captures, access outcomes, hashes and transport status remain in the current claim packages and external-source registries. Source prestige is not evidential authority; each source is used only in its registered role. The Lean source-binding gate checks custody and identity, not empirical truth by reputation.

Successor machine-identity appendix

- Preceding source: publications/successors/quantum_computation/THE_QUANTUM_FOLD_MACHINE_PAPER_001_V1_4.md
- Preceding source SHA-256:
sha256:1ef0a0e6908d486d8573092f1378ae99f5dfce493093c859b456aa0a00825ca7
- Lean report: generated/lean4_validation/reports/whole_model_validation.json

- Lean report SHA-256:
sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf
- Ordered census SHA-256:
sha256:f3f540f77a9b677bf7ddb9f5c6ea1ea41f08e57bb07163e25f385fd4d1dcde71
- Execution manifest SHA-256:
sha256:517fd288f4484f43fdc0343b17fcabd06ea9e12c977a4512d4a891dd038fce41
- Protected engine seal:
sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a
- Verification-authority seal:
sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8
- Publication authorised: false
- Remote actions performed: false

Lean 4 independent verification update

Verified result

The pinned Lean toolchain `leanprover/lean4:v4.32.0` compiled the verification project and the executable verifier returned `PASS`. Its immutable report identity for this successor is
sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf.

Verified surface	Result
Ordered census claims	2,777
Proof-bearing accepted claim gates	2,777
Source-bound claims	2,777
Candidate records	898,902
Decision records	898,902
Passed controls	11,108
Registered branches	17
Issues	0

This paper's registered branch surface contributes **290** of the accepted claims.

Native theorem proof and artifact verification are different layers

`SFTValidation/Root.lean` formalises the exact registered two-member operational grammar: `unpresentedAbsence` and `presentedOccurrence`. The decision function rejects the former and accepts the latter. Lean proves existence by the accepted constructor and uniqueness by exhaustive case analysis over the two constructors. `#print axioms` reports no imported or user-declared axioms for the exported root theorems. This is a kernel-checked proof of the unique survivor inside the exact registered operational grammar.

`SFTValidation/Gates.lean` defines twelve Boolean acceptance obligations and can construct an inhabitant of `ClaimGate.Accepted gate` only when all twelve equal `true`. The runtime verifier then parses the complete ordered census and execution manifest, recomputes identities, cardinalities, one-to-one decision coverage, survivor count, controls, closure fields, empirical boundaries, certificate bindings and authoritative receipts, and calls that proof-bearing constructor for every current claim.

The exact scientific statements of the other claims are not all re-expressed as 2,776 separate Lean propositions in this version. They are validated as the repository's complete registered machine artifacts through Lean-executed checks. The source-manifest gate uses a read-only Python bridge solely to instantiate the already registered source factories; Lean consumes the result. This layer does not replace the protected admission engine, does not write receipts and does not substitute for source experiments or empirical replication.

Fail-closed custody event

The first whole-model run halted on six source-capture byte hashes. The discrepancy was traced to line-ending normalisation in the working tree, not to a changed scientific target or a failed theorem. The registered source bytes were restored, exact-byte Git attributes were added for those six captures, all 385 registered external source bindings were re-audited with no mismatch, and the unchanged Lean verification was rerun to `PASS`. The preserved halt demonstrates that source drift is detected rather than silently forgiven.

What the result supports

The result materially strengthens the model's case for formal coherence, exhaustive current-census coverage, unique-survivor enforcement, dependency and provenance integrity, cross-branch consistency and reproducibility by an implementation outside the admission engine. It does not by itself establish that every empirical claim is true in nature, that the registered grammar is the only conceivable grammar outside its stated boundary, or that SFT is superior to every rival theory. Those questions remain governed by the papers' declared experiments, falsification conditions and comparative tests.

Successor conclusion

Version 1.5.0 preserves the preceding paper's derivations, evidence classes, corrections, adverse and unresolved records, ownership limits and open frontier, while integrating the independent Lean 4 result and every later current claim in its declared scope. This paper contributes 290 current claim(s) within `quantum_computation` to the total dependency spine.

The new formal result establishes that the registered two-class operational root has one Lean-proved survivor and that all 2,777 current claims satisfy the proof-bearing artifact gates. The major conflation resolved by this update is between native theorem proof and whole-repository artifact verification: both are valuable, but they are not the same evidential act. No empirical status is strengthened merely because the artifact graph passes, and no historical halt or correction is erased.

The current evidence therefore establishes formal coherence, current-census completeness, unique-survivor enforcement and intact source, dependency, certificate and receipt custody for this version. Field-specific empirical conclusions remain exactly those authorised by their current records. Lawful discovery, falsification, stronger evidence and versioned extension remain open, and publication itself remains pending Maria Smith's explicit confirmation.